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Computer Studies Form 1

Chapter 1

INTRODUCTION TO COMPUTERS

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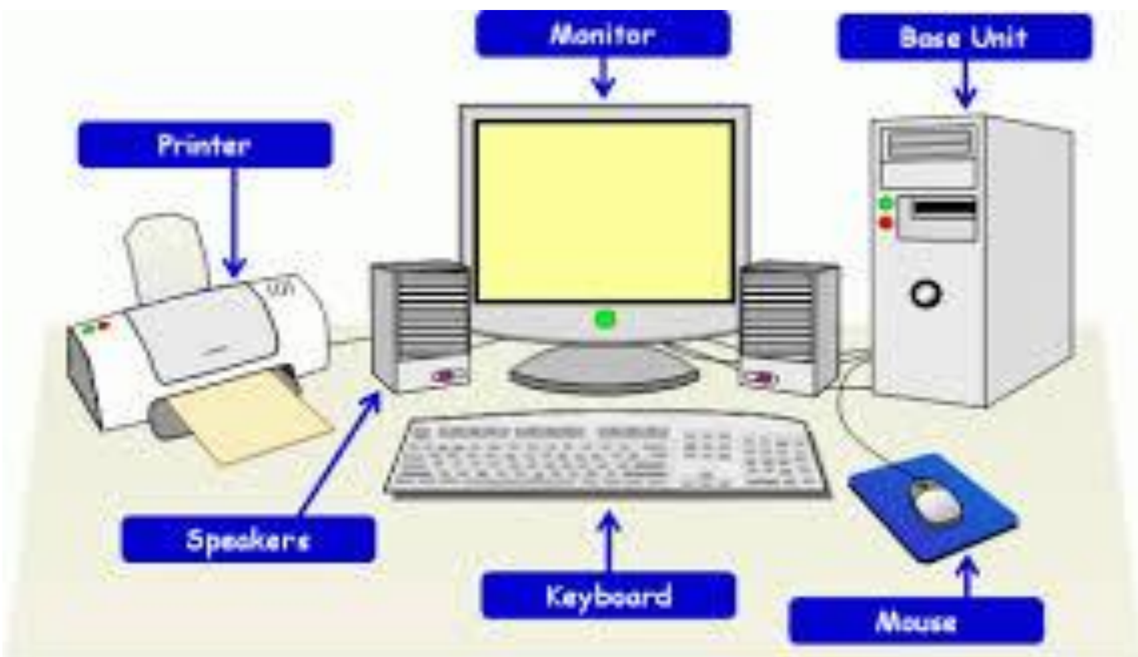
CHAPTER 1: INTRODUCTION TO COMPUTERS

Definition of key terms

1. **A computer** is an electronic device that process data to information. It is said an electronic because it utilizes electrical signals to process data and uses instructions called programs to complete data.

Input (Data) → Process (Data) → Output (Infomation)

Computers are designed in different sizes and designs but the most common is a computer referred to as **Personal Computers** (PC) which are mostly used in offices, schools, business premises and homes.



- A typical computer shown above is basically made up of a system unit and peripheral devices such as monitor, keyboard and a mouse.
- The system unit is the part that houses Unit namely **Tower** and **Desktop.**

2. **Data and Information:**

- Data is fact figures that do not have much meaning to the user and may include numbers, letter and symbols. Information is the product of data that is meaningful to the user.

3. **Information Technology (IT)**:

- It refers to the use of hardware, software and their technologies to collect, organize, process, secure, store exchange information. Such hardware includes Computer, PDAs, **Smart** Phones and Printers... information may be informs of text and graphics, sound or video.

4. **Communication Technology:**

- It refers to the use of devices and communication channels to transmit information correctly, effectively and cost effectively.
- Such devices include radio, transmitters and receivers, telephones, satellites, fax machines and communication channels include telephone lines and radio waves.

5. **Information and Communication Technology (ICT)**:

- It refers to the integration of communication technologies and information for the purpose of acquiring, processing, storing, standardizing and disseminating information for public consumption.

6. **Information System**:

- Refers to the set of components namely persons, procedures or hardware and software resources that collect, process and deliver information in a given organization.

7. **Garbage in Garbage out (GIGO):**

- This is a phrase which implies that if erroneous data is entered in a computer and command to process it is given, the computer will output erroneous results.
- It means a computer does not do things on its own users must enter data correctly.

8. Program

- Computer program is defined as set of instructions that tell computer to perform a certain task.
- It is also referred to computer software

History of computer development

Computers have undergone some evolution processes from non- electronic computing devices to electronic computing devices

Non-electronic computing devices.

Non- Electrical computing devices are tools that were used to perform arithmetic computations manually or Non-Electronic. These include **sticks, stones, abacus, bones etc.**

a) **Abacus:**

- It was invented by the Chines and it was meant for counting instrument dates back to 300 BC.
- The Abacus has bead – like parts that move along rods. Each bead above the middle bar stands foe five units while a bead below stands for one unit.

b) **Napier's bones:** Napier `Bones was developed by John Napier, a Scottish mathematics in the 17th Century. It was used for performing multiplication and division.

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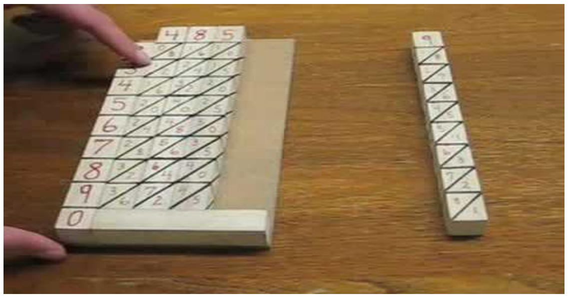
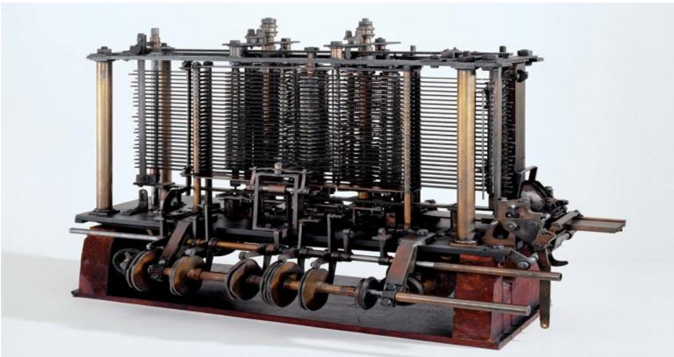


c) **La Pascale Machine: is** a counting machine that was made by Blaise Pascal in the 17^{th} Century. It was used for addition and Subtraction.

d) **The Analytic Engine:** The Analytic engine was designed by Charles Babbage in 1932 used for English and Mathematics. Die technological limitation, Babbage never implemented it. The engine is recognized as the first real computer and Babbage as the father of computing.

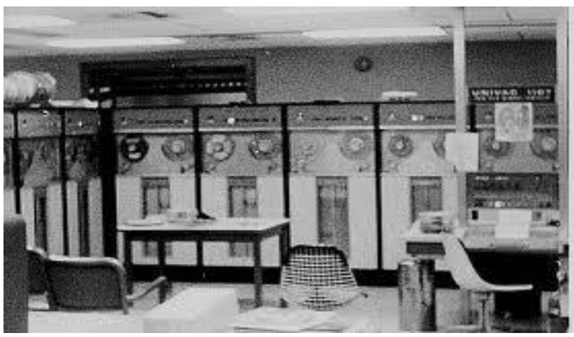


Generations of electronic computers

The age of modern Electronic Computers can be tracked back to 1940s. And they are classified into five generations.

a) **First Generation Computers (1940`s to 1958):**

- Characterised by the use of **vacuum tubes**
- it was very large physically, and
- generated a lot of heat hence constantly broke down.



b) **Second Generation Computers (1958 – 1964)**

- Characterised by the use of tiny solid state electronic devices called **transistors**
- Transistors were much smaller than the vacuum tubes.
- These computers produced less heat, were much faster, smaller in sizes and more reliable than made of vacuum tubes.
- Examples of second generation computers include IBM 1402 and 7070, **UNIVAC 1107, ATLAS LEO Mark iii** and **Honeywell 200.**

c) **Third generation computers (1964 – 1970):**

- It used devices called **integrated circuits (IC`s)**
- IC`s has thousands of small transistors circuits packed on semiconductor called a silicon chip.
- It emit less heat, were small in sizes, easier to program, **use** and program, and maintain compared to their predecessors.
- Examples of third generation include smaller and less expensive minicomputers such as IBM 360 and ICL 19000 series.

d) **Fourth Generation Computers (1970 to Present)**

- Characterised by the use of Microprocessors (Silicon chip)
- The silicon chip designed into very smaller size.

- are characterized by very low emission of heat, are small in size and easier to use and maintain.
- Example of fourth generation include IBM 370 and 4300, Honeywell DPS-88 and Burroughs 7700.

e) **Fifth Generation Computers:**

- 5th Generation computers are characterized by **Artificial Intelligence**, connectivity to internet, superior hardware and software.
- It will be able to work without human intervention and help managers to make decisions.
- Are computers coming up today, with very high processing power and speed than their predecessor.
- Very small in size but powerful.
- They work with special programs **Artificial Intelligence.**

Uses of first computers

Generation	Characterised by	Purpose
1^{ST} Generation Computers	**Built 1^{st} World War (WW1) using vacuum tubes**	The initial purpose of 1^{st} generation computers was computation of very large mathematical and scientific figures. ENIAC was developed during the 1^{st} ww1 to make certain calculation of a hydrogen bomb.
2^{nd} Generation Computation	Built using **transistors**. Had tape storage, printer and operating system and storage programs	**The 2^{nd} Gen**eration computers such as PDP 1 and IBM 1400 series were programmable computers that were mainly used for scientific, business application and compute games.
3^{rd} Generation Computers	Built in **integrated circuits** and semiconductors.	They used to process more than one task. They have more application programs such as word processors, calculators and business applications.
4^{th} Generation Computers	Built in very large integrated circuits characterized by **microcomputers**	They were affordable and used most application. Financial applications such as VisiCalc and networks

		particularly the internet became common
5th Generation Computers	Today`s Computers **characterized** by massive processing power and use of **Artificial Intelligence**	Most modern computers are used for large numbers of applications in particular expect system used in decision making.

Uses of computers

Computers are used in each and every day just because they are more efficient and accurate. Computers are used in some of these areas:

- Supermarket – are used for stock control.
- Offices – promote efficiency in offices by reducing the time and effort needed to access and receive information.
- Hospitals – are used in life support machines and keeping patient`s records.
- Transport – computers are used to monitor vehicle traffic in busy town and aircraft navigation and making reservation.
- Communication – computers has made easy sending of information very fast and efficient.
- Law Enforcement – Law enforces are able to carry out their criminal investigations through the use of machine which can trace finger prints, images and other identifications features.
- Bank – for processing cheques and withdrawing money through ATM`s
- Home – for internment and family budget
- School – analyzing academic data
- Industry – for manufacturing process control
- Police station - matching fingerprints
- Communication – instant messaging, e-mail services, video conferences.
- Education – e-learning, research, communication and records management.
- Domestic and entertainment – computers are used for recreational such as watching movies, playing music and computer games and storing personal information.

- Library Services – it enable library personnel to easily access and keep updated records of books and catalog processing.

Types of computers

Computers are categorized based on the following criteria

- Based on data processed or functionality
- Based on size
- Based on purpose

Classification based on data processed or Functionality

Computers can be categorized as Digital, analog or hybrid

a) **Digital Computers:** are computers that process data which discrete in nature also known as binary digits. Mots home appliances such as TV`s, Microwaves, Wall Clocks are digital in nature.

b) **Analog Computers:** are computers that process data which is continuous in nature also known as waves.
Analog computers are used to manufacture process control like monitoring and regulating furnace temperature and pressure.

Analog Computer **Digital Computer**

c) **Hybrid Computers:** are designed to process both analog and digital data.

Comparison of digital and analog computers.

- Digital computers works on discrete signal which is in form of on and off while analog computers works on continuous signal
- Digital computers are simpler to develop than analog computers.
- Digital computers are more reliable since they are fast than analog computers
- Digital computers consume less power compared to analog computers.
- Digital computers are easy to use than anlog computers
- Digital computers are easy to store data since uses 0s and 1s which data storing in an analog computer is quit difficult

Classification based on size

- Computers are classified into four main elements according to its sizes and capacity and data storage
- These are Micro-computers, mini computers, main frame computers and super computers

a. Micro – computers

- It is a type of computer which uses microchips as its CPU
- Micro - computers are generally known as Persona Computers (PC)
- Micro – Computers are known as PC because they are designed to be used by one person at time.
- Micro – Computers are generally used for general purpose in places like:
  - Schools
  - Home
  - Business

Examples of micro – computers

- **Desktop Computers:** are computers designed for use on a desktop in an office environment and home Personal Computers (PC) are desktop computer.

- **Laptop Computers:** Are PCs sufficiently small and light for user comfort use his or her lap. Laptop operates on mains electricity or by rechargeable batteries and are also known as pocket PCs.


- **Personal Digital Assistant (PDA):** are PCs which uses touch sensitive screens of which image choices re made by pressing the image button using a special pen known as **STYLUS**.

b. Mini computers

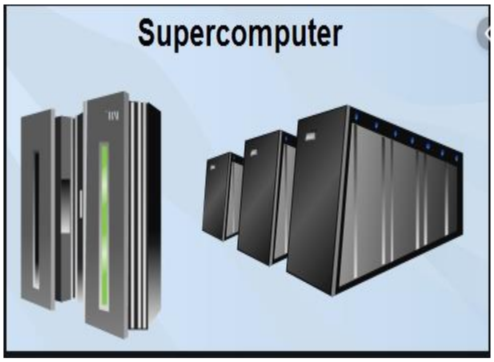
- Mini – computers are large multi – user computers
- They are smaller and less powerful than mainframes.
- Mini Computers are used to link other computers on a local area networks.

c. Main frame computers

- ❖ These are large and powerful computers that has the capacity of handling data processing many users at the same time.
- ❖ Users submit their tasks for processing to the mainframe by using dumb terminals.
- ❖ It is also used as central information storage.
- ❖ Most mainframe computers act as server in computers connected on the internet, of which it controls hundreds of computers on the network.

d. Super computers

- ❖ These are most powerful computers on the classes of computers.
- ❖ It is basically a mainframe computer that has its power increased.
- ❖ Super computers work on respective tasks such as weather of the global and nuclear bomb detonations.



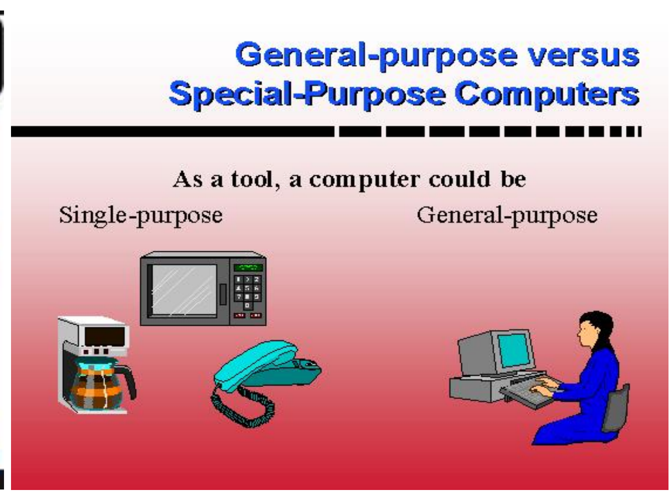
Classification based on according to purpose.

a) **General purpose:**

- ❖ are designed to perform a variety of task when loaded with appropriate programs.

b) **Special Purpose**:

- ❖ designed to perform or serve a specific purpose or to accomplish one particular task. They perform the task quickly and very efficiently because they are dedicate to a single task.



Units of data representation

Computer hardware perceives data items using binary system. The following terminologies are used to quantify the data in a computer

Bit: smallest unit of data which can either be 0 or 1

Byte: A group of bits (often 8) used to represent a single character in the computer memory. A Byte is the basic unit for measuring memory.

Nibble: A Nibble is half a byte, which is usually a grouping of 4 bits.

Word: Two or more bytes make a word. The term word length is used as a measure of the number of bits in each word. Data is usually read or written to memory in words. A word can have 16 bits, 32 bits, 64 bits etc.

Kilobyte: A Kilobyte is exactly 2^{10} bytes = 1024 bytes. Roughly speaking a kilobyte is approximately equal to 1000 bytes (one thousand bytes).

Megabyte: A megabyte is exactly 2^{20} bytes **= 1024 x 1024** bytes = 1,048,576 Bytes

- Roughly speaking, a megabyte is approximately 1,000, 000 bytes (one million bytes) or (one thousand kilobytes).

Gigabyte: A gigabyte is exactly 2^{30} bytes = 1024 x 1024 x1024 x 1024 bytes = 1,099,511,627,776 bytes.

- Roughly speaking, a gigabyte is approximately 1,000,000,000,000 byte (one trillion bytes) or (one thousand gigabytes).

Converting between data units

When making conversion between various data units, we use the following approximations.

1. 1 byte = 8 bits = 1 byte

2. 1 kilobyte = 1000 bytes = 1 thousand bytes.
3. 1 megabyte = 1000 megabytes = 1 billion bytes
4. 1 Gigabyte = 1000 megabytes = 1 billion bytes
5. 1 Terabyte = 1000 Gigabytes = 1 Trillion bytes

Character sets

Data is entered in the computer in form of characters. Each character, be a number, alphabet or symbol is recognized by the computer because of its unique sequence of O1s and 1`s. a word can be made of one or more characters. Therefore, when you Press character A on a keyboard, the character set converts a binary number / code to its equivalent human readable form and vise – versa. It is the character set for example that marks a binary code to the letter A on the keyboard.

The two widely used character sets are:

1. **American standard Code for Information Interchange (ASCII**).

❖ It uses seven bits to encode characters in computes, communication equipment and other devises. It is used to mainly encode text characters.

Dec	Hex	Octal	Binary	Char	Description
0	00	000	0000000	NUL	null character
1	01	001	0000001	SOH	start of header
2	02	002	0000010	STX	start of text
3	03	003	0000011	ETX	end of text
4	04	004	0000100	EOT	end of transmission
5	05	005	0000101	ENQ	enquiry
6	06	006	0000110	ACK	acknowledge
7	07	007	0000111	BEL	bell (ring)
8	08	010	0001000	BS	backspace

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<u>Table of an extract for ASCII character set encoding</u>

2. Extended Binary Coded Decimal Interchange Code (EBCDIC).

- It uses eight bits to encode characters
- mainly in IBM computers.

NB: ASCII and EBCDIC tables are widely available on the internet.

EBCDIC

	_0	_1	_2	_3	_4	_5	_6	_7	_8	_9	_A	_B	_C	_D	_E	_F
0_	NUL 0000	SOH 0001	STX 0002	ETX 0003	SEL	HT 0009	RNL	DEL 007F	GE	SPS	RPT	VT 000B	FF 000C	CR 000D	SO 000E	SI 000F
1_	DLE 0010	DC1 0011	DC2 0012	DC3 0013	RES/ENP	NL 0085	BS 0008	POC	CAN 0018	EM 0019	UBS	CU1	IFS 001C	IGS 001D	IRS 001E	IUS/ITB 001F
2_	DS	SOS	FS	WUS	BYP/INP	LF 000A	ETB 0017	ESC 001B	SA	SFE	SM/SW	CSP	MFA	ENQ 0005	ACK 0006	BEL 0007
3_			SYN 0016	IR	PP	TRN	NBS	EOT 0004	SBS	IT	RFF	CU3	DC4 0014	NAK 0015		SUB 001A
4_	SP 0020										¢ 00A2	. 002E	< 003C	(0028	+ 002B	| 007C
5_	& 0026										! 0021	$ 0024	* 002A	) 0029	; 003B	¬ 00AC
6_	- 002D	/ 002F									¦ 00A6	, 002C	% 0025	_ 005F	> 003E	? 003F
7_										` 0060	: 003A	# 0023	@ 0040	' 0027	= 003D	" 0022
8_		a 0061	b 0062	c 0063	d 0064	e 0065	f 0066	g 0067	h 0068	i 0069						± 00B1
9_		j	k	l	m	n	o	p	q	r						

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Chapter 2

COMPUTER HARDWARE

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Chapter 2: **Computer Hardware**

Introduction

They are two major components of a computer system and these are:

- ✓ Hardware System
- ✓ Software System

Hardware system

- ❖ Hardware systems are physical components of the computer that apple to the sense of touch.

Hardware has five major components, namely

- ❖ Input devices
- ❖ Output devices
- ❖ Central processing Unit
- ❖ Storage devise
- ❖ Memory

Input devices

Are devices that get data into the computer system? Example of input devices are:

- ❖ Keyboard
- ❖ Microphone
- ❖ Digital camera
- ❖ Digital video camera
- ❖ Pointing devices such as mouse, joystick

Key board

- Keyboard is a board with a set of keys used to send command into the computer system.

Functions of a Keyboard.

- Keying text and numbers for processing
- Sending commands into computer for execution.
- Navigating through a document
- For playing computer games.

Disadvantages of Using a Keyboard

- It is easy to type data wrongly from source document.
- It is time consuming compared to other data capturing methods.
- Data entry operations are prone to repetitive strain injuries (RSI), such as – aches and pains swelling and difficult movement of fingers, wrist, arms and neck.

Pointing Derives:

- Pointing devices whose major function are choosing commands from on screen menu and drawing of graphics. Examples of pointing devices are

- **Mouse** : a Mouse is an input device commonly used in mot computers
- A mouse has a movement sensitive ball and sensitive rollers inside.
- The movement of rollers inside transmits the signals through the computer via a cord.

Types of a mouse

1. **Traditional Mouse:** it has a ball and rollers inside.
2. **Optical Mouse:** it uses radio waves when communication with the computer.
3. **Touch Pad:** it is a pointing device commonly found in laptop computers. The on screen pointer is moved by sliding a finger a long for game controlling.
4. **Track ball**: a track ball is one of type of mouse with a movement located ball on top. The ball is rolled using fingers. The left and right buttons are usually located in the sides of the devise.
5. **Touch Screen / Slide Rule:** Touch sensitive screens are the kind of screens the user can make choices and press button images using **STYLUS.** Computers like Personal Digital Assistant (PDA) are having touch screens.
6. **Graphic Tables**: a graphic tablet consist of an electronic writing area and a special "PEN" that work with it. It is used to create graphic images. The pen included to a graphical tablet is pressure sensitive.
7. **Track Point:** it is a small rubber projection embedded between the keys of the keyboards scanners.

Functions of pointing devices.

Pointing and clicking on icons to send commands or opening window dialog boxes

- Cutting icons or text
- Pasting icons or text
- Drawing Graphics
- Dragging Icons

SCANNERS

Scanners are used for a wide range of data input. Most popular scanners are used to enter picture on a computer.

Types of Scanners

- **Optical Mark Recognition (OMR)**
  - It is used to read special shape of line on cards.
  - OMR is commonly used for making multiple choices system
- **Optical Character Recognition (OCR)**
  - It is used to read handwriting or printed text into the computer system.
- **Electrical Point of Sale (EPOS)**
  - It is used to read bar cords on goods and use the data collected to produce customer`s bills and updates stock records for the shop.
- **Electronic Fund Transfer at Point of Sale (EFTPOS)**
  - Are used to read information inform of magnetic strips of card.
  - The card is swiped through a scanner, then the scanner transfer account information into the Auto – Teller Machines (ATM) to read information from the card.

Out Put Devices

- Are devices used to give out results of the processed data.
- The following diagram illustrates thre common examples of out pur devices

Forms of output devices

- Soft copy
- Hard Copy

Soft Copy

- A soft copy is a kind of input which is not tangible but can be seen or heard.
- Soft copy can be audio through speakers or visual through monitors

Visual Display Unit (VDU)

Visual display Unit is also known as a Monitor or Screen which is used to display data inform of text, picture and Video.

- It is known as a monitor because it enables a user to see what is going on in the compute.

Types of VDU or Monitors

- Cathode Ray Tube (CRTs) Monitors
- Liquid Crystal Display (LCD) Monitors
- Gas Plasma Displays

Screen Resolution

Screen resolution is the number of pixels a monitor is able to display image on the screen are made up of a collection of small dots called pixels. Pixels are short form for picture "**pix"** elements **"el"** therefore pix + el form pixels e.g. 400 x 800 pixels. A screen is divided into rectangular.

Data Projectors

- Data projectors are used to display output from computer into a plain white screen like a wall or whiteboard.
- Output form a data projector is a creative way of presenting computer output to the audience.

Hardcopy Output Devices

A hardcopy is a tangible output that can be felt such as printed paper. Example of output devices are:

- Printers
- Plotters

Printers

- **Are used to produce information on a piece of paper**
- The quality of a hard cop depends on the printer`s printing mechanisms.

Categories of Printers

- Impact Printers
- Non – Impact Printers

Impact Printers

- Impact printer, they form a character by hammering the shape of a character into a carbon or ink ribbon placed against the paper. Examples are:
- Dot matrix printers
    a. Dot Matrix printers has got a head with needles which prick on the ribbon to produce a partten on a paper.
- Daisy wheel Printers

Disadvantage of Impact Printers

- They are generally slow
- Print quality is not good
- They cannot print color text graphics.

Ink Jet Printers

- It forms the image of the page by spraying tiny droplets of ink on the paper.
- They are common for home and office use

Advantages of Ink Jet printers

- out quality is good
- they are cheap
- they can print colour graphics

Disadvantages of Ink – Jet printers

- they are slow
- They use expensive toner.

Laser Jet Printers

- they operate on the same technology as a photocopier
- It uses a drum coated with photosensitive materials which are charged to write an image using LEDs
- The drum rolls through the tonner and make it charged
- The tonner is then deposited into the paper and then fused into the paper with heat.
- Most printers are monochrome (prints one color), but there are some expensive laser printers which can print in colour.

Thermal Printers

- Thermal Printers burn dark spots heat sensitive paper.
- They are cheap but they use expensive thermal paper.

Factors to consider when purchasing a Printer.

- The choice of purchasing a printer depends on a number of factors of which include:

1. **Print Quality** : Dot Matrix are good for bulky printing of draft of draft document while Laser is good for printing official document and thermal are good for checkout counter receipts.
2. **Initial Coast.** Though the prices of printers have come down, laser and thermal printer are still expensive compared to inkjet printers.
3. **Running Cost**. The cost of maintaining an inkjet printer is higher than of maintaining a laser printer.
4. **Speed**. **The** speed of a printer is measured in terms of number of pages it can print per minute.
5. **Colour Printing**. **Most printers** support black and colour printing but colour printers especially laser are relatively more expensive.

Plotters

- They are commonly used to print geographical architectural and engineering drawings such as maps, advertisement posters to be placed on bill boards.

The Central Processing unit (CPU)

- The central processing Unit (CPU) also known as processor is the most important component of the computer.
- CPU is actually regarded as the as the brain of the computer, because processing activities are carried out inside the processor.
- The CPU is mounted on a circuit board known as the motherboard or system board

Components of the CPU

- The control unit (CU)
- The Arithmetic and Logic Unit (ALU)
- Register or memory
- Basic input / output (I/O)

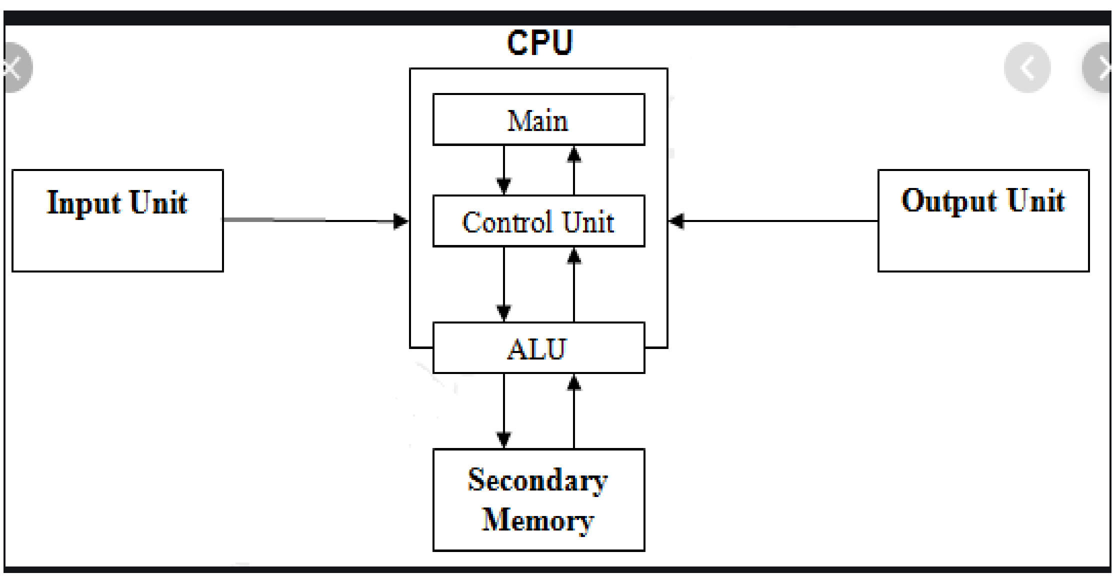
Control unit (CU)

- It controls various components of the computer
- CU is also known as control system or central controller/
- CU is responsible for reading and interpreting the instructions in the program one by one.

Arithmetic Logic Unit (ALU)

- It is capable for performing mathematical and logical operations

OVERALL FUNCTIONAL ORGANIZATION OF THE CPU



Register or memory

- It read information stored in digital computer inform of numbers such as 0^s and 1^s

Types of memory

- Random Access Memory (RAM)
- Read Only Memory (ROM)
- RAM - is the temporary storage within the computer and it is volatile memory
- ROM – is the permanent storage within and outside the computer.
- RAM – is also known as primary storage and ROM known as secondary storage.
- Memory is one of the components that determines the capability of a computer in terms of speeds and handling complex task.

TYPES OF RAM

There are two types of RAM and these are:

1. Static RAM (SRAM) is a fast type of memory mostly located inside a microprocessor and it is used for special purpose memories such as cache memory which is used to enhance the processing speed by holding data and instructions that are instantly required by the processor.
2. **Dynamic RAM (DRAM)** is a relatively slower type of RAM compared to **SRAM**. DRAM is needed periodically for storage purpose to maintain data storage in the computer system.

SPECIAL PURPOSE MEMORIES

These are memories included inside the microprocessor or input and output devices in order to enhance the performance of the computer system. These type of special memories are:

1) **Cache Memory** which is a fast type of RAM. They are three types of cache memory namely :
   - **a)** **Level 1** also known as primary cache is located inside the microprocessor
   - **b)** **Level 2** also known as external cache it may be inside the microprocessor or mounted on the motherboard.
   - **c)** **Level 3** is the latest type of cache that works with Level 2 cache to optimize the performance.

2) **Register** it holds one piece of data at a time and is inside the CPU. Examples of registers are:
   - **A) Accumulator,** it temporally holds the results of the last processing step of the ALU.
   - **B) Instructor Register** it temporally holds an instruction just before it is interpreted into a form that the main memory
   - **C) Address Register,** it temporally holds the next piece of data waiting to be processed.
   - **D) Storage Register,** it temporarily holds a piece of data that is on its way to and from the CPU and the main memory.
3) **Buffer,** are special memory that are found in input and output devices. The input data is hold in the output buffer and the output is hold in the buffer. For example computer printers have buffers where they can store massive documents sent by the CPU for printing for the CPU to continue processing other task while the printer continues to print in the background.

Memory capacities

Memory and storage capacity is measured in special units called bytes. A Byte is equivalent to single character. Characters can be numbers from 0 to 9 or letters from A to Z and symbols including spaces between characters. Bytes are formed from bits. For example 8 bits make 1 byte. Memory quantities can be expressed in:

- **Kilobytes (KB)** are approximately one thousand bytes, but the actual size is 1024 bytes, because the computer uses base 2 system (0 and 1 : 2^n)
- **Megabytes (MB)** are approximately one million bytes, but the actual size is 1048575 bytes
- **Gigabytes (GB)** are approximately one billion bytes but the actual size is 1073741824 bytes.
- **Terabytes** are approximately one trillion bytes but the actual size is 1099511627776 bytes

Computer performance

- Computer performance can be determined by the following factors
- The speed of the CPU, there are different types of processor e.g. / Pentium and power PC. The speed is measured in (MHz). The greater the number of MHz, the better the performance.
- The amount of RAM, most computer are using 32 MB of ram but can be upgraded to 128 MB. The more RAM improvement the higher performance of the computer system.
- Hard Disk speed and capacity. Hard Disk Speed varies. It is always advised to buy large hard disk capacity to control the program in the computer.

Storage devices

- Storage devices are devices that help to keep processes data from the computer system
- Storage devices are also known as backing storage, because they help to keep the copy of original file from the computer system
- The following diagram illustrates various examples of storage devices

Examples of storage devices

Type of storage devices

- Magnetic media
- Solid state media
- Optical media

Magnetic media

- They store data in magnetic fields, examples of magnetic media are :
    a. Floppy diskettes
    b. Zip Diskettes
    c. Jazz Disk

Floppy Disk

- **They** consist of a flexible plastic disk coated with metal oxide that can be magnetized.
- The disk is enclosed inside a plastic jacket to protect from dust, sunlight and moister.

Advantages of a Floppy Disk

- It is cheap

- Floppy drives are available in most computers which is making it easy to use.

Disadvantages of a Floppy Disk

- It holds very little data
- It can easily get damaged when exposed to sunlight, moister, and dust.

Zip Diskette

- They are working on the same technology with floppy diskettes and the way they store data.

Parts of a Floppy diskette

- **Read/ Write Window**, which allows the retrieval of Data and entry of Data in the computer system
- **Metal Shutter,** its function is to cover the read / write window to avoid dust and direct sunlight to the magnetic tapes.
- **Write protected notch,** its function are to lock and unlock the floppy diskette (for data protection)
- **Index Hole,** it facilitates the rotation of the Diskette.
- **Hub,** it holds the magnetic diskette to remain within its position.
- **Disk Label,** it holds the name of data stored within its position.
  a. **Zip Diskette** has small read / write head which allow it to store more data than floppy diskette.

Care of a Floppy Diskette

- Always store disks carefully
- Keep the disks away from any magnetic.
- Keep the disks away from direct heat e.g. radio to or sunlight.
- Do not touch the exposed recording surface.

Solid State Media

- They store data on flash memory microchips
- Data in solid state media can be erased and re-written for several times.
- Some solid state media has a write – protect switch. examples of solid state media are
    - **b.** USB flash disk
    - **c.** Key drives
    - **d.** Thumb drives
    - **e.** Pen drives

Advantages of solid-state drives

- They are fast
- They have large storage capacity.
- They can be re –written to

Optical Media

- “Optical” means something to do with light.
- Optical media use a laser beam light to read or write on a disk.

Types of Optical Media:

- CD - ROM
- CD - R
- CD - RW
- DVD
- VCD
- CD

Compact Disk - Recordable (CD - R)

- They allow users to record or changed
- Once data is recorded cannot be deleted.

Advantages of CD - ROM

- They are cheap to produce in large quality
- They have large storage capacity.

Disadvantages of CD-ROM.

- Data cannot be deleted or changed
- Data retrieval is slower than a hard disk.

Compact disk- recordable (cd-r)

- They allow users to record data using CD-R drives.
- Once data is recorded cannot be deleted.

Advantage of CD-R

- They store large amount of data.
- Data cannot be destroyed easily hence it deals with backing up data.

Disadvantage of CD-R

- Some CD-R software has poor writing capacities
- Some CD-R has poor storage capabilities

Compact disk rewritable (CD-R)

- It allows users to write data for several times.
- Data can be changed or deleted.
- It has large storage space.

Disadvantages of CD-RW

- Only the CD-R recorder can record data to disk.

- Some CD-RW does not work in other CD- players.

Video Compact Disk (VCD) and Digital Versatile Disk (DVD)

- They can store data up to 17 GB.
- Data once recorded cannot be changed or deleted.

Disadvantage of VCD/DVD

- They have large storage capacity to all the CD'
- DVD player can read CD-ROM

Disadvantage of VCD/VD

- They are expensive
- DVD's cannot be read in other CD drives

Communication Devices

- Communication devices components used to connect computer on a network.

Types of communication devices

- Modulation Demodulation (**MODEM)**
- Network bub or switch.

- Network bridges

Modem

- Modems are devices used to change signal from analogue to digital and from digital to analogue
- Modulation is the process of converting digital signal to analogue signals
- Demodulation is the process of converting analogue signals to digital signals

Types of modems

- Internal Modems
- External Modems
- Internal modems are built inside the computer systems
- External modems are connected using telecommunication line.

Advantages of modems

- It is cheap
- It can be used anywhere provided there is a telephone line
- It is easy to use hence is plug and play

Disadvantages of modems

- Data transfer using modems are relatively slow as a result much faster internet connections where developed such as.
- ISDN (Integrated Service Digital Network)
- ADSL (Asymmetric Digital Subscriber line)

Advantages OF ISDN AND ADSL

- Data transfer is much faster than a modem
- It has the capacity of handling large amounts of data.
- It enables video conferencing (user can talk to and see each other).

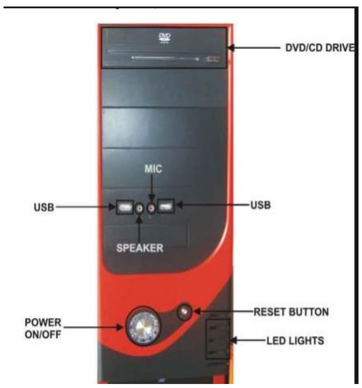
Network switch/hub

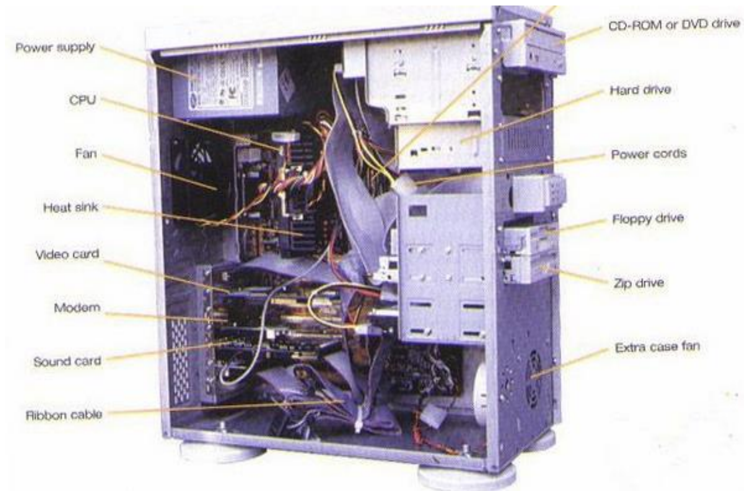
- Are devices used to connect computers on a local Area network (LAN)
- The computer on a network can access each other through a hub or switch.

- Network switch or hub they create direct access to information on a network.

Components of the system unit

Computer system unit is made up of several parts that work together to form a complete unit





- ❖ The power button
- ❖ The mother board
- ❖ Intergraded circuits (IC) chips
- ❖ The processor
- ❖ Power cables
- ❖ The LED lights
- ❖ The hard disk
- ❖ The ports
- ❖ Memory slots / Chips
- ❖ Power box

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Computer Studies
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Chapter 3

COMPUTER SOFTWARE

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Compiled By: Eliot Kalenga

Chapter 3: Computer Software

Introduction

They are two major components of a computer system and these are:

- ✓ Hardware System
- ✓ Software System

Computer Software

Software refers to set of instruction that help computer to do something

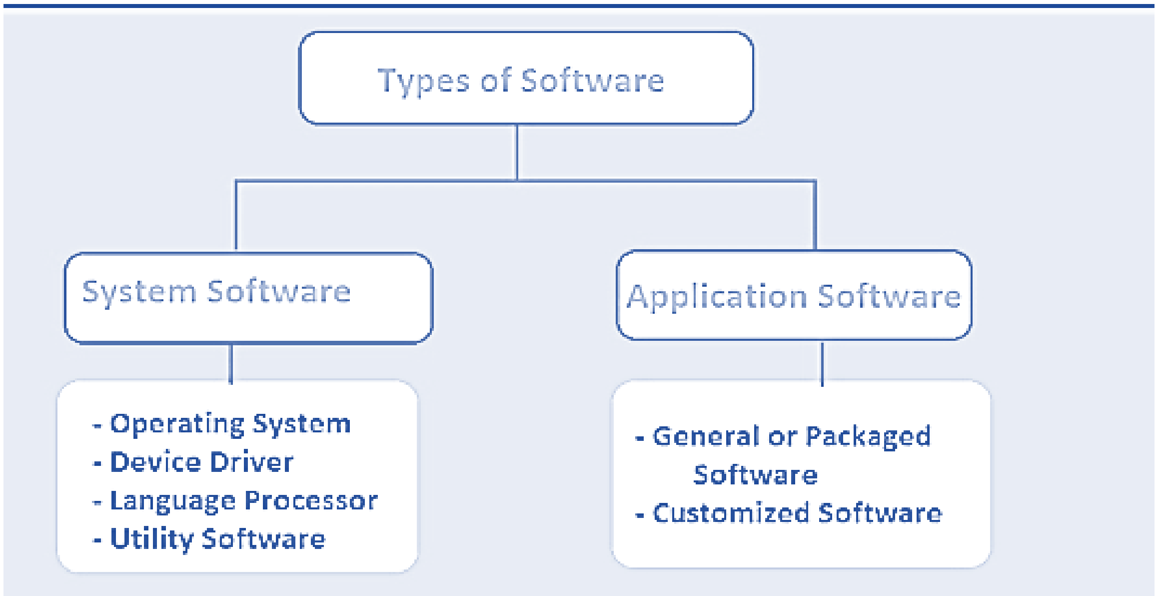
Examples of computer software

- ❖ Microsoft windows
- ❖ Linux
- ❖ Unix
- ❖ Microsoft office
- ❖ A dope Photoshop
- ❖ Adobe illustrator
- ❖ Quick books
- ❖ SPSS
- ❖ Open office
- ❖ Vlc
- ❖ Sage line50
- ❖ STATA

Types of computer software

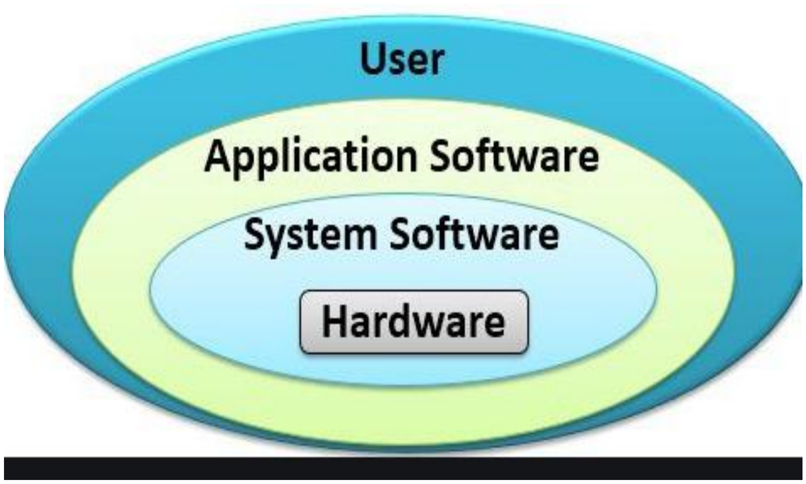
There are two main types of computer software

1. System software
2. Application software



System software

- System software is a type of computer program that is designed to run a computer's hardware and application programs.



Functions of system software

- ❖ Booting the computer and making sure that all the hardware elements are working properly

- Performing operations such as retrieving, loading, executing and storing application
- Storing and retrieving files
- Performing a variety of system utility functions.

Division of system software

- **Operating systems** – control computer resources
- **Firmware** – are permanent instructions stored in electronic chips
- **Utility software** – performs commonly used task in system – level utility and application level utility
- **Language processor (translator)** – it translate the instructors such assemblers, interpreter and compilers

Application software

Are programs that are designed to help the user accomplish specific task, application software are also known as application packages.

SOFTWARE	USES	EXAMPLES
Word processor	Typing documents like letters	Word, lotus, WordPro, open office writer
Spreadsheet	Calculating budgets	MS excel, lotus 1-2-3
Desktop publishing	Designing publications like newspapers and books	Adobe page maker, Ms publisher, adobe InDesign
Computer aided design	Technical drawing	AutoCAD
Databases	Keeping records and files	Ms access, MySQL, fox base paradox
Graphics software	Designing and manipulation graphics	Corel draw, Photoshop

Classification of application according to acquisition

- **In-house developed software** – are developed to meet a particular need
- **Vendor off-the-shelf software** – are designed and made available to purchase and are developed address many tasks.

Advantages of standard software

- The can be easily be installed and run.
- They are cheaper than in-house developed software
- They are readily available for almost any task
- They have minor or no errors since they are thoroughly tested
- They can easily be modified to meet a user's needs

Disadvantage of standard software

- They may have some features not needed by user, which may take extra storage space
- They may require users to change to change processes and hard ware for compatibility which may inurn be expensive
- They may lack some features required by the user

NB: Software can also be classified according to end-user License (EUL) and general public license (GPL)

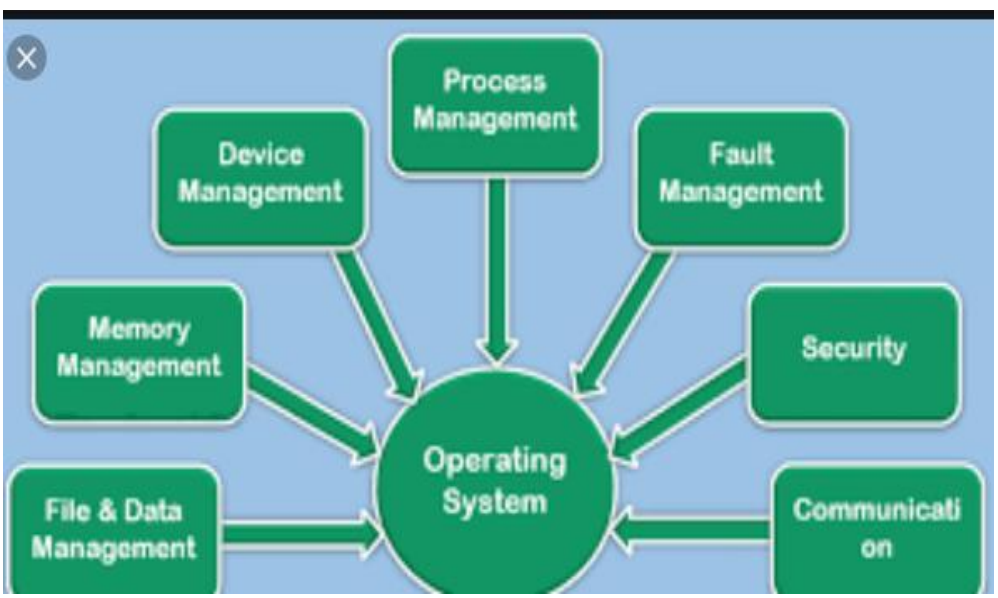
Comparison of system software and application software

Subject	System Software	Application Software
Usage	Runs to operate the computer hardware	Runs to meet a specific user requirement
Installation	Installed on the computer when operating system is installed.	Installed according to user's requirements.
Dependency	can run independently. It	can't run independently.

	provides platform for running application softwares.	They can't run without the presence of system software.
User Interaction	Usually runs in the background with indirect user interaction	Runs in the foreground with direct user interaction
Programming Languages Used	Low level languages are used to write the system software.	High level languages are used to write the application software.

Operating systems (OS)

- Operating systems is a group of computer programs that help to manage computer resource
- Operating system acts as the interface between the computer and applications



Importance of operating system

Operating system is an important software for computer because in controls and runs all the activities of computer system it is also known as System software.

The following are points summarises the importance of operating system.

- ❖ It manages all the resources of computer and also users and requests of users that's why it is also called **resource manager.**
- ❖ It manages CPU input/ output devices memory and internet to be used in more efficient and convenient way.
- ❖ It also provides a platform for application software to run without operating system we cannot use computer.
- ❖ It manages user accounts and users.
- - Creating, editing and deleting of user accounts
- ❖ It manages files
- - It manages/ organizes all the files stored in computer.
- ❖ It manages memory and memory allocation.

Types of operating system

Operating systems are classified based on the following criteria

- ❖ Number of users
- ❖ Number of task
- ❖ User interface

<u>Types of operating systems according to number of users</u>

- o Single user operating system
- o Multi-user operating system

Single user operating system

- ❖ It requires the user participation when loading and executing computer programs
- ❖ Single user operating system is also known as **interactive** program
- ❖ Example of single user operating system is disk operating system (DOS)

Multi-user tasking operating system

- ❖ It allows a number of users to access the same peripheral device at a time, such as printers on a network.
- ❖ It also allows a numbers of work stations to access the mainframe without their own processing ability
- ❖ Example of multi-user OS includes windows and Novell.

Types of operating systems according to number of tasks

- Single tasking operating system
- Multi- tasking operating system

Single tasking operating system

- Single-task operating systems allow one user to do one thing at a time.
- An example of single-task operating system is the operating system used by personal digital assistants (PDAs), also known as handheld computers.

Multi – tasking operating system

- It allows user to perform several tasks running simultaneously and to switch between them
- Example of multi-tasking includes windows UNIX operating system

Types of Operating systems according to user interface

User interface

- it is means of communication operating system provide between the computer and the user

Types of user interface

- command line interface
- menu driven interface
- graphic interface

Command line interface

- It uses functions (commands) with special syntax which the user and computer can understand
- It requires a user to type the exact commands for further instructions

```
Microsoft Windows [Version 10.0.10240]
(c) 2015 Microsoft Corporation. All rights reserved.

C:\Users\Eliot Kalenga>cd..

C:\Users>cd eliot kalenga

C:\Users\Eliot Kalenga>cd..

C:\Users>cd..

C:\>cd users

C:\Users>cd training
The system cannot find the path specified.

C:\Users>
```

Menu driven interface

- ❖ It displays a list of menu for the users to select the program they want to use
- ❖ It was inverted to address the problem of memorizing commands
- ❖ It displays a lot of menu and submenus which is making it difficult for the user to select the programs they want.

Graphic User Interface (GUI)

- ❖ It is the most user friendly interface
- ❖ It provide the user with the following services
  - Windows
  - Icons
  - Menus
  - Pointers
  - Abbreviated as WIMP (**W**indows, **I**cons, **M**enus, **P**ointers)

- ❖ GUI is making it very easy to communicate with the computer

Functions of operating systems

Operating system has a number of tasks to perform and some of these includes;

- **For job sequencing**; it controls the amount of time allocated to task by the central processing-unit

- **For controlling functions**; it controls peripheral devices such as keyboards, printers and scanners and controlling the amount of disk space used
- **For activating computers** from standby mode
- **For error handling ;** OS detected any error in the loading and executing of programs instructions
- **Memory management;** it controls the amount of memory used for each application.
- **Device scheduling;** in multi-user environment OS is responsible for allocating the accessibility of device to each user on a network
- **Interrupt handling** – break from a normal sequential processing of instructions in a program

Factrors `to consider when choosing an operating system

- The hardware configuration of the computer such as the memory capacity, processor speed and hard disk capacity.
- The type of computer in terms of size and make. For example, some earlier apple computer would not run on Microsoft operating systems
- The application software intended for the computer.
- The document available
- The cost of operating system
- Reliability and security provided the operating system.

Information organization

- It is the major function of operating system to organize information
- It is important to organize data in the computer environment because of the following reasons;
  - It enables easy access to information
  - It is easy to transfer information from one location to another
  - Easy sorting of information

Features of different operating systems

Disk operating system

- DOS is a command driven interface
- It allows a chain of commands to be entered into a computer to customize the operating system for a particular task
- The user has to type commands in exactly the correct syntax to perform any operation.

Contents of DOS

- DOS has a number of features of which some of them are;
  - Command interpreter of processor
  - External commands
  - Command or DOS prompt

Command interpreter or processor

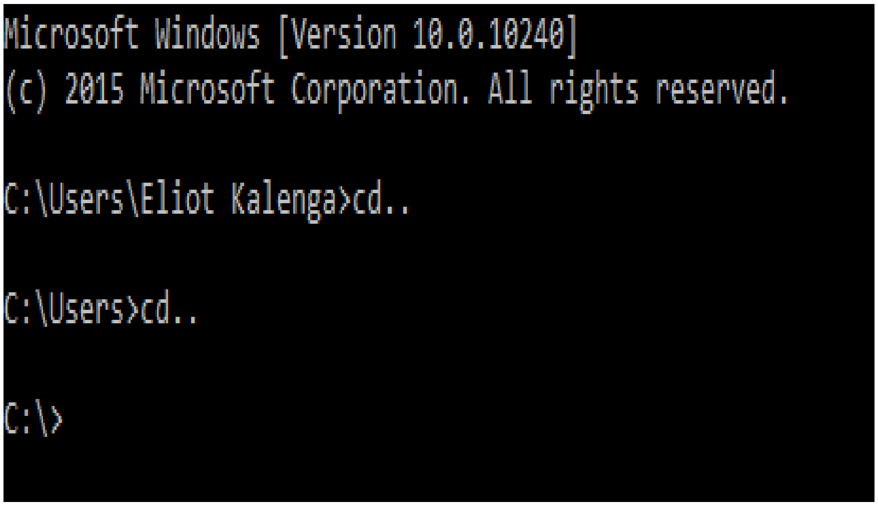
- It is a file which holds instructions for interpreting the commands used in DOS
- It interprets the commands which are frequently used
- Commands interpreter are also known as internal commands e.g. COPY, DIR and CLS

External commands

- It is a file which holds instructions that are not frequently used and they are kept in floppy diskettes
- The commands are run when required

Command line or DOS prompt

- It displays the current drives such as C:\> waiting for instructions after booting the computer
- It allows a chain of complex command to communicate with the computer system



Information organisation in DOS

- It is helpful to organise information in good manner for easy retrieval
- DOS uses its own DOS file instructions to organise information
- Information in DOS are organised in the following area

1. Drives of which each drives is represented by letter such as
   A. For first floppy drives
   B. For second floppy drives
   C. For hard drives
   D. For hard drives
   E. For flash disk
   F. For CD/DVD drives

- In DOS the drive letter is followed by a colon, for example A:

2. Files which hold related information, a file is similar to a paper document

3. Directions where all files are kept. A directory is like a container for keeping files

DOS file management systems

- DOS uses specific commands to work with files
- When working in dos it shows a command prompt with a cursor blinking after it. For example C:\>

Commands used in dos and their meaning

i. **CD**….. or CD means changing drives from the current drives to another drives
ii. **DIR** means viewing the contents of the directory page by page
iii. **DIR/P** means viewing the contents of the directory page by page
iv. **CD** means returning to the root directory
v. **MKDIR** or MD means making a new directory
vi. **RMDIR** for removing or deleting the directory
vii. **DEL** for deleting file in a directory
viii. **COPY** for copying directory from one location for another
ix. **COPY*.*** for copying all files in the directory
x. **REN** for renaming files in the directories
xi. **CLS** means clear screen

Disk management (formatting a disk)

- A disk cannot be used in a computer if it is not formatted
- Formatting is a process in which the computer arranges the surface of a disk into a form it can be recognized and used to store data
- When formatting a disk in DOS, **FORMAT** command is used. Type FORMAT to the prompt system followed by drive letter
- **FORMAT** command is used to prepare a disk making it ready to be used.
- **DISKCOPY** are external commands used to copy the content of a disk while **COPY** is internal command

Wildcard character

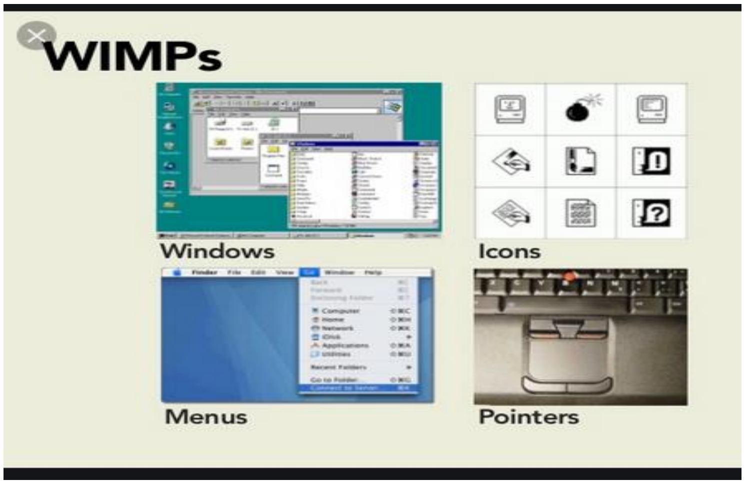
- A wildcard character is a character that can be used to present more than one letter in a directory contents
- Examples of wild card character are (*) (?) and (.)
- A question mark in a file or extension stands for a single character in the position it is allocated
- For example if all files has to be deleted in directory type DEL??????????????? In the prompt system
- Asterisk in a file name means any character or combination of characters in the position has to be deleted. Asterisks are short hands for series of questions marks. For example DEL*.*

Windows operating system

- It uses graphic user interface to facilitate communication between the computer and the user
- GUI is the most user friendly window because it has a number of sub programs

Contents of windows

- Windows consist of GUI which has four major elements, such as
  - **W**indows
  - **I**cons
  - **M**enus
  - **P**ointer
    The elements are abbreviated as WIMP



Windows

- Windows organises a group of related tasks
- Some windows are known as dialog boxes
- Each window provides menus and icons which can be used execute commands relevant to the task in the window
- Almost every window in the windows environment has some basic buttons for **closing, restoring the window, minimizing and maximizing**
- The dialog box window has only the button for closing

Icons

- Icons are buttons with small meaningful pictures or symbols representing the computer programs easy memorizing their functions
- Icons are used as shortcuts for opening the items or sending commands they represent

Types of icons

- Programs icons
- Folder icons
- Toolbar icons

- Program, file and file icons are displayed in a dialog box or the desktop

- ❖ A program icon consist of symbols foe the program they represent and small caption under the icon.
- ❖ Folder icons they consist of yellow boxes with name of the folder
- ❖ File icons are just like program icons with a symbol for the program that was used to create the file and the name of the file under it
- ❖ Icons in toolbars have small meaningful pictures without any caption

Menus

- ❖ Menus are indicated in two major WAYS;
  - By name
  - By a small black triangle on the right of a list box or an icon

Pointers

- ❖ Pointers are tools use
- ❖ For pointing and clicking on icons

Advantages of windows over dos

- It is more user friendly because of the GUI
- It is faster than DOS
- It is easy to maintain than DOS
- Windows has many utility programs than DOS
- It is task oriented than DOS
- It can handle several task at a time than DOS

Linux operating system

Linux is one of popular version of UNIX operating System. It is open source as its source code is freely available. It is free to use. Linux was designed considering UNIX compatibility. Its functionality list is quite similar to that of UNIX.

Examples of linux distibutions

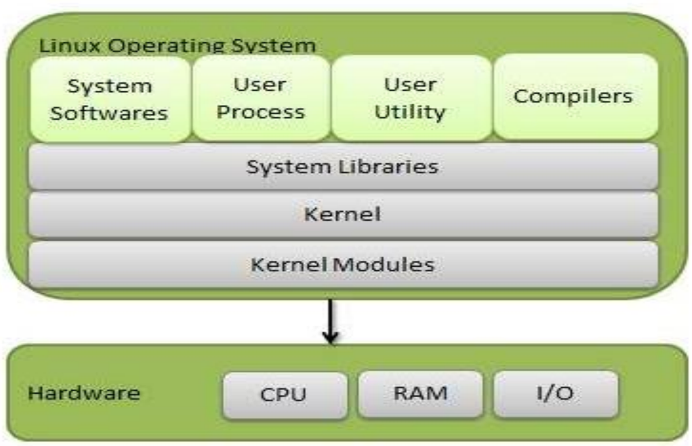
- Ubuntu linux

- Centos
- Suse linux
- Fedora
- Kali

Components of Linux System

Linux Operating System has primarily three components

- **Kernel** – Kernel is the core part of Linux. It is responsible for all major activities of this operating system. It consists of various modules and it interacts directly with the underlying hardware. Kernel provides the required abstraction to hide low level hardware details to system or application programs.
- **System Library** – System libraries are special functions or programs using which application programs or system utilities accesses Kernel's features. These libraries implement most of the functionalities of the operating system and do not requires kernel module's code access rights.
- **System Utility** – System Utility programs are responsible to do specialized, individual



level tasks.

Basic Features of Linux operating system

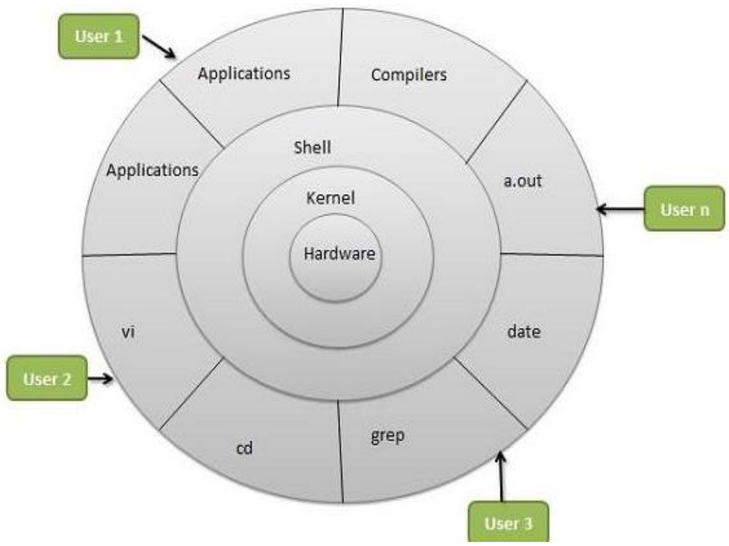
Following are some of the important features of Linux Operating System.

- **Portable** – Portability means software can works on different types of hardware in same way. Linux kernel and application programs supports their installation on any kind of hardware platform.
- **Open Source** – Linux source code is freely available and it is community based development project. Multiple teams work in collaboration to enhance the capability of Linux operating system and it is continuously evolving.
- **Multi-User** – Linux is a multiuser system means multiple users can access system resources like memory/ ram/ application programs at same time.
- **Multiprogramming** – Linux is a multiprogramming system means multiple applications can run at same time.
- **Hierarchical File System** – Linux provides a standard file structure in which system files/ user files are arranged.
- **Shell** – Linux provides a special interpreter program which can be used to execute commands of the operating system. It can be used to do various types of operations, call application programs. etc.
- **Security** – Linux provides user security using authentication features like password protection/ controlled access to specific files/ encryption of data.

Architecture of Linux operating system

The architecture of a Linux System consists of the following layers :

- **Hardware layer** – Hardware consists of all peripheral devices (RAM/ HDD/ CPU etc).
- **Kernel** – It is the core component of Operating System, interacts directly with hardware, provides low level services to upper layer components.
- **Shell** – An interface to kernel, hiding complexity of kernel's functions from users. The shell takes commands from the user and executes kernel's functions.
- **Utilities** – Utility programs that provide the user most of the functionalities of an operating systems.

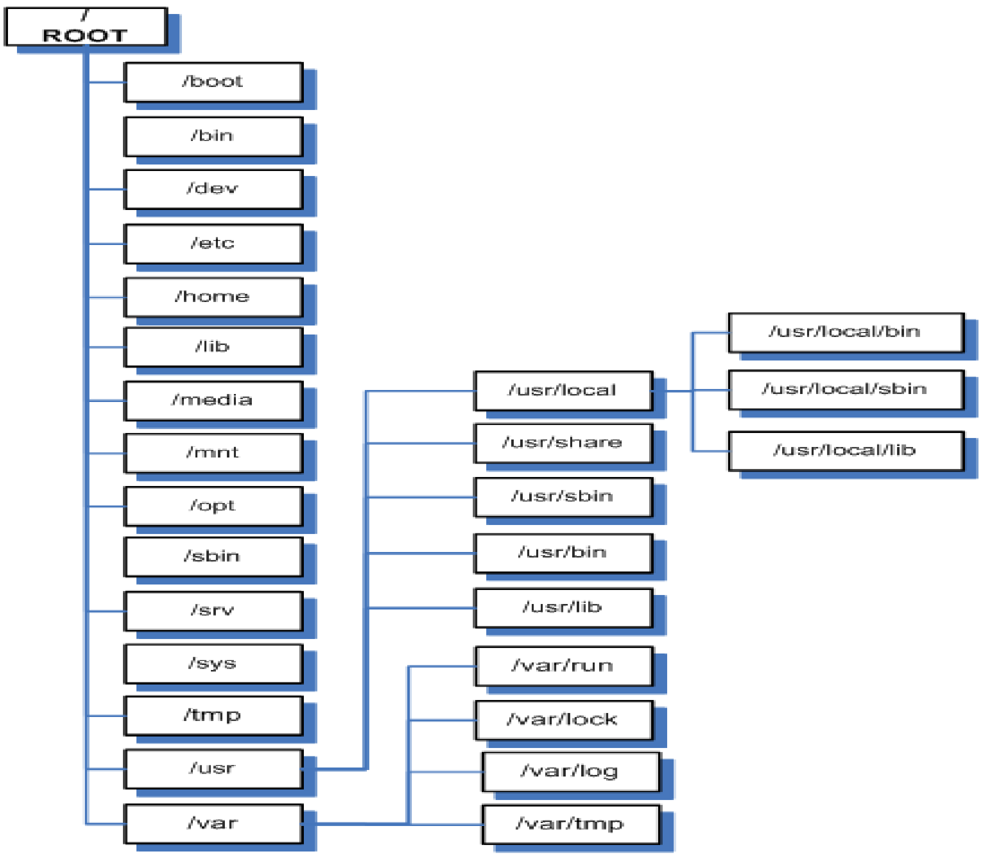


Linux file system

In Linux everything is a file. Linux organizes its files in a hierarchical directory structure called a tree. The first directory in the filesystem is called the root directory. The root directory contains files and subdirectories, which contain more files and subdirectories, and so on.

In windows each storage device has a separate filesystem. Linux has always one filesystem tree no matter how many drives or storage devices are attached to the computer. In Linux, a storage device is mounted at various points on the tree.

The tree in Linux is like an upside-down tree with the root at the top and the various branches descending below.



This structure could vary from distribution (distro) to distribution and this is a very generic Linux directory structure. The table below describe what these directories hold.

Directory	Description	Comments
/	Root	The root directory. Every single file and directory starts from the root directory. Only root user has write privilege under this directory. /root and / are different. /root refers to root user's home directory
/bin	User Binaries	Contains binary executables for all users of the system and commands required in single user mode e.g. cat, ls, cp, ping, grep etc.
/boot	Boot Loader	Contains the Linux kernel, initial RAM disk image (for drivers

	Files	needed at boot time) and Boot loader files. /boot/vmliuz – the Linux kernel /boot/grub/grub.conf or menu.lst – configure the boot loader
/dev	Device Files	Here is where the kernel maintains a list of all the devices it understands. These include terminal devices, usb, or any device attached to the system.
/etc	Configuration Files	Contains configuration files required by all programs. They may be for the system or network. It also contains startup and shutdown shell scripts used to start/stop individual programs. e.g. /etc/resolv.conf, /etc/logrotate.conf /etc/crontab, a file that defines when automated jobs will run /etc/fstab, a table of storage devices and their associated mount points /etc/passwd, a list of the user accounts
/home	Home Directories	each user is given a directory in /home. Ordinary users can write files only in their home directories.
/lib	System Libraries	Contains the library files for all the binaries held in the /sbin & /bin directories
/media	Removable Media Devices	On modern Linux systems the /media directory will contain the mount points for removable media such as USB drives, CD-ROMs, etc. that are mounted automatically at insertion.
/mnt	Removable Media Devices	On older Linux systems, the /mnt directory contains mount points for removable devices that have been mounted manually.
/opt	Optional add-on Applications	This is mainly used to hold commercial software products that may be installed on your system.
/proc	Process Information	The /proc directory is not a real filesystem in the sense of files stored on your hard drive. Rather, it is a virtual filesystem maintained by the Linux kernel.
/root		This is the home directory for the root account.
/sbin	System Binaries	contains binary executables for use by system administrators, for

		system maintenance purpose.g. iptables, reboot, fdisk, ifconfig, swapon
/tmp	Temporary Files	Contains temporary files created by system and users. Files under this directory are deleted when system is rebooted. There is also a /var/tmp directory which holds temporary files too. the only difference between the two is that /var/tmp directory holds files that are protected at system reboot(not flashed).
/usr	User Programs	Contains binaries, libraries, documentation, and source-code for second level programs. /usr/bin - contains binary files for user programs e.g at, awk, cc, less, scp /usr/lib - shared libraries for the programs in /usr/bin. /usr/local – contains programs that are not included with your distribution but are intended for system-wide use are installed. Programs compiled from source code are normally installed in /usr/local/bin. /usr/sbin - Contains more system administration programs.e.g. atd, cron, sshd, useradd, userdel /usr/share - /usr/share contains all the shared data used by programs in /usr/bin. /usr/share/doc – contains documentation files organized by package
/var	Variable Files	Contains files hat are expected to grow e.g. databases, spool files, user mail, etc. /var/log - contains log files, records of various system activity

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Chapter 4

USING COMPUTERS

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CHAPTER 4: USING COMPUTERS

The keyboard

- A keyboard is board with a set of keys used for entering data in a computers system
- Command from the keyboard is what tells the what to do
- A key is a button on keyboard which performs specific tasks

Types of keyboard

- There are two main types of keyboards, namely
    - Standard keyboard
    - Extended keyboard

  Standard keyboard has three different types of keys of which includes
    - Alphabet keys
    - Digit keys
    - Control keys
    - Function keys
    - Cursor keys
- While ***extended keyboard*** has only alphabetic keys and control keys with one set of numerical keys

Alphabet keys

- Are used for entering text data
- They arrangement of the keyboard are based on the standards type writer known as **QWERTY**. The **QWERTY** from is proved by counting keys from Q to Y which make the term **QWERTY**

Digit keys

- Digit keys are used to enter numbers

- Digit keys are in two sets the first set is aligned on top of alphabet key and the second set is located to the far right of the keyboard and is known as numerical keypad

Function keys

- Are used to enter specific commands into the computer
- These are from F1 to F12 on standard keyboard and F1 to F10 on extended key board

Cursor keys

- Are used to move the cursor. Cursor keys include the four arrow keys between the numeric keypad and the control keys such as, HOME, END, PAGEDOWN and PAGE UP

Control keys

- Are keys used for editing and key combination to send shortcuts commands into the computer
- The control keys surround the QWERTY key board and example of control keys are ENTER, CTRL, CAPSLOCK, ALT, DELETE, SPACEBAR, TAB, BACKSPACE and SHIFT

Functions of the control keys

- Back space: are used to delete backwards
- Enter ; are used for starting anew line or creation a paragraph
- Shift : are used for typing the symbol which is on top of the key such symbol like:<,> ?, &, *, (,), ^, %, $, €, ‟, !, ;¬+, ~, @< :, {, }.
- Shift is also used for typing upper case and lower case
- DELETE: are used to delete forward to use delete key you need to highlight the text to be deleted
- Caps lock: for typing upper case only this is done when the caps button is on

Typing

- Refers to insertion of text in word processor or any application program and text box
- Typing is done by pressing the desired keys on the key board
- Some keys have two symbols on them, to type the top symbol hold SHIFT + the key which is holding the symbol

Typing upper case (capital letter)

- Hold down SHIFT together with desired alphabetical letter
- Switch on CAPS LOCK by pressing on caps lock button. Then all characters will be in upper case

Blind typing

- It refers to the process of pressing hands dangled at the middle of the keyboard while fingers press other keys
- Fingers are used to type a letter or symbol close to it
- Thumb finger used to place space bar. All the eight finger including thumbs rest in the Centre of the keyboard

Finger layout and hand position

- The left hand with four fingers must be pressed on keys ASDF and its thumb on space bar
- The right hand finger should lay on JKL and thumb on space bar

Good keyboard posture

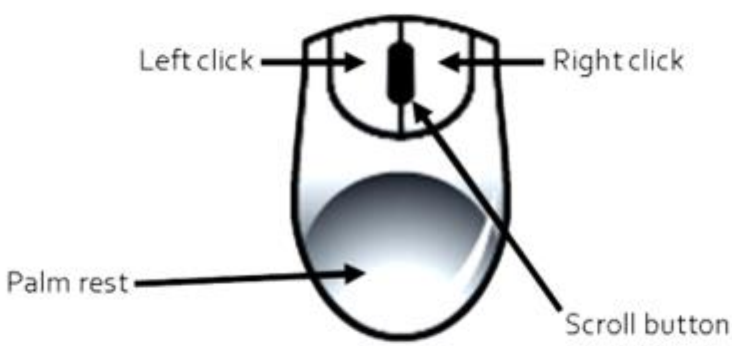
- Sit upright with both feet firmly on the ground maintaining an alert posture
- Place the material to be typed on your left, in a position you can read without straining

- ❖ Rest both hands on the keyboard with fingers resting on the home row in readiness to press other keys
- ❖ Always return fingers to the home row position after striking other keys
- ❖ Start typing slowly first. Do not look on your fingers when typing. If you do so o
- ❖ You will nervy to type quickly and accurately.

Mouse or mice skills

- A mouse is a device which is used to input data in the computer system
- A mouse is the main tool which is used to interact with graphical interface (GUI)
- A standard personal computer mouse has two buttons while a standard Macintosh mouse has one button.

Parts of a mouse



Mouse terms

Left click

- **Single click (or Click)**: Press and release the left mouse buttons (left button). This is used to select options or items
- **Double click**: Quickly press and release the left mouse button twice. Mainly used for opening documents

- **Drag and drop**: Point over an object using a pointer, click and hold un object and move it to the desired location and release the button

Right click

- **Right click**: press and release the right mouse button. This usually provides a list of shortcuts. To exit the list simply click anywhere else way a from the list

Scroll

This is the rotating button between the two click buttons and is used to scroll up and down pages. In some cases, the rotating button between the two click buttons can pressed for other actions.

Types of computer mouse

Computer mouse are classified into many categories as follows

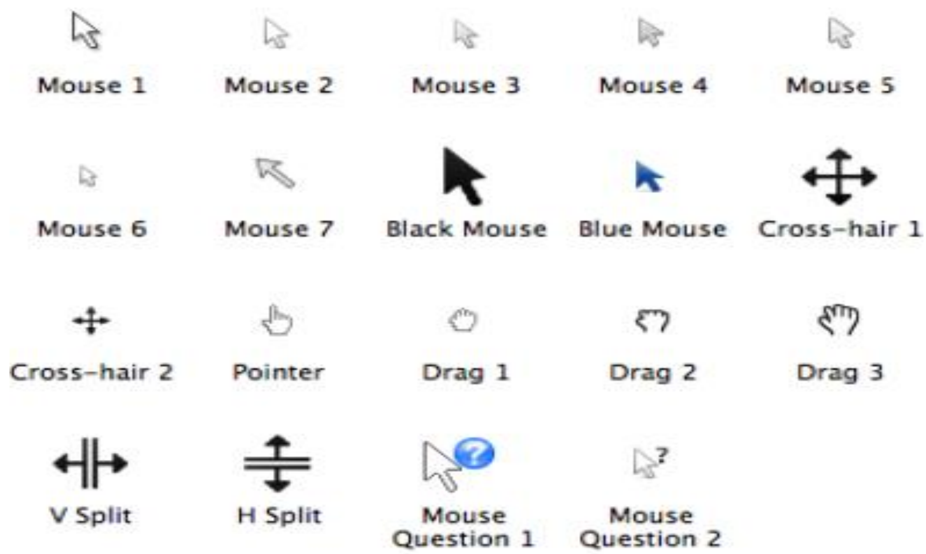
- **Mechanical mouse:** Houses a hard rubber ball that rolls as the mouse is moved. Sensors inside the mouse body detect the movement and translate it into information that the computer interprets.
- **Optical mouse:** Uses an LED sensor to detect tabletop movement and then sends off that information to the computer for merry munching.
- **Infrared (IR) or radio frequency cordless mouse:** With both these types, the mouse relays a signal to a base station wired to the computer's mouse port. The cordless mouse requires power, which comes in the form of batteries.
- **Trackball mouse:** Like an upside-down mouse. Rather than roll the mouse around, you use your thumb or index finger to roll a ball on top of the mouse. The whole contraption stays stationary, so it doesn't need a lot of room, and its cord never gets tangled.
- **Stylus mouse:** Another mouse mutation enjoyed by the artistic type is the stylus mouse, which looks like a pen and draws on a special pad.

- **Cordless 3-D mouse:** This kind of mouse can be pointed at the computer screen like a TV remote.

Mouse pointer

The mouse pointer is the little indicator that moves when you move the mouse. Sometimes, the mouse pointer has a different shape, depending on where it's currently resting or to show that a computer is processing something so you must wait.

The following diagram illustrate various shapes of mouse



Functions of a mouse

- Mouse is used to perform the following
  - Selecting text or graphics
  - Drawing auto shapes
  - Pointing on menus
  - Displaying menus
  - Inputting data

Good mouse use

- Place the mouse on a flat surface
- Gently hold the mouse with your right hard, using the thumb and the two fingers
- The index finger should rest on the left button while the middle rest on the right button

Advantages of using a mouse

- It is inexpensive
- It is easy and convenient to use

- Most computers software accommodates the use of a mouse

Disadvantages of using a mouse

- It cannot be used to input text
- It is slow when selecting menu options
- It requires a flat surface to operate
- It is not very accurate for drawing purpose

Practical hands on skills

Turning on the computer (Booting)

- This is also called booting of a computer
- **Booting** is the process when the computer transfers the application program from the main memory to the application areas or working area.

Types of booting

Cold Booting - when the computer is being switched on for the first time

Warm Booting - when the computer is being re-started while it was on operation.

To boot your computer system follow the steps:

1. Switch on the power button on the system unit (cold Boot)
2. Press the button to turn on the monitor
3. Wait for your computer to load operating system\

Logging in a computer system

- Open the log in window by pushing **Ctrl**, **Alt** and **Delete** buttons simultaneously.
- If your user name is visible:
  - Type in your password into the **Password** field.

- Click the arrow.

- If some other user name is visible:
  - Click the **Switch User** button.
  - Choose **Other User**.

  - Type your username and password into the respective fields.
  - Click the arrow.

Logging off

- Click the **Start** button in the lower left corner.
- Click the arrow button which can be found next to the **Shut down** button.

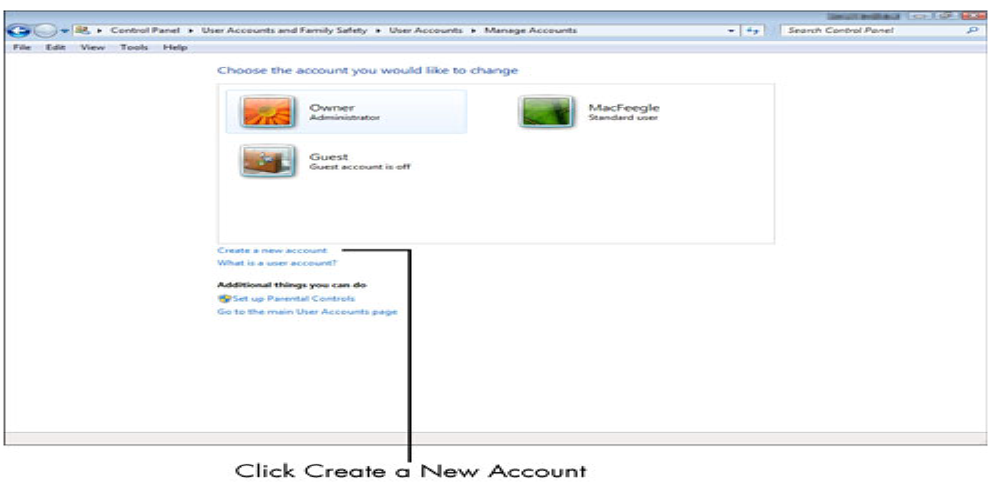
- Choose **Log off**.

Creating user accounts

To create a new user account:

1. Choose Start→Control Panel and in the resulting window, click the Add or Remove User Accounts link.

   The Manage Accounts dialog box appears.



2. Click Create a New Account.

   The Create New Account dialog box appears.

The click on create account

Changing user Account types

To change account type follow the following steps

- Open Control Panel.
- Click the **Change account** type option.
- Select the **account** that you want to modify.
- Click the **Change** the **account** type option.
- Select either Standard or Administrator depending on your requirements.
- Click the **Change Account** Type button.

Creating a new user with a User Passwords

- Select the Windows Start menu button.
- Select Control Panel .
- Select User Accounts .
- Select Manage another account .
- Select Create a new account .

- In the New Account Name text box, type a name for the new account.
- Click Create Account .
- The new user now appears in the Choose the account you would like to change box.
- "Make changes to your **user account**", click Set a **password**.
- In the "New **password**" and "Confirm new **password**" fields, enter the **password**.

Using the mouse

1. Roll the mouse over the screen to move the pointer
2. One left click for basic functions
3. Two left clicks for opening programs
4. One right click for various help various help functions

Navigating through windows operating system

Recall that the windows operating system has a WIMP based GUI where W is for Windows. Some examples of windows that you encounter on a windows operating system are the following:

- Desktop window
- Folder window
- Application window

Folders of a computer

In **computers**, a **folder** is the virtual location for applications, documents, data or other sub-**folders**. **Folders** help in storing and organizing files and data in the **computer**. The term is most commonly used with graphical user interface operating **systems**.

Some examples of common folder in a windows operating system are

- Desktop folder
- Documents folder
- Music folder
- Downloads folder
- Pictures folder

Windows desktop folder (desktop window)

The desktop- The desktop is the onscreen work area analogous to a physical desktop. Everything else you see on the screen is actually resting on top of this virtual desktop. from the moment you start your computer, to the moment you turn it off, the desktop is always there. The desktop has two parts – The **work area**, and the **taskbar**. The work area has desktop icons. The taskbar has the start button, taskbar buttons/icons, toolbars and notification area.

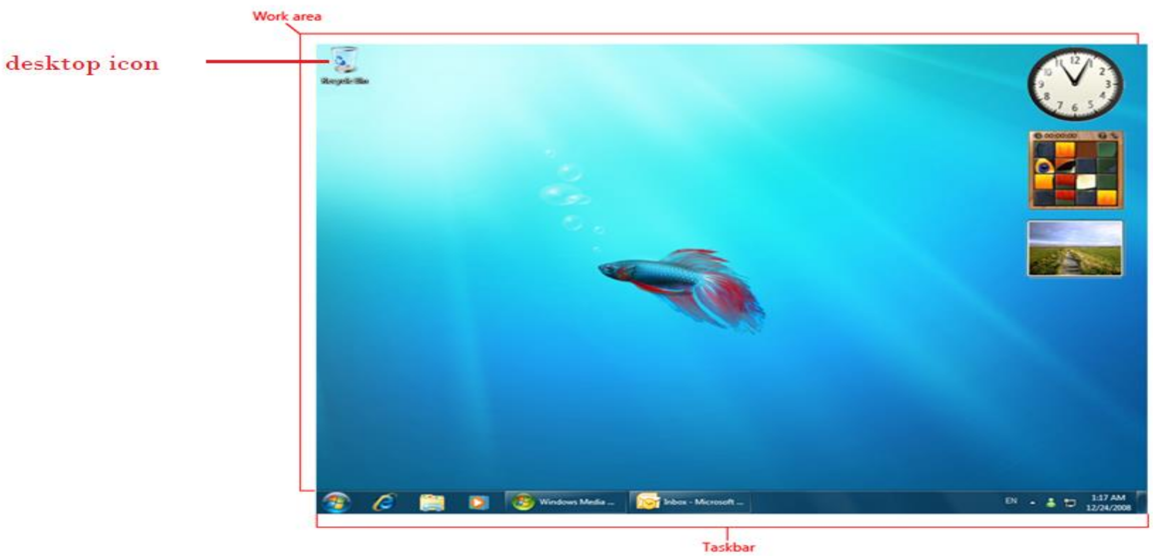


Figure x.x: windows desktop

Work area. The onscreen area where users can perform their work, as well as store programs, documents, and their shortcuts. Although in technical terms the desktop refers both to taskbar and work area, people loosely take the work area as the desktop and they use it interchangeably. If you right click the work area you get desktop properties. The appearance of the desktop can be changed using the following desktop properties: Display settings and Personalize.

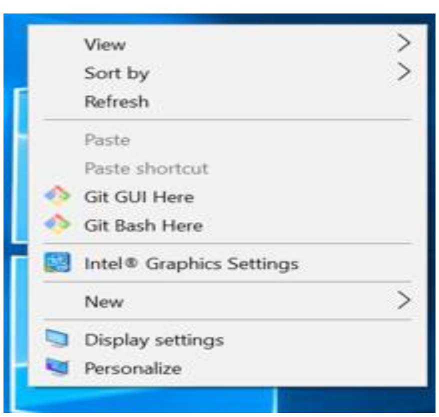


Figure x.x: Desktop properties

The desktop icons-Each little picture on the desktop is an *icon*. Each icon, in turn, represents some program you can run, or some location on your computer where things are stored. The Recycle bin icon is one of the most common icon that sits on the desktop- the recycle bin keeps deleted items. If items need to be permanently deleted, the recycle bin must be emptied. If you want to use an item from the recycle bin then it needs to be restored. A restored item returns to the previous location where it was deleted from.

The taskbar - This is a special toolbar that is unique to the desktop. You can use the taskbar for such tasks as switching between open windows and starting new applications. The taskbar includes the following:

- Start menu
- Quick Launch bar (Windows Vista and earlier only)
- Taskbar buttons
- Toolbars (optional)
- Notification area

Figure x.x: windows Taskbar

Start Button - The start button is where you can start any program on your computer. When you click the Start button, the **Start menu** opens.

The **Start** menu contains commands that can access programs, documents, and settings. Examples of some contents (commands) of the start menu are accessories- paint, calculator, notepad, snipping tool; Microsoft office programs – word, excel, access, and so on; file explorer – my documents, my pictures, my computer (This PC), and so on; **Control Panel**; **Games**; **Help and Support**; **Shut down**; **Search programs and files**.

Figure x.x: Windows Vista start menu

The Notifications area - The Notifications area contains the icons that keep you posted as to the status of various programs or services running on your computer.

Folder window

Each window can be divided into several sections: folder content, Navigation pane on the left, the Preview pane on the right, the Search pane above, and the Details pane

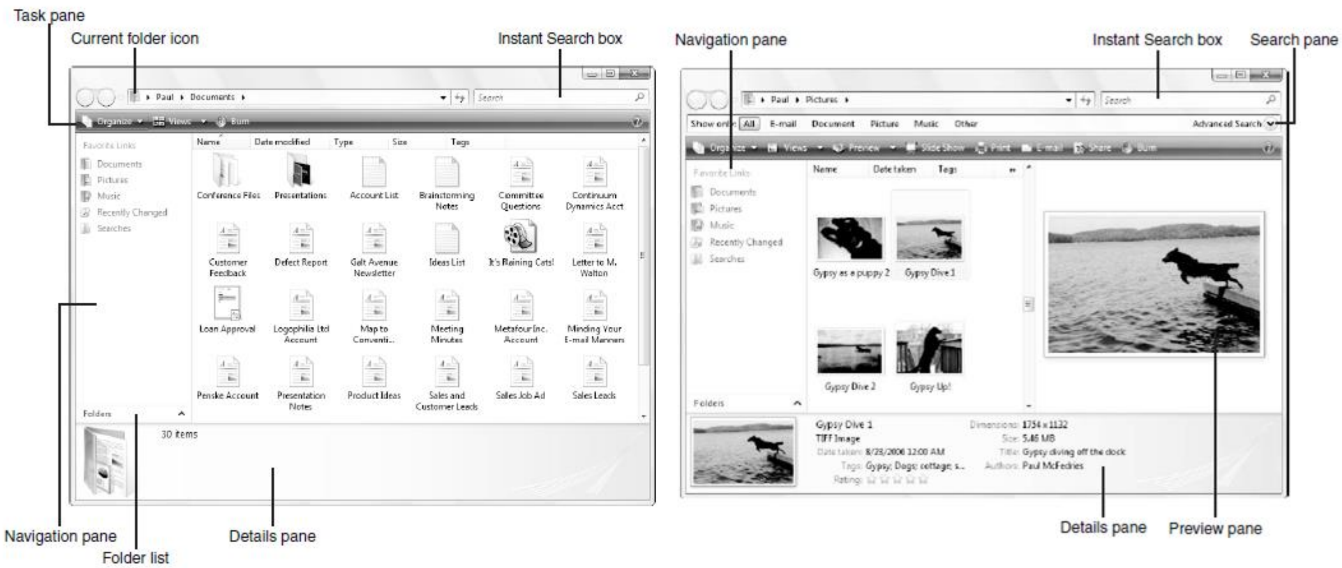


Figure x.x: folder window

Application window

Most windows have certain elements in common, such as a title bar and a menu bar. Not all windows, however, have every element.

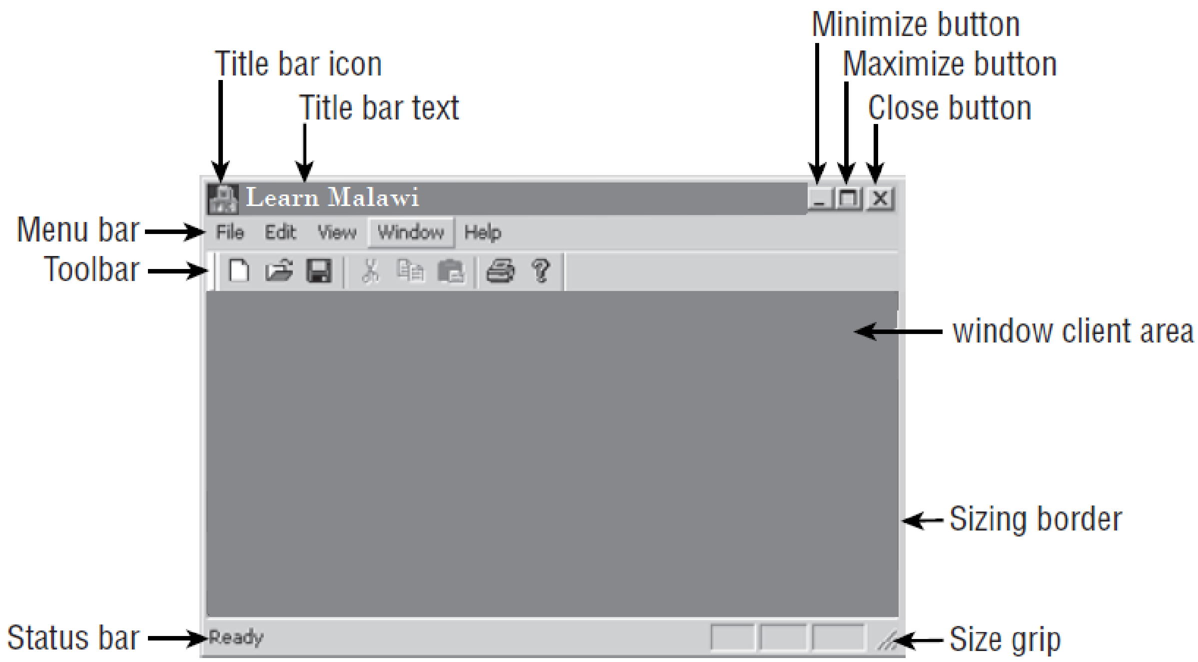


Figure x.x: elements of an application window

A menu bar displays commands or options in drop-down menus. The **Close button** in the upper right corner of a window closes the window. The **Maximize button** in the upper right corner of a window enlarges the active window so that it fills the entire desktop. The **Minimize button** in the upper right corner of a window reduces the active window to an icon on the Windows taskbar.

Menus

Menus are hierarchical lists of commands or options available to users in the current context. Commands are actions users can take while using your application.

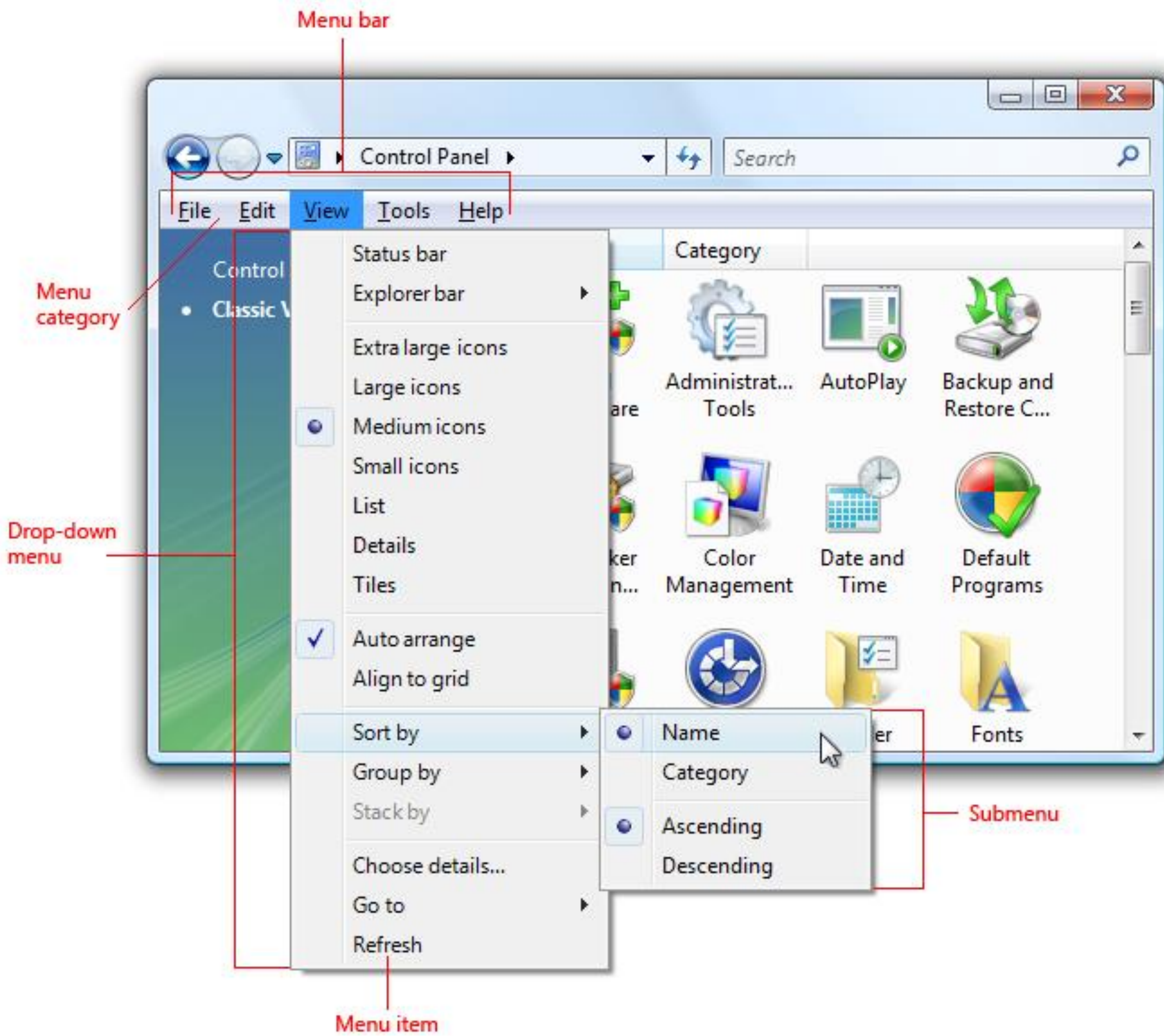


Figure x.x: Menu example. Menu items are the ones we are calling commands or options.

Toolbars are a way to group commands for efficient access. Toolbars can be more efficient than menu bars because they are direct (always displayed instead of being displayed on mouse click), immediate (instead of requiring additional input) and contain the most frequently used commands (instead of a comprehensive list). In contrast to menu bars, toolbars don't have to be comprehensive or self-explanatory just quick, convenient, and efficient.

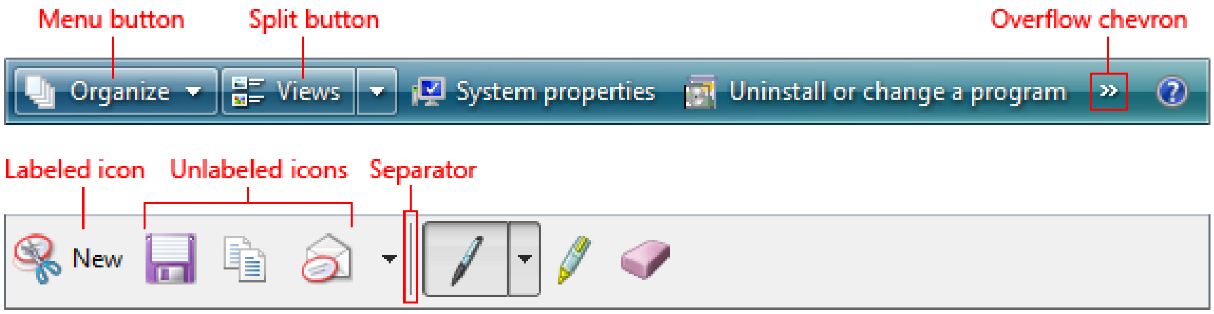
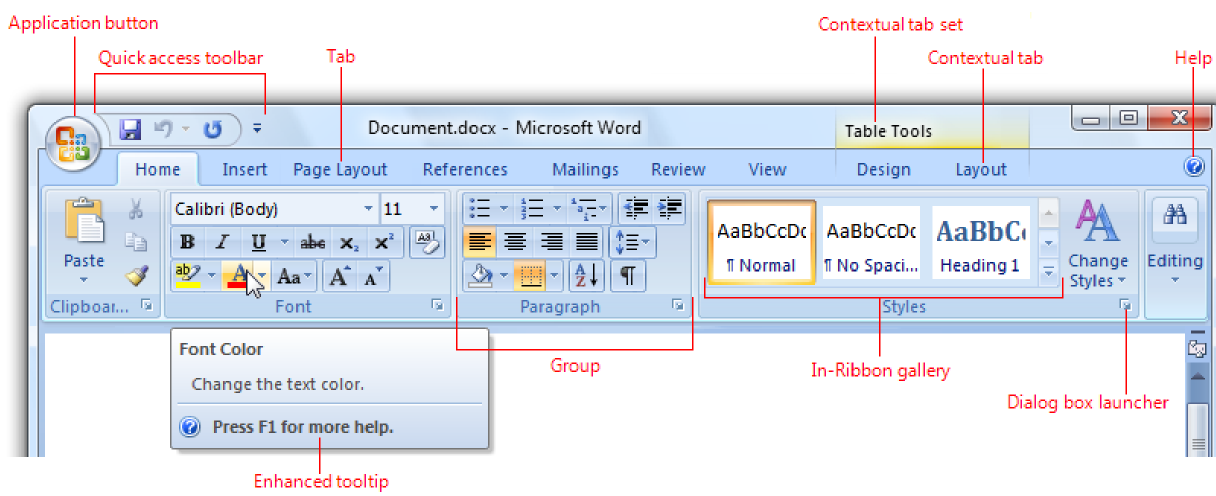


Figure x.x: toolbar example

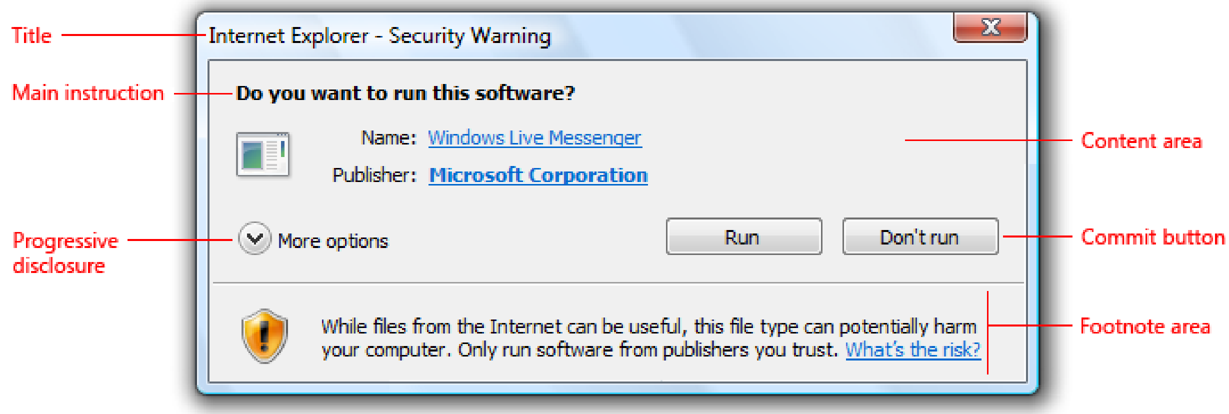
A ribbon is a command bar that organizes a program's features into a series of tabs at the top of a window. A ribbon can replace both the traditional menu bar and toolbars.



Ribbon example

Dialog Boxes

A dialog box is a secondary window that allows users to perform a command, asks users a question, or provides users with information or progress feedback. There are two types of dialog boxes – modal and modeless. **Modal dialog boxes** require users to complete and close before continuing with the owner window. **Modeless dialog boxes** allow users to switch between the dialog box and the owner window as desired.



Dialog boxes consist of a title bar (to identify the command, feature, or program where a dialog box came from), an optional main instruction (to explain the user's objective with the dialog box), various controls in the content area (to present options), and commit buttons (to indicate how the user wants to commit to the task).

Standard Icons

Standard icons are the error, warning, information, and question mark icons that are part of Windows.

The standard icons are notable because they are built into many Windows application programming interfaces (APIs), such as task dialogs, message boxes, balloons, and notifications. They are also commonly used on in-place messages and status bars. The standard icons have these meanings:

- **Error icon.** The user interface (UI) is presenting an error or problem that has occurred.
- **Warning icon.** The UI is presenting a condition that might cause a problem in the future.
- **Information icon.** The UI is presenting useful information.
- **Question mark icon.** The UI indicates a Help entry point.

Windows Search facility

Windows provides search at several levels for example a search facility is provided along the start menu. A search facility is also provided along the folder window. The Microsoft search allows supports searching via tags, comments, and document metadata.

Turning off the computer

1. Go to the start icon on the button of screen
2. Left click once
3. Left click once on turn off computer or Shutdown
4. Left click turn off computer. Wait until computer turns off then off screen and power

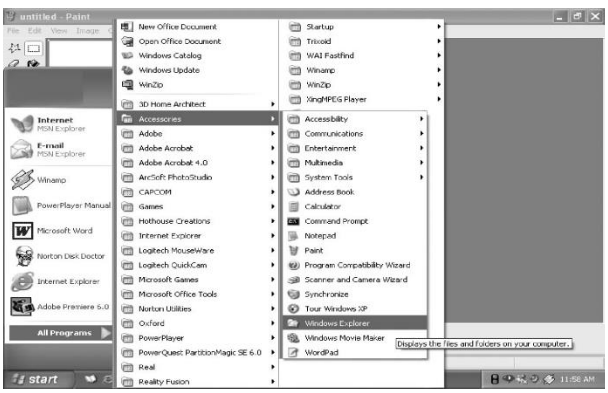
File Management in a computer system

- File management in windows can be done through Windows explorer or My Computer.

- Windows Explorer displays the hierarchical list of files, folders, and storage drives (both fixed and removable) on your computer.
- It also lists any network drives that have been mapped to as a drive letters on your computer.

Using Windows Explorer

Windows offer another utility "Windows Explorer" which helps you in working with files and folders on your computer.

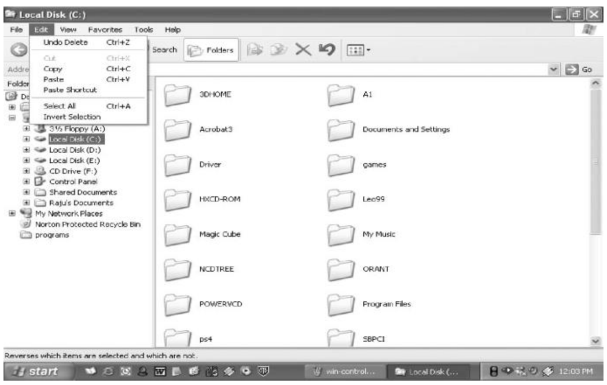


To open Windows Explorer,

- Click on Start,
- Point to All Programs,
- Point to Accessories, and then click on Windows Explorer

The left pane of the Explorer window shows a hierarchy of all the drives, folders and desktop items on your computer. A drive or folder that contains other folders has a plus sign to the left of the icon. Click the plus sign to expand it and see the folders inside.

Opening drives and folders



Two drives nearly all computers have are a floppy drive (drive A:) and a hard drive (drive C:). If you have more than one drive, then they are named D:, E: and so on. If you have a CD drive or a DVD drive, it also is named with a letter.

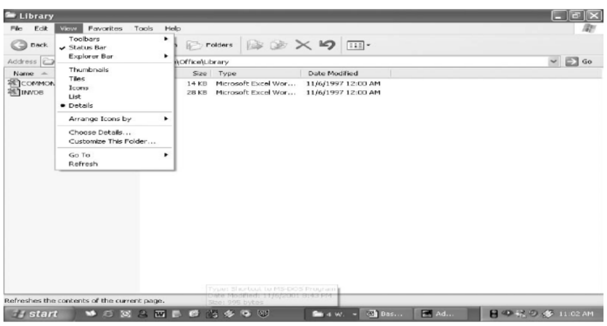
- Opening a hard drive is easy. Just double click the icon representing the drive you want to open. Files and folders contained in the drive are now shown in the opened window. Now for opening a folder, double click its icon.

Coping or Moving a file or Folder using My Document

- Click on Start, and then click on My Documents.
- Click the file or folder to be copied. More than one file or folder can be copied at a time.
- To select more than one consecutive files or folders, click the first file or folder, press and hold down CTRL key, and then click the last files or folders.

- To select non-consecutive files or folders, press and hold down CTRL key, and then click each of the files or folders to be copied.
- Under Edit menu, select Copy.
- Select the target drive or folder to which you want to copy the files
- Under Edit menu, select Paste to copy the desired file or folder to the target drive.

View file details



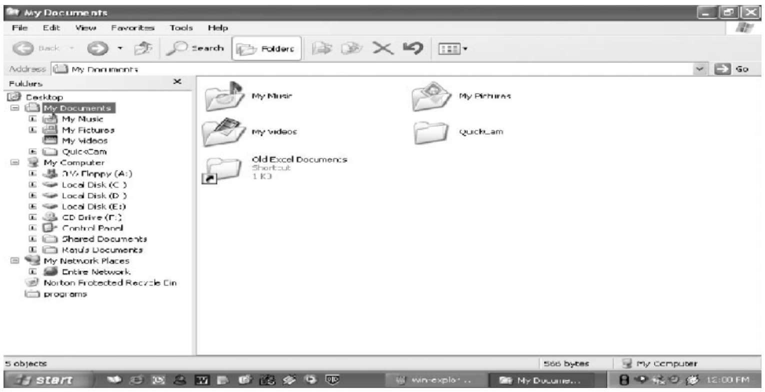
1. Click on Start, and then click on My Documents.

2. Double-click the folder that contains the files to be viewed.

3. On the View menu, click Details.

4. It will display all the details about the files such as Name, Type, size etc.

Copying and moving files using Explorer



1. Click Start, point to All Programs, point to Accessories, and then click Windows Explorer.
2. Make sure the destination for the file or folder you want to move is visible.
3. Drag the file or folder from the right pane and drop it on to the destination folder in the left pane to move the file or folder there.
4. If you drag an item while pressing the right mouse button, you can move, copy, or create a shortcut to the file in its new location.
5. To copy the item instead of moving it, press and hold down CTRL while dragging.
6. If you drag an item to another disk, it is copied, not moved. To move the item, press and hold down SHIFT while dragging.
7. Dragging a program to a new location creates a shortcut to that program. To move a program, right-click and then drag the program to the new location.

Create a new folder

- Folders help you to organize your files. You can create a folder either by using My Computer window or through Windows Explorer.
- You can create a Folder in any existing disk drive or folder or on the windows desktop. The steps for creating a folder are:

1. Click on Start, and then click on My Documents

2. Under File menu click New and select Folder.

3. A new folder is displayed with the default name, New Folder.

4. Type a name for the new folder, and then press ENTER.

5. A new folder can also be created directly on the desktop by right-clicking a blank area on the desktop, pointing to New, and then clicking Folder.

Rename a file or folder

1. Click on Start, and then click on My Documents

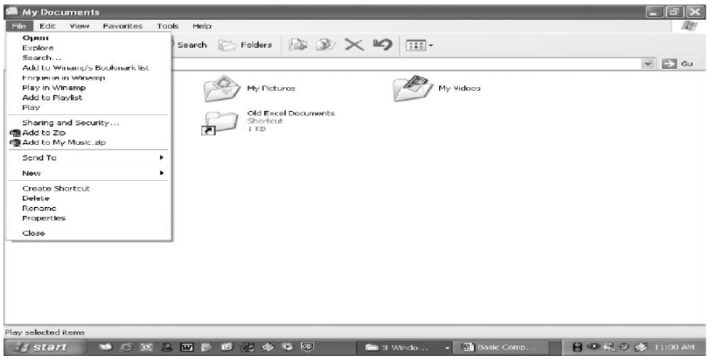
2. Click on the file or folder you want to rename.

3. Under File menu click on Rename.

4. Type the new name, and then press ENTER key.

5. Alternately file or folder can also be renamed by right-clicking it and then clicking on Rename.

Delete a file or folder



1. Click on Start, and then click on My Documents

2. Click on the file or folder you want to delete.

3. Under File menu click on Delete.

4. Files or folders can also be deleted by right-clicking the file or folder and then clicking Delete.

5. Deleted files or folders are stored in the Recycle Bin, till they are permanently removed from the Recycle Bin.

6. To retrieve a deleted file, double-click the Recycle Bin icon on the desktop. Right-click on the file to be retrieved, and then click Restore.

7. To permanently delete a file, press and hold down SHIFT key and drag it to the Recycle Bin.

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Computer studies form 1

Chapter 5

SAFE USE OF COMPUTERS

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CHAPTER 5. SAFE USE OF COMPUTERS

Protection of computers and users

Computers are delicate devices that need to be handled carefully. To protect the computers the users do the following:

- Security of computer hardware and soft ware
- Powering the computer on/off
- Protecting users from hazards.
- Safe disposal of computers components.
- Protecting against malware, viruses and worms.

Hazards to system

The following are hazards to computer system

a. **Power surge/cut**

- Power surges or unexpected power cuts can not only cause instant loss of data but can also fry a processor rendering it useless.
- It is not just power from the grid that causes problems either, lightning can surge through cables (even phone wires) frazzling your system and a build up of static can cause similar results.

b. **Dirt/Dust**

- dusty environment will clog a computer and block cooling vents causing a computer to overheat.
- Dust can also contain conductive material and particles can stick to circuit boards and cause a short circuit.
- A cloud of dust can also explode if it comes into contact with a source of ignition (like a computer).
- Even home computers if not properly cleaned can succumb to problems caused by too much dust.

c. **Water/fluids**

- Of course computers are electrical and with all electrical equipment, computers and water do not mix, just one spilt cup of coffee could see the end of your PC.

d. **Heat**

- Processors can run exceptionally hot and if a computers cooling system is inadequate (because the machine has been upgraded, overclocked or just clogged up with dust and grime) it will only be a matter of time before it packs up for good.

e. **Cold**

- Just as with heat, computers don't enjoy the cold too much either.
- Processors will not operate at all if the operating temperature is too cold as condensation inside the machine can freeze and expand damaging the processor and electronics.

f. **Knocks/bangs**

- Computers are sensitive machines, simply moving a PC to another room can cause havoc, disrupting the delicate circuitry and hard drives.
- Dropping a computer or severe knocks and bangs will permanently damage the circuits and processors or dislodge wiring.

Hazards to computer users

The following are some of hazards to computer users

Musculoskeletal Problems

- This includes areas of your body such as your back, neck, chest, arms, shoulders and feet.
- Having sore muscles and complaints of the muscles being tired are common.

- Numbness may occur in the arms and hands.
- These troubles may occur because the posture you assume when using the computer is most likely incorrect.
- You may find that you are sitting in an uncomfortable chair, or that you have a workstation that is not ergonomically correct for your body.

Tips to Consider

- Find a correct height for both your desk and chair so that your computer screen is at eye level or slightly lower.
- Sit with your back straight, legs at 90 degree angles to the floor, and feet resting flat on the floor.
- ALWAYS take small breaks from your computer work to stretch your muscles, keep your blood flowing, and to rest your eyes.

2. Vision Problems

- Computers are notorious for their bright lights, glare and flickering images that can cause strain on your eyes.
- Finding that you constantly focusing on the screen with delays in blinking can result in drying out your eyes.

Tips to Consider

- Make sure to adjust the brightness on your computer screen so that your eyes are not as strained.

- For example, if you are sitting in a dark room your computer screen will most likely be very bright and cause your eyes to strain, so to save your eyes you should lower the brightness.
- Tilt your screen to decrease any glare.
- Maintain a proper vision distance from the screen, and do not forget to blink.

3. Repetitive Stress Injuries

- You may notice pain in your neck, shoulders, or really anywhere from the shoulders to your fingers related to repetitive muscle use.
- Using the computer may cause you to use your muscles in an odd way that may cause increased stiffness, pain, or swelling in any of those areas.
- One of the most common conditions related to repetitive use of your muscles when using the computer is carpal tunnel syndrome.

Tips to Consider

- Place your mouse at a location next to the keyboard that will require you to move your whole arm to get to it rather that just twisting your wrist outward to reach it and move it.
- Type gentle to decrease the stress put on each of your fingers.
- Keep your wrists flexible when typing; avoid keeping them fixed in a certain position; keeping them flexible will avoid repetitive, strenuous stress.

- Relax your arms and try to get a few stretches in when you are not typing or using your mouse.

4. Headaches

- Headaches are common and may occur because of the increased muscle tension or from pain in the neck.
- Any vision problems, or continued strain on the eyes can also cause headaches.

Tips to Consider

- Attend regular eye exams in order to work toward correcting any vision problems.
- Try your best to keep your neck straight in front of the computer and take breaks.

5. Obesity

- Prolonged use of computers may lead to an overall sedentary lifestyle that lacks adequate physical activity and/or exercise.
- In children prolonged use of computers, or electronics in general, is a major contributing factor to obesity.

Tips to Consider

- Set limits for using electronics.
- Encourage outdoor play or a certain hobby that may take away time spent using electronics in order to lead a more active lifestyle.
- As for adults, if your occupation requires computer use for up to 8 hours daily, you should not use a computer again when you get home…you should take a break and try to squeeze in some exercise until you go back to work.

6. Stress Disorders

- Technology impacts our behaviors and emotions.
- Prolonged use of computers may be accompanied by poor health and increased pressure placed on you in your workplace environment, which could both lead to stress.
- The longer your stress occurs and is left untreated, the greater your chances are of contracting more serious health problems.
- Stress can lead to decreased attention span, lack of concentration, dizziness and becoming easily burned out.

Tips to Consider

- Promote your own health and prevent future health conditions or worsening the ones you already have by seeking treatment options for any stress that you may encounter.
- Try things from yoga, to natural remedies, to medications as prescribed by a medical provider to combat your stress.

7. Laptop Use Injuries

- Laptop injuries fall into a category of their own; there is a growing use of laptops that continues to cause more pain and strain among those individuals who use them.
- Laptops are designed for short periods of use for those who do not have access to desktop computer.
- In present day individuals choose to use laptops over desktops more frequently, due to convenience.
- The problem is this: the screen and keyboard are very close together and there is really no right way to use a laptop because if you position the screen at the right height for your back and neck, it will cause you to have to lift your arms

and shoulders too high to use it and vice versa…no matter what it will probably cause a problem for you somewhere.

Tips to Consider

- Use a desktop computer that is set up ergonomically-correct for you as frequently as possible; only use a laptop intermittently.
- Use separate laptop equipment, such as a wireless mouse or keyboard or a laptop stand.
- As always, take frequent breaks.
- If you have to take your laptop with you, make sure to carry it in a backpack or luggage; otherwise it may cause extra strain on your muscles from carrying it.

8. Sleeping Problems

- Artificial lighting that is given off from computer screens can actually trick your brain and suppress its release of melatonin – the substance that assists your sleeping patterns.

Tips to Consider

- Refrain from using a computer right before going to bed.
- Resort to reading a book or something to that degree prior to going to bed, so falling asleep may come more easily for you.

9. Hearing Loss From Headphones

- At times you may be required to use headphones in order to better concentrate on something or maybe because the background noise level is too high.
- Frequently individuals will turn the volume up very high, when actually it would not even need to be close to that volume to hear the audio effectively.
- Listening to audio with headphones on a consistent basis and using a volume that is too high can result in hearing impairment.

Tips to Consider

- Keep the volume of your headphones down to a tolerable level, one that blocks out any extra noise but that is just loud enough for you to hear.
- Listening to your headphones at approximately 80 decibels is recommended; if you are unaware of what that sound level is it can easily be researched.

10. Increased Risk of Blood Clots

- Being immobile and not allowing your blood a chance to get moving around your extremities may cause it to pool, creating build-up of blood cells that will eventually clot (or stick together) due to not being able to be circulated around.
- Blood clots can be life-threatening if they break away from where they are lodged and travel to another area, such as your lungs.

- Sitting in one position for too long (especially if your legs are crossed), generally over a period of over 4 hours, can greatly increase your risk for this.

Tips to Consider

- Avoid crossing your legs when using a computer for an extended period of time.

- Take many breaks and stretch your legs to get the blood flowing to decrease the chance of it pooling in your extremities.
- If you do have to sit for an extended period, make sure to bend and move your extremities even while sitting because any little bit will he

Measure that protect hardware and software

- Burglar proofing the room- It includes fitting grills on doors, windows, and roof to control forceful entry into a computer room.
- Installing fire prevention and control equipment such as smoke detectors, non-liquids based fire extinguishers.
- The room should be well laid out with enough space for movement and setup of computers.
- Powering the computer on/off, the computer should be always be switched off using the correct procedures to avoid loss of data and destruction of software.
- Dust and dump control, the room should be installed with good windows curtains and an air conditioning system that filters dust particles.
- Insulating power cables, cables and power sockets should be well installed and be of the correct power rating to avoid short circuit.
- Avoid taking meals in the compute laboratory, food particles may fall in moving computer parts like keyboards.

Data and software loss

Data loss is defined as unexpected missing of data files from a computer system

Factors that can lead to software and data loss.

There are many factors that may lead to data loss or software loss. Some of them have been described below

1. **Deleting files accidentally**

- Data files deleted from the computer unintentionally by the user

2. **Viruses and damaging malware**

- Data files may be damages by viruses
- There are numerous new viruses which attack computers every day.
- Being connected to worldwide network has many advantages; however, it opens computers to many serious risks. may differ greatly but the majority of viruses affects operational software, misuses Internet connection and damages stored data.

3. **Mechanical damages of hard drive**

- Hard drives of computers are the most fragile parts of computers; they break down more often than any other device connected to computing.
- There are so many moving parts inside of hard drives that it is no wonder they break down so easily.

4. **Power failures**

There are two adversary effects of power failures.

- When you are halfway through writing a long article and you have not saved it yet, then in case of power going out you lose your data. This is perhaps the simplest example but imagine working with sophisticated databases or creating detail-rich graphic illustrations...you get the idea what can happen if you lose power during working.
- Another, even deeper problem may arise when power failures affect operation

systems or hardware of computers. Shutting computer down suddenly without proper shutdown procedures may cause problems with rebooting operation system later.

5. Theft of computer

- It is a real tragedy to lose both computer and data at the same time .
- There is always the danger of burglars breaking in to your home and stealing electronic devices. While traveling, you may leave your laptop unattended; lose it in an airport, conference centers, coffee shops or any other crowded place.

6. Spilling coffee, and other water damages

- Since laptop usage has been growing during past few years, the damages caused by spilling drinks on to the computers have become more often as well.
- Average laptop does not have extra covers to protect internal parts from getting soaked
- Liquids cause short circuit of important electronic components and they are really hard to recover afterwards.

7. Fire accidents and explosions

- Explosions happen rarely but fire most probably completely destroys both you computer and data saved on it.

- Fire is also dangerous to the backups that are stored in the same house. For example, having weekly backups stored on an external hard drive which is kept in same building does not help much if the building burns down.
- In this case, both computer and backup drive will be destroyed and data will be completely lost. The safest practice against fire is to make regular backups and keep them in other (different) locations.

Measures against loss of data and software

1. Up – to- date antivirus program must be installed in the computer.
2. Make sure you scan all external storage media and e-mail for viruses
3. Take regular backup of all the important data and software and store in a safe place.
4. Connect the computers to a UPS to prevent data and software loss during power surge
5. All data must be properly saved and the computer must be safely shut down.
6. Enforce physical security to avoid theft of computer and computer accessories
7. Use trusted software to a void corrupting the operating system.
8. Write protect the storage media so that only the trusted users can save content on the storage media.
9. Handle the storage devices and media with care to avoid damage due to dropping, dust heat, water or magnetic effects.

Safe disposal of computers components

Computers that you no longer need should be disposed of with great care. The data on your computer can easily be accessed whether you sell, scrap, give away or donate it, and even 'deleted' data can be retrieved with relative ease by criminals.
The following are **risks** associated with careless disposal of computers

1. The personal information stored in files on your computer can be accessed and used for criminal activity.
2. Any passwords stored on your computer could give access to secure websites holding your personal and financial information.

3. Any browsing history stored on your computer can be accessed.
4. Emails stored on your computer can be accessed.
5. Disposing of your computer without having retrieved the information you may need in the future may cause inconvenience or disruption.

Guidelines for Safe Disposal of computers

1. Copy all of the data you will need in the future, on to your new PC or storage device, or back it up in the cloud.
2. Fully erase the hard disk(s) so that any personal information is completely deleted. Simply deleting files is not enough to permanently erase them.
3. Ensure that any **CDs** or **DVDs** which contain your data are removed from the computer.
4. Don't forget that your **CDs, DVDs**, memory cards, **USB** sticks and other **USB** connected devices may also contain your sensitive data and should be disposed of with equal care.
5. If the computer equipment is at the end of its life and you do not intend to sell it or give it away, take it to a proper disposal facility, which will ensure that is dismantled and the components recycled correctly and responsibly

6. Donate, resell, or recycle.

- You can't just throw your computer away. There are various components within a computer that are either environmentally toxic, such as mercury and lead, or valuable, such as copper, aluminum, and gold.
- Because of this, there are many programs available to help take your old computers off your hands.

Computer Viruses

A computer virus is a type of computer program that, when executed, replicates itself by modifying other computer programs and inserting its own code.

Types of Computer Viruses

The following are the common types of computer viruses

1. Boot sector virus

This type of virus can take control when you start — or boot — your computer. One way it can spread is by plugging an infected USB drive into your computer.

2. Web scripting virus

This type of virus exploits the code of web browsers and web pages. If you access such a web page, the virus can infect your computer.

3. Browser hijacker

This type of virus "hijacks" certain web browser functions, and you may be automatically directed to an unintended website.

4. Resident virus

This is a general term for any virus that inserts itself in a computer system's memory. A resident virus can execute anytime when an operating system loads.

5. Direct action virus

This type of virus comes into action when you execute a file containing a virus. Otherwise, it remains dormant.

6. Polymorphic virus

A polymorphic virus changes its code each time an infected file is executed. It does this to evade antivirus programs.

7. File infector virus

This common virus inserts malicious code into executable files — files used to perform certain functions or operations on a system.

8. Multipartite virus

This kind of virus infects and spreads in multiple ways. It can infect both program files and system sectors.

9. Macro virus

Macro viruses are written in the same macro language used for software applications. Such viruses spread when you open an infected document, often through email attachments.

<u>What are the signs of a computer virus?</u>

A computer virus attack can produce a variety of symptoms. Here are some of them:

- **Frequent pop-up windows.** Pop-ups might encourage you to visit unusual sites. Or they might prod you to download antivirus or other software programs.
- **Changes to your homepage.** Your usual homepage may change to another website, for instance. Plus, you may be unable to reset it.
- **Mass emails being sent from your email account.** A criminal may take control of your account or send emails in your name from another infected computer.

- **Frequent crashes.** A virus can inflict major damage on your hard drive. This may cause your device to freeze or crash. It may also prevent your device from coming back on.
- **Unusually slow computer performance.** A sudden change of processing speed could signal that your computer has a virus.
- **Unknown programs that start up when you turn on your computer.** You may become aware of the unfamiliar program when you start your computer. Or you might notice it by checking your computer's list of active applications.
- **Unusual activities like password changes.** This could prevent you from logging into your computer.

Ways in which viruses are spread

- Sharing of infected removable storage media
- Downloading of untrusted freeware and shareware
- Through e-mails
- Accepting software updates from invalidated sources

Ways of preventing the spread of virus viruses

- Avoid sharing of infected removable storage media
- Avoid downloading untrusted freeware and shareware.
- Install and update trusted antivirus software

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Computer Studies Form 2

Chapter 1

COMPUTER AND SOCIETY

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CHAPTER 1: COMPUTERS AND SOCIETY

Areas where Computers are used

There are so many fields in which computers are used some of these areas are as follows

A. Financial systems
B. Accounting system
C. Banking systems
D. Sales and marketing
E. Retail system
F. Reservation of epos terminal
G. Education system
H. Communication system
I. Industrial system
J. Scientific and research systems
K. Scientific and research systems
L. Library systems
M. Entertaining and multimedia systems
L. Transport systems
M. Law enforcement systems
N. Computers in agriculture
N. Human resource system
O. Communication system

A. Financial systems

- In financial systems computers enables organization to manage their financial
- The financial institutions include:

1. Accounting systems

2. Banking systems
3. Payroll processing systems

B. Accounting systems

- Accounting systems are popular in business management
- Computers in accounting systems are used to enter customer's order and billing sup systems records for generating customers' orders
- For keeping of the items in stock and help in management of items reorder
- Computers also help in hook keeping and produce reports in data being processing.

Business accounting activities

Here are the six accounting business;

1. Customer order and billing orders and delivery.
2. Inventory management – for keeping track of items in stock and to know items to be reordered
3. General ledger accounting – for summarizing the financial transactions.
4. Accounts receivable for keeping track of record owned by each other such as taxes
5. Accounting payable – keep track of record business owes others such as taxes
6. Cash book – for recording daily cash transaction in the business

C. Banking systems

- The banking industry is one of the earliest customers of information and communication technology.
- The computers in banking services perform the following tasks.

1. Process customer transactions

  - Computers in bank are used for easy carrying out financial transactions such as

- Customers deposits
- Withdrawing money through ATM machines
- Calculating interest on savings and loans
- Creating reports for each customer account

2. Cheque clearing and processing

- Computerized cheques clearing and processing is made possible due to special characters on printed using ink containing magnetic particles
- Cheques are processing using magnetic ink character reader which enters all details on a computer of processing.

3. Electroning funds transfer (eft)

- EFT is the movement of money between different accounts using a computerized system
- EFT transfer funds from the payers account by crediting the payee's account electronically in the same or different bank.

4. Internet banking

- It enables bank users to access their bank accounts through the internet.
- Bank customers can query, account statement, pay bills and transfer fund electronically
- Internet banking is also known as e – banking

5. Mobile banking

- It is the process where account holder in different banking system performs transaction using mobile phones.

6. Payroll processing systems

- Computers in pay roll systems are used to process accurate information about employee incomes including gross pay, deductions and the net pay.
- Payroll systems are designed to produce reports that can be analyzed to meet specific organizational information needs such as pay slips and reports that shows a breakdown of payroll expenses against production and income.

D. Sales and marketing

- Without proper marketing a business cannot survive in a competitive environment
- Computers in sales and marketing are being used in a number of ways to promote sales and marketing.

Computers in sales are used in the following areas

1. Electronic commerce or e-business
2. Electronic presentation
3. Advertising

E. Retail systems

Computers are becoming more popular in retail stores as :

- Supermarkets
- Distribution outlets.
- Computers are used in stores for stock control and transaction handling.
- Shops owners processing data to be ordered and reordered
- Most retail transaction are handled using and electronic point – sale (**EPOS)**
- **EPOS** is a computer terminal used in retail stores to input and output data.
- **EPOS** terminal has all the normal facilities of cash register, direct data capture devices such as bar reader, card reader, a monitor and a receipt printer.
- Transaction at the point of sale terminal my involve the following steps;
  - The bar code leader is passed over the items bar code to generate the item number.
  - Using the number, the computer searches for the item with corresponding number in the product file.
  - Once record is found its description and price is used for processing sales.

Advantages of epos terminal

1. Correct price are used at the checkout counter.
2. They are fast because the attendant does not have to enter details manually.

F. Reservation of epos terminal

Computer in reservation systems are used mainly to make bookings in areas such as

- Airlines
- Hotels
- Car rental and theaters
- Bookings are made from a remote terminal connected to a centralized computer Known as server.
- To access a server a client makes enquires via the remote terminal referred to as client computer or using mobile phone

G. Education systems

Most educational institutions uses computers for administrative task such as :

- Compiling examinational reports
- Writing memos
- For accounting purposes.
- Computers are also playing an increasingly important role in educational institutions in the following ways

1. Computers aided tutorials , where by computers act as teachers through electronic programmers

2. Problems solving through the use of electronic programs for mathematical and other science researches.
3. E-learning is where by people pasue different courses through the internet electronic.

H. Communication systems

- Communication refers to the distribution of data information from one person to another.
- Effective and efficient data communication is achieved by use of high – speed electronic devices such as computers.

- Examples of communication systems are:

1. The internet

- It facilitates the transfer of communication from one location to another location services available on the internet transfer;
- World wide web (www)
- Electronic mail (e-mail)

2. Facsimile (fax)

- Facsimile in short (fax) machine is a telecommunication device used to send documents via telecommunication channel.
- A document is placed in the machine, scanned and converted into analogue from then transmitted over a telephone line.
- The receiving fax machine converts the analogue data into the original softcopy and prints a hardcopy.
- To send fax over the internet a special MODEM, called fax MODEM is required and is being attached to the sending and receiving computers.

3. Video conferencing

- Video conferencing refers to the use of computers, a digital video cameras, audio capturing devices and communication networks to enable people in different location to see and talk to each other.
- Each participant's computer is attached with web camera (webcam) speakers and a microphones and appropriate communications software.
- Video conferencing is popular in TV broadcasting stations where field reporters interact with new casters.

4. Telecommuting

- Telecommuting it is the term used to refer to a situation where an employee work usually at home using computer connected to the workplace network.
- Telecommuting has reduced unnecessary travel to the work place and also reduced travel expenses as well as less due to commuting inconveniences such as traffic jams.

I. Industrial systems

- The application of computers technology in industrial or manufacturing process has become one of the most effective methods of automated production which has improved the productivity.
- Computers are used in some of the following industrial plants;
- Motor vehicle manufactures
- Mining plants
- Chemical plants
- Refineries
- Computers in industries are used in a number of ways of which some includes
- Process control
- Industrial simulation

- Computers aided design and manufacturing

J. Scientific and research systems

- A computer has a wide variety of applications in science, research and technology some of which are;

1. Weather forecasting – through the use of geographical information systems (GIS)
2. Medical research
3. Military and space exploration science
4. Social researches.

K. Library systems

- Library use computerized system for a number of task such as;
- **Lending system** : the library lending system manages the issuance and return of borrowed reading materials.
- **Inventory;** it involves use of computer to manage stock, which includes checking for books currently in stock and those on high demand.
- **Cataloguing system**; it is a collection of cards with information about each book reference found in the library.

L. Entertaining and multimedia systems

- The advancement in multimedia technology has produced computers that can be used in recreation and entertainment.
- The term multimedia refers to text, video and sound multimedia computers are computers that not only processes text but can also process video and sound.
- Multimedia computers should have the following elements.

1. A high resolution monitor (screen)
2. A sound card that processes sound

- Multimedia computers can be used in the following areas;

1. Games
2. Music and video

- Multimedia computers are used in recording synthesizing, editing and adding special effects to music.
- In video industry computers are used to produce highly simulated and animated movies, general scenes and actors.

O. Transport systems

- Computers are also highly used in transport systems in the following areas;

1. Road traffic control
2. Air traffic control – through the use of geographical position system (GPS)
3. Shipping control whereby computers are used to guide the paths or direction taken by spaceships and water vessels as they travel to distant land geographical position systems (GPS) is also used.

P. Law enforcement systems

- Crime has become more sophisticated hence very difficult to deal with
- Immediate and accurate information is very crucial detection.
- Computers in law enforcement are used for biometric analysis
- Biometric analysis refers to the process or study, measurement and analysis of human biological characteristic such as fingerprints, skin colour, retina of the eye and voice.
- Barometric devices are attached to the computers to perform its tasks in identifying people by recognizing one or more specific attributes such as fingerprints like iris colour.

Q. Computers in agriculture

- In agriculture computers are used for;

1. **Agriculture** – for improving resistance of crops
2. **Agribusiness** – for marketing product.
3. **Agricultural administration** – for record keeping.

R. Human resource system

- Human resource information systems (HRIS) have become a new way of processing all aspects of human resource management and some of them are;
  - Recruitment
  - Placement
  - Monitoring
  - Monitoring and appraisal
  - Leave management
  - Payment processing

Implications of using computers

The use of ICT offers a different set of opportunities and challenge in our society. Some of the effects of ICT in our society are;

1. Legal issues
2. Economic issues
3. Environmental issues
4. Effect on employment
5. Effects on automated production
6. Issues of workers health
7. Ethical issues

1. Legal issues

- The internet and remote communications means that exchange of information is no longer paper based/face but electronic between remote individuals. But fraud and validity is taking a Centre stage.
- Government has to pass new registration to support electronic communications e.g. a mobile SMS and e-mail need to be recognized as legal documents.

2. Economic issues

The use of computers provides both benefits and challenge economically

Benefits

- More efficiency means better utilization of resources.
- Accountability and transparency in economic planning and management will enhance development.
- Individuals can transfer money and do e-commerce hence enhancing economic development.

Challenges

- Electronic fraud i.e scam money laundering
- The needs to enforce security
- ICT equipment are expensive to buy, install and maintain.

3. Environmental issues

- Emission of heart into the environment and other forms of electromagnetic radiations.
- Cadmium oxide found in mobile phone and laptop batteries could leak into underground water if proper disposal is not done
- Emission of greenhouse gases during manufacture of ICT components

Electromagnetic emission – low emission ICT device has to be used.

Energy consumption and radiation – the environment protection agency (EPA) launched energy star policy to encourage minimal use of power by electronic devices

Environmental pollution – IT components have contributed to environment pollution because of huge garbage damps of dead computer parts, printers, ink, toner cartridges and monitors.\

4. **Effects on employment**

- Computers at work place has resulted in creation of new jobs replacement of computer illiterate workers and displacement of jobs that were formerly manual.

5. **Automated production**

- Number of manufacturing industries such as vehicle assembly, oil refineries and food processing use computers.

Advantages of using automated production

- It has increased efficiency due to balancing of workload and production.
- It has improved customer services. Adequate and high quality goods are produced
- Efficiency utilization of resources such as raw materials, personnel and equipment hence less expense incurred.

Disadvantages of automated production

- High initial cost of setting up an automated system. E.g the cost of one industrial robot is high than employing a human resource.
- Automated production may lead to unemployment in areas that are labour intensive one person can do the work of twenty people.

6. **Issue of workers health**

It has resulted to

- repetitive strain injuries,
- eye strain and headaches,
- electromagnetic emissions and
- environmental issues.

7. **Ethical issues**

- Ethical in the society is both an informal and formal process of placing values on actions, classifying them as either good or bad,
- computer has changed the way people used to communicate, work and relate.
- This has resulted to ethical dilemma due to the use of internet, social network and mobile phones.

Disadvantages of computing ethics

- It has promoted corruption and fraud in offices.
- It has increased

Cultural effects

- Most of moral intergrity has been compromised through some networking platforms such as
- Hacking
- Eavesdropping
- Piracy
- This is due to the exposure to websites on the internet

Safeguarding computers

- Computer resources are expensive, therefore need to be safeguarded from
  a. Theft

b. Breakage
c. Virus infection and
d. Unauthorized access

- **Safeguarding** your **computers** requires protecting your hardware against damage or theft, protecting **computer** systems against malware and protecting valuable data from being accessed by unauthorized personnel or stolen by disgruntled staff.

Ways of safeguarding computers

- **The following are some of ways for safeguarding of computers**

i. Use of firewall

- Set up between an internal computer network and the Internet.
- Filter out unwanted intrusions

ii. Content filtering ;

- Exclude or deny the access to some web pages on the internet

iii. Use of passwords

- used to authenticate a user to access a system.
- Good password must be combination of letters, numbers and characters(avoid the obvious)

iv. Use of up to date anti-virus

- To protect computer and data against infection by viruses

Physical and Logical security layers

Physical security is the **protection** of personnel, hardware, software, networks and data from physical actions and events that could cause serious loss or damage to an enterprise, agency or institution.

Logical Security consists of software safeguards for an organization's systems, including user

- identification and password access,
- authenticating,
- access rights and
- authority levels.

These measures are to ensure that only authorized users are able to perform actions or access information in a network or a workstation.

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Chapter 2

USING WORD PROCESSOR

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CHAPTER 2: USING WORD PROCESSOR (MS WORD)

Word processors

Definition of word processing

Word processing is the process of creating, formatting editing and deleting text and graphics

What is an electronic word processor?

It is an application program used to create, format, edit and delete text and graphics.

Purpose of a Word Processor

- In general word processors have the following purposes

a. Creating text and graphics

- It allows different input devices such as keyboard, mouse and joystick which can be used to key in text and draw graphics in word processor.

b. Formatting text and graphics

- It allows user to change text and graphic into different styles such as bold, italic underline, Arial and time new roman.

c. Editing text and graphics

- It involves corrections of errors and adding of text and graphics

d. Deleting of text and graphics

  - It also allows the user to remove the text and graphics completely in a document.

EXAMPLES OF WORD PROCESSORS PROGRAM

- Microsoft word
- Notepad
- Word pad

  NB: Before electronic word processors, Manual word processors were used such as type writer

A typewriter

ADVANTAGES OF WORD PROCESSOR OVER A TYPEWRITER

- Easy formatting of text and graphics in word processor than a typewriter .
- Mistake can be corrected before printing in a word processor than a type writer.
- Many copies can be processor than a typewriter.
- Computer keyboard do not requires pressing hard in order to key in text than a typewriter.
- It is easy to insert pictures and other graphics within the stretch of text while type writer requires manual insertion of pictures.

CONTENTS OF WORD PROCESSOR

The following are some of the features of WP.

1. **Cursor**
   - This is blinking vertical line that shows the user where next to type.

- You can only be able to type exactly at the position of the cursor.
- Once you type, the cursor then moves to the next available space.

2. **Formatting and Editing**
   - Formatting is changing the appearance of text by selecting font typeface or hand writing style), colour, alignment, indentation (moving text away from the margins), bolding, changing size, underline, italic and making bulleted list.
   - Editing is correcting errors and ensuring clarity and accuracy like removing spelling and grammar mistakes.
3. **Spelling and grammar check**
   - WPs have the ability to check the spelling and the grammar of a document
   - This can be set to indicate such mistakes by showing different colours on them. For example in Ms-Word, spelling mistakes are underlined in red and grammar are underlined in green.
4. **Word Wrap**
   - This is the ability to take an incomplete word to the next line automatically without pressing the enter key. This is known as word wrap.
5. **Thesauras**
   - This is a Greek word for storehouse. It stores synonyms and antonyms of different words.
   - The user of the WP has the option to obtain different words which mean the same or opposite as the word to be changed.
6. **Auto-correct**
   - Word which is frequently used with WP can be made to automatically be corrected or be completed by WP in the process of typing, hence the name auto-correct.
7. **Undo and redo**
   - Any action done by the user of the WP, can be undone by these two features.

- If the user types some words which was not intended, one has the option to click undo button for the action to be omitted or if one had deleted some word or sentence or paragraph by mistake, one simply presses the redo button to have the same deleted items back to the right place.

8. **Mail Merge**

- This is the ability of WP to create a common letter, e-mail or labels and add different addresses or particulars to each letter for different people at the same time.
- It has three main parts:
  a. Creating main letter,
  b. Creating addresses and
  c. Merging the addresses to the letter.

9. **Dictionary**

- WP has a dictionary where certain words can be added, to make them accepted by WP such that it will not show the red colour symbolizing that it is a spelling mistake or a non English word.
- You can add nouns from a different language and WP will recognize them as part of the English language.
- Depending on the default language used by the computer, certain words may not be accepted though they are English words.
- You need to add such words to the dictionary of WP. For example, British English will accept labour, but American will only accept labor and so on.

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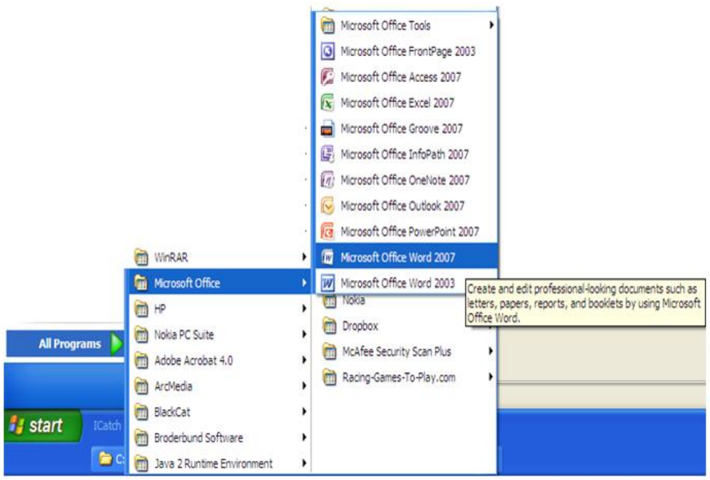
Opening Microsoft word (MS word)

- Click the start button which is at the bottom corner the of the screen.
- Select programs with the mouse pointer
- Click Microsoft word.

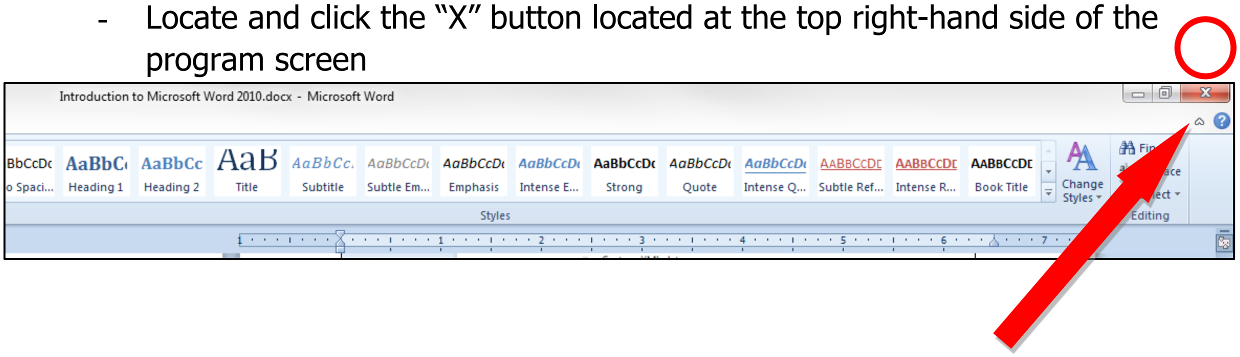


The Microsoft Word window appears and your screen looks similar to the one shown here.

Closing the program

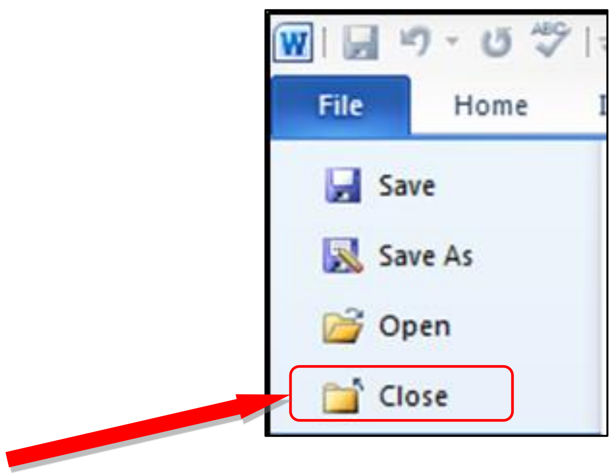
- Closing the program will allow the program to shut down, saving the computer's resources for the other programs you'd like to run.

- Locate and click the "X" button located at the top right-hand side of the program screen



CLO

- Click on the File tab and click "Close" from the menu

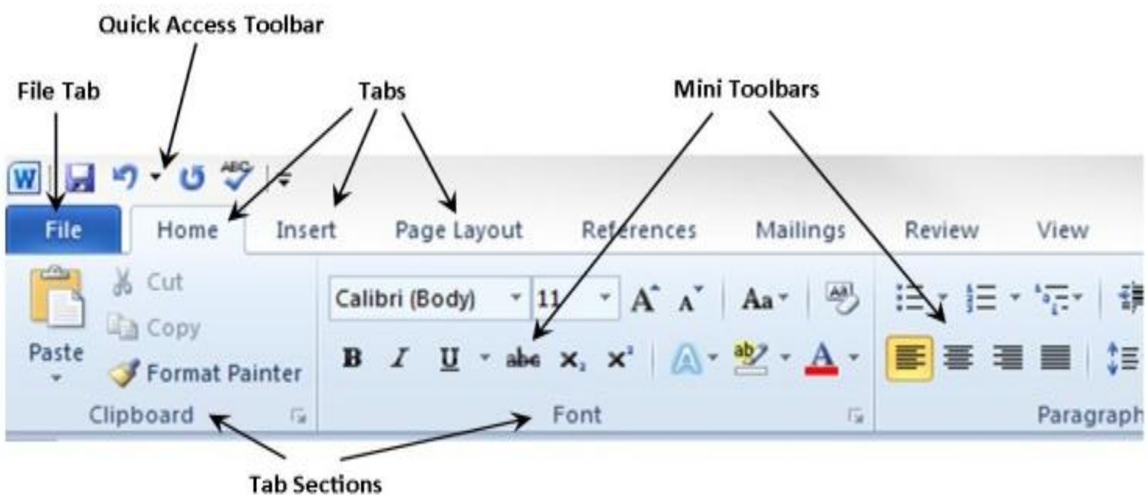


CLO

Tabs and ribbons

- **Tabs and Ribbons** contain all of the key tools for operating Microsoft Word 2010 smoothly and efficiently.

- **Ribbons** are located near the top of the program. Locate the "Home" ribbon and note the different tabs and tab sections.



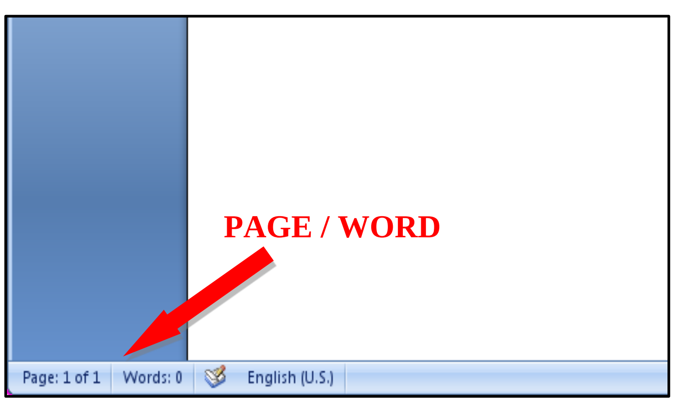
DEFINITIONS:

- **TAB –** A broad set of tools needed to perform a specific type of job for your document
- **RIBBON –** The area of Microsoft Word 2010 that contains the operating tools
- **TAB SECTION –** A *specific* set of tools needed to perform a more *specific* job for your document

Page & word count

REASON: Page and word count helps you track how many pages in your document as well as how many total words it contains.

- Locate the page and word count feature in the lower left portion of your program screen



Changing of margins

- Click file menu
- Click page set up to display page setup dialog box.
- Click margins tab
- Types in the measurement of the margins
- Click ok

Viewing a document window

- The word document window can display its contents in different viewers such as;

a. **Normal view**

- This is a default view for a document

b. **Draft View**

- Draft view is the most frequently used view. You use Draft view to quickly edit your document.

c. **Web Layout**

- Web Layout view enables you to see your document as it would appear in a browser such as Internet Explorer.

d. **Print Layout**

- The Print Layout view shows the document as it will look when it is printed.

e. **Reading Layout**

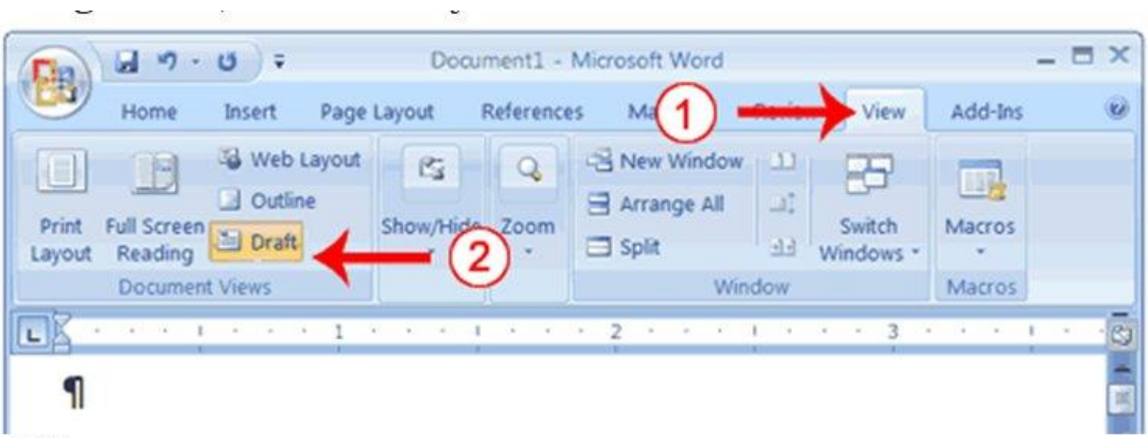
- Reading Layout view formats your screen to make reading your document more comfortable.

f. **Outline View**

- Outline view displays the document in outline form.

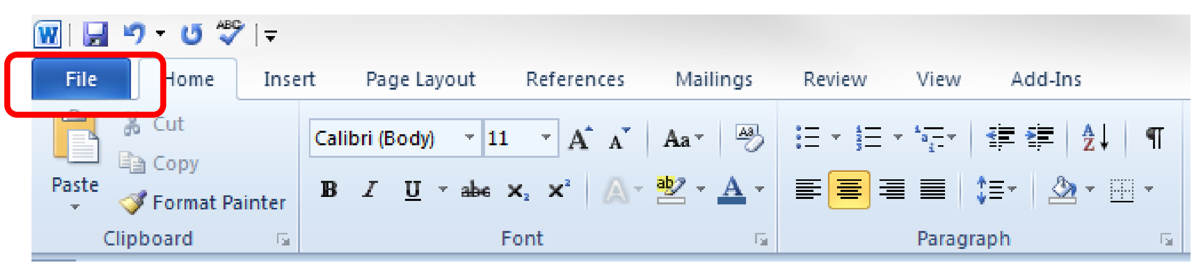
To change from one view mode to another;

- Click view menu
- Click the view that is required.

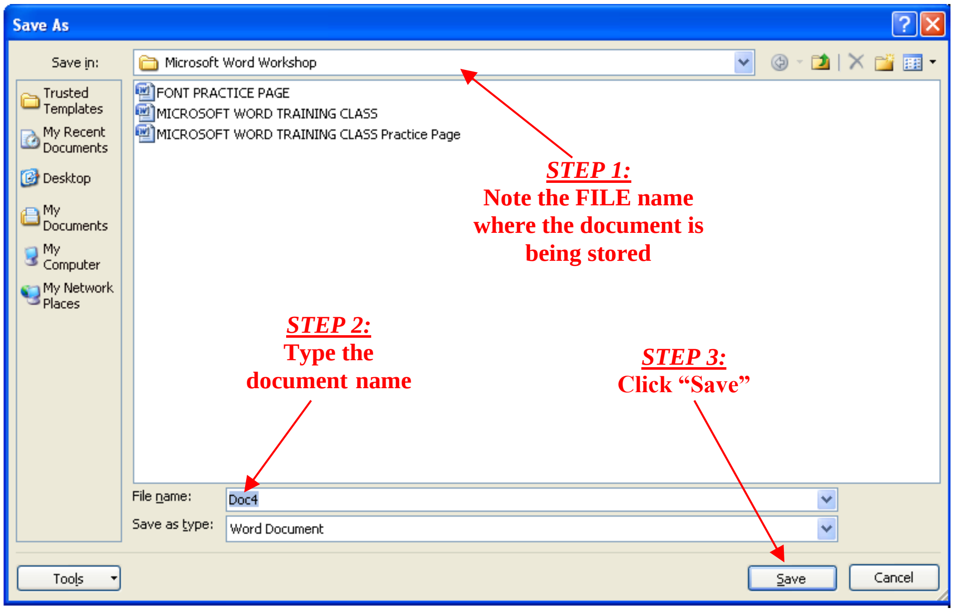


Saving a document

- Once a document has been created it has to be saved so that it is stored in the computer.
- To save the document for the first time, do the following;



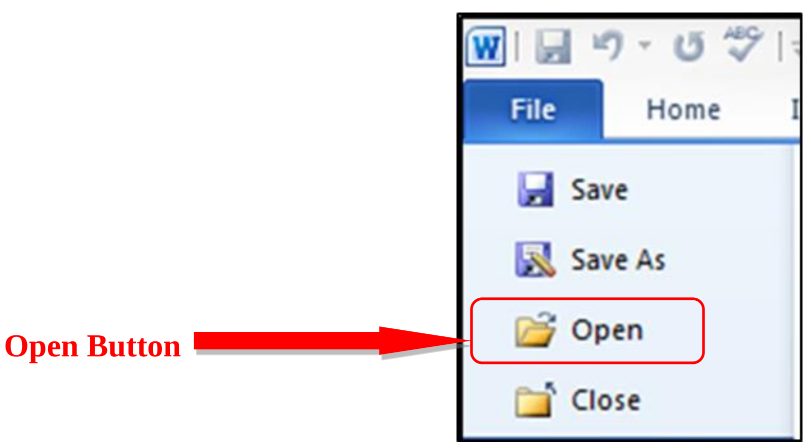
- Click file menu
- Click save as, then the save as dialog box appears
- Type in the file name in the text box.
- Select the location where the document has to be saved.



- Click save

Opening a saved document

- Click file menu then open dialog box appears.
- Open the file or holder where the file is saved.
- Click open, them the document window will open.

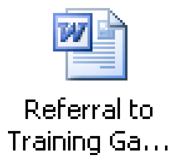


DEFINITIONS:

- **File –** a document or item that can be opened with a specific program
- **Folder –** a small place where you store files

- **Drive –** a large place that stores many folders and programs

ICONS (BUTTONS):



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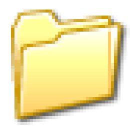


FOLD

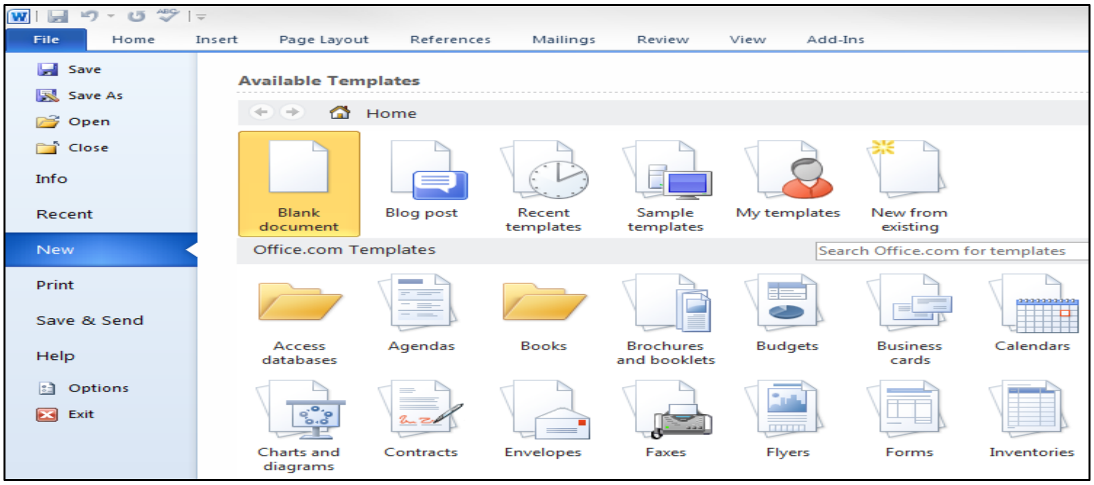
DRIV

Opening a New document

- Click on the "File" tab and locate the "New" button
- Click on the word "New"

DEFINITIONS:

- **Template –** a pre-formatted document that allows you to simply "fill in the blanks" rather than create a similar document from scratch
- **Sample templates –** a template that is already installed on your computer and ready to use
- **My templates –** additional templates that typically come standard with microsoft word
- **New from existing –** a template that you previously built and saved yourself
- **Office.com templates –** templates found online, typically through the microsoft website



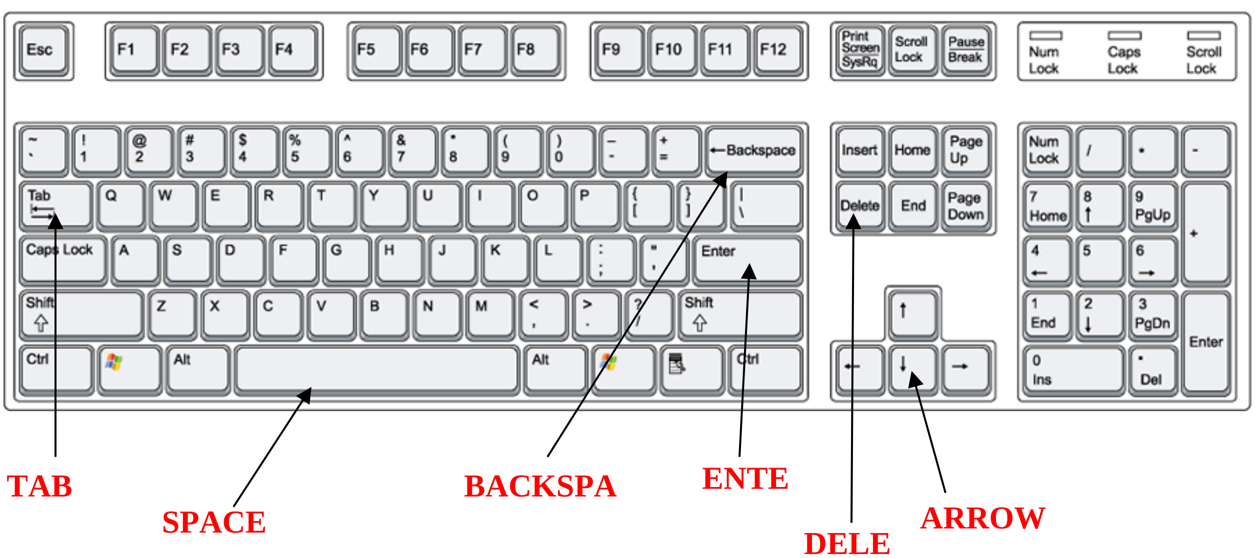
Typing a document

The keyboard

New

- The SPACE BAR, RETURN, and ARROW keys are the main keyboard tools used to navigate through a Microsoft Word 2010 document.

- Locate the following keys on your keyboard



- Open the existing document file labeled "Keyboard Practice" located on the C: drive in the folder labeled "Training Class Files – DO NOT REMOVE"
- Follow the instructions at the top of the document to complete the activity

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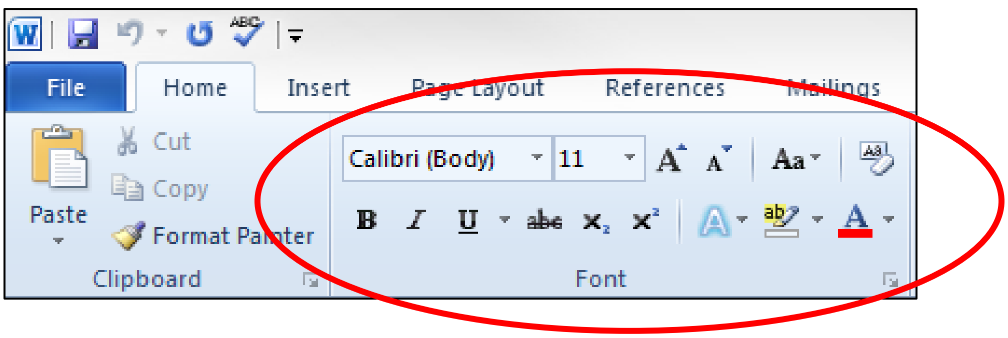
Font options

- Using the FONT options will allow you to customize your text and affect the overall presentation of the document

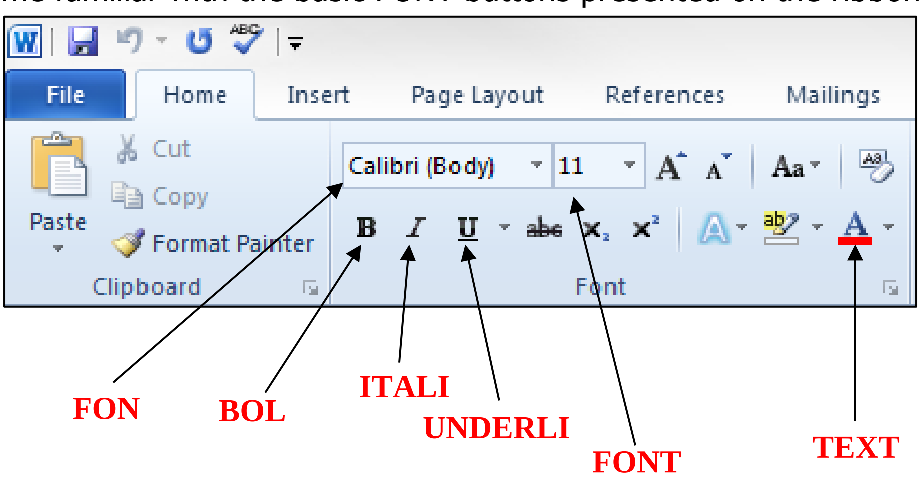
DEFINITIONS:

➢ **FONT –** The style and typeface in which the text of a Microsoft Word document is presented

NB : Locate the FONT ribbon on Microsoft Word



➢ Become familiar with the basic FONT buttons presented on the ribbon



Indentations & bullets

- Indentations & bullets allow you to shape your document so it is easier to scan and read complex information. These tools are used frequently in the business world.

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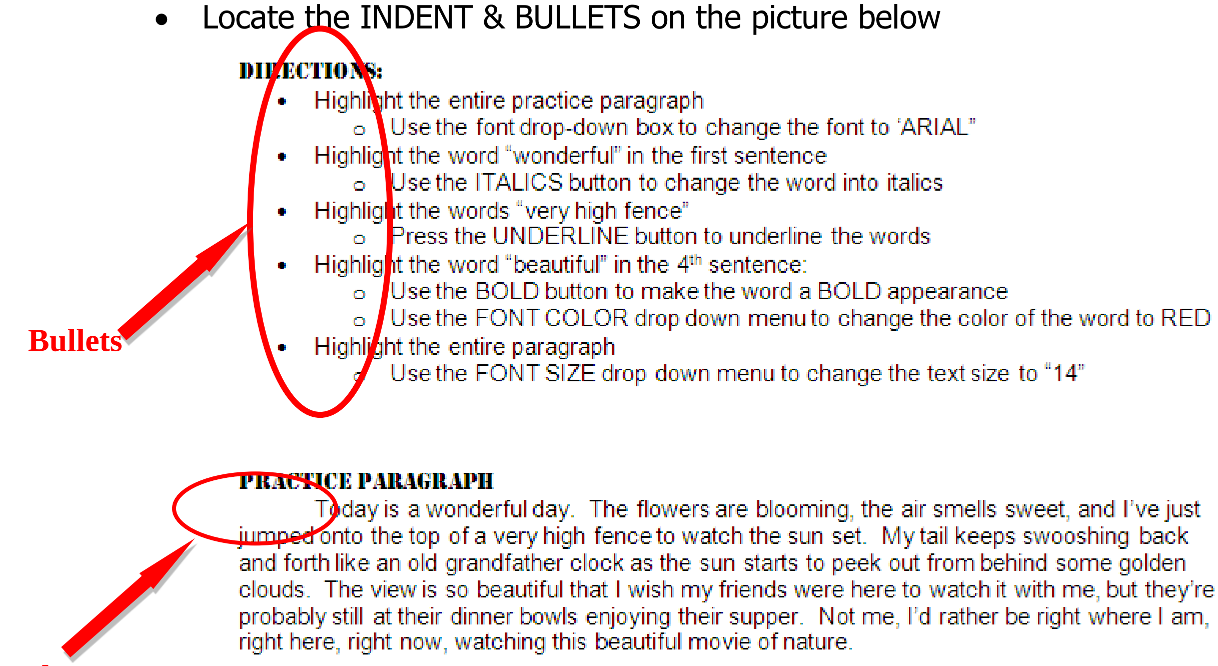
DEFINITIONS:

- **INDENT** - set in from the margin; "Indent the paragraphs of a letter" – Performed by using the "Tab" key on the keyboard
- **BULLET** - A symbol or used to introduce items in a list – Performed by using the "Bullets" button on the "Paragraph" tab section

- Locate the INDENT & BULLETS on the picture below



DIRECTIONS:

- Highlight the entire practice paragraph
  - Use the font drop-down box to change the font to 'ARIAL"
- Highlight the word "wonderful" in the first sentence
  - Use the ITALICS button to change the word into italics
- Highlight the words "very high fence"
  - Press the UNDERLINE button to underline the words
- Highlight the word "beautiful" in the 4th sentence:
  - Use the BOLD button to make the word a BOLD appearance
  - Use the FONT COLOR drop down menu to change the color of the word to RED
- Highlight the entire paragraph
  - Use the FONT SIZE drop down menu to change the text size to "14"

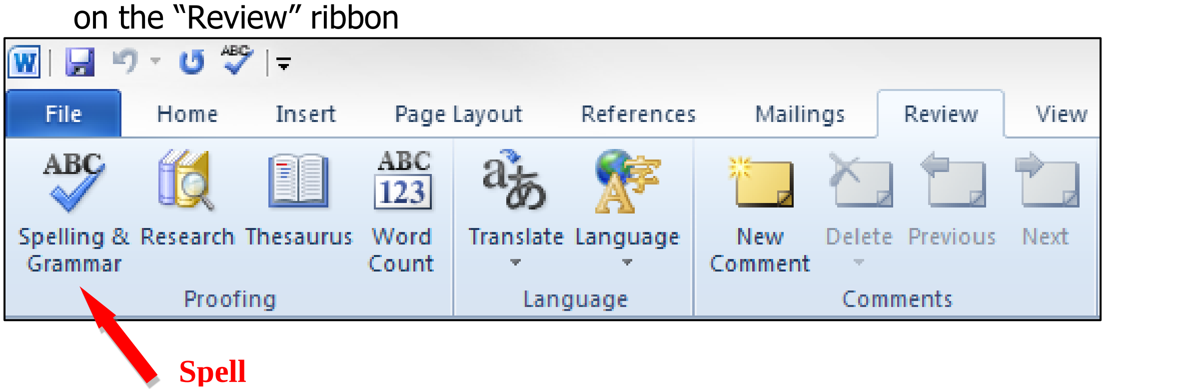
PRACTICE PARAGRAPH

Today is a wonderful day. The flowers are blooming, the air smells sweet, and I've just jumped onto the top of a very high fence to watch the sun set. My tail keeps swooshing back and forth like an old grandfather clock as the sun starts to peek out from behind some golden clouds. The view is so beautiful that I wish my friends were here to watch it with me, but they're probably still at their dinner bowls enjoying their supper. Not me, I'd rather be right where I am, right here, right now, watching this beautiful movie of nature.

Using spell check

- Using Spell Check helps to ensure that your document is free of any spelling or grammatical errors. This is an important feature that is used often by professionals of all areas.

- Locate the SPELL CHECK button on the picture below. NOTE: It is located on the "Review" ribbon



Formatting a document

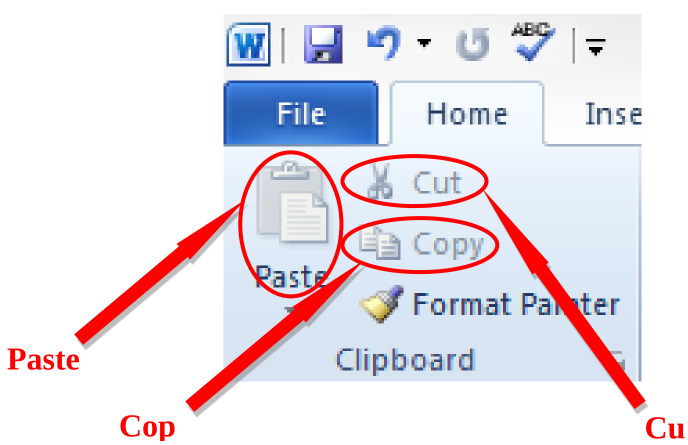
CUT, COPY & PASTE

- The Cut, Copy, & Paste features are a vital part of most everyday business applications of Microsoft Word, as well as many other programs.

DEFINITIONS:

- **Clipboard** – This is the area where items being cut, copied, and pasted are temporarily stored. Think of it just as the name implies, as an imaginary clipboard to hold the items needed for the project at hand
- **Cut** – The process of removing a text, picture, or other object from the document and placing it on the CLIPBOARD
- **Copy** – The process of making a copy of a text, picture, or other object from the document and placing it on the CLIPBOARD
- **Paste** – The process of removing something from the CLIPBOARD and adding it to the current document

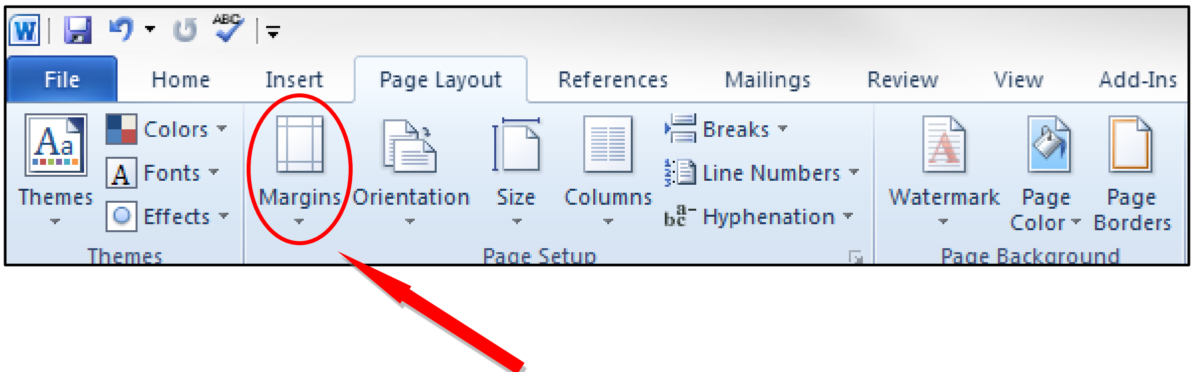
- Locate the "CUT", "COPY" & "PASTE" buttons located in the "Clipboard" Tab Section of the "Home" tab



Changing margins

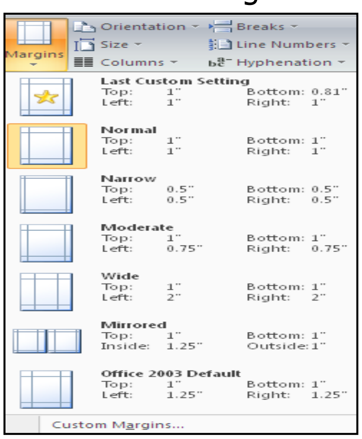
- Margins allow you to further shape your document by setting the outer limits of where text can and cannot be typed

- Locate the "Margins" menu found in the "Page Setup" Tab Section of the "Page Layout" tab



Margi

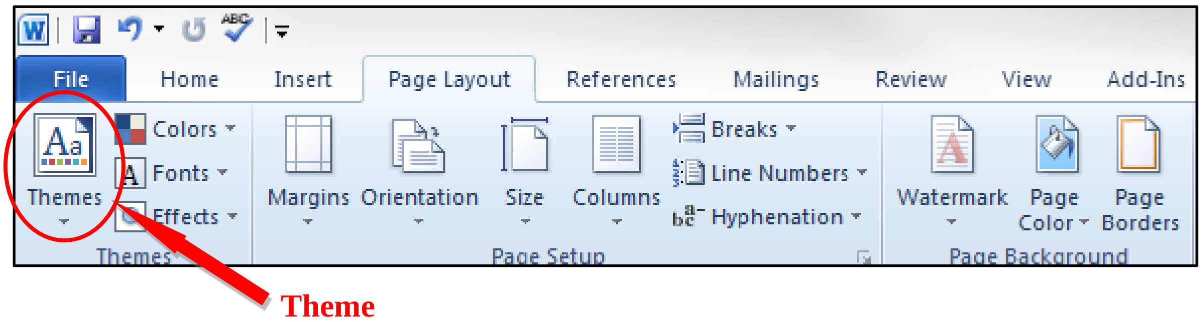
- Click on the "Margins" menu button and view the different options



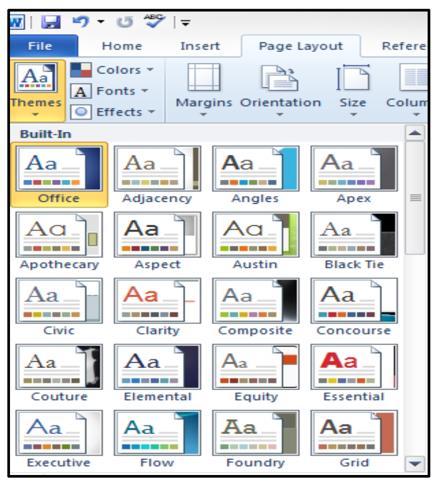
Using themes

- Themes can help to add a little extra flair to your document by changing colors and layouts at the touch of a button.

- Locate the "Themes" menu found in the "Themes" Tab Section of the "Page Layout" tab



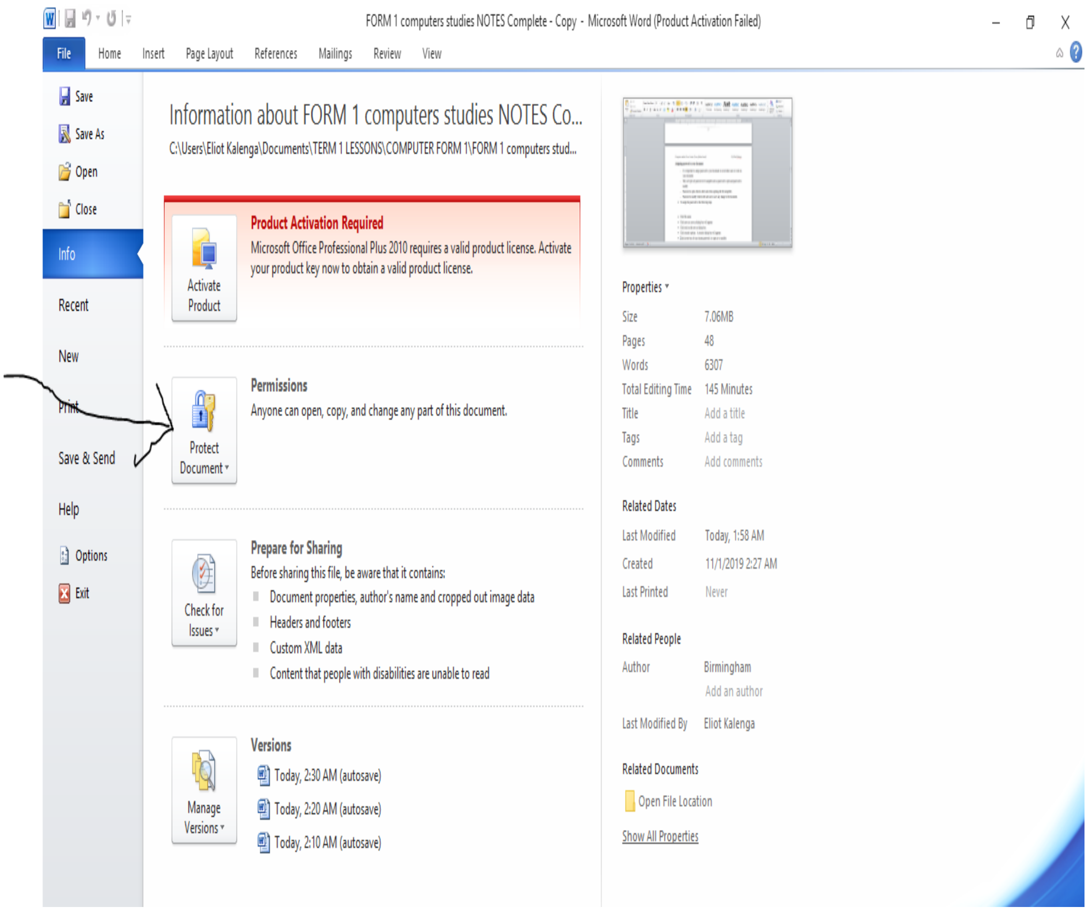
Click on the "Themes" menu button and view the different options



Assigning password to your document

- It is important to assign password to your document to avoid other users to work on your document.
- They are types of password to be assigned such as password to open and password to modify.
- Password to open restricts other users from opening the file altogether.
- Password to modify restricts the users not to save any change to the document.

- To assign the password to the following steps



- Click file menu
- Click save as a save as dialog box will appears
- Click tools on the save as dialog box.
- Click protect document A security dialog box will appears.
- Click in text box of your chosen password (to open or to modify)
- Click ok. You will be prompted to type the password again click ok.

Merging

- Merging is the process of bringing text and graphic together. They are two main types of merging and these are;
  a. Mail merge

b. Text merge

a. Mail merge

- involves bringing information from a data source such as addresses and labels to be come one with text which is typed.
- Mail merge helps to simplify the tasks to type one letter and attach it with many addresses at once.

b. text merge involves bringing together text typed and saved in different files to be come one element. Text merge compares the two documents and highlighting the differences which can be edited accordingly.

MERGING MAIL

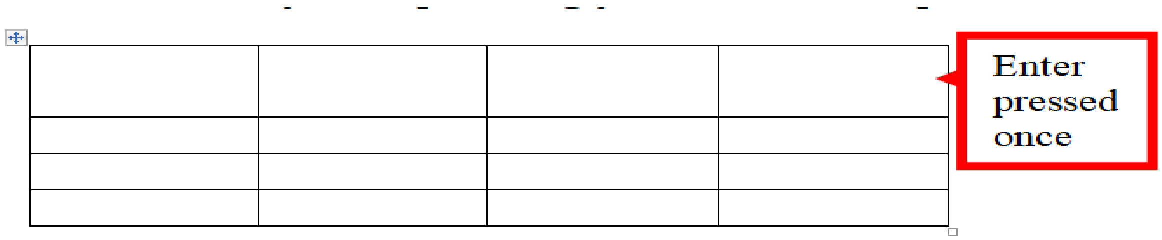
- When creating mail merge a list of addresses for the recipients is required. To create a mail merge from a new mailing list do the following in Ms word window;
  - Click **maillings**
  - Select **start mail merge**
- In the task pane follow the follow all the six steps as follow;
  - Select the document type e.g letters, e-mail messages and labels.
  - Starting document e.g select current document in the task pane.
  - Selecting recipients by clicking type new list. Then type the new recipients address in the new address list window, and click save and OK to confirm.
  - Write your letter by adding the address block and greetings line on top of your letter.
  - Preview your letter and make some changes if required
  - Complete the merge by either printing or editing individual letter. In every step remember to click next in the dialogue box.

Creating and Editing a Table

- Open **Word** or the document where you wish to **put a table**. You can **insert tables** into any version of **Word**.
- Position the cursor on the area where you want the **table** to be **inserted**. Click the "**Table**" button that is located under the "**Insert**" tab. ...
- Choose your method of **inserting** your **table**.

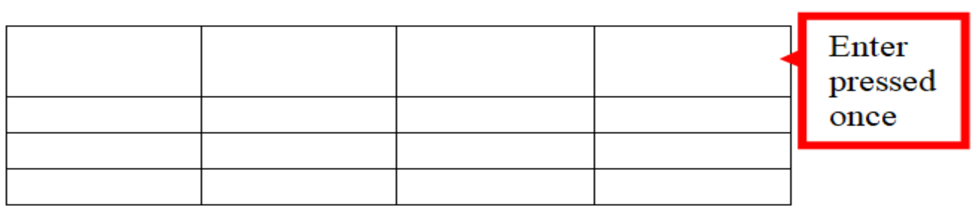
Editing tables

Tables can be resized, new columns/rows inserted, rows/columns can be merged and they can also be split.

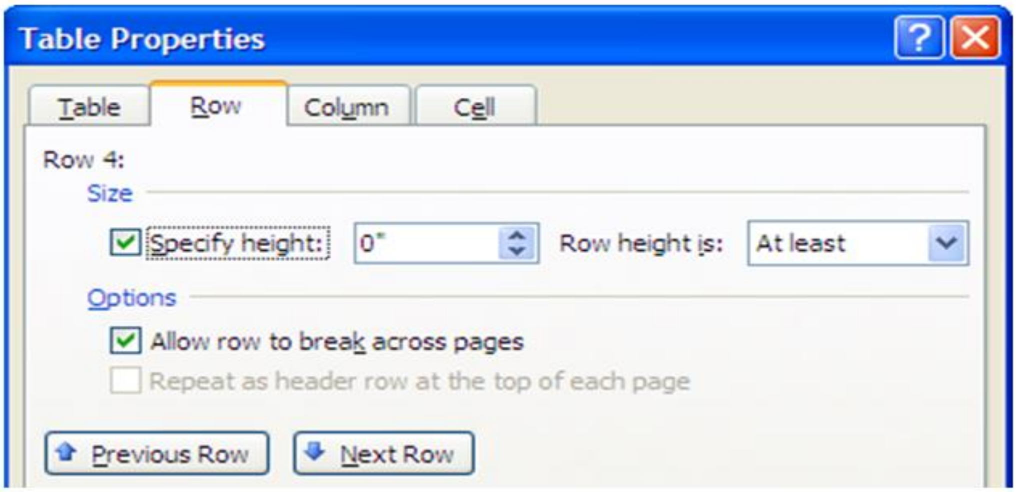


- Resizing rows/columns

The easier way of expanding your rows is to place the cursor within a row and press enter.

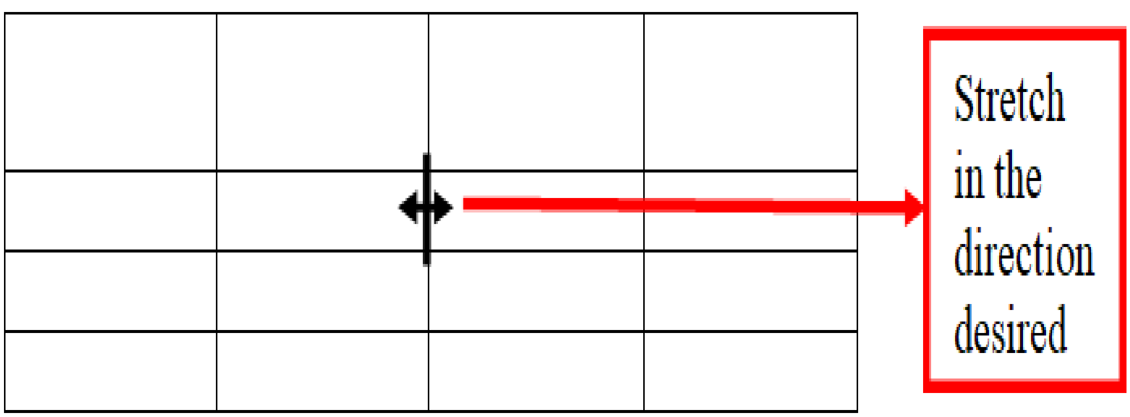


Hint: Alternatively, select the row to expand and right click. A pop-up window will appear, choose table properties then click the row tab. The window below will appear. On size, specify height.



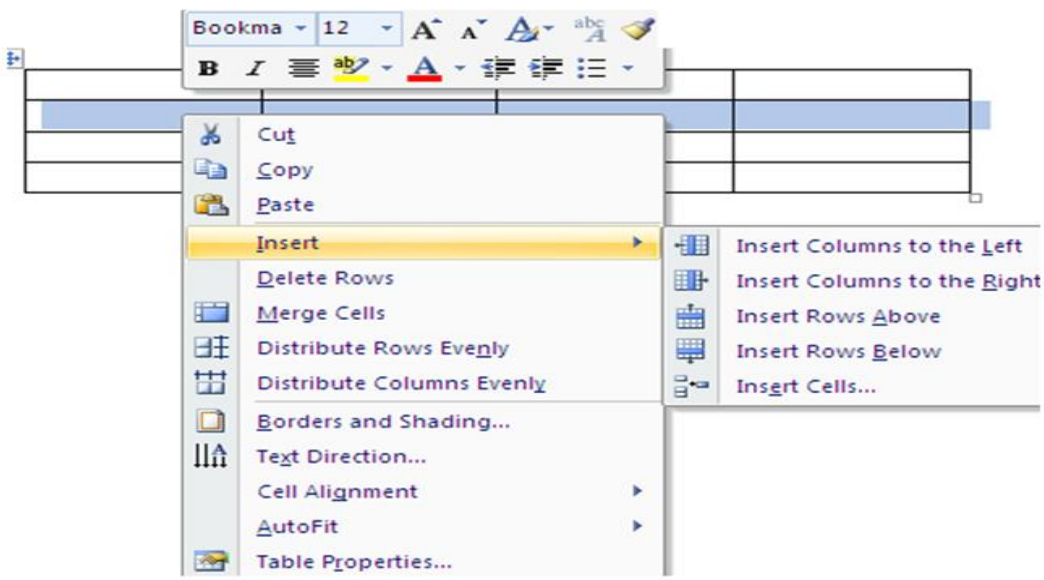
Resizing columns

Place the cursor on the boundaries of the column you intend to expand. A double headed pointer as shown below will appear. Columns are resized by moving the double headed pointer in the direction you want to expand the table. This is illustrated below.



Inserting rows and columns

To insert a row or a column, simply select the row or column. Right click on it and click insert, then choose row/column and the position of insertion desired



Deleting rows/columns

• Highlight the row/column to delete.

• Right click on it

• On the pop-up menu, click either delete rows or delete columns depending on what you selected to delete.

Merging rows/columns

• Highlight the row/column to merge

• Right click on the same

• On the pop-up window click merge cells.

• The cells will be merged.

Entering data in a table

- Simply place your cursor in the cell you wish to enter data and type the usual way.

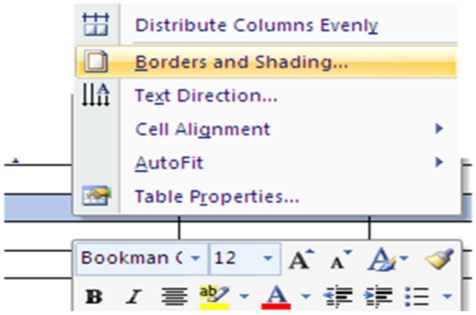
Formatting Tables

- Formatting a table is to make it appealing to the reader.
- You can put borders and shading on it to enhance the contents.

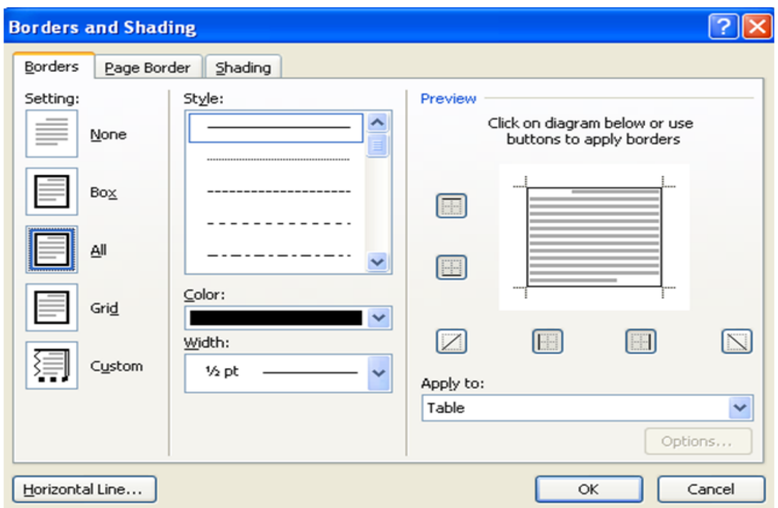
Steps:

- Highlight the entire table or row/column as desired.
- Right click and choose borders and shading from the pop-up menu.

When you click on borders and shading, the following window pops-up.



When the borders tab is clicked, the following is seen.



To apply the borders

- Highlight where to apply border,
- Click a setting e.g. Box, grid
- Choose the style you want,
- Choose colour
- Choose width to apply to the borders
- On Apply to box, click where to apply the border

Exercise

- Create a table of 4 rows and 5 columns
- Apply the grid borders to the table
- Apply colour of your desire Apply appropriate width.

Exercise

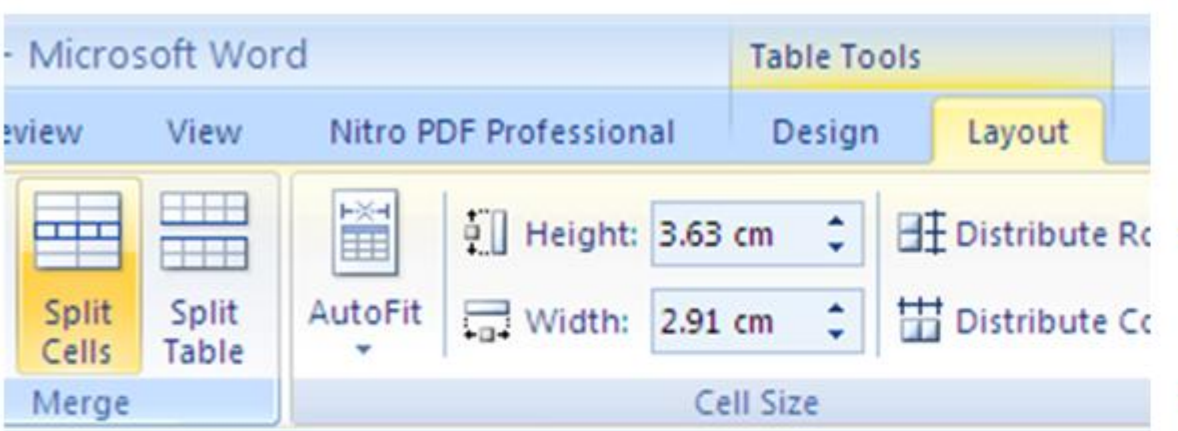
Using shading tab, apply shading on the first row of the table in on the exercise above. Make sure to choose, the style, the colour and where to apply the shading.

Split cells

Click in a cell, or select multiple cells that you want to split.

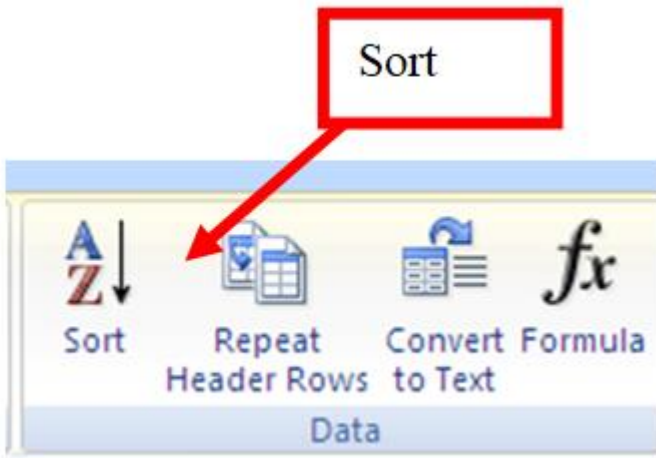
Under Table Tools, on the Layout tab, in the Merge group, click Split Cells.

Hint: Once you highlight the table tools will appear on top of the menus.

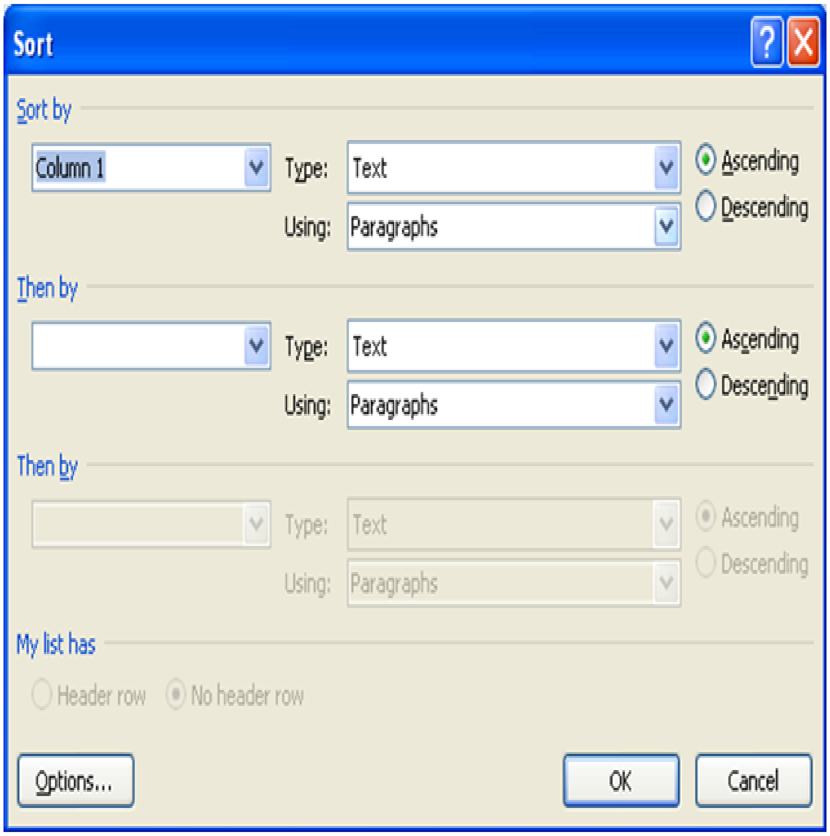


Sort the contents of a table

In Print Layout view, move the pointer over the table until the table move handle appears. Click the table move handle to select the table that you want to sort.



Under Table Tools, on the Layout tab, in the Data group, click Sort. In the Sort dialog box, select the options that you want.



Sort a single column in a table

- Select the column that you want to sort.
- Under Table Tools, on the Layout tab, in the Data group, click Sort.
- Under M list has click Header row or No header row. Click Options. Under Sort options, select the Sort column only check box. Click OK.

Sort by more than one word or field inside a table column

- To sort the data in a table that is based on the contents of a column that includes more than one word, you must first use characters to separate the data including data in the header row.

- For example, if the cells in a column contain both last and first names, you can use commas to separate the names.

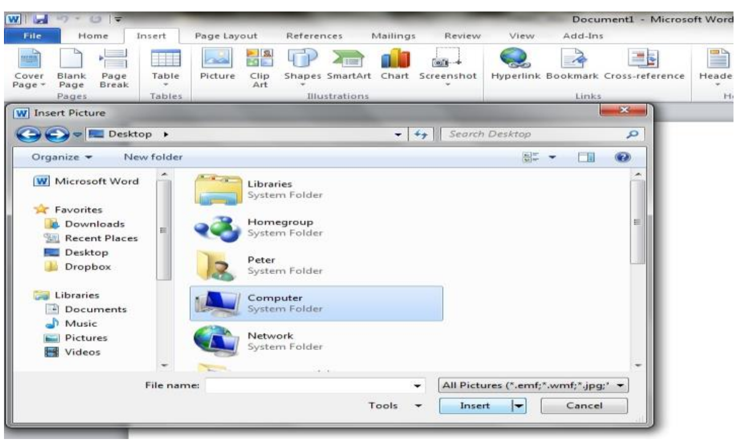
- Select the column that you want to sort.

- Under Table Tools, on the Layout tab, in the Data group, click Sort.

Inserting images in a document

- To insert an image into your document

- Select the "Insert" menu. Locate and press the "Picture" button in the "Illustrations" section.
- In the Dialog box that appears, browse to the photo you wish to insert and press the "Insert" button
- .



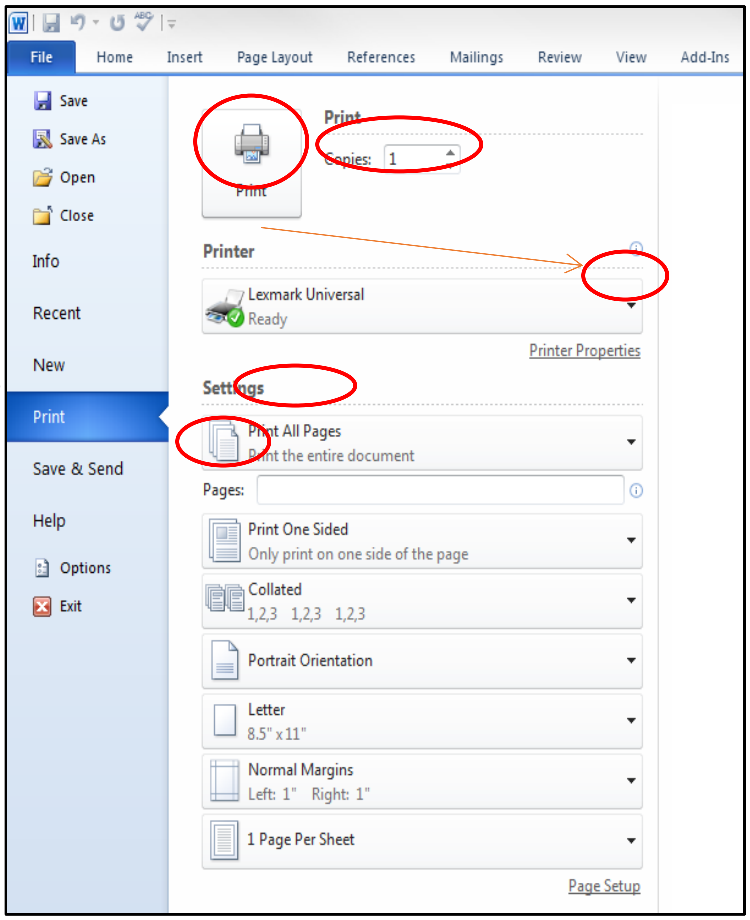
Printing a document

PRINT PREVIEW / PRINT

- Print preview allows you to see your document as it will appear when printed.
- This is important to check for any last minute issues before printing. Microsoft Word 2010 has a built in Print Preview from the Print Menu.

- Locate and left-click the "Print" menu button located in the "File" tab
  - Notice the different options to customize your print

Print



- Become familiar with the following areas of this menu
  - **Printer** – Selects the printer that you want to print the document
  - **Print All Pages/Pages** – Decides how much of the document to print
  - **Copies** – Select how many copies of each printed sheet you would like

  - **Print Button** – Starts the printing process
- Printing can be done in many ways in application programs such as Ms Word, excel and Ms Access, to print your work do the following;
  - Click on the file menu
  - Click page setup to set page orientation either **portrait** or **landscape**
  - Click print preview.
  - On file menu select print then on the print dialogue box specify the number of copy needed and the page need either **Odd or Even** only, but by default **ALL** is selected to print all the work
  - Click OK

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Chapter 3

PRESENTATION SOFTWARE

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CHAPTER 3: PRESENTATION SOFTWARE

Introduction: business executives, managers, marketers, teachers and many other categories of people often find themselves faced with the challenge of talking to an audience concerning a particular topic when trying to sell new ideas. Presentation software helps to simplify the work of managers.

Presentation software

What is presentation graphics

- **An electronic presentation** is a collection of slides that may contain text, pictures, drawings, tables sound and video.
- **A slide** is the work area in a presentation where information such as text, graphics or multimedia content is placed.
- The presentation collection can run automatically or can be controlled by a presenter.
- **A presentation software** is an application software used to create an electronic presentation that communicates ideas, messages and other information to the audience**.**

Types of presentation soft ware

There are many types of presentation software available as stand alone or as a component of a suite. Example of presentation software are;

- Microsoft PowerPoint
- Lotu's freelance Graphics
- Corel presentations
- Harvard Graphics
- Open Office Impress e.t.c

Purpose of using presentation software

Some of the areas where electronic presentation has become handy include;

- **presentation facts and figures –** examples, sales people often have to give presentation to customers or managers just to let managers the sales performances and the customers the latest product.
- **Teaching –** presentation software is regularly used as teaching.
- **Reporting research findings -** researchers and students present their project reports through electronic presentation.
- **Conferences –** presentation software is regularly used during workshop and seminars speakers use software to drive their point home.

Advantages of using presentation software

- It is easy to learn.
- It has large library of background templates and custom layout.
- Multimedia effects can easily be added to the presentation.
- Presentation are easy to edit.
- It can easily produce output in different format e.g. interactive whiteboard and handouts.
- Excellent for summarizing facts using charts or diagrams to an audience.
- Can be used to produce a set of hangouts for people to write on whilst presentation is being given.
- Let' the speaker face audience and make eye contact rather than facing the screen.

Features of presentation software

- They have large range of readily available **pre-designed templates** that one can easily use to create the presentation.
- It has a selection of **layouts** used to create slides.

- It has **master slide** used to set up content that presenter could wish to appear on every page such as footers or page number.
- It has **animation and transition** effects used to add emphasis to presentation.
- It allows the users to run on-screen shows and upload the presentation on the internet and print the handouts that can be distributed to the audience to help them easily follow the presentation.

Presentation design guidelines

To produce an effective presentation one should adhere to four P's that is

- Plan
- Prepare
- Practice
- Present

The following factors should be considered when designing a presentation;

- You must have the objective of the presentation.
- You must consider the length of the presentation.
- You must consider the targeted audience.
- You must consider the type of presentation e.g business or academic
- You must consider the content of the presentation e.g should meet subject matter.
- You must consider the organization e.g should have opening slide, main body and the conclusion

Starting PowerPoint 2010/ 20013

1. On the programs menu , point on Microsoft Office
2. Click Microsoft office PowerPoint 2003 / 2007. The PowerPoint application window with a default blank slide will appears. Check the figure below;

Features of the PowerPoint application window

When working with Ms PowerPoint observe the following features;

- **Microsoft office button** which will help you to perform command like save as when saving your work e.t.c
- **Ribbon,** it has tabs like Home, insert, Design, Animation, Slide Show, review and view in Ms Office 2007.
- **Quick access button** contains frequently use commands like undo or redo.
- **Work area** contains panes to insert and graphics.
- **Status bar,** which displays the current actions in the slides.
- **Mini toolbar** it display common formatting tools, such as bold, Italic, fonts, Font size and Font color.
- **Note pane**, used to add more notes that could not fit on the slide.

Working with Presentation Software

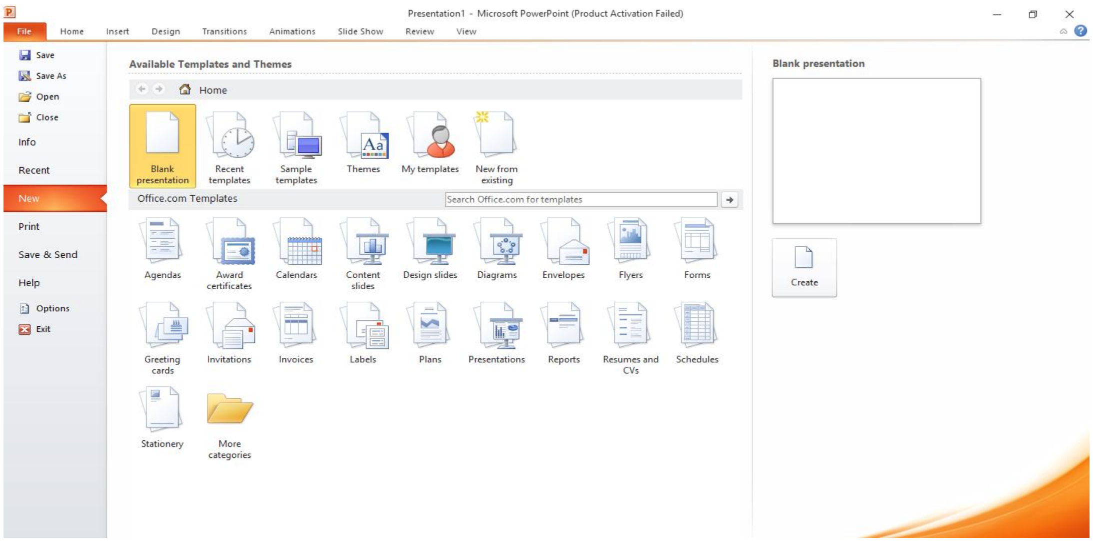
Creating a new presentation

There are three ways of creating new power point presentation

i. From blank slide
ii. From template
iii. From existing presentation

To create a new presentation from a blank slide proceeds as follow;

1. Click Microsoft office button.
2. Click new then a new presentation dialog box appears (Ctrl + N)4we
3. Click blank presentation



To create a new presentation from template

1. In the new presentation window, click installed templates or press Shift + Ctrl + P)
2. In the installed template section, choose the desired template.

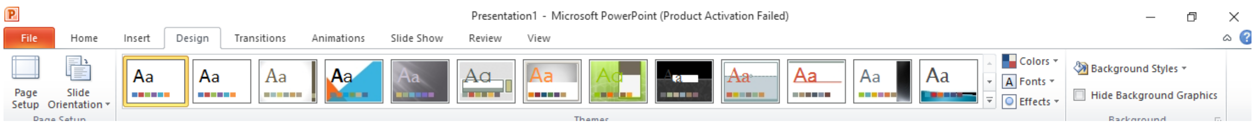
To create a presentation from existing presentation:

1. In the new presentation window, click new from existing.
2. On the displayed dialog box, select the file you want to use. The file will open in slide master view.
3. Modify the presentation as desired then save it with a different name.

Choosing appropriate layout

To apply a slide layout, proceed as follows:

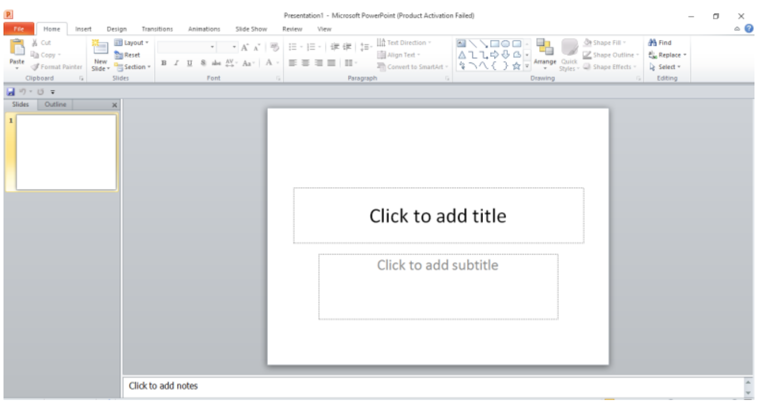
1. On the home tab, in the slide ground, click layout icon.
2. Select a layout style from the displayed thumbnails as shown in the figure below.

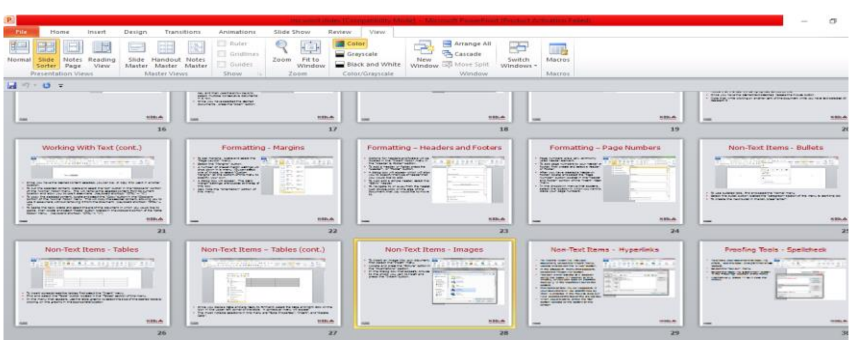


Choosing a slide view

There are three main views of slides in a presentation

1. **Normal view** – is the main editing view and it has four working areas, outlines, slide tab, slide pane and notes pane.



2. **Slide sorter view**
   - Displays all the slides
   - One is able to open any slide of his choice
   - 
3. **Slide show**
   - A full screen view

Adding a new slide

To add a new slide proceeds as follows:

1. Select the slide immediate before where you want the new slide to be inserted.
2. Click new slide layout that fits the contents.
3. Add contents to the slide.

Formatting a presentation

For the presentation to appear to the audience do the following;

- Make text large enough
- Keep it simple
- Make use of pictures where possible
- Use good color background
- Use good background not distractive one.

Formatting drawing

To format a shape :

1. Select the shape.
2. Drawing tools tab appears.
3. Click on formatting tab to display the ribbon.

Format the selected shape as follows;

- Fill shape with patterns click shape fill to change the fill color.
- Change shape outline, click shape outline to change the outline colour, width and style.
- Shape effects, click shape effects to apply visual effects such shadow, Bevel and 3-D rotation
- Group objects, click on group button arrange group
- Align objects, click on align button in arrange group
- Rotate objects, click on rotate button in arrange group.

Setting up preparation for a presentation

To make the presentation happen do the following setup;

1. Rehearse and time the delivery of presentation before hand.
2. Be sure to test the slide show using the projector if one will be used during the preparation.
3. Decide whether the presentation is to be set up to run continuously on a screen so that passersby can stop and watch.
4. Decide handout can be given to the audience for them to make their own notes.
5. Ask whether you will use data projector or overhead projector.
6. Ask if the presentation software are available in the PCs to be used and burn the presentation software onto a CD/DVD

Setting up the show

To setup the show type, proceed as follows:

1. Click the slide show tab.

2. On the slide show ribbon, click setup slide show. A dialog box is displayed as shown below.
3. In the dialog box, do the following:

- Click presented by a speaker (full screen) to deliver the presentation before a live audience.
- Click browse by a an individual (window) to enable your audience to view the presentation on the CD or internet.
- Click show scrollbar check box to allow your audience to scroll through a self – running presentation from unattended computer. Click browse to kiosk. A kiosk is a terminal located in an area frequently visited by many people.

4. In the show options tab, specify how you want sound files narrations or animation in your presentation E.g select loop continuously until ESC.
5. Use the options in the advance slides sections to specify how to move from one slide to another e.g. click manually to advance to each slide manually during the presentation.
6. Click OK to apply the settings and close the dialog box.

Printing a presentation

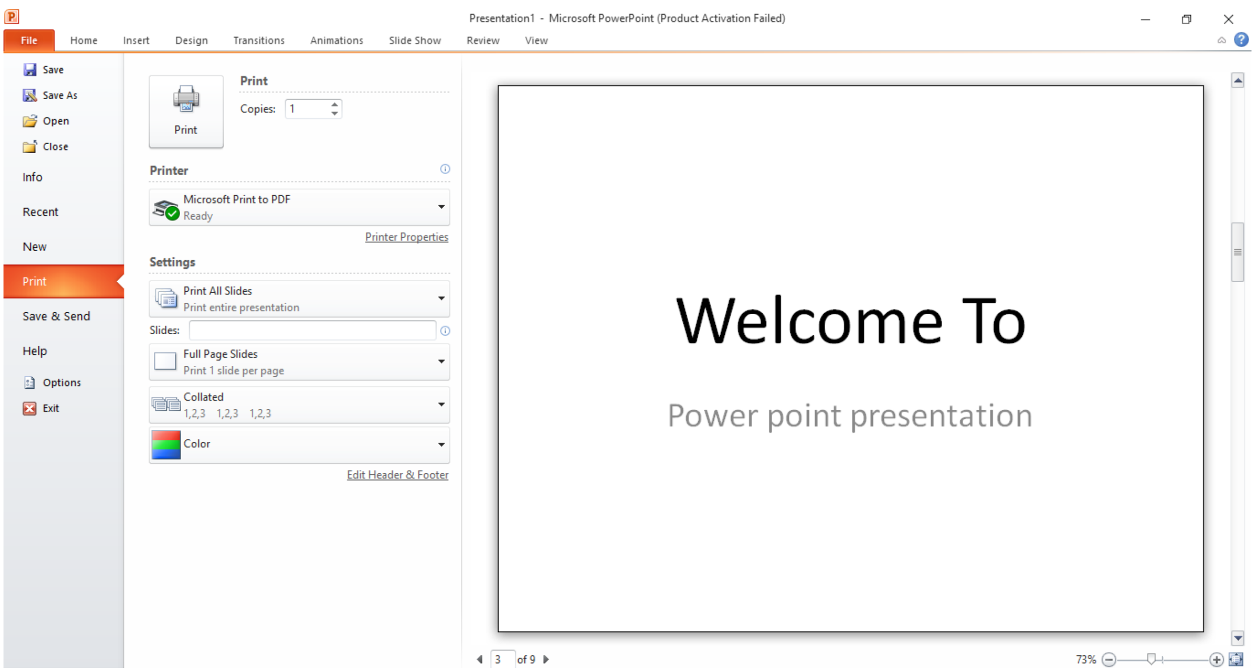
Check the four options below when printing a presentation:

- **Slide –** this option lets you prints one slide per page.
- **Handout –** this option allows for more slides (1,2,3,4,6, or 9) per page.
- **Notes page –** prints slides that includes the speaker notes.
- **Outline view –** used to print the outline of the presentation.

To setup the print options proceed as follow

1. Click the Ms Office Button.
2. Click print. A print dialog box is displayed.

3. In the print dialog box, click the arrow next to print what. Choose what you want to print.
4. Click the preview button to see how the hardcopy will look like
5. Specify other print options and click OK.



Presenting

To start the slide show proceed as follows:

1. Simply press F5 or click the slide show tab
2. On the show ribbon, select any of the start show options.
3. If advancing to the next slide is not automatic, you may use the arrow keys to move forward or use a mouse. Press Escape (ESC) key to end the slide show

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Chapter 4

MANAGEMENT OF COMPUTERS

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CHAPTER 4: MANAGEMENT OF COMPUTERS

Introduction: computers are made up of separate parts which are assembled together a working unit. So it is important for us to know how to assemble a computer, configure its hardware and software and read system information.

Hardware Installation

Setting up computer is very important before carrying out any activity. The following precautions should be observed;

- All devices should disconnect from power source before starting to work on them.
- You must be guided by your ICT trainer working on peripheral devices.
- Never work alone because you may need help in case of emergency.
- You must discharge any static electricity that might have built in the hands by touching the earthed metallic object and then wearing an antistatic wrist member. The human body can hold 200 volts of static charge that can damage sensitive components on the mother bold.

Tools and other requirements when connecting computers hardwares

The tools required include:

- Screw drivers
- Antistatic wrist member
- Pliers
- Device manuals
- Device software (drivers)

Mounting internal devices

External devices are connected to the motherboard through **ports** and internal devices are connected through **SLOTS** and SOCKETS . study the device manual carefully before connecting the internal and external devices.

When connecting the internal components note the following;

- **Power supply** – it receives AC power from the wall socket and converts it to 12 volts DC power cables needed by computer components.
- **Power cables** – provide power to system unit components.
- **Optical disk drive** – used to read / write data from or to optical disk.
- **Hard disk** – used to store computer programs and data for a long time.
- **Data cables** – enable exchange of data between components.
- **Motherboard** – used to interconnect all devices, chips and components with copper circuit drawn on it.
- **Expansion lots** – helps when adding new device to the computer such as TV card, network card E.t.c
- **Interface ports** – enables connection of peripheral devices such as mouse, keyboard e.t.c
- **Memory slots** – enables installation of RAM chips on the motherboard.
- **Chips** – have processing logic and firmware needed for correcting functioning of the computer.

Mounting hard drives and optical drives

Special ribbon cables are used to correct computer components on the motherboard. Hard disk and optical drive are connected on the motherboard through **interface connectors** referred to as **controllers.**

Types of controllers

- Enhancement Integrated Drive Electronic **(EIDE).**
- Serial advanced Technology Attached **(SATA).**

- Small Computer System Interface **(SCSI)**

Hard drive

NB: SATA has replaced EIDE, ATA and PATA. SATA is more efficient and support hot-swapping. Hot-swapping means that the drive can be removed or inserted while the computer is still on. SATA and EIDE are the common controllers used in most computers.

Steps to be followed when mounting an EIDE or SATA Hard drive.

To mount and EIDE or SATA Hard drive proceeds as follows;

- Wear antistatic wrist member to discharge any static charge on the body
- Determine which drive will be master and use the drive information to determine which jumper settings to use for a master or a slave
- Check that a free drive bay is available, slide a drive into the bay and screw it into place.
- Ensure that there is a free power connector from the power supply unit and connect it to the drive. Notice that it is designed to fit in its socket in only one direction.
- Identify pin 1 as labeled on the drives socket and match it with the red or brown continuous line of the ribbon cable. Most cables will only fit in one direction.
- Connect the interface cable to the drive, then into the controller slot on the motherboard.

- If installation is complete replace the casting cover.
- If installation is complete replace the casing cover.

Installation floppy drives

Floppy drives are installed the same way the EIDE drives but floppy drives have no master or slave configuration. It is possible to attach two floppy drives on the ribbon cable and one floppy drive will automatically be assigned by letter A and the other on the motherboard will be assigned by letter B

Connecting external devices

You must check the port and interface cable of the device being connected to the computer system. When connecting external device proceed as follows;

Gently and carefully connect the interface cable of each device to the correct port and to the device if it is not already fixed permanently.

Connect the computer to the power source and switch it on.

Observe boot up information on the screen to see whether power On – Self Test (post) display any error message.

A successful boot means that the computer was properly set up. Then it also means that new programs must be installed in the hard disk drive.

Device Management

Hardware and software have to be configured for easy management using system information.

System Information refers to Hardware and software configurations of a computer

System information

System information is not limited to the following;

a. The hardware device present on the computer including used space and empty space.
b. The type of processor on the computer and its speed.
c. The Random Access Memory (RAM) installed on the computer and its size.
d. The width of the system bus.
e. The size of the hard disk on the computer including used space and empty space.
f. The type of operating system installed

Checking system information

System information can be accessed through the control panel of the operating system (for windows user) when the computer is on.

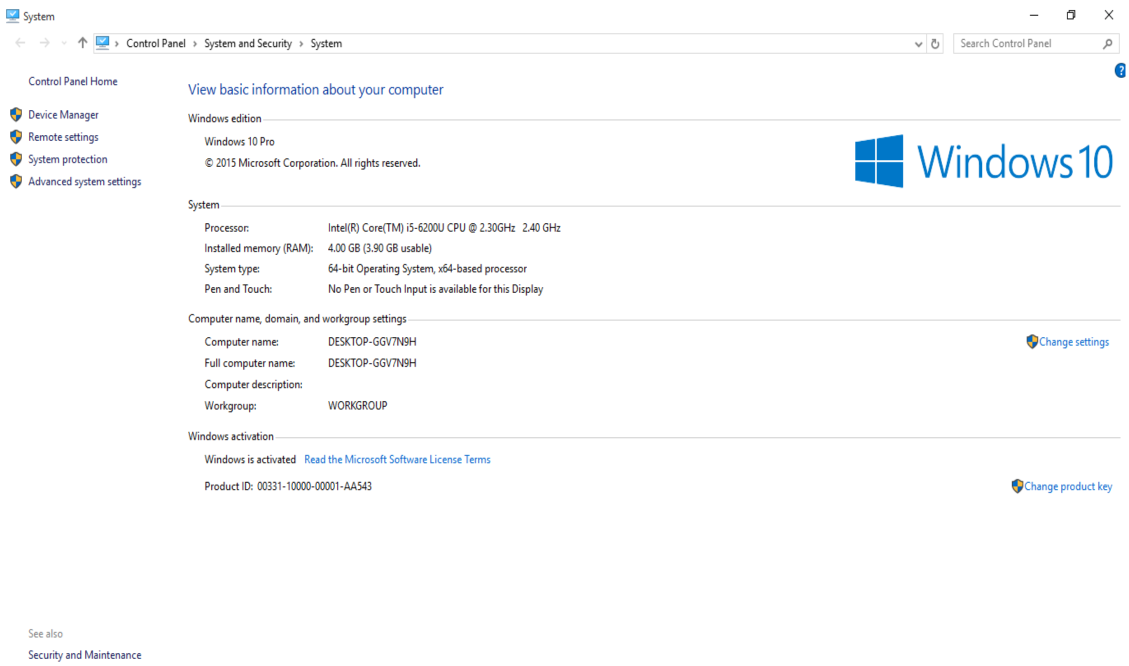
- It is possible to check all these when the computer has booted properly. In windows,

Following the steps below;

- Click start then select control panel.
- In the control panel windows, click system.
- The system windows open as shown below.

Alternatively you can go to

- my computer in windows 7 environment then
- right click any where
- click on properties
- showing system information.

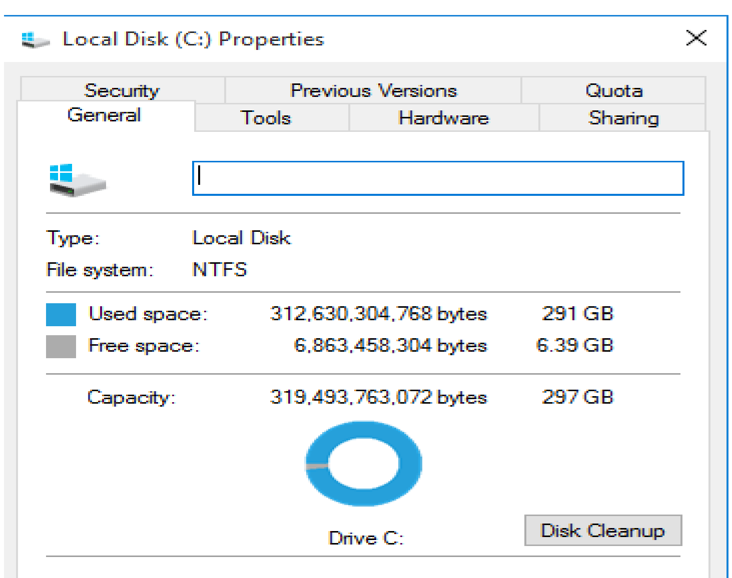


Showing system information.

CHECKING THE HARD DISK SIZE

To check hard disk sizes do the following;

- Double click my computer icon on the desktop
- In my computer window, right click the icon associated to your hard disk (s) and then select properties. Check the details shown below;



Importance of checking system properties

It is important to check system information before buying a computer or loading any software in an existing one because the system information determines the cost, performance and the software that can run in the computer.

The following properties are generally true:

- The larger the Random Access Memory (RAM), the better the performance of a computer the and higher the cost .
- The larger the hard disk size of a computer, the larger the data the computer can store and the larger the software that can be installed on the computer. Larger hard disks also imply better performance and higher cost.
- The higher the cache memory of a computer, the faster the response time of the computer. Cache memory also has implications on the cost of the computer with higher cache implying higher cost.

For good functioning of the computer all the system properties must be balanced

The table below summarises the effect of system information on the performance and cost of a computer

System information	**Effect on performance**	**Effect on cost**
Large RAM size	Fast	High
Large Disk space	Fast	high
High processor speed	Fast	high
High cache memory	Fast	high
	Fast	high

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Chapter 5

TROUBLESHOOTING COMPUTERS

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CHAPTER 5: TROUBLESHOOTING OF COMPUTERS

Introduction

There are number of problems associated with computers, someone would want to diagnose and fix such problem.

Computer troubleshooting: is a process of diagnosing and fixing computer related problems

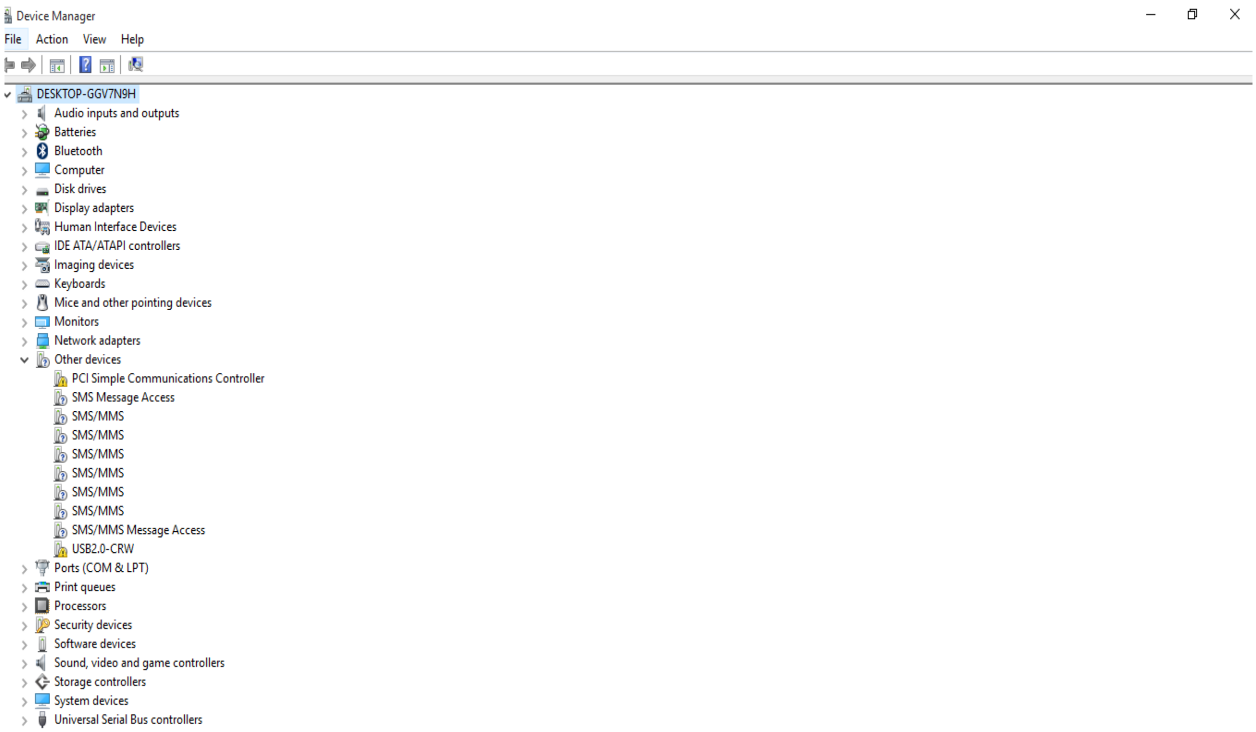
Tracing Hardware Problem

Most hardware problems are caused by the following:

1. **Loose cable connections** or improper fitting in the motherboard slots.
2. **Spoilt and damage hard ware device** like disconnected circuits, cable breaks or burned chip on the motherboard .
3. **Lack of appropriate drivers** or corrupted software drivers for a specific hardware. Drivers are special firmware that enables hardware components to function properly.

   Problem 1 and 3 can be fixed easily by loose points and installing drivers respectively but 2 my require replacement of component's.

It is possible to find out which hardware devices are functioning properly and which are not via device managers' window and drivers indicated and those are not working Properly or those have missing drivers have a **!** or **?**



Possible solutions to hardware problems

1. If the device was bought separately form rest of the computer, them it came with an installation disk except for plug and play devices that has the relevant drivers. However, most devices nowadays have drivers somewhere on the internet which are downloadable.
2. Allow windows to detect the device and search for a driver on the hard disk or internet. This is usually one of the easiest ways of solving the problems.

If 1 and 2 does not then personnel for further advice,

Some hardware related problems and non diagnostic troubleshooting

Hardware related problem	Suggested cause	Suggested solution
Number key pad does not work	**Number lock key is active**	**Press number lock key**
Monitor is blank	• Monitor not switched on • Cable not connected • Bad/damaged monitor • Computer on stand by	• Switch on the monitor • Connect cable • Swap monitor • Press power button of computer
Computer cannot switch on	• Power supply damage • Power surges due on and off of electricity • Bad power cable damaged • Switch not working	• Replace power supply • Disconnect all cables and wait for 10 minutes then connect the cables and switch on • Replace the power cable • Call for a technician
Monitor showing no data	• Data cable not connecting monitor and mother board • Bad/damaged video card • Wrong /damage memory module	• Connect data cable properly • Replace video card • Replace RAM

Computer beeping /cannot start	• Wrong/damaged memory • Memory not installed properly • Hard drive damaged	• Install right memory • Install memory correctly • Replace hard drive
Computer shows message 'low disk space'	• Hard drive is full • Attack by virus • Non compress video , clips/files, films and audio music	• Delete some files and empty recycle bin • Run a fresh anti-virus • Compress or remove video and music files

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CHAPTER 1: Using spreadsheet

Objectives

By the end of this chapter, the student should be able to:

Contents

- Meaning of a spreadsheet
- List types of spreadsheets
- Exploring features of a spreadsheet
- Benefits of spreadsheets
- Differences between a workbook and a worksheet
- identifying cells, columns and rows
- Difference between labels and values
- Differences between relative cell references and absolute cell references
- Launching a spreadsheet programme
- Creating a worksheet and a workbook
- Saving a workbook
- Closing a worksheet
- Opening a workbook
- Entering data in a worksheet
- Editing a worksheet and a workbook
- Formatting a worksheet and a workbook
- Protecting workbooks
- Using shortcuts and commands in a worksheet
- Using relative and absolute cell addressing in a worksheet
- Freezing panes
- Hiding and unhiding columns
- Using basic formulae

- Using basic built-in functions in a worksheet (eg SUM, AVERAGE, MIN, MAX, COUNT, IF)

Introduction

People across the global work with numbers and manipulate them using formulas. spreadsheets have become the one of best approaches for data organization, manipulation and analyzing

Definition of spreadsheet

- This is a book which organizes data in roles and columns

Types of spreadsheet

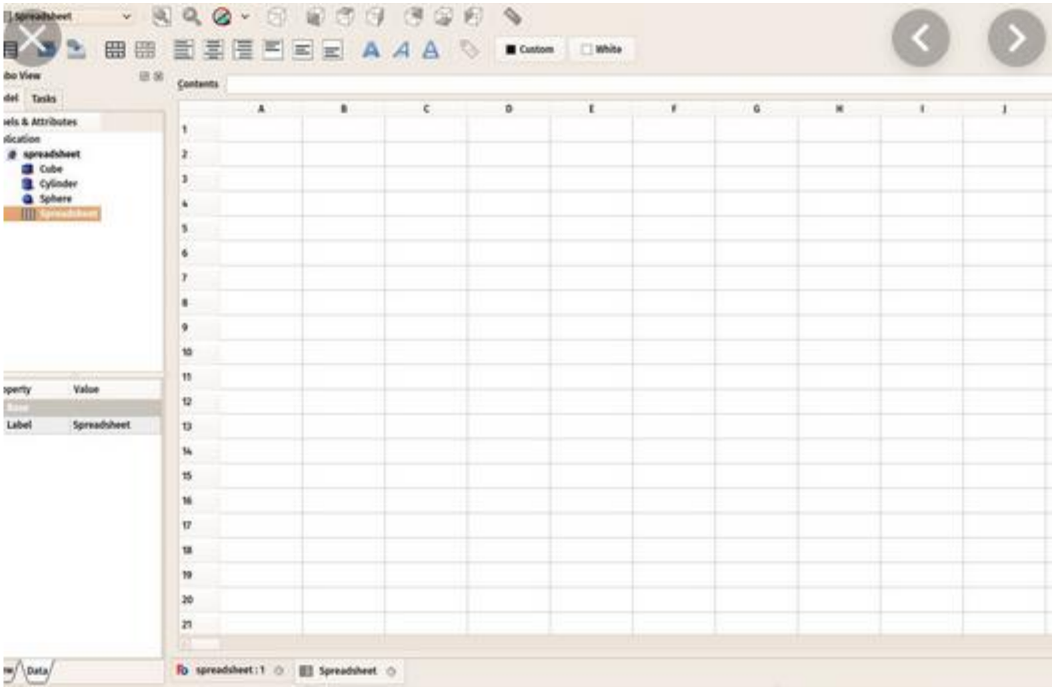
- **Spreadsheets are classified into two**
  a. Manual or traditional spreadsheets
  b. Electronic spreadsheets

Manual spreadsheet

- This is made up of sheet of papers divided into rows and columns on which numerical data is entered manually

Electronic spreadsheet

This is application software consisting of rows and columns used to organize, calculate and analyse numerical data electronically



Examples of electronic spreadsheets

a. Lotus 1-2-3
b. Microsoft Excel
c. LibreOffice/OpenOffice/NeoOffice
d. Quattro pro
e. Lotus symphony spreadsheets
f. PlanMaker
g. Google Sheets - (online and free).

Uses of electronic spreadsheets

1. Storing data

- Data can be stored in orderly manner due to the arrangement of rows and columns

2. Analyzing data

- Various analytical actions can be performed on data entered in electronic spreadsheet

3. Data presentation

- Electronic spreadsheet provides many ways of presenting data entered, including tables, graphs and charts

4. Future planning

- Due to visualization aspect, one is able to have an insight of present and future trend of data hence ability to calculate the potential effects of a mode;

5. Information modeling

- Able to include calculations that are automatically processed when when the values in a cell are edited

6. Data manipulation

- Allow users to enter custom formulas with a range of common inbuilt functions for data manipulation.

7. Decision making

- Ability to carry out performance measurement, e.g sales projections analysis

Advantage of electronic spreadsheet over manual spreadsheet

1. Accuracy and speed

- Complex calculations in electronic spreadsheets can be perfomed very quickly and accurately than using manual spreadsheet

2. Professional- looking results

- Data in electronic spreadsheets look professional due to the ability to quickly edit and format, perform calculations, create graphs and print the spreadsheet

3. Ability to edit data

- Data entered in electronic spreadsheet can be edited and revised using program commands
- Cell entries can be erased, moved or copied

4. Ability to format data

The appearance of data entered in electronic spreadsheets can be enhanced in many ways

5. Auto recalculations

- Dada in a cell where formula has been used can change and updated automatically if cell contents has been edited

6. Chart creation

- Ability to create chart for the data hence enhance data visualization

7. Easy to store and to retrieve data

- Ability to hold vast amount of data than traditional spreadsheets with tools for easy retrieval and sorting of said data

8. Ability to use multiple worksheets and files

- Has ability to open and use multiple files and able to create multiple worksheets

9. Forecasting

- Predictions can be made about what is likely to happen during next time

10. Modeling scenarios

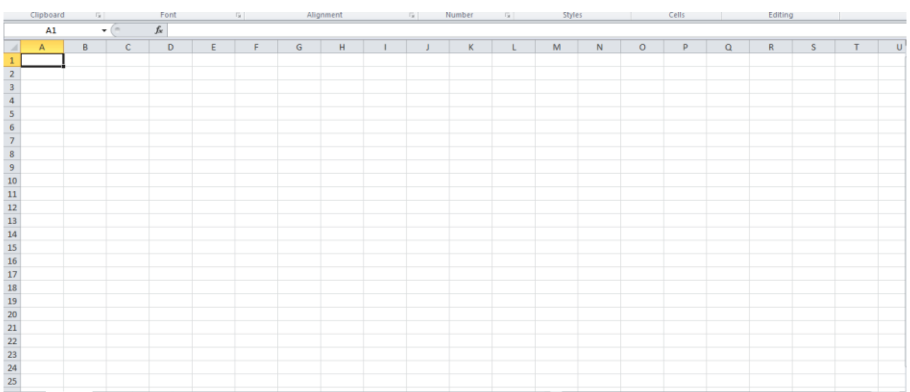
- Has ability to model different scenarios to find out what is likely to happen when a variable is changed

Components of spreadsheet

- Any Spreadsheet has the following components

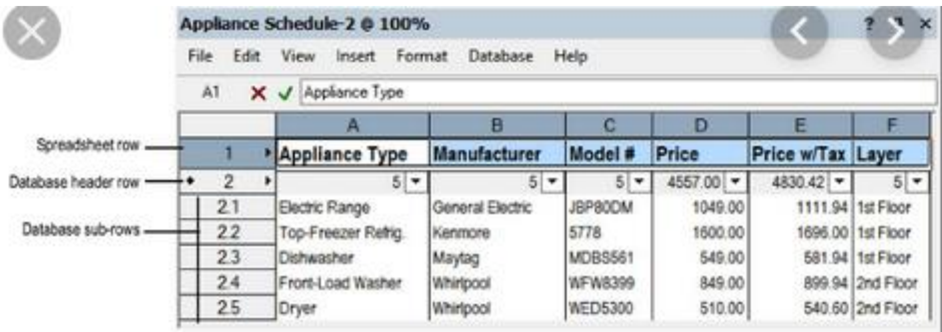
i. Worksheet

- Work area that has rows and columns where data is entered



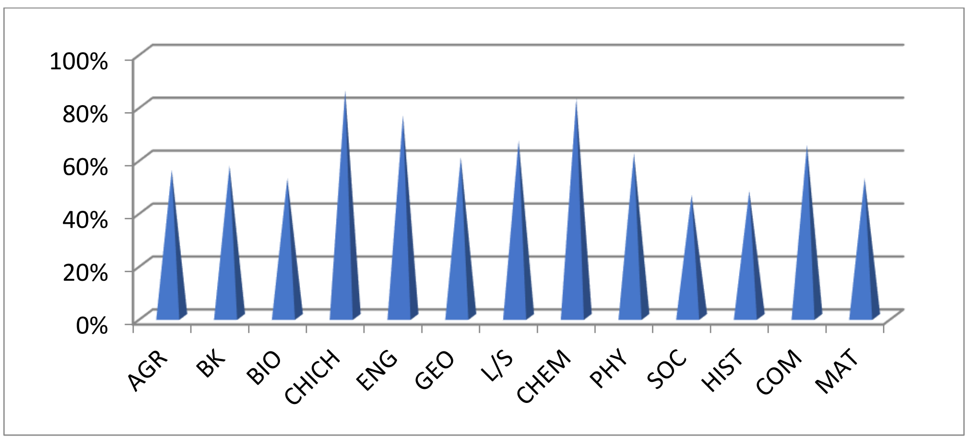
ii. Database

- Collection of related data items organized for rapid search and retrieval of data



iii. Charts/graphs

- Visual representation of worksheet data
- Helps users to easily pick out pattern and trends of data



Using spreadsheet (MicroSoft Excel 2007)

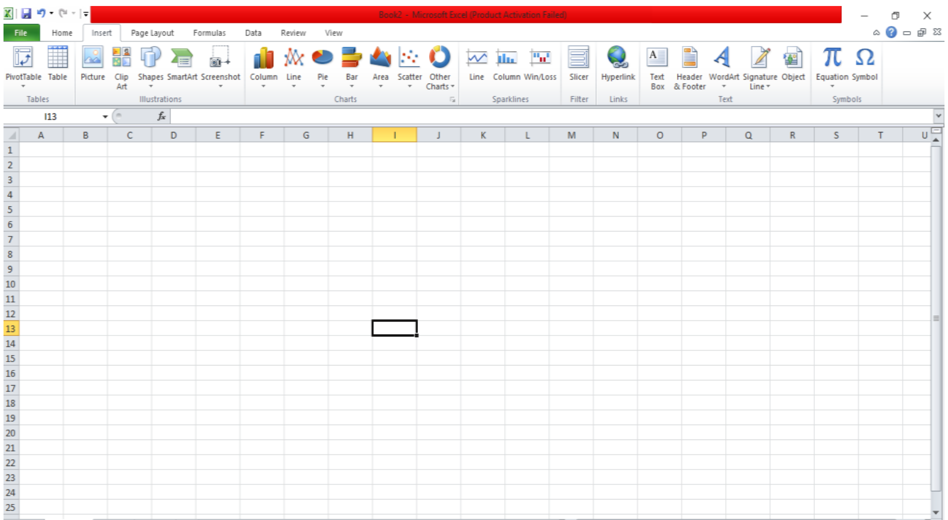
This is one of package of Microsoft Office application software

Starting Excel

To start Microsoft excel window spreadsheet follow the following steps

1. Click the start button on the lower left corner of your computer
2. Click the All programs arrow at the bottom left of the start menu

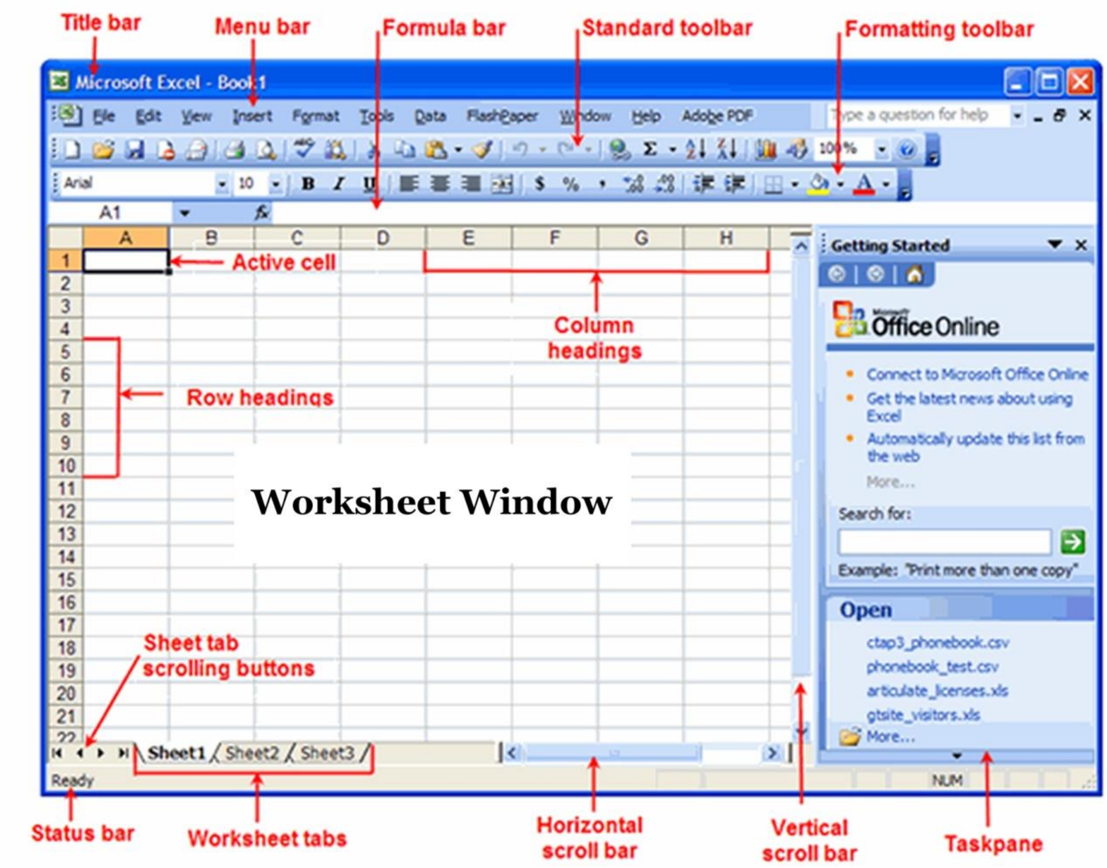
3. Click the Microsoft Office folder on the start menu. This will open the list of Microsoft office application
4. Click the Microsoft Excel 2007 option



The above excel window is displayed

Features of Excel Spreadsheet

The following diagram illustrates various features of excel spreadsheet window



User Interface of Spreadsheet Program

1. Rows

- This is horizontal arrangement of cells in a worksheet
- Rows are labeled using numbers 1,2,3, ….

2. Columns

- A column is vertical arrangement of cells in a worksheet
- Columns are labeled using letters A, B, C …. In a worksheet

3. Cell

- A cell is the intersection between a column and a row

- A cell pointer indicates the current active cell and usually highlighted with a bold outline

4. Worksheet

- This is working area made up of rows and columns where data is entered

5. Cell address

- This is a name of cell
- Also called cell reference
- Cell address is formed by combining column position and row position , for example cell **A1** is formed by intersecting column **A** and Row **1**

6. Workbook

- This is excel file that contains one or more worksheets
- Excel will assign file name to the workbook such as Book1, Book2, Book3 and so on, depending on how many workbooks are opened.

7. Worksheet tabs

- Worksheet tabs identify the various worksheets in a workbook and allow you to move from one worksheet to another
- A workbook initially contains three worksheets which are saved in a single file

8. Title bar

- This displays the name of the program as well as the name of the current workbook if it has been saved

- It the workbook has not been saved, it is identified by a number for example Book1

9. Scroll bars

- There are vertical and horizontal scroll bars
- They are used to scroll the workbook window vertically or horizontally through a worksheet

10. Status bar

- It displays helpful information as you use the program
- The ready indicator that appears lets you know that the program is ready for data input

11. Name box

- This identifies an active cell by indicating its cell address
- An active cell is a cell with a black highlighted outline/boarder which is ready for data entry

12. Formula bar

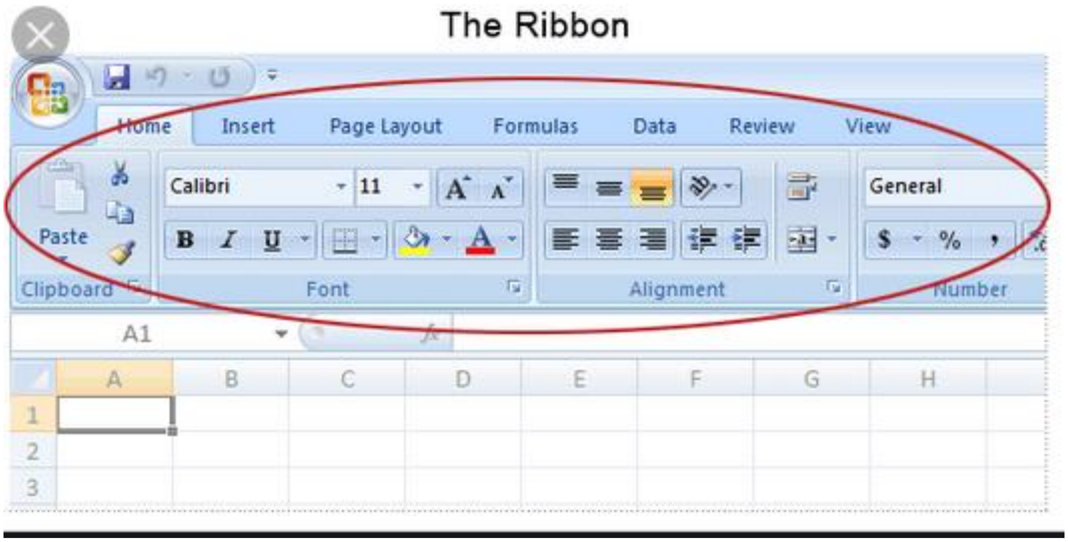
- This displays the contents of the active cell, if any

13. Quick access toolbar

- This is a customizable toolbar that contains a set of commands that are independent of the tab that is currently displayed.

14. Ribbon

- Strip of buttons and icons located above the work area
- Above the ribbon are a number of tabs such as Home, Insert and page layout



Navigating through the excel worksheet

There are many ways of how one can navigate through excel worksheet

1. Using arrow keys of a keyboard
   - Up and down arrows to move up or down of the active cell
   - Left and right arrows to navigate into a cell to the left or right of an active cell
2. Clicking a cell where you want to navigate into
3. Using tab key
   - To navigate into cell to the right of an active cell

Entering data in a worksheet

There are four types of data that can be entered into an excel worksheet

1. Labels

- These are alphanumerical characters entered into a cell

- For example names of students in form three at Kings Foundation can be entered as labels
- Sometimes numerical values can be formatted so that they are used as labels

2. Values

- This is numerical data that can be manipulated mathematically in a worksheet
- Examples of such values include currency, number (0-9) and date

3. Formula

- This is mathematical expression that can be used to manipulate data in a worksheet
- For example formula =B1+B3 adds the contents of cells B1 and B3.

4. Function

- This is inbuilt formula that can be quickly used instead of having to create a new formula each time.
- For example a function =SUM(B1:B7) adds up the contents of cells B1 up to B7

Process of entering data in Excel

- To **enter data in Excel**, just select a cell and begin **typing**.
- You'll see the text appear both in the cell and in the formula bar.
- To tell **Excel** to accept the **data** you've typed, press **enter**.

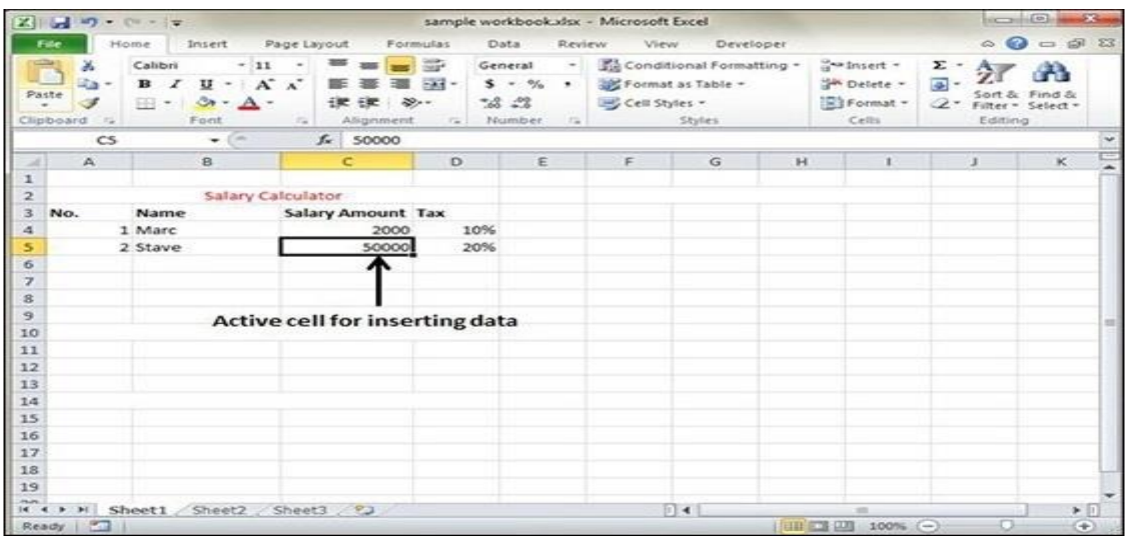
- The information will be **entered** immediately, and the cursor will move down one cell.

Editing a worksheet

This involves changing the appearance of the data being handled in an excel worksheet

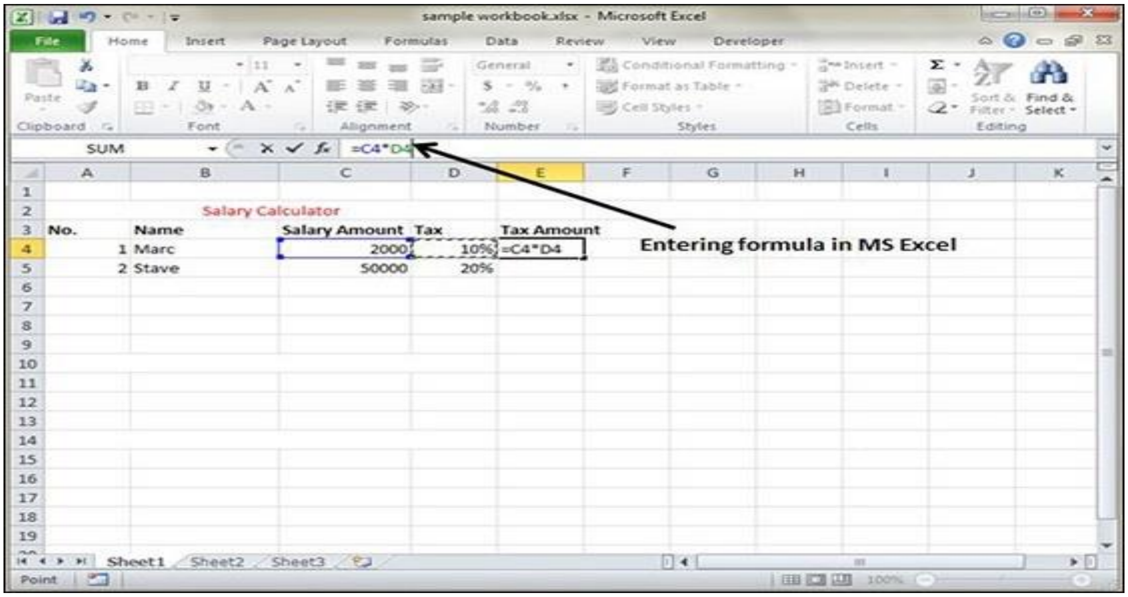
Inserting Data

For **inserting data** in **MS Excel,** just activate the cell type text or number and press enter or Navigation keys.



Inserting Formula

For inserting formula in MS Excel go to the formula bar, enter the formula and then press enter or navigation key. See the screen-shot below to understand it.



Modifying Cell Content

For modifying the cell content just activate the cell, enter a new value and then press enter or navigation key to see the changes. See the screen-shot below to understand it.

Select Data in a work sheet

There are three ways of doing this

i. Select with Mouse

- Drag the mouse over the data you want to select. It will select those cells as shown below.

ii. Select with special

- If you want to select specific region, select any cell in that region. Pressing **F5** will show the below dialogue box
- Click on **Special button** to see the below dialogue box. Select **current region** from the radio buttons. Click on **ok** to see the current region selected.

Delete data in a worksheet

There are two ways

i. Delete with a mouse

- Select the data you want to delete.
- **Right Click** on the sheet. Select the **delete option**, to delete the data.

ii. Delete with a delete key

- Select the data you want to delete. Press on the **Delete Button** from the keyboard, it will delete the data.

iii. Selective delete for rows

- Select the rows, which you want to delete with **Mouse click + Control Key.** Then right click to show the various options. Select the **Delete option** to delete the selected rows.

Move data in a worksheet

To move data in a worksheet follow the following steps.

Step 1 – Select the data you want to Move. **Right Click** and Select the **cut option.**

Step 2 – **Select the first cell** where you want to move the data. Right click on it and **paste the data**.

Copy and paste

- **MS Excel** provides **copy paste** option in different ways. As follows:

  i. Using a shortcut

- To copy and paste, just select the cells you want to copy. Choose **copy option** after right click or press **Control + C**.Select the cell where you need to paste this copied content. Right click and select paste option or press **Control + V**.

In this case, **MS Excel** will copy everything such as values, formulas, Formats, Comments and validation. MS Excel will overwrite the content with paste. If you want to undo this, press **Control + Z** from the keyboard.

  ii. Using office Clipboard

- When you copy data in MS Excel, it puts the copied content in Windows and Office Clipboard. You can view the clipboard content by **Home** → **Clipboard**.

- View the clipboard content. Select the cell where you need to paste. Click on paste, to paste the content.

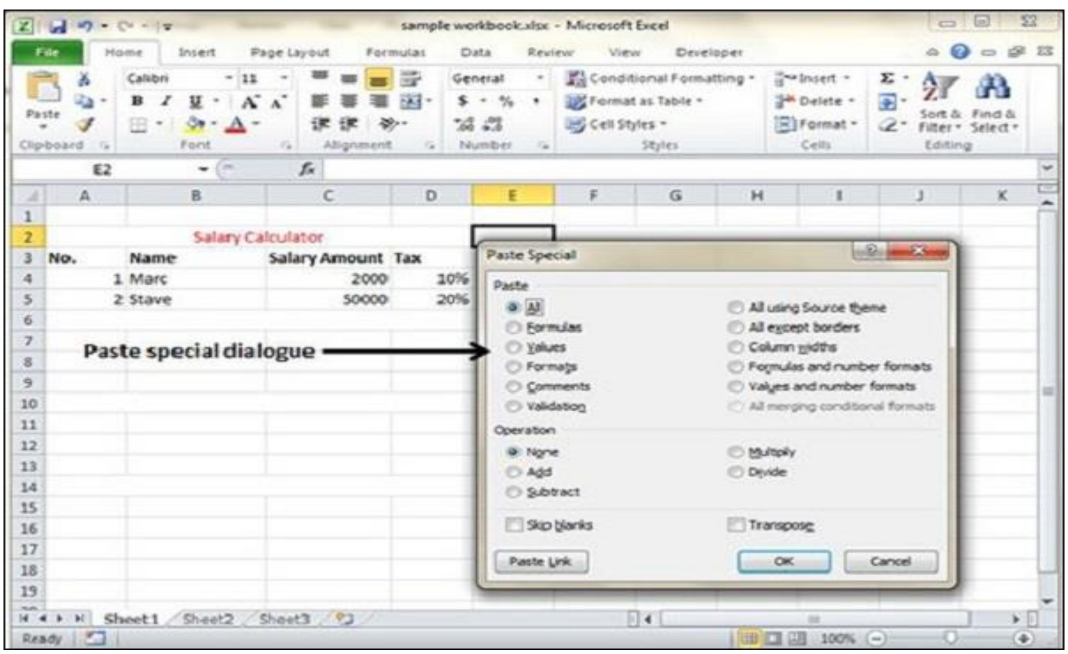
iii. Copy and paste I a special way

- You may not want to copy everything in some cases. For example, you want to copy only Values or you want to copy only the formatting of cells. Select the paste special option as shown below.

Below are the various options available in paste special.

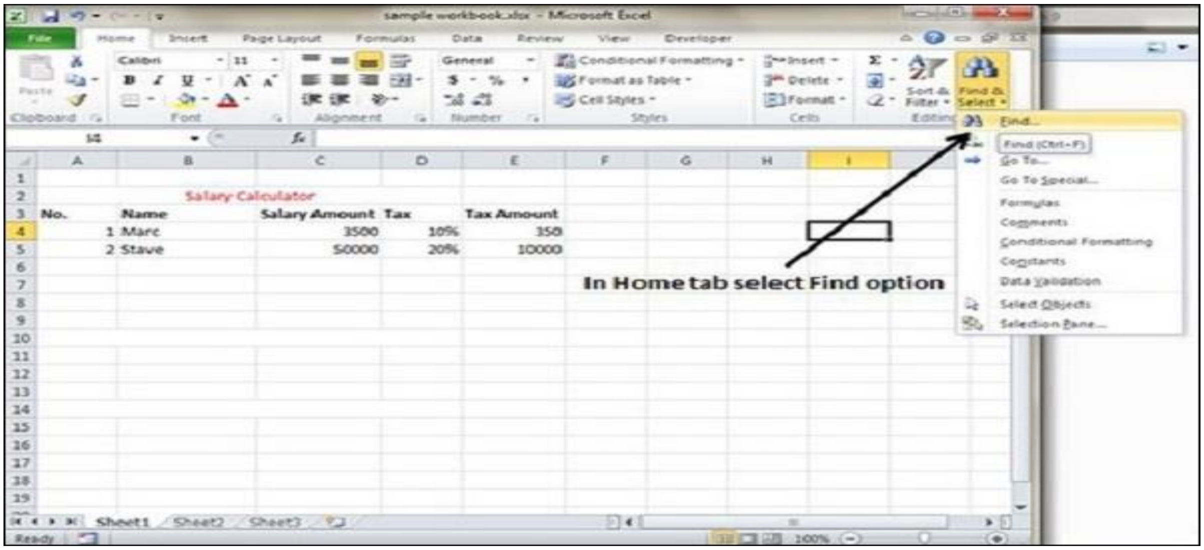
- **All** – Pastes the cell's contents, formats, and data validation from the Windows Clipboard.
- **Formulas** – Pastes formulas, but not formatting.
- **Values** – Pastes only values not the formulas.
- **Formats** – Pastes only the formatting of the source range.
- **Comments** – Pastes the comments with the respective cells.
- **Validation** – Pastes validation applied in the cells.
- **All using source theme** – Pastes formulas, and all formatting.
- **All except borders** – Pastes everything except borders that appear in the source range.
- **Column Width** – Pastes formulas, and also duplicates the column width of the copied cells.
- **Formulas & Number Formats** – Pastes formulas and number formatting only.

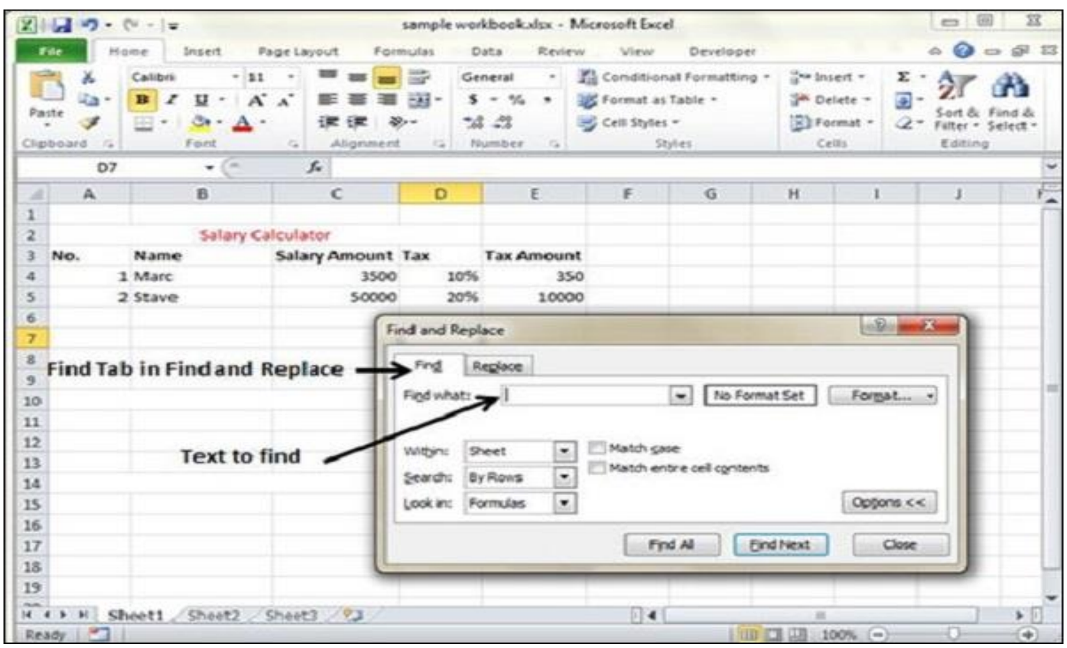
- **Values & Number Formats** – Pastes the results of formulas, plus the number.
- **Merge Conditional Formatting** – This icon is displayed only when the copied cells contain conditional formatting. When clicked, it merges the copied conditional formatting with any conditional formatting in the destination range.
- **Transpose** – Changes the orientation of the copied range. Rows become columns, and columns become rows. Any formulas in the copied range are adjusted so that they work properly when transposed.



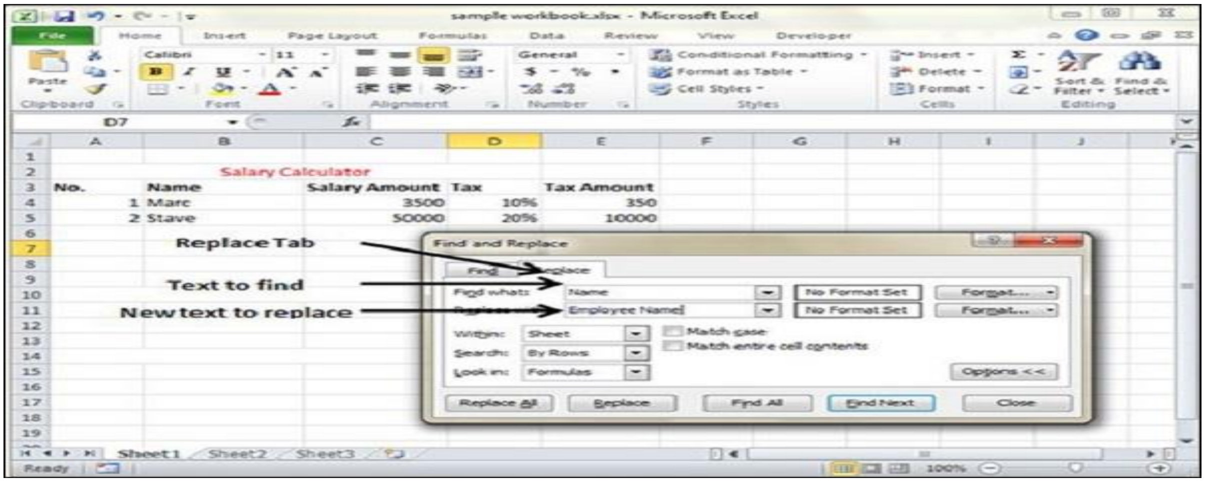
Find and Replace

To access the Find & Replace, Choose **Home** → **Find & Select** → **Find** or press **Control + F Key**. See the image below.





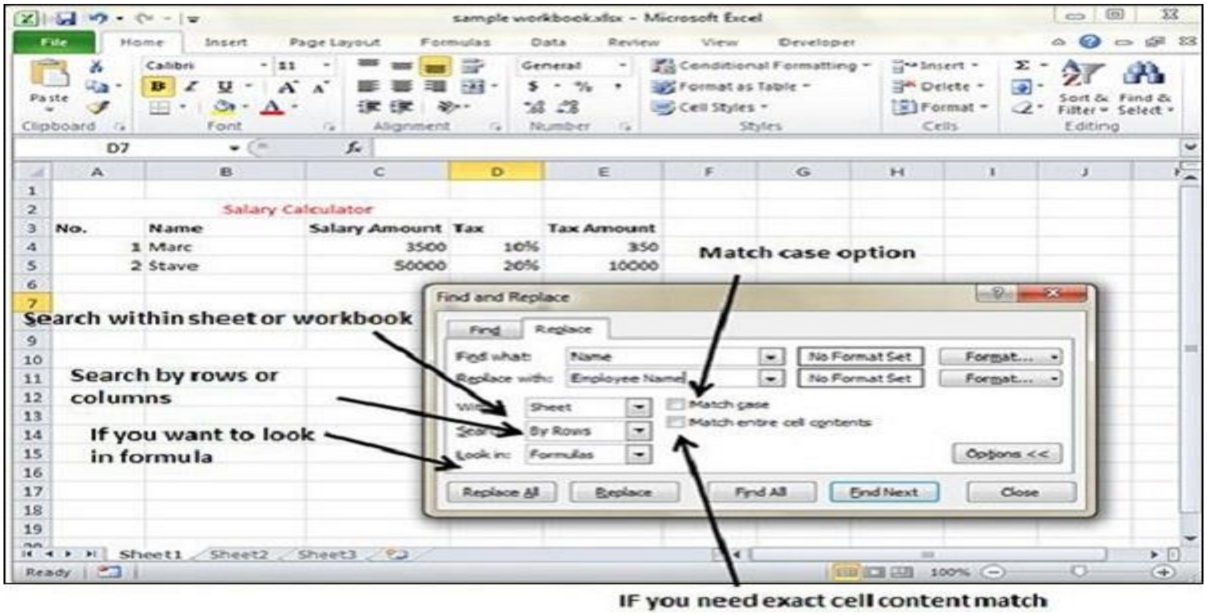
You can replace the found text with the new text in the **Replace tab**.



Exploring Options

Now, let us see the various options available under the Find dialogue.

- **Within** – Specifying the search should be in Sheet or workbook.
- **Search By** – Specifying the internal search method by rows or by columns.
- **Look In** – If you want to find text in formula as well, then select this option.
- **Match Case** – If you want to match the case like lower case or upper case of words, then check this option.
- **Match Entire Cell Content** – If you want the exact match of the word with cell, then check this option.



Spell Check

Let us see how to access the spell check.

- To access the spell checker, Choose **Review** ⇨ **Spelling** or press **F7**.
- To check the spelling in just a particular range, **select the range** before you activate the spell checker.
- If the spell checker finds any words it does not recognize as correct, it displays the **Spelling dialogue** with suggested options.

The following are various options available in **spell check** dialogue.

- **Ignore Once** – Ignores the word and continues the spell check.

- **Ignore All** – Ignores the word and all subsequent occurrences of it.
- **Add to Dictionary** – Adds the word to the dictionary.
- **Change** – Changes the word to the selected word in the Suggestions list.
- **Change All** – Changes the word to the selected word in the Suggestions list and changes all subsequent occurrences of it without asking.
- **AutoCorrect** – Adds the misspelled word and its correct spelling (which you select from the list) to the AutoCorrect list.

Zooming in and Zooming Out

- By default, everything on screen is displayed at 100% in MS Excel.
- You can change the zoom percentage from 10% (tiny) to 400% (huge).
- Zooming doesn't change the font size, so it has no effect on the printed output.

Zoom In

- You can zoom in the workbook by moving the zoom slider to the right.
- It will change the only view of the workbook. You can have maximum of 400% zoom in

Zoom Out

- You can zoom out the workbook by moving the slider to the left.
- It will change the only view of the workbook.
- You can have maximum of 10% zoom in

Undo and Redo Changes

You can reverse almost every action in Excel by using the Undo command. We can undo changes in following two ways.

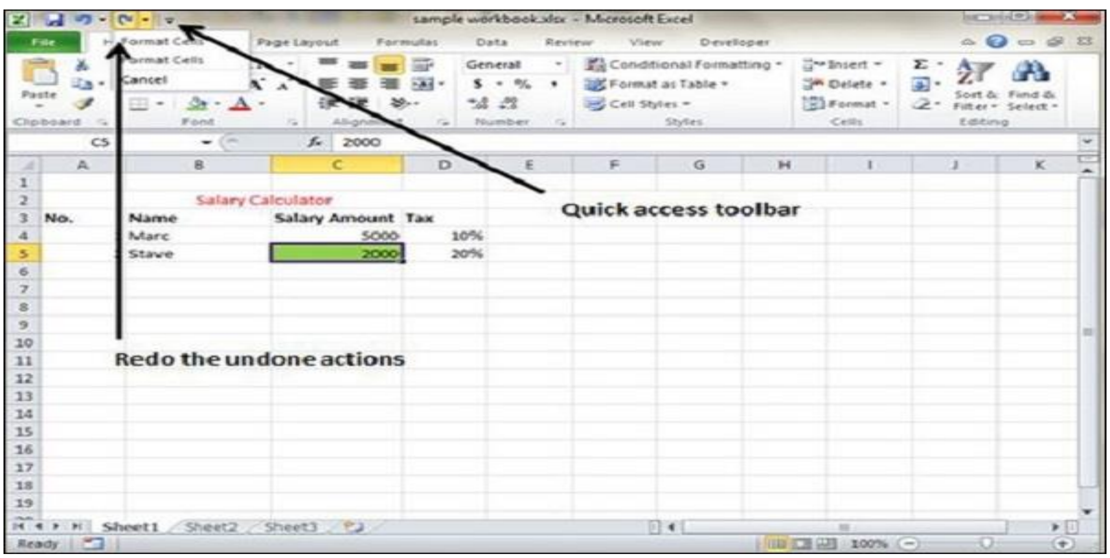
- From the Quick access tool-bar » Click Undo.

- Press Control + Z

- You can reverse the effects of the past 100 actions that you performed by executing Undo more than once.
- If you click the arrow on the right side of the Undo button, you see a list of the actions that you can reverse. Redo Changes

You can again reverse back the action done with undo in Excel by using the Redo command. We can redo changes in following two ways.

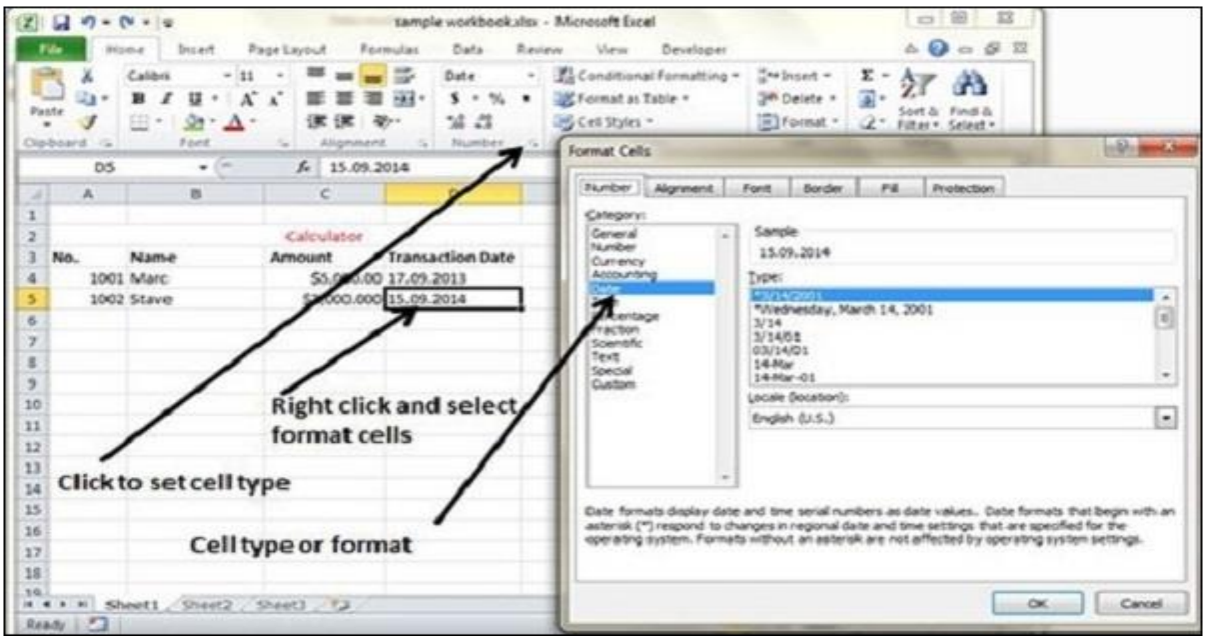
- From the Quick access tool-bar » Click Redo.
- Press Control + Y.



Formatting Cell

- MS Excel Cell can hold different types of data like Numbers, Currency, Dates, etc.
- You can set the cell type in various ways as shown below –

- Right Click on the cell » Format cells » Number.
- Click on the Ribbon from the ribbon.



Various Cell Formats

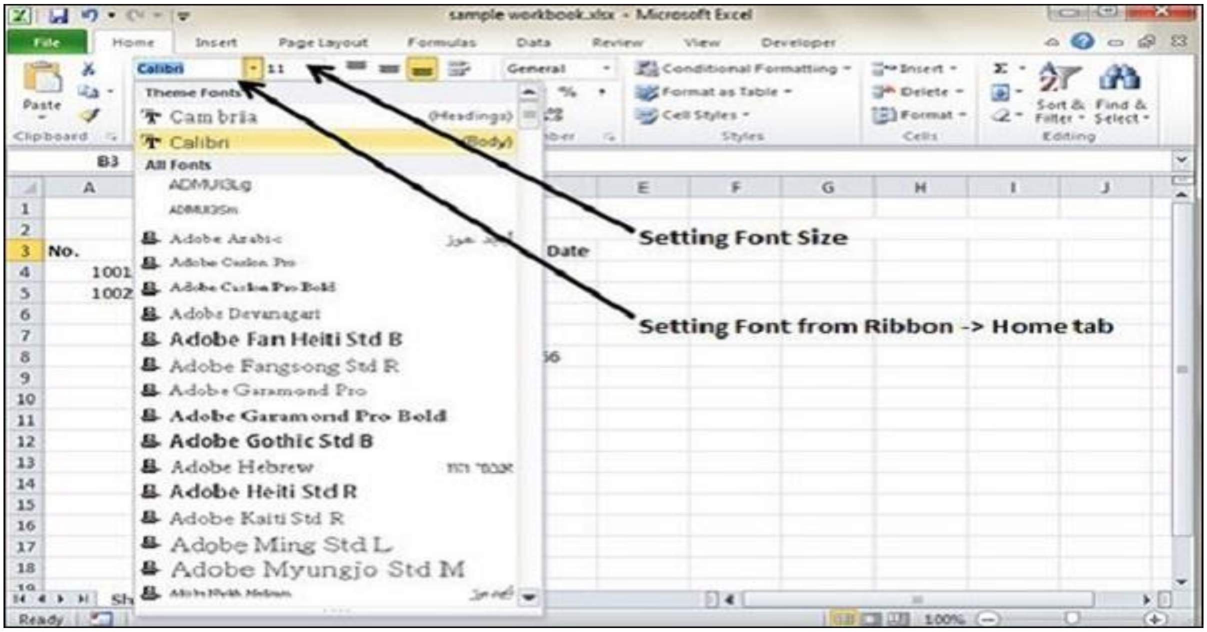
Below are the various cell formats.

- **General** – This is the default cell format of Cell.
- **Number** – This displays cell as number with separator.

- **Currency** – This displays cell as currency i.e. with currency sign.
- **Accounting** – Similar to Currency, used for accounting purpose.
- **Date** – Various date formats are available under this like 17-09-2013, 17^{th}-Sep-2013, etc.
- **Time** – Various Time formats are available under this, like 1.30PM, 13.30, etc.
- **Percentage** – This displays cell as percentage with decimal places like 50.00%.
- **Fraction** – This displays cell as fraction like 1/4, 1/2 etc.
- **Scientific** – This displays cell as exponential like 5.6E+01.
- **Text** – This displays cell as normal text.
- **Special** – Special formats of cell like Zip code, Phone Number.
- **Custom** – You can use custom format by using this.

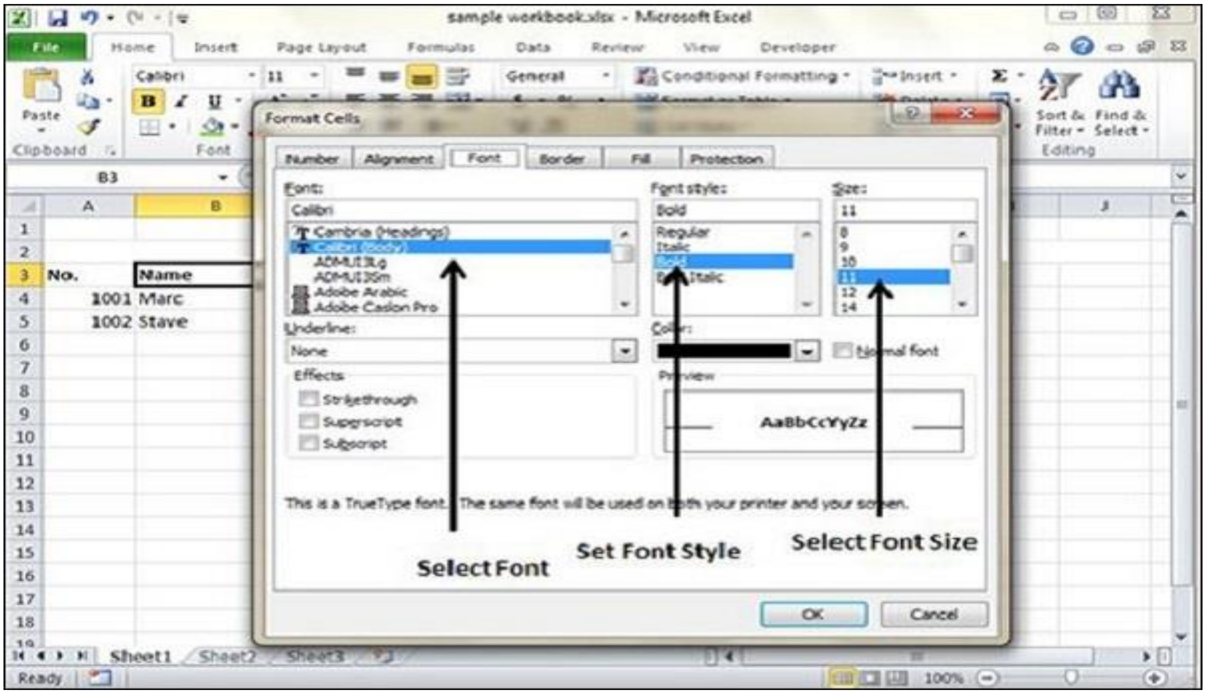
Setting Font from Home

- You can set the font of the selected text from **Home » Font group » select the font**.



Setting Font From Format Cell Dialogue

- Right click on cell » Format cells » Font Tab
- Press Control + 1 or Shift + Control + F



Text Decoration

Various options are available in Home tab of the ribbon as mentioned below.

- **Bold** – It makes the text in bold by choosing **Home » Font Group » Click B** or Press **Control + B**.
- **Italic** – It makes the text italic by choosing **Home » Font Group » Click I** or Press **Control + I**.
- **Underline** – It makes the text to be underlined by choosing **Home » Font Group » Click U** or Press **Control + U**.

- **Double Underline** – It makes the text highlighted as double underlined by choose **Home » Font Group » Click arrow near U » Select Double Underline**.

- There are more options available for text decoration in Formatting cells » Font Tab » Effects cells as mentioned below.

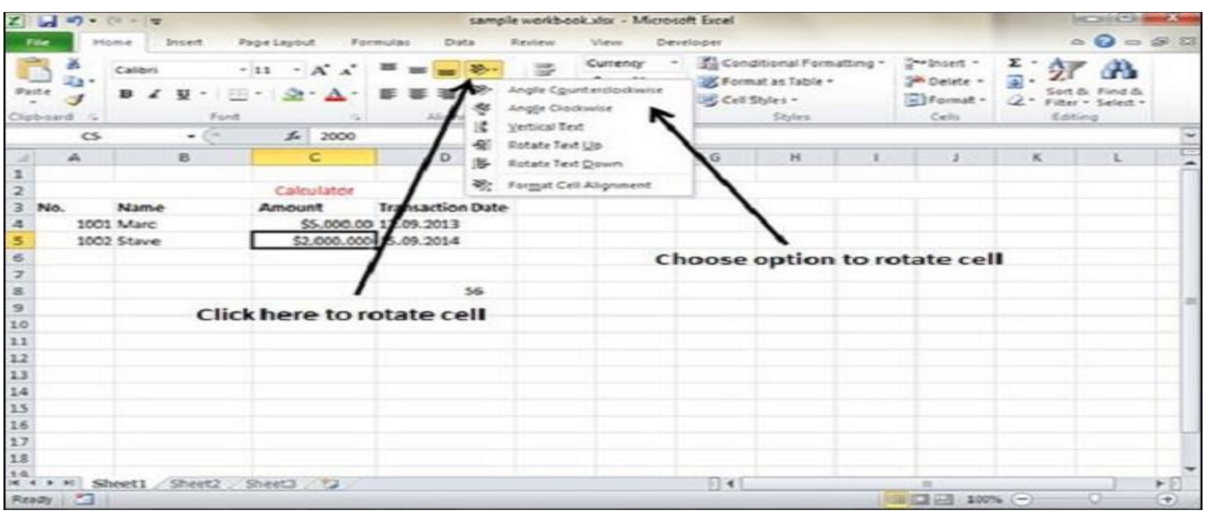
- **Strike-through** – It strikes the text in the center vertically.
- **Super Script** – It makes the content to appear as a super script.
- **Sub Script** – It makes content to appear as a sub script.

Rotate Cells

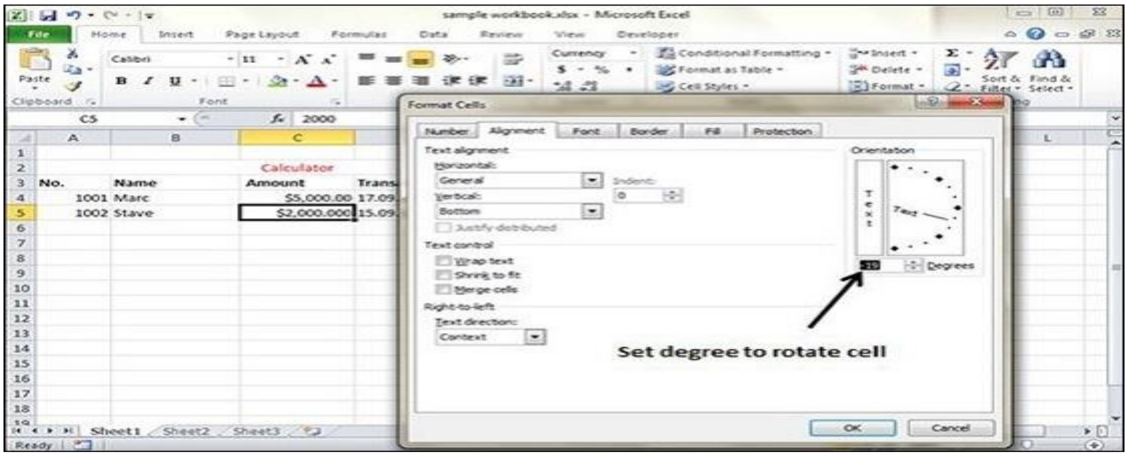
Rotate cells

There are several ways to do this

1. Click on the **orientation** in the **Home tab**. Choose options available like Angle CounterClockwise, Angle Clockwise, etc.

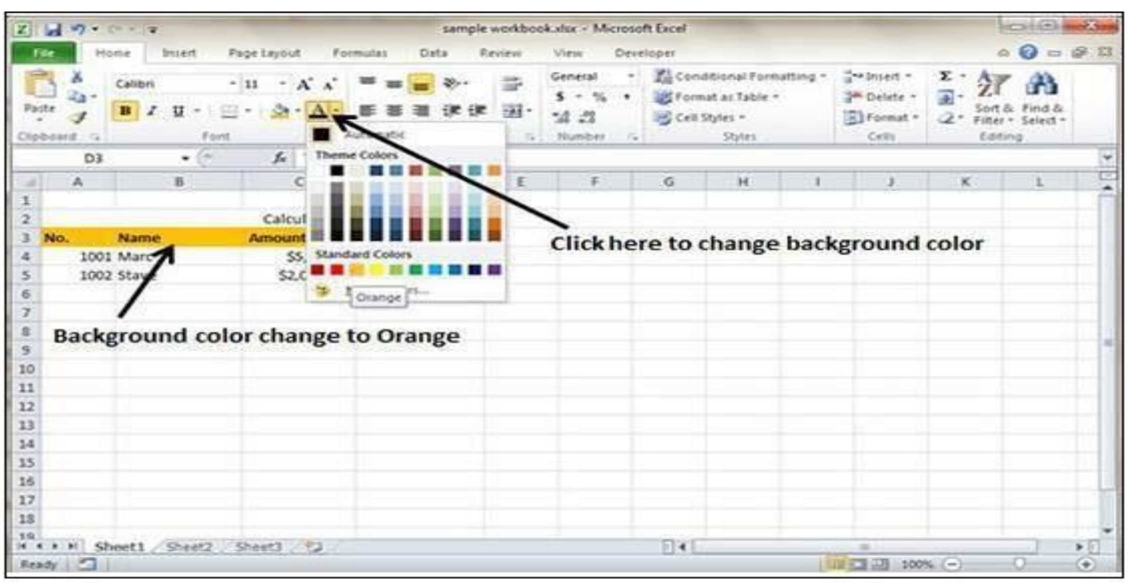


2. Right Click on the cell. Choose Format cells » Alignment » Set the degree for rotation.



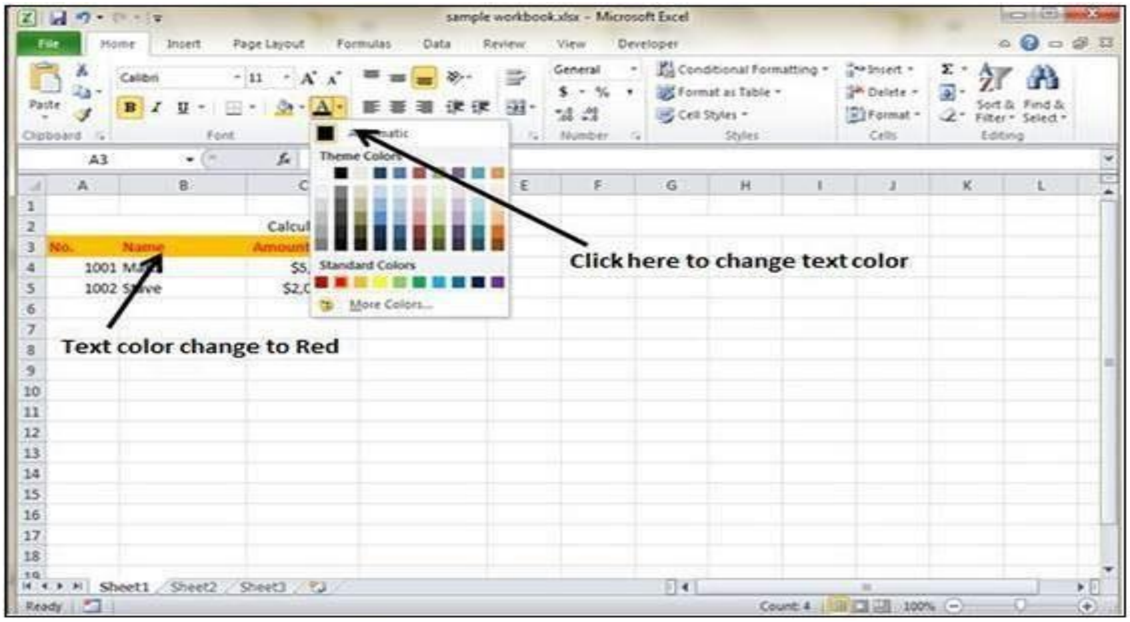
Changing Background Color

- By default the background color of the cell is white in MS Excel.
- You can change it as per your need from **Home tab » Font group » Background color**.

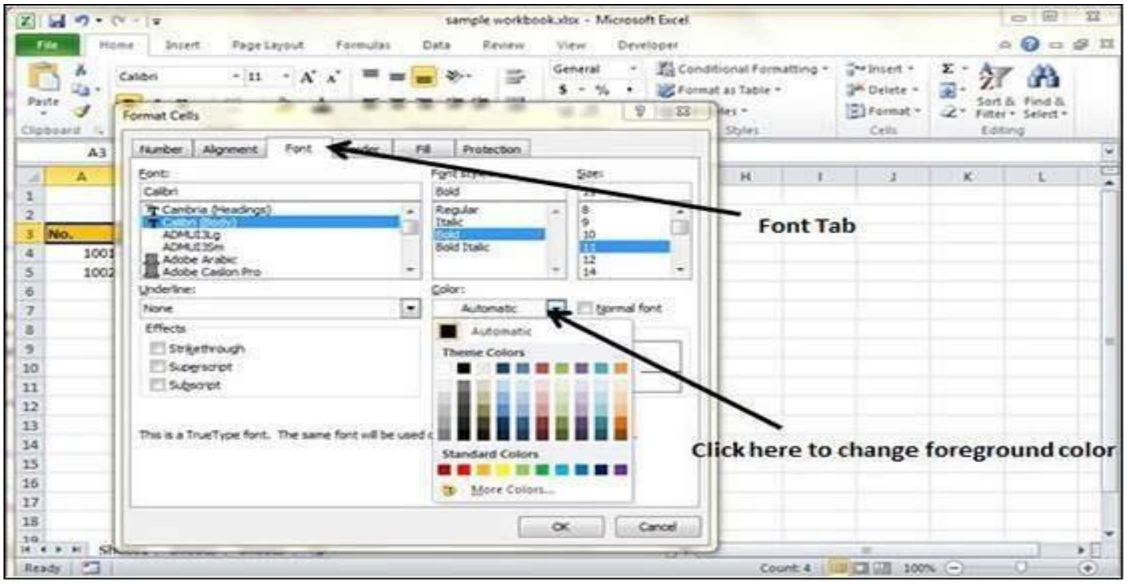


Changing Foreground Color

- By default, the foreground or text color is black in MS Excel. You can change it as per your need from **Home tab » Font group » Foreground color**.



Also you can change the foreground color by selecting the cell **Right click » Format cells » Font Tab » Color**.



Change Alignment

This involves making changes on the orientation of data in a cell
Can be done in two ways

1. Using Home Tab

- You can change the Horizontal and vertical alignment of the cell.
- By default, Excel aligns numbers to the right and text to the left.
- Click on the available option in the Alignment group in Home tab to change alignment.

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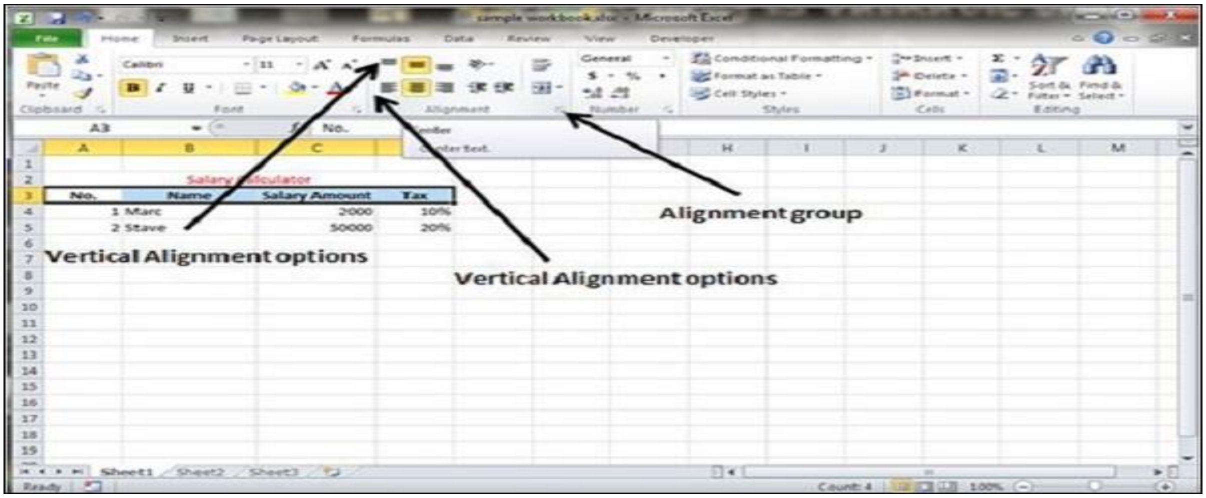
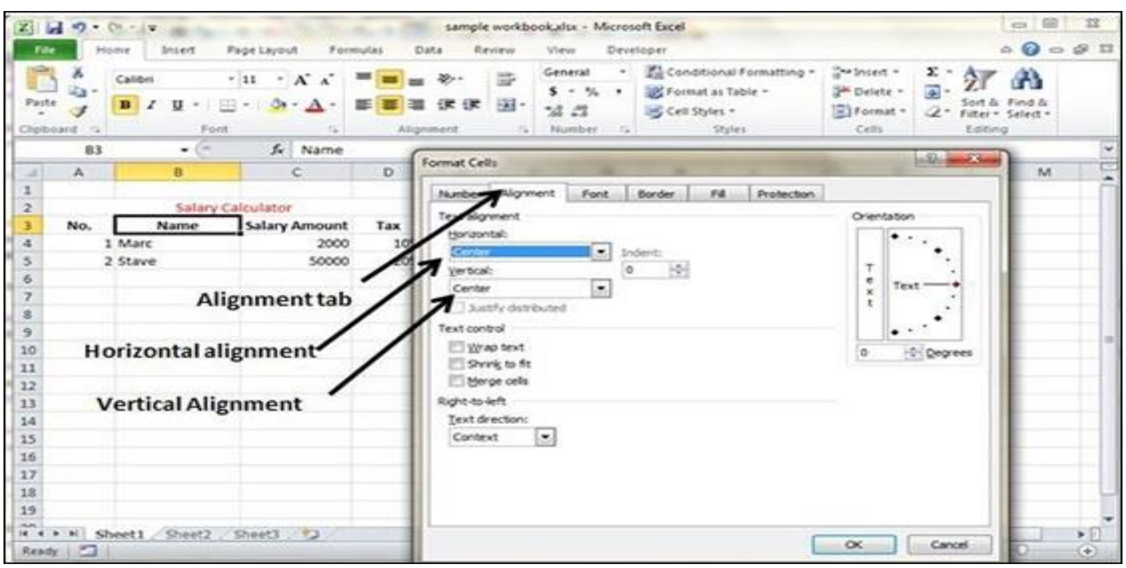


2. Using Format Cells

- Right click on the cell and choose format cell. In format cells dialogue, choose **Alignment Tab**. Select the available options from the Vertical alignment and Horizontal alignment options.



Exploring Alignment Options

1. Horizontal Alignment – You can set horizontal alignment to Left, Centre, Right, etc.

- **Left** – Aligns the cell contents to the left side of the cell.
- **Center** – Centers the cell contents in the cell.
- **Right** – Aligns the cell contents to the right side of the cell.
- **Fill** – Repeats the contents of the cell until the cell's width is filled.
- **Justify** – Justifies the text to the left and right of the cell. This option is applicable only if the cell is formatted as wrapped text and uses more than one line.

2. Vertical Alignment – You can set Vertical alignment to top, Middle, bottom, etc.

- **Top** Aligns the cell contents to the top of the cell.
- **Center** Centers the cell contents vertically in the cell.
- **Bottom** Aligns the cell contents to the bottom of the cell.
- **Justify** Justifies the text vertically in the cell; this option is applicable only if the cell is formatted as wrapped text and uses more than one line.

Merge Cells

- MS Excel enables you to merge two or more cells.

- When you merge cells, you don't combine the contents of the cells.
- Rather, you combine a group of cells into a single cell that occupies the same space.

You can merge cells by various ways as mentioned below.

1. Choose **Merge & Center control** on the Ribbon, which is simpler. To merge cells, select the cells that you want to merge and then click the Merge & Center button.
2. Choose **Alignment tab** of the Format Cells dialogue box to merge the cells.

Additional Options

The **Home » Alignment group » Merge & Center control** contains a drop-down list with these additional options –

- **Merge Across** – When a multi-row range is selected, this command creates multiple merged cells — one for each row.
- **Merge Cells** – Merges the selected cells without applying the Center attribute.
- **Unmerge Cells** – Unmerges the selected cells.

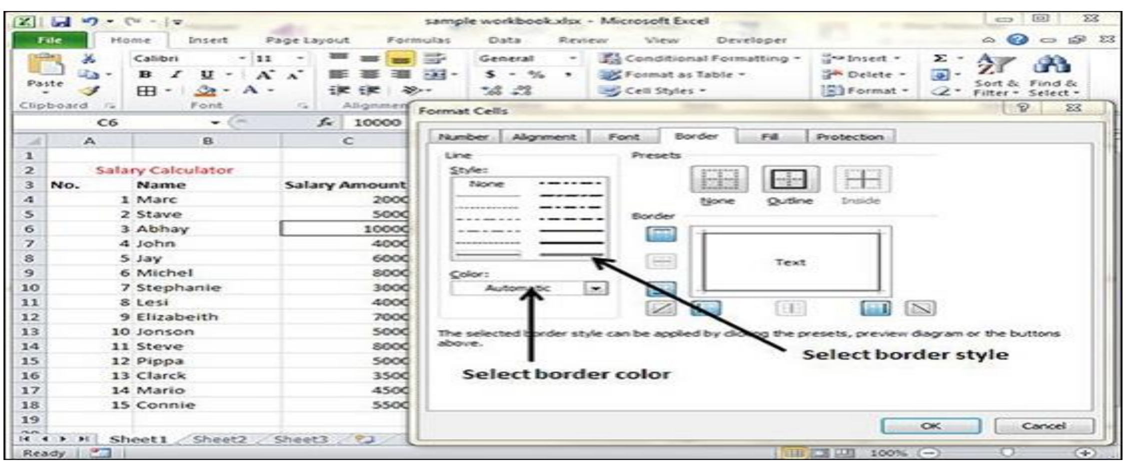
Wrap Text and Shrink to Fit

- If the text is too wide to fit the column width but don't want that text to spill over into adjacent cells, you can use either the Wrap Text option or the Shrink to Fit option to accommodate that text.

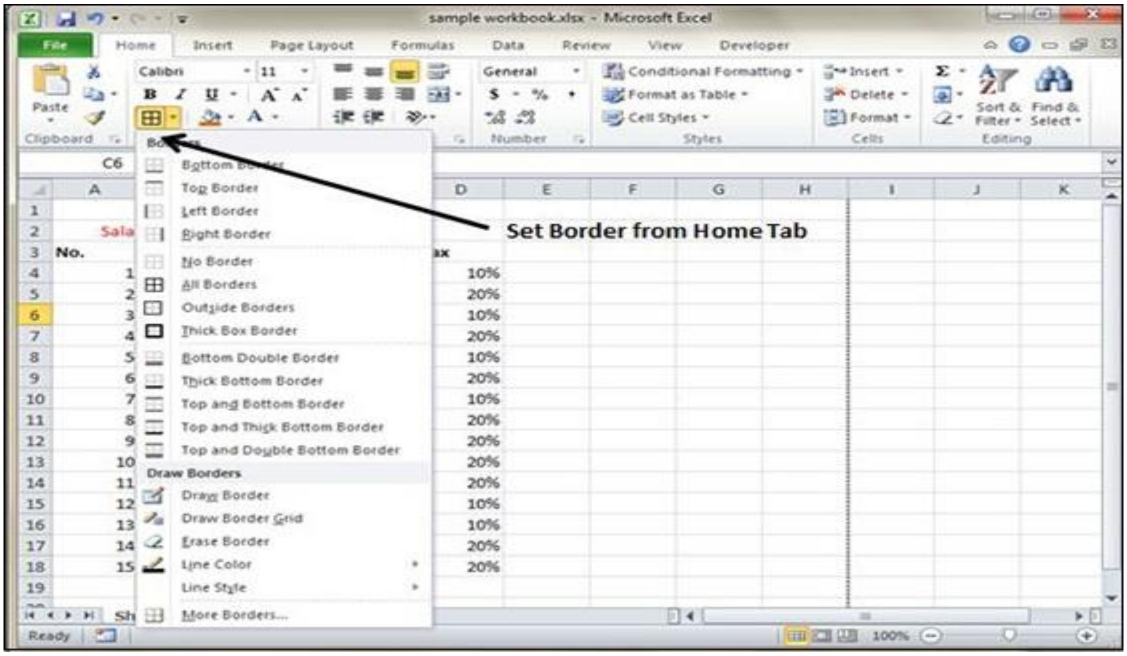
Boarders and shades

Apply Borders

- MS Excel enables you to apply borders to the cells. For applying border, select the range of cells **Right Click » Format cells » Border Tab » Select the Border Style**.

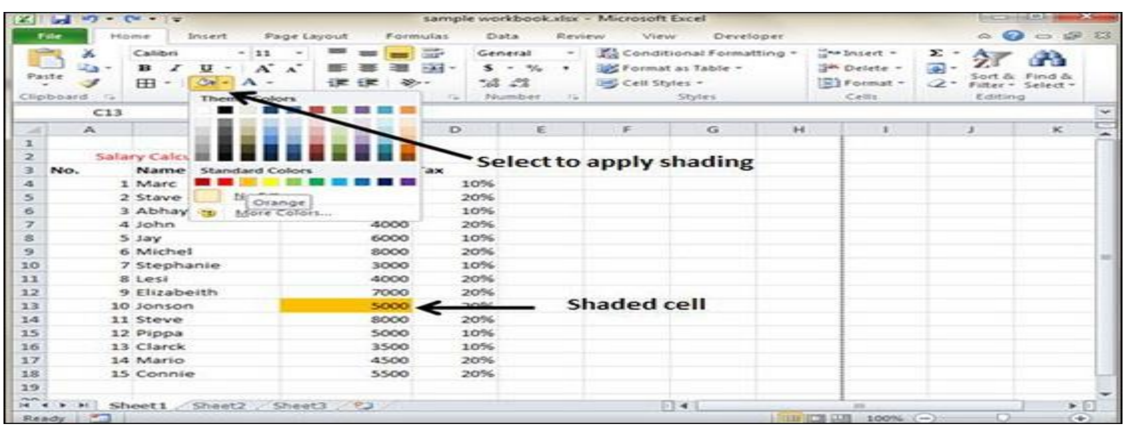


- Then you can apply border by Home Tab » Font group » Apply Borders.



Apply Shading

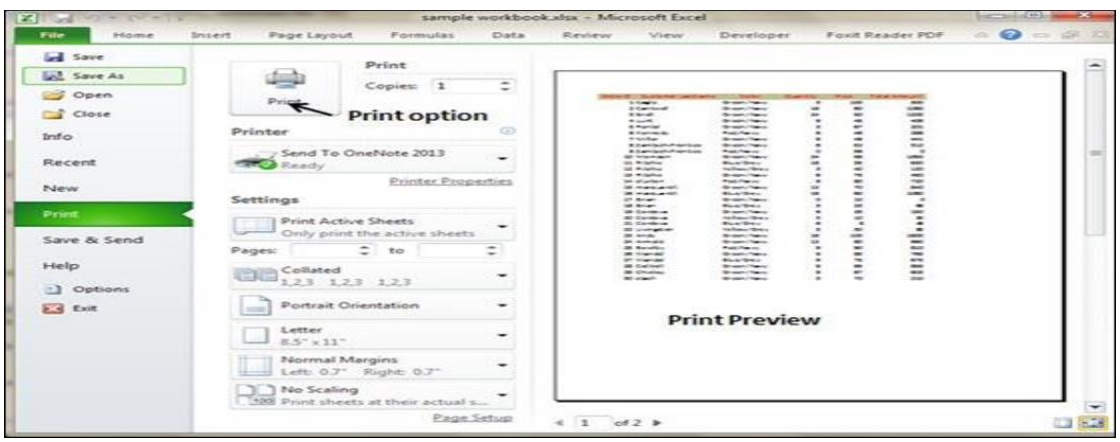
- You can add shading to the cell from the **Home tab » Font Group » Select the Color**.



Printing a worksheet

- If you want to print a copy of a worksheet with no layout adjustment, use the **Quick Print** option.
- There are two ways in which we can use this option.

1. Choose **File » Print** (which displays the Print pane), and then click the Print button.
2. Press Ctrl+P and then click the Print button (or press Enter).



Adjusting Common Page Setup Settings

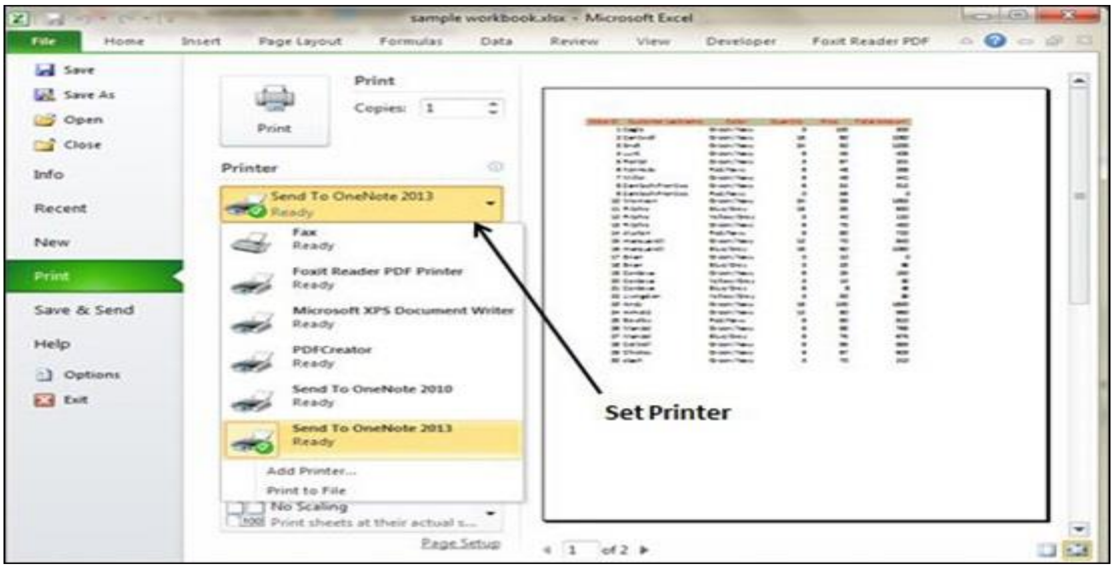
- You can adjust the print settings available in the Page setup dialogue in different ways as discussed below.
- Page setup options include Page orientation, Page Size, Page Margins, etc.

1. The Print screen in Backstage View, displayed when you choose **File » Print**.

2. The **Page Layout tab** of the Ribbon.

Choosing Your Printer

- To switch to a different printer, choose **File » Print** and use the drop-down control in the Printer section to select any other installed printer.

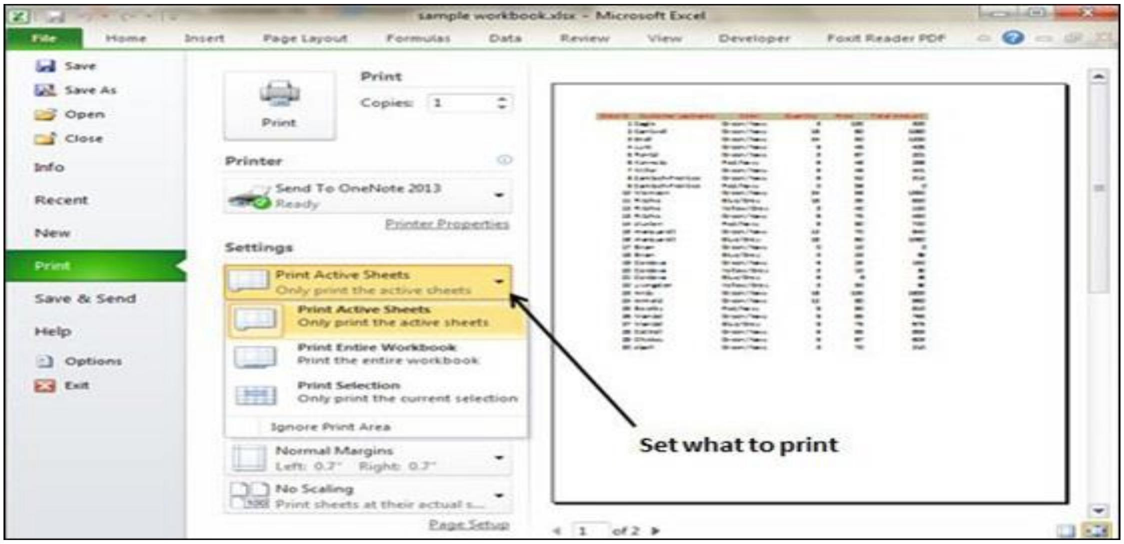


Specifying What You Want to Print

- Sometimes you may want to print only a part of the worksheet rather than the entire active area. Choose **File » Print** and use the controls in the Settings section to specify what to print.

- **Active Sheets** – Prints the active sheet or sheets that you selected.
- **Entire Workbook** – Prints the entire workbook, including chart sheets.

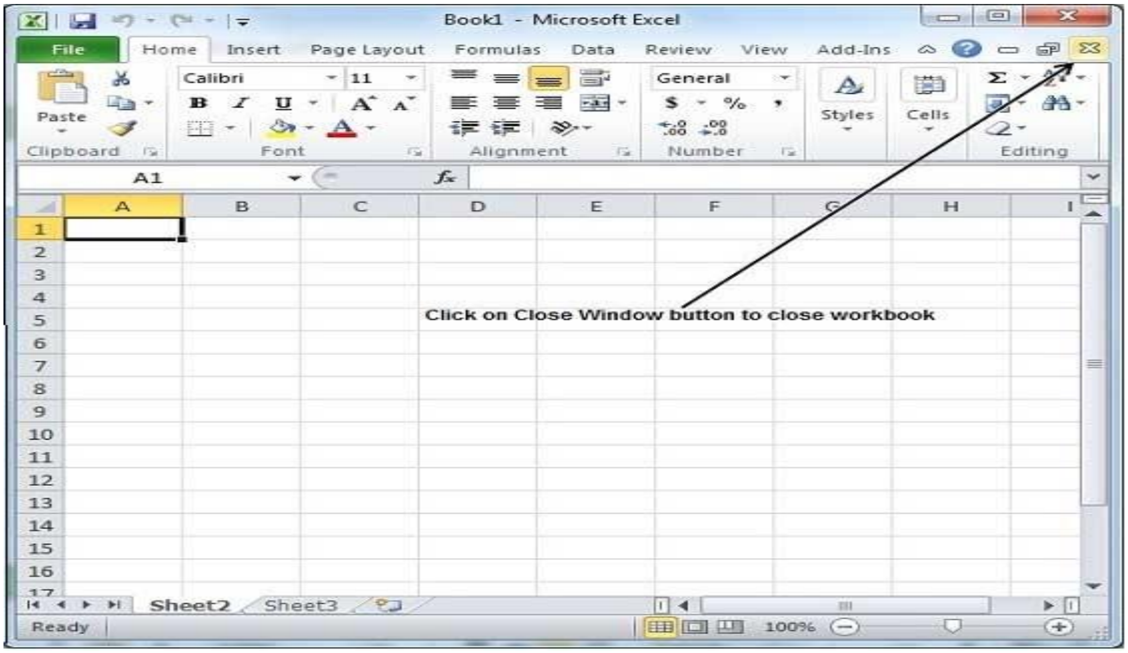
- **Selection** – Prints only the range that you selected before choosing **File » Print**.



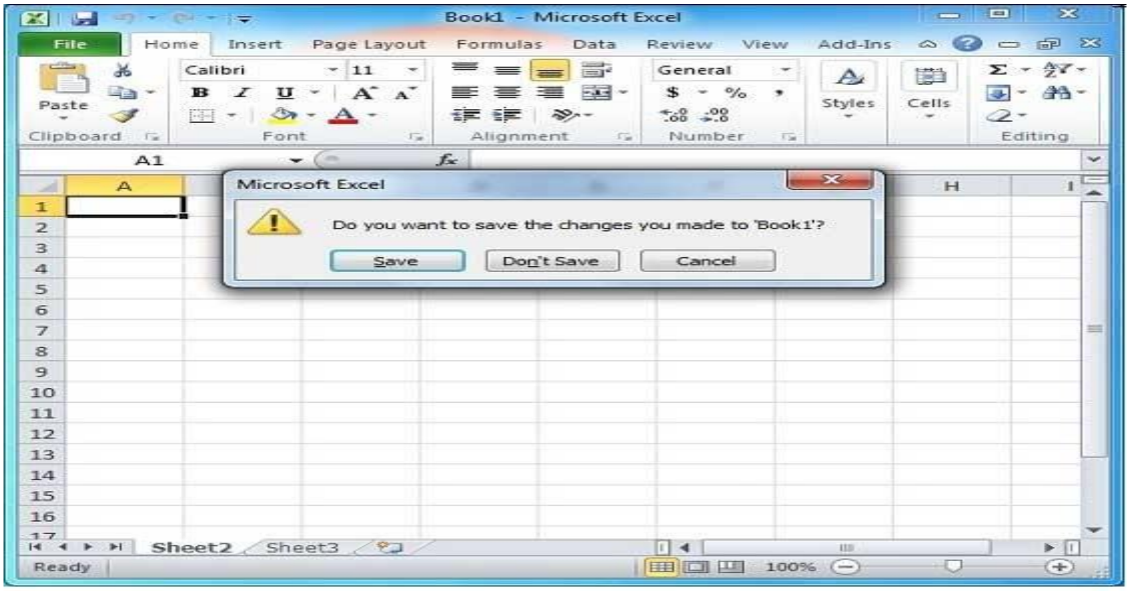
Close, save and Open a Workbook

To close a workbook follow the following steps

Step 1 – Click the **Close Button** as shown below.



You'll see a confirmation message to save the workbook.



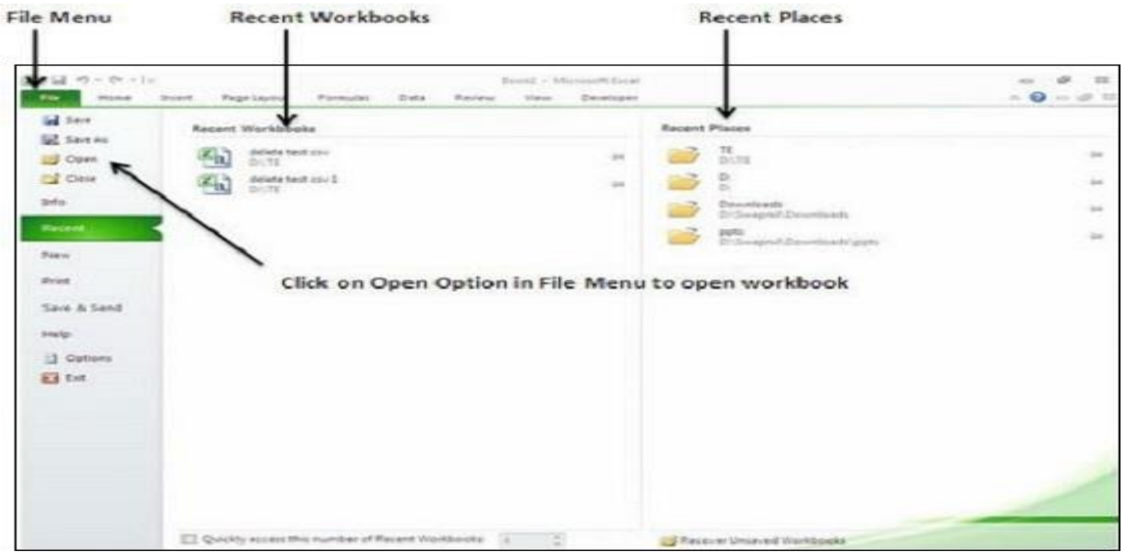
Step 2 – Press the **Save** Button to save the workbook

Opening a worksheet

The following are the steps for opening excel workbook

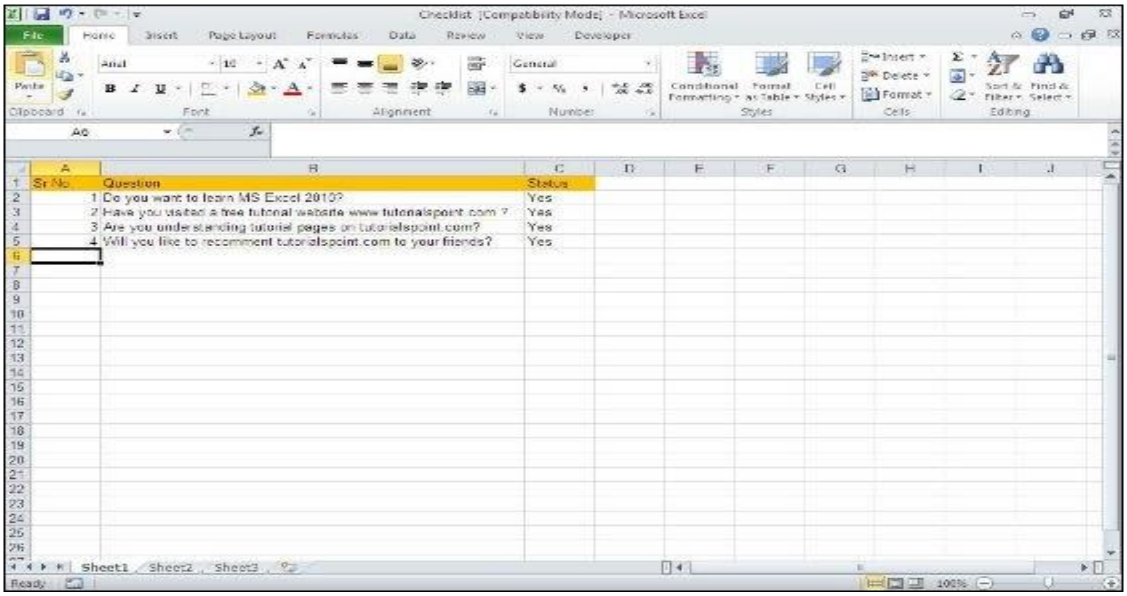
Step 1 – Click the **File Menu** as shown below. You can see the **Open option** in **File Menu.**

- There are two more columns Recent workbooks and Recent places, where you can see the recently opened workbooks and the recent places from where workbooks are opened.



Step 2 – Clicking the **Open Option** will open the browse dialog as shown below. Browse the directory and find the file you need to open.

Step 3 – Once you select the workbook your workbook will be opened as below –



Cell referencing

- A **cell reference** refers to a **cell** or a range of **cells** on a worksheet
- Can be used in a formula so that Microsoft Office Excel can find the values or data that you want that formula to calculate.
- In one or several formulas, you can use a **cell reference** to refer to: ... Data contained in different areas of a worksheet.

Types of cell reference

iv. There are three types of cell references

1. ***Relative cell references:***

- It does not contain dollar signs in a row or column, e.g. A2.
- Relative cell references type in excel change when a formula is copied or dragged to another cell.
- In Excel, cell referencing is relative by default, it is most commonly used cell reference in the formula.

2. ***Absolute cell references***

- Absolute Cell Reference in Excel contains dollar signs attached to each letter or number in a reference, e.g. B4,
- Here if we mention a dollar sign before the column and row identifiers, it makes absolute or locks both the column and the row i.e. where Cell reference remains constant even if it copied or dragged to another cell.

3. ***Mixed cell references in Excel:***

- It contains dollar signs attached to either the letter or the number in a reference. E.g. $B2 or B$4**.** It is a combination of relative and absolute references (mixed reference type in excel)

Formulae and functions

Definitions

A **formula** is mathematical expression which is used to perform calculation and return a result in excel worksheet

A **functions** are predefined (inbuilt) formulas that perform certain types of calculations automatically and return the result.

Entering Formulas

v. A formula in excel always begins with an equal sign (=) which defines it as numerical entry

Formulas use the following arithmetic operators to specify the numerical operation to perform

Operator	**Description**	**examples**	**Comment**
+	Addition	=B2+B3 =45+67	Adds the contents of cells B2 and B3 Adds 45 and 67
vi.	Subtraction	=B2+B3 =67 - 45	Subtract the contents of cells B2 and B3 Subtract 45 from 67
*	Multiplication	=B2+B3 =45+67	Multiply the contents of cells B2 and B3 Multiply 45 and 67
/	Divison	=B2/B3	Divides the contents of Cell B2 by

		=45/67	the contents of cell B3 Divide 45 by 67
^	Exponentiation (raising to a power)	B2^2 45^2	Square the contents of cell B2 Square 45

Order of Operations

If more than one operator is used in a formula, there is a specific order that Excel will follow to perform these mathematical operations. This order of operations can be changed by adding brackets to the equation. An easy way to remember the order of operations is to use the acronym:

BEDMAS

The Order of Operations is:

Brackets
Exponents
Division
Multiplication
Addition
Subtraction

How the Order of Operations Works

Any operation(s) contained in brackets will be carried out first followed by any exponents.

After that, Excel considers division or multiplication operations to be of equal importance, and carries out these operations in the order they occur left to right in the equation.

The same goes for the next two operations – addition and subtraction. They are considered equal in the order of operations. Which ever one appears first in an equation, either addition or subtraction, is the operation carried out first.

Therefore to enter a formulas in a cell follow the following steps

1. Type an equal sign
2. Click or type a cell location, if using constants type a number
3. Type a mathematical operator
4. Click or type the cell location
5. Use parenthesis where necessary to control the order of operations
6. Press the Enter

Using Excel Function

Excel functions must also start with an equal sign followed by *function name* and a*rguments.*

The following is a general syntax for writing excel functions

*=Function Name(Argument**1**, Argument**2**, …., Argument **n**)*

Function Name – identifies the type of calculationto be performent

Argument – This is the data the function uses to perform the calculation

Arguments can either be numbers or names or range or cell references that contains numbers. For example **=SUM(B2,B3,B4)** adds the contents of cells B2, B3 and B4

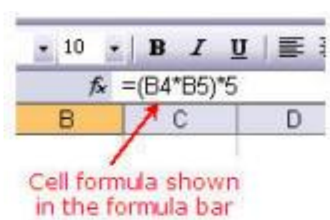
Entering a function in a cell

There are two ways to get the function wizard.

i. If you look at the **Standard Toolbar**, the function wizard icon looks like this:

ii. The other way to get to the function wizard is to go to the Menu **INSERT** -- down to **FUNCTION**.

vii. Either way you get there, at this point Excel will list all of the functions available.



viii. Upon choosing the function, Excel will prompt you for the information it needs to complete the function. Mini descriptions are available for each of the cells.

ix. It is often necessary for you to understand the functions in order to be able to figure out these descriptions. Once you've learned the functions, though, it is faster to type the basic function in from the keyboard as opposed to going through the steps of this tool.

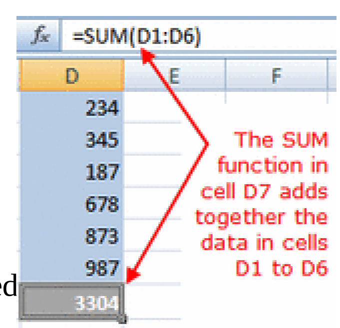
Some commonly used function

The following are some of commonly used function in excel spreadsheet

1. SUM
2. PRODUCT
3. MULTIPLY
4. DIVIDE
5. AVERAGE
6. MAX
7. MIN
8. COUNT
9. IF

SUM FUNCTION

- The Sum function takes all of the values in each of the specified cells and totals their values.



- The syntax is: =SUM(first value, second value, etc)

- Blank cells will return a value of zero to be added to the total.
- Text cells can not be added to a number and will produce an error.

Let's use the table here for the discussion that follows:

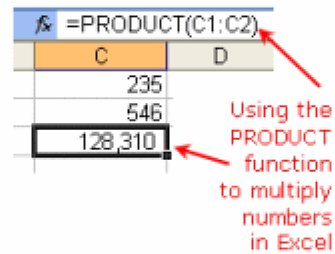
We will look at several different specific examples that show how the typical function can be used! Notice that in A4 there is a TEXT entry. This has NO numeric value and can not be included in a total.

	A
1	25
2	50
3	75
4	Test
5	

Example	Cells to ADD	Answer
=sum(A1:A3)	A1, A2, A3	150
=sum(A1:A3, 100)	A1, A2, A3 and 100	250
=sum(A1+A4)	A1, A4	#VALUE!
=sum(A1:A2, A5)	A1, A2, A5	75

PRODUCT FUNCTION

An example of multiplying two numbers, such as 235 and 546, using the PRODUCT function would be:



=PRODUCT(235 , 546)

The answer of 128,310 will appear in the cell where you type the function.

While this approach to using the PRODUCT function works, it limits the usefulness of the function. A better way of using the function is to type the numbers you are multiplying into cells on the spreadsheet and then enter those cell references (the address of the cells) into the function.

For example, if we enter the numbers 235 and 546 into cells C1 and C2, we would write the function as:

=PRODUCT(C1:C2)

The answer is still 128,310, but the advantage of this approach is that if the numbers ever change, you only need to change the numbers in cells C1 or C2 and the function automatically updates the answer.

For example, if you find that the number in C1 wasn't 235 but 230, simply type 230 in cell C1 and the function updates the answer to 128,580.

This approach works well for instances where you have constantly changing numbers – say on a monthly income statement where the income amounts get multiplied by set numbers to find deduction rates for taxes, pensions, or medical benefits.

DIVISION

To divide two numbers in Excel you need to create a formula. Important points to remember about Excel formulas:

- formulas in Excel always begin with the equal sign (**=**)
- the equal sign always goes in the cell where you want the answer to go
- the division symbol is the forward slash (**/**)

Use Cell References in Formulas

Even though you can use numbers directly in your division formula, it is much better to use the references or addresses of the cells containing your data. If you use the cell references [A1, B1, F2] rather than the actual data, later, if you need to change the data in either cell, the results of the formula will update automatically without you having to rewrite the formula.

Setting Up the Division Formula

As an example, lets create a formula in cell E1 that will divide the contents of cell C1 by cell D1.

Our formula:

=C1 / D1

Our data:

- place the number 20 in cell C1
- place the number 10 in cell D1

Division Formula Steps

To divide 20 by 10 and have the answer appear in cell E1:

1. Type an equal sign in cell E1.
2. Click on cell C1 with the mouse pointer.
3. Type the division sign (**/**) in cell E1.
4. Click on cell D1 with the mouse pointer.
5. Press the **ENTER** key on the keyboard.
6. The answer 2 should be present in cell E1.
7. Even though you see the answer in cell E1, if you click on that cell you will see our formula in the formula bar above the work area.

To expand your formula to include additional operations - such as subtraction or addition - just continue to add the correct mathematical operator followed by the cell reference containing your data.

AVERAGE FUNCTION

- The average function finds the average of the specified data. *(Simplifies adding all of the indicated cells together and dividing by the total number of cells.)*
- The syntax is as follows.

  =Average (first value, second value, etc.)

- Text fields and blank entries are not included in the calculations of the Average Function.

Let's use the table here for the discussion that follows:
We will look at several different specific examples that show how the average function can be used!

	A
1	25
2	50
3	75
4	100
5	

Example	**Cells to average**	**Answer**
=average (A1:A4)	A1, A2, A3, A4	62.5

=average (A1:A4, 300)	A1, A2, A3, A4 and 300	110
=average (A1:A5)	A1, A2, A3, A4, A5	62.5
=average (A1:A2, A4)	A1, A2, A4	58.33

MAX FUNCTION

This will return the largest (max) value in the selected range of cells.

- Blank entries are not included in the calculations of the Max Function.

x. Text entries are not included in the calculations of the Max Function.

- Let's use the table here for the discussion that follows.
We will look at several different specific examples that show how the Max functions can be used!

	A
1	10
2	20
3	30
4	Test
5	

Example of Max	**Cells to look at**	**Ans. Max**
=max (A1:A4)	A1, A2, A3, A4	30
=max (A1:A4,	A1, A2, A3, A4 and 100	100

100)		
=max (A1, A3)	A1, A3	30
=max (A1, A5)	A1, A5	10

MIN FUNCTION

This will return the smallest (Min) value in the selected range of cells.

- Blank entries are not included in the calculations of the Min Function.

Text entries are not included in the calculations of the Min Function. Let's use the table here for the discussion that follows.

We will look at several different specific examples that show how the min functions can be used!

	A
1	10
2	20
3	30
4	Test
5	

Example of min	Cells to look at	Ans. min
=min (A1:A4)	A1, A2, A3, A4	10
=min (A2:A3, 100)	A2, A3 and 100	20
=min (A1, A3)	A1, A3	10
=min (A1, A5)	A1, A5 (displays the smallest	10

	number)	

COUNT FUNCTION

- This will return the number of entries (actually counts each cell that contains number data) in the selected range of cells.

- Blank entries are not counted.
- Text entries are NOT counted.

Let's use the table here for the discussion that follows.
We will look at several different specific examples that show how the Count functions can be used!

	A
1	10
2	20
3	30
4	Test
5	

Example of Count	**Cells to look at**	**Answer**
=Count (A1:A3)	A1, A2, A3	3
=Count (A1:A3,	A1, A2, A3 and 100	4

100)		
=Count (A1, A3)	A1, A3	2
=Count (A1, A4)	A1, A4	1
=Count (A1, A5)	A1, A5	1

COUNTA FUNCTION

xi. This will return the number of entries (actually counts each cell that contains number data **OR** text data) in the selected range of cells.

- Blank entries are not Counted.
- Text entries **ARE** Counted.

Let's use the table here for the discussion that follows.
We will look at several different specific examples that show how the CountA functions can be used!

	A
1	10
2	20
3	30
4	Test
5	

Example of CountA	Cells to look at	Answer
=CountA (A1:A3)	A1, A2, A3	3
=CountA (A1:A3, 100)	A1, A2, A3 and 100	4
=CountA (A1, A3)	A1, A3	2
=CountA (A1, A4)	A1, A4	2
=CountA (A1, A5)	A1, A5	1

IF FUNCTION

The IF function will check the logical condition of a statement and return one value if true and a different value if false. The syntax is

- =IF (condition, value-if-true, value-if-false)
- value returned may be either a number or text
- if value returned is text, it must be in quotes

Let's use the table here for the discussion that follows. We will look at several different

	A	B
1	Price	Over a dollar?
2	$.95	No

specific examples that show how the IF functions can be used!

3	$1.37	Yes
4	comparing #	returning #
5	14000	0.08
6	8453	0.05

Example of **IF** typed into column B	Compares	Answer
=IF (A2>1,"Yes","No")	is (.95 > 1)	No
=IF (A3>1, "Yes", "No")	is (1.37 > 1)	Yes
=IF (A5>10000, .08, .05)	is (14000 > 10000)	.08
=IF (A6>10000, .08, .05)	is (8453 > 10000)	.05

PMT FUNCTION

The PMT function returns the periodic (in this case monthly) payment for an annuity (in this case a loan). This is the PMT function that was used for the car purchase in the first example. There are a few things that we must know in order for this function to work. To calculate the loan we must know a combination of the following

- (rate) interest rate per period
- (NPER) number of payments until repaid
- (PV) present value of the loan (amount we are borrowing)
- (FV) future value of the money (for saving or investing)
- (type) enter 0 or 1 to indicate when payments are due.

=PMT(rate, NPER, PV, FV, type)

	A	B	C
1		**computer ledger**	
2			
3		car loan	$12,000.00
4		interest	9.60%
5		# of payments	60
6			
7		Monthly Pmt.	$252.61

equation goes into c7 **=PMT(C4/12,C5,-C3)**

C4 is the yearly interest and since it's compounded monthly we divide by 12

C5 is the number of months (# of payments)

-C3 is the amount of money we have (borrow - negative)

Note that the rate is per period. If we have an annual interest rate of 9.6% and we are calculating monthly payments, we must divide the annual interest rate by 12 to calculate the monthly interest rate.

SIN, COS. TAN FUNCTIONS

Excel has most of the math and trig functions built into it. If you need to use the **SIN**, **COS**, **TAN** functions, they can be typed into any cell. If you wanted to find:

angle	Sin	cos	Tan
REF	=sin(REF)	=cos(REF)	=tan(REF)
0	0.00	1.00	0.00
30	0.50	0.87	0.58
45	0.71	0.71	1.00
90	1.00	0.00	
180	0.00	-1.00	0.00

format for **degrees** formula = sin (angle * pi()/180) the argument angle is in degrees

format for **radians** formula = sin (angle) the argument angle is in radians

To calculate trig functions in degrees you must convert them - otherwise excel will calculate them in radians.

You can type in either an actual number for the REF or you can also type in a reference from the excel spreadsheet (like A2).

COPYING A FUNCTION OR A FORMULA

Sometimes when we enter a formula, we need to repeat the same formula for many different cells. In the spreadsheet we can use the copy and paste command. The cell locations in the formula are **pasted** relative to the position we **Copy** them from.

	A	B	C
1	5	3	=A1+B1
2	8	2	=A2+B2
3	4	6	=A3+B3
4	3	8	=? + ?

Cells information is copied from its relative position. In other words in the original cell **(C1)** the equation was **(A1+B1)**. When we paste the function it will look to the two cells to the left. So the equation pasted into **(C2)** would be **(A2+B2)**. And the equation pasted into **(C3)** would be **(A3+B3)**.

FILL DOWN THE FORMULA OR A FUNCTION

If you have a lot of duplicate formulas you can also perform what is referred to as a FILL DOWN.

Often we have several cells that need the same formula (in relationship) to the location it is to be typed into. There is a short cut that is called Fill Down. There are a number of ways to perform this operation. One of the ways is to

1. select the cell that has the original formula
2. hold the shift key down and click on the last cell (in the series that needs the formula)
3. under the **edit** menu go down to **fill** and over to **down**

	A	B	C
1	5	3	=A1+B1
2	8	2	fill down
3	4	6	fill down
4	3	8	fill down

Cells information is copied from its relative position. In other words in the original cell **(C1)** the equation was **(A1+B1)**. When we paste the function it will look to the two cells to the left. So the equation pasted into **(C2)** would be **(A2+B2)**. And the equation pasted into **(C3)** would be **(A3+B3)**. And the equation pasted into **(C4)** would be**(A4+B4)**.

Chapter 2: Introduction to Desktop publishing

Objectives

By the end of this chapter, the student should be able to:

Contents

- Definition
- Benefits of desktop Publishing
- Examples of desktop publishing software
  - corel draw
  - photoshop
  - publisher
  - page maker
- Types of Desktop Publishing Software
  - graphical based
  - layout based
- Features of Desktop Publishing Software
- Working with desktop publishing software
  - creating, formatting, editing, saving, printing, opening, closing, graphics, inserting objects, exiting
- Designing publications
  - Types, orientation, layout, margins, text boxes, page size, manipulating graphics, moving, resizing, reshaping, flipping and rotating, cropping, exporting and printing publications

Definition of desktop publishing

- Desktop publishing (DTP) refers to the process of designing publications of professional quality
- **Examples of publicatipns include**

i. Newspapers
ii. Invitational cards
iii. Posters,
iv. Fliers
v. Journals
vi. Books

Desktop publishing software

- This is special purpose software which is used to create publications of professional quality.

Examples of publishing software

i. Adobe pageMaker
ii. Microsoft publisher
iii. quarkXpress
iv. Adobe InDesign
v. Adobe Illustrator
vi. CorelDraw
vii. SerisPageplus
viii. Apple page

Benefits of desktop publishing

i. Every item on a page is contained in a frame hence can be edited and formatted independently
ii. DTPs provide mote control on how text and graphics can be arranged and formatted

iii. Frames containing text or graphics need not flow in logic sequence, For example, a story on page 1 may be continued on page 8
iv. In DTPs, provide master used to set a common layout which may be repeated on several pages.
v. In DTPs, publications can be printed in a form suitable for commercial, digital or offset printing using color separations.
vi. Most DTPs have predefined templates such as brochures, booklets, posters and business cards
vii. Multiple stories from different authors can be handled with ease
viii. DTPs ensures files print properly in their true colour, fonts and measurements

Types of DTP software

- DTP software can be classied into two broad categories namely
  i. Graphic-based DTPs
  ii. Layout-based DTPs

Graphic-based DTPs

- These DTPs are specifically used to edit and format graphics objects such as pictures and vector drawings

- Vector drawings are freehand drawings such as those drawn by fine arts

  Examples of Graphic-based DTPs are

    i. Adobe photoshop
    ii. Adobe illustrator

Layout-based DTPs

- These types are specifically used to design page layout for text and graphic
- Examples of layout-based DTPs includes
  i. Adobe page maker
  ii. Microsoft publisher
  iii. Adobe InDesign

Purpose of Desktop Publishing software

- The following are the purpose of DTPs

i. Graphics Design

- DTPs lets the user to manipulate graphics such as pictures

ii. Page layout design

- DTPs are used to design a page layout by setting consistent picture and object locations, diving a page in a number of newspaper columns and adding layers.
- A layer can be viewed as the arrangement of objects on top of each other with each object being on its layer.

iii. Printing

- DTPs offer more flexibility in printing, like in image color separation.

Features of DTP software

- The following are the common tools of in most DTP software

i) Select Tool

- Used to select, move and resize images and text

ii) Text tool

- Used to draw text frames, insert and manipulate text

iii) Shapes tool

- For drawing basic shapes like rectangles and for importing objects

iv) Zoom tool

- For magnifying publications view

v) Rotate tool

- For rotating text or graphics

Getting started with Microsoft publisher

To launch Microsoft publisher 2010, follow the following steps

1. On start menu click Microsoft office

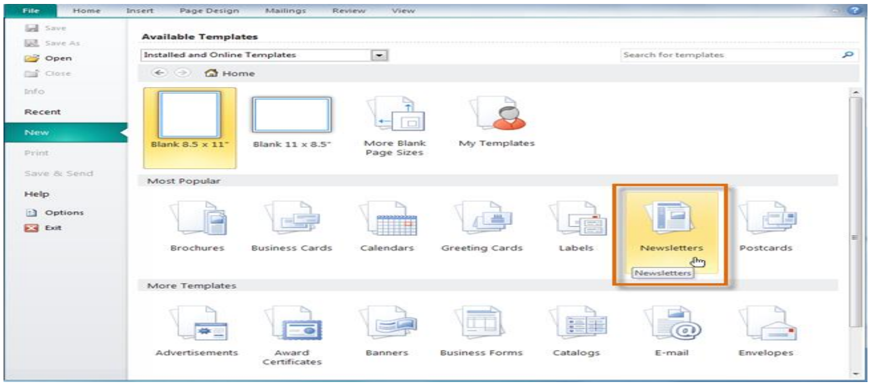
2. Click on Publisher 2010
3. Choose whether to to use blank design or template.

Creating a new publication

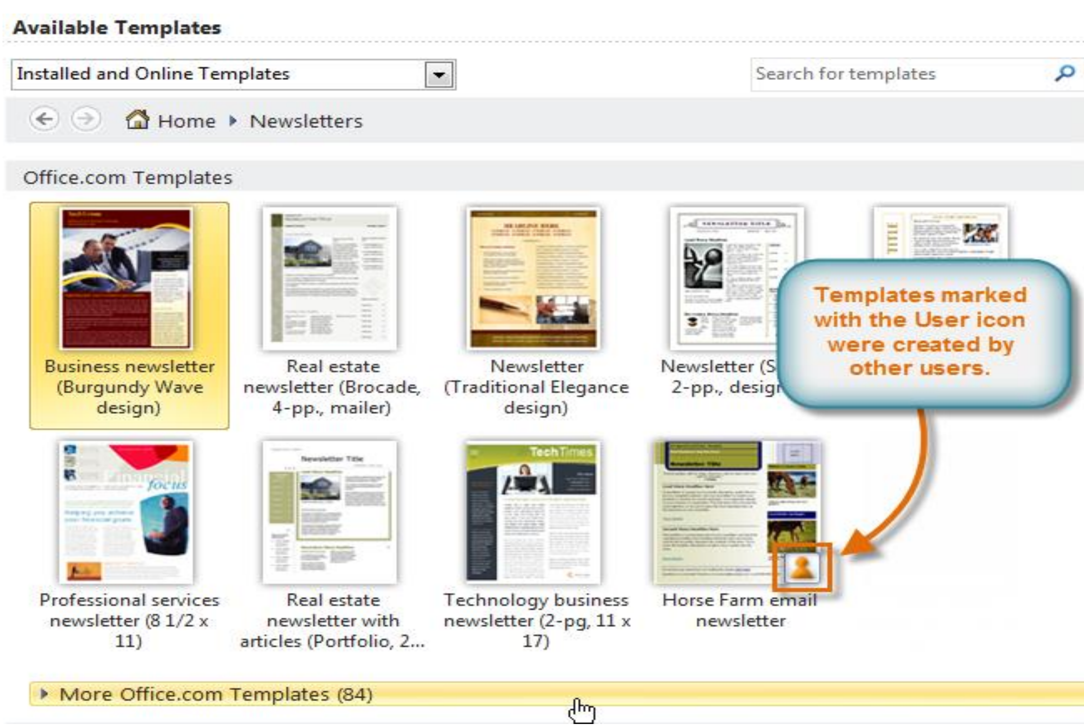
- Publisher offers dozens of templates for almost any type of publication you would want to create, including
  i) **brochures**,
  ii) **newsletters**, and
  iii) **greeting cards**.
- Of course, if you can't find a template you like you can always **modify** one to suit your needs or even create a publication from a blank page.
- Understanding Publisher's templates and layout tools will help you create publications that look the way you want.

To create a new publication from a template:

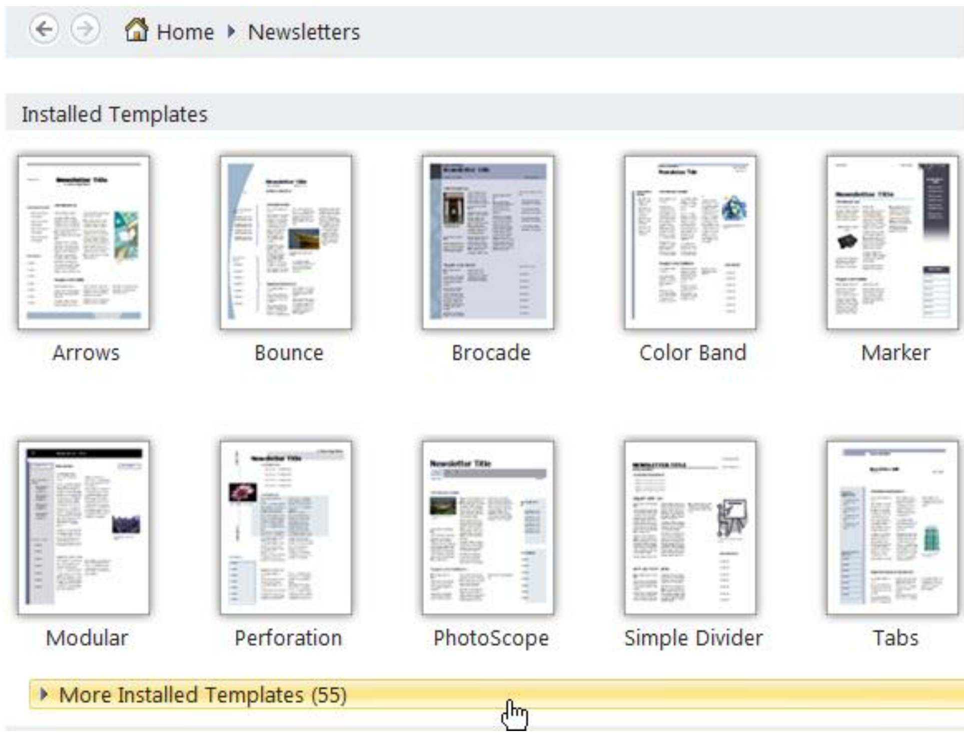
1. Click the **File** tab to go to **Backstage view**, then select **New**. The **Available Templates** pane will appear.
2. Select the **type** of publication you wish to create.



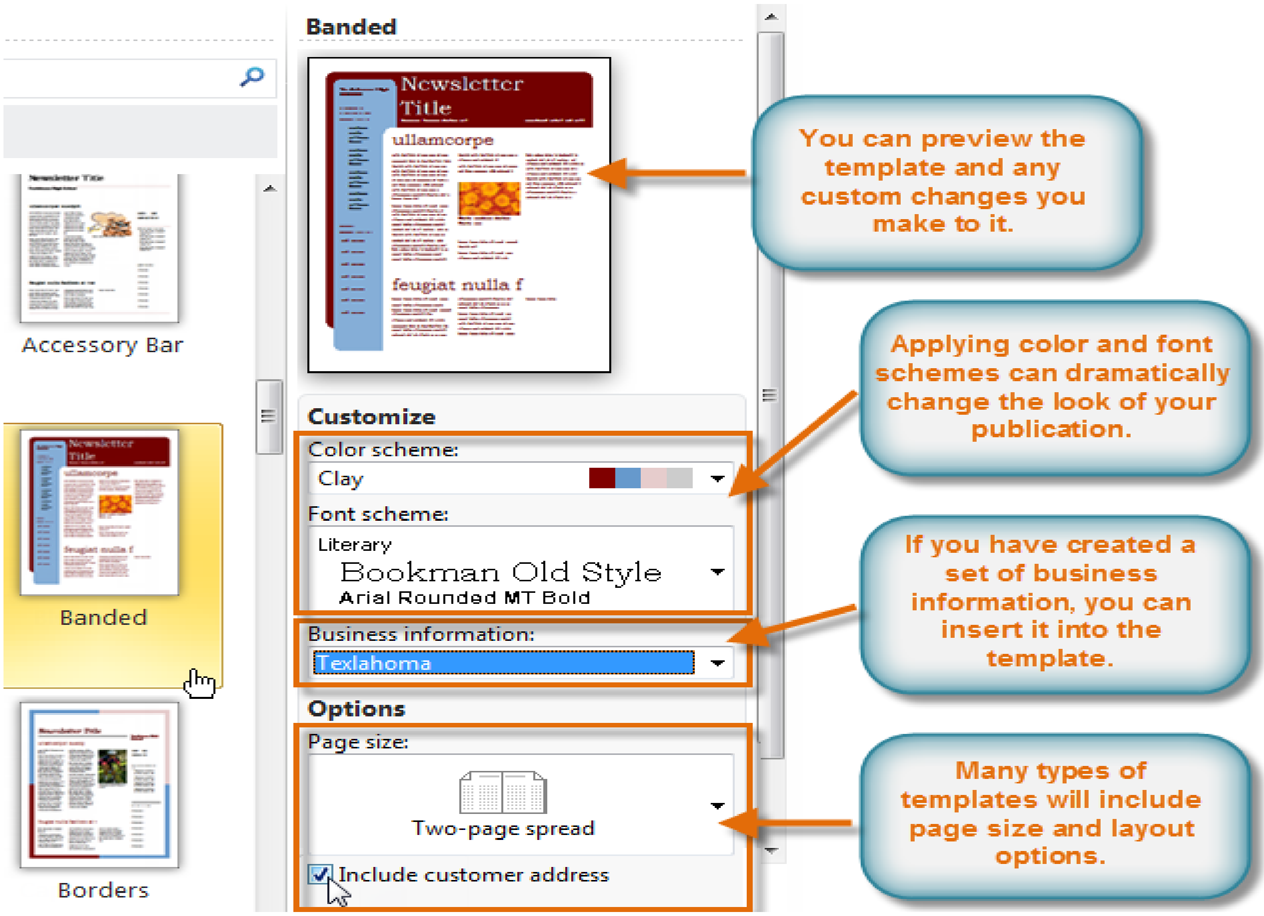
3. A selection of templates will appear in the Available Templates pane. Choose from one of two categories:
    - **Office.com templates**, which include templates created by **other users**. User-created templates are indicated with a **User icon** .
    - .



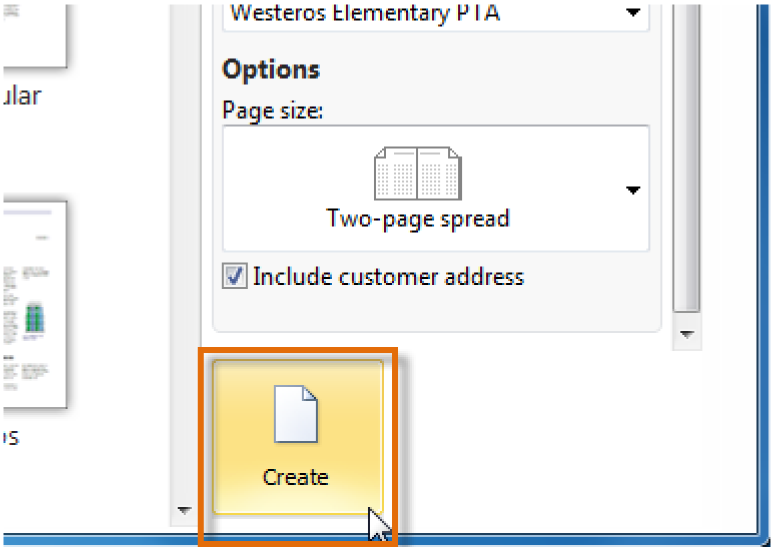
- **Installed templates** created by Microsoft.



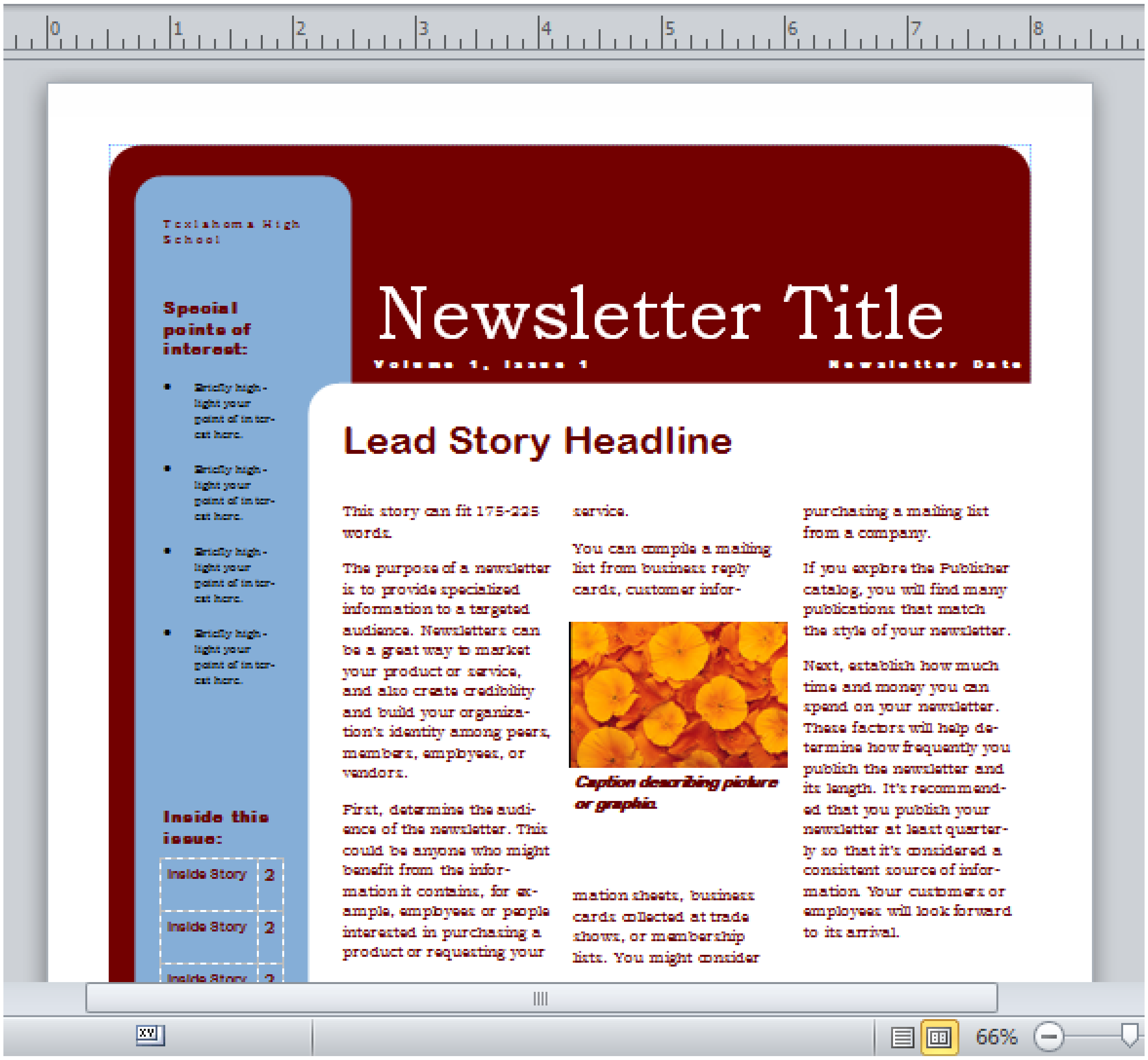
4. A preview of the selected template will appear in the **Preview pane** on the right. Review the template, and modify template **options** as desired.



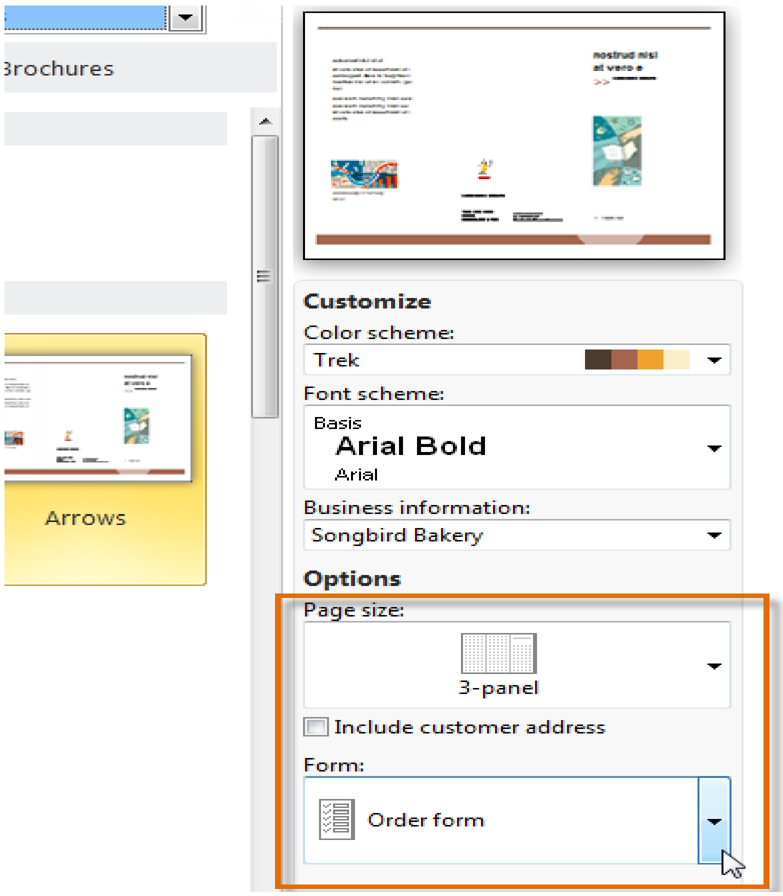
5. When you are satisfied with the template, click **Create**.



6. The new publication will be created.



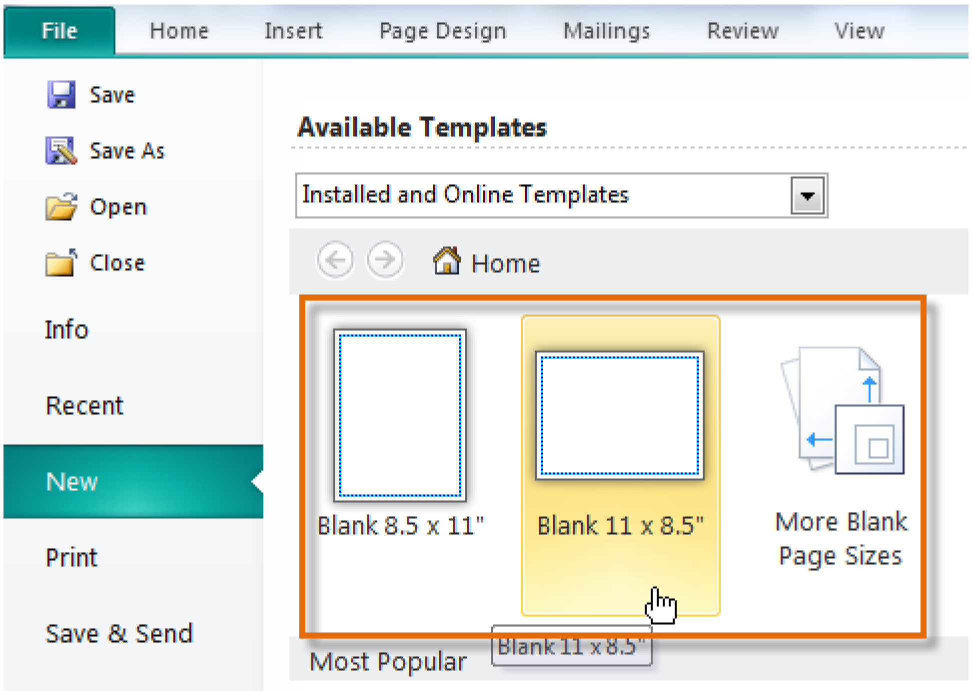
- Depending on the type of publication you create, your template may offer more **customization options** than shown in the example above.



Creating blank publications

- If you don't want to use a template or can't find a template that suits your needs, you can create a **blank publication**.
- Remember, when you create a blank publication you will have to set up page margins, add guides, and make all layout and design decisions on your own.

- To create a blank publication, click the **New** tab in Backstage view, then select a **blank page size** in the Available Templates pane.



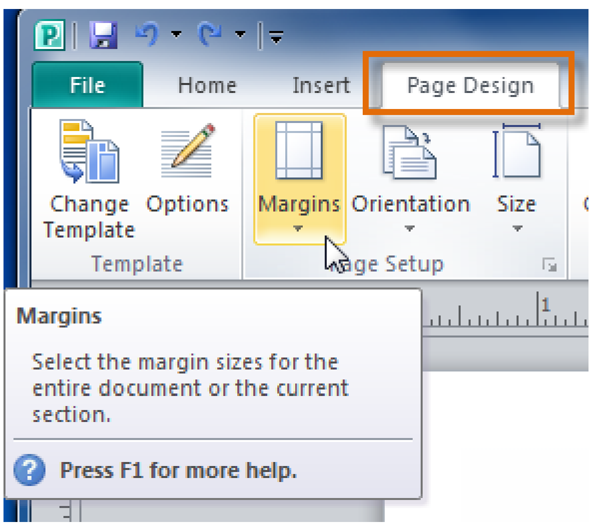
Customizing your publication layout

- Whether you chose to create a publication from a template or from a blank page, you may decide to change the publication **layout**.
- Three components you can change are
    - i) **margins**,
    - ii) **size**, and
    - iii) **orientation**.

- Although you can modify these settings at any time, you should be careful if your publication already contains objects like **text**, **images**, and **shapes**, as you'll have to adjust them to fit the new layout.

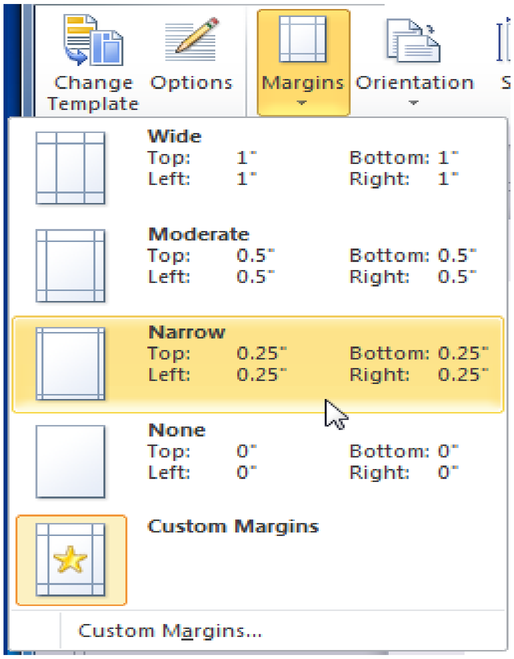
To modify page margins:

1. On the Ribbon, select the **Page Design** tab, then locate the **Page Setup** group.
2. Click the **Margins** drop-down command.

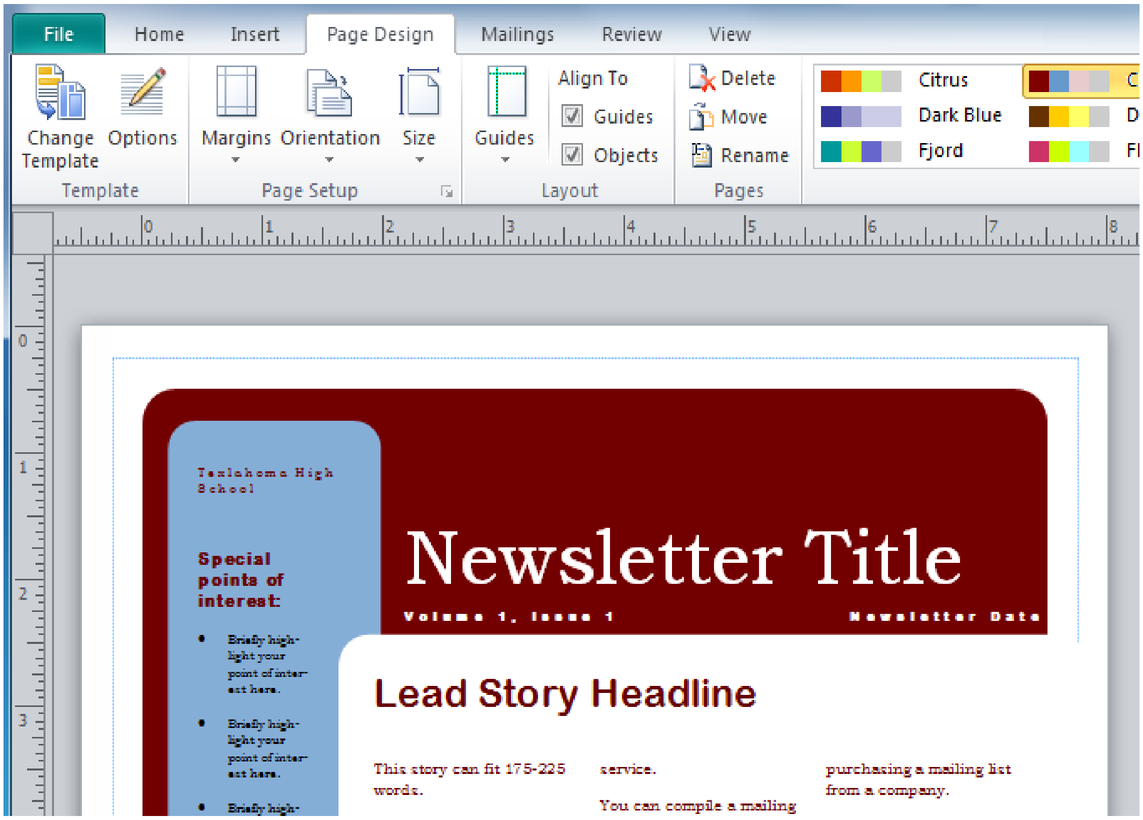


3. A drop-down list will appear. Select the desired margins.

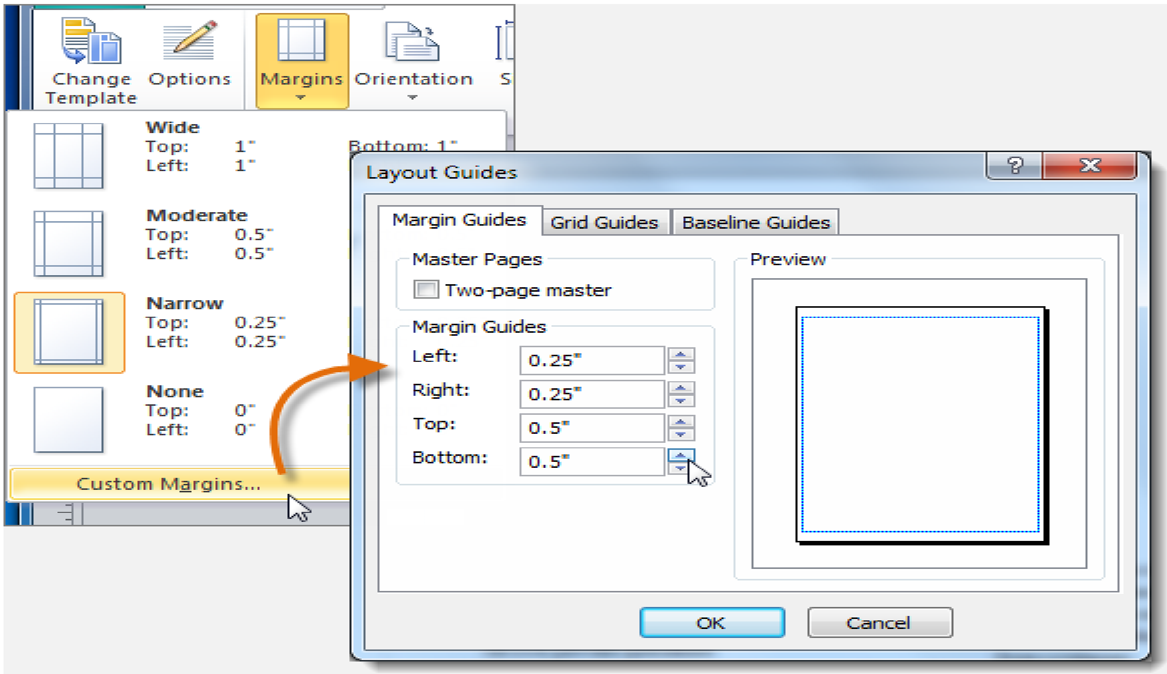
learnmalawi



4. The new margins will be applied.

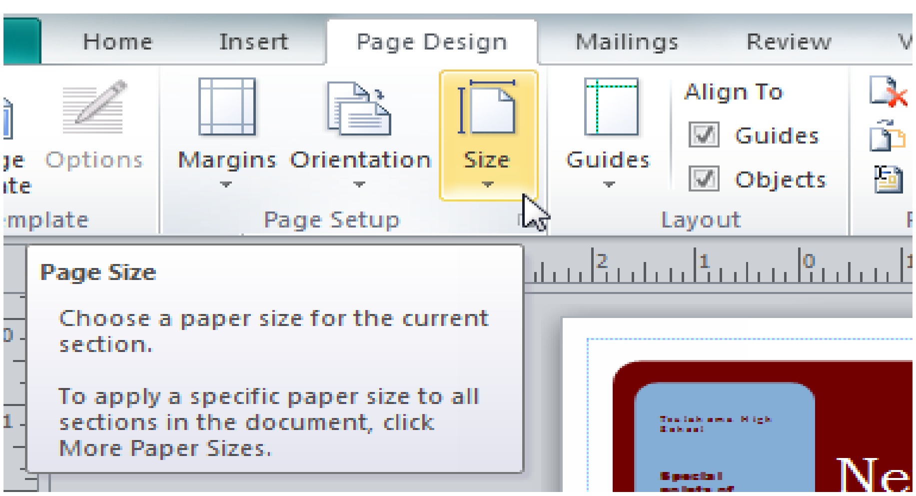


- If you are not satisfied with any of the margin options, select **Custom margins...** to open the **Layout Guides** dialog box. There, you can specify margin widths.

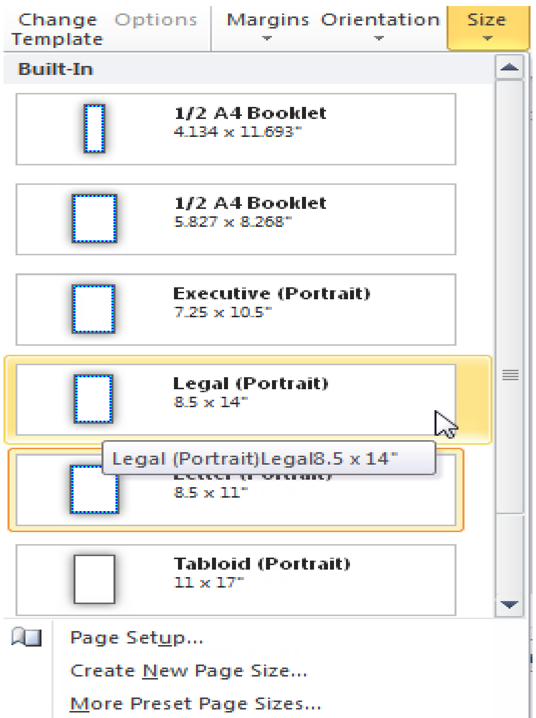


To change page size:

1. On the Ribbon, select the **Page Design** tab, then locate the **Page Setup** group.
2. Click the **Size** drop-down command.

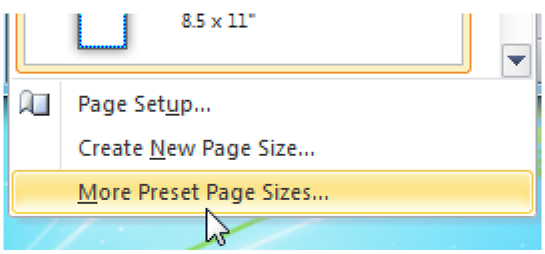


3. Select the desired page size from the drop-down list that appears. Remember, you should make sure your printer is capable of handling paper that size.



4. Your publication's page size will be changed.

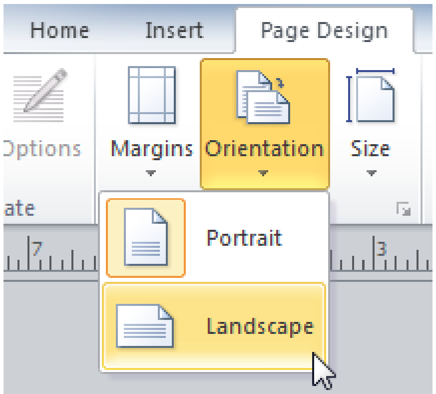
- If the desired page size isn't included in the drop-down list, select **More Preset Page Sizes...** to view a larger list of page sizes.



To change page orientation:

1. On the Ribbon, select the **Page Design** tab, locate the **Page Setup** group, then click the **Orientation** drop-down command.

2. Select **Portrait** orientation to make your publication taller than wide or **Landscape** to make it wider than tall.



3. Your publication's page orientation will be changed.

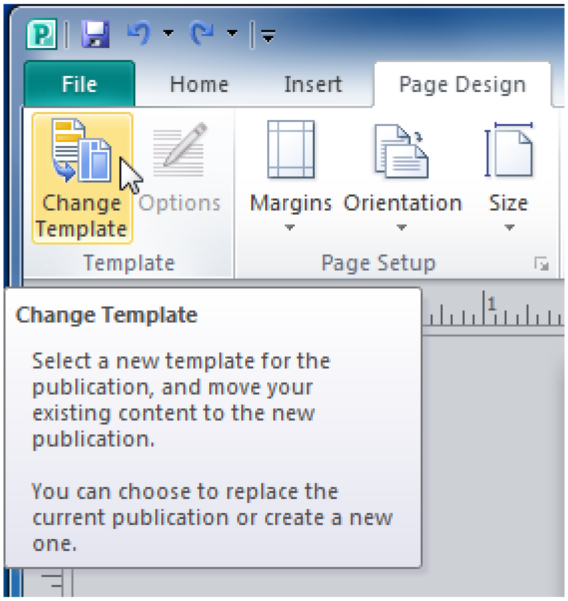
- Depending on the template you chose, changing the page orientation may have a **negative effect** on your presentation.
- While some templates work equally well in both orientations, others do not.

Changing or adding a template

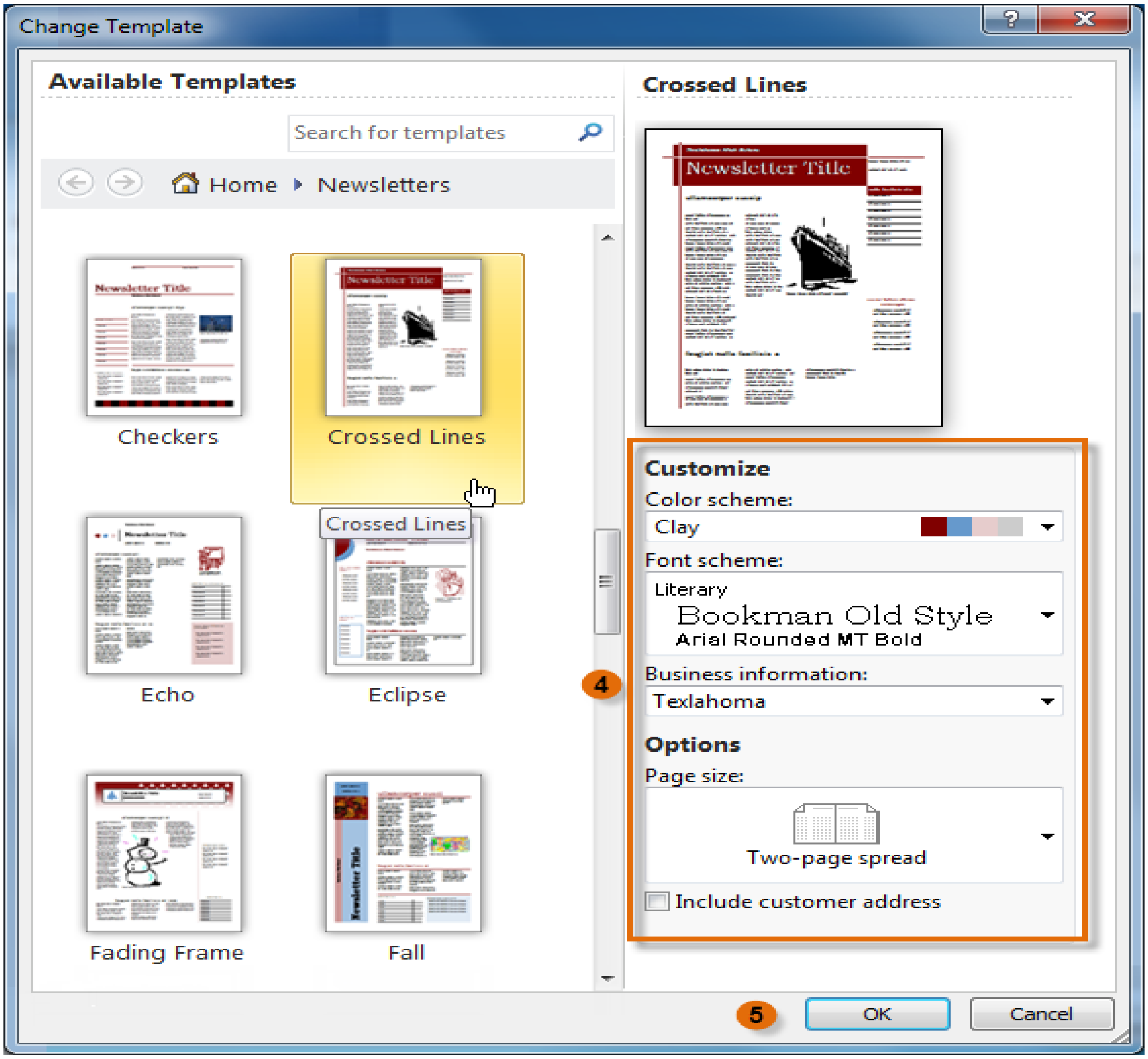
- If you create a publication from a template and later decide the chosen template doesn't quite suit your needs, you can always **change** it.
- You can also apply templates to publications that were originally created from blank pages.

To apply a new template to an existing publication:

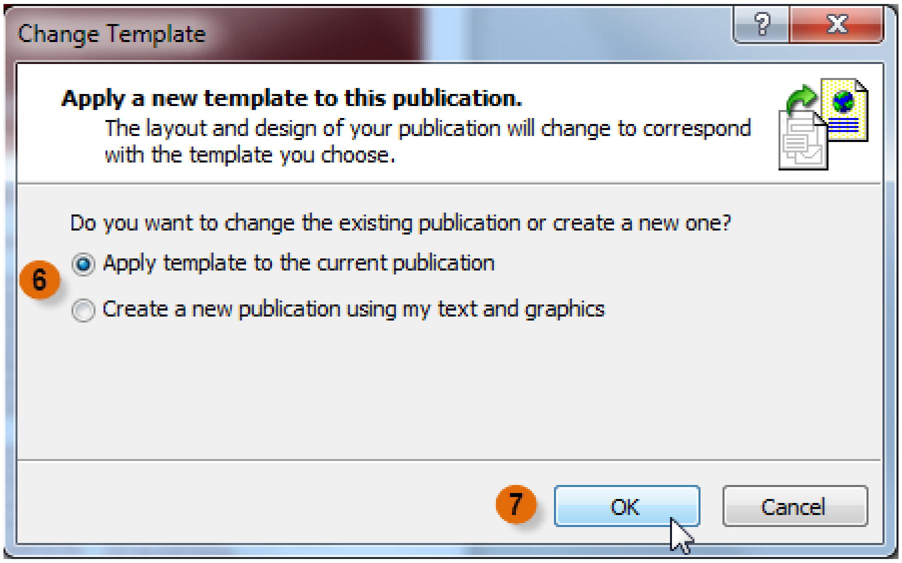
1. On the Ribbon, select the **Page Design** tab, then locate the **Template** group.
2. Click the **Change Template** command.



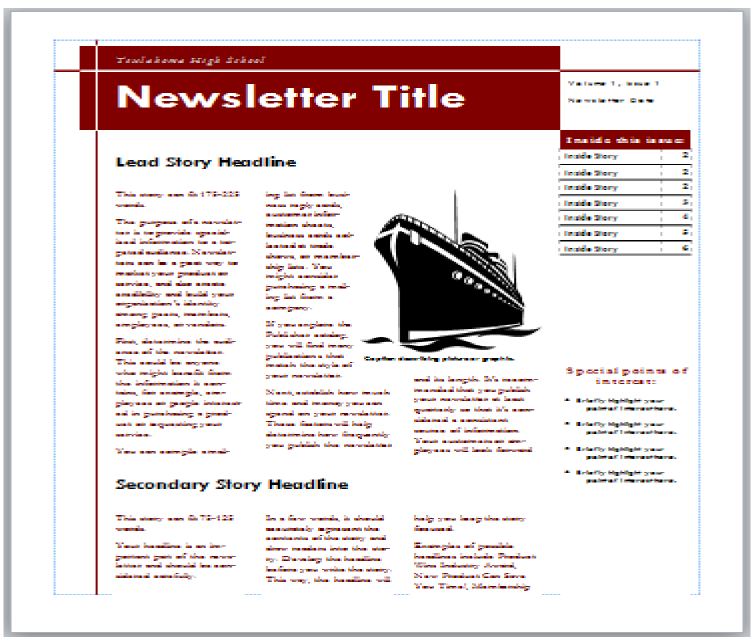
3. The **Change Template** dialog box will appear. Select a template to preview it in the **Preview** pane.
4. Modify template options as desired.
5. When you are satisfied with the new template, click **OK**.



6. A dialog box will appear asking you how you wish to use the template. You can either:
    - Apply the template to the **current** publication
    - Create a **new** publication that includes the text and images you have added
7. Click **OK**.



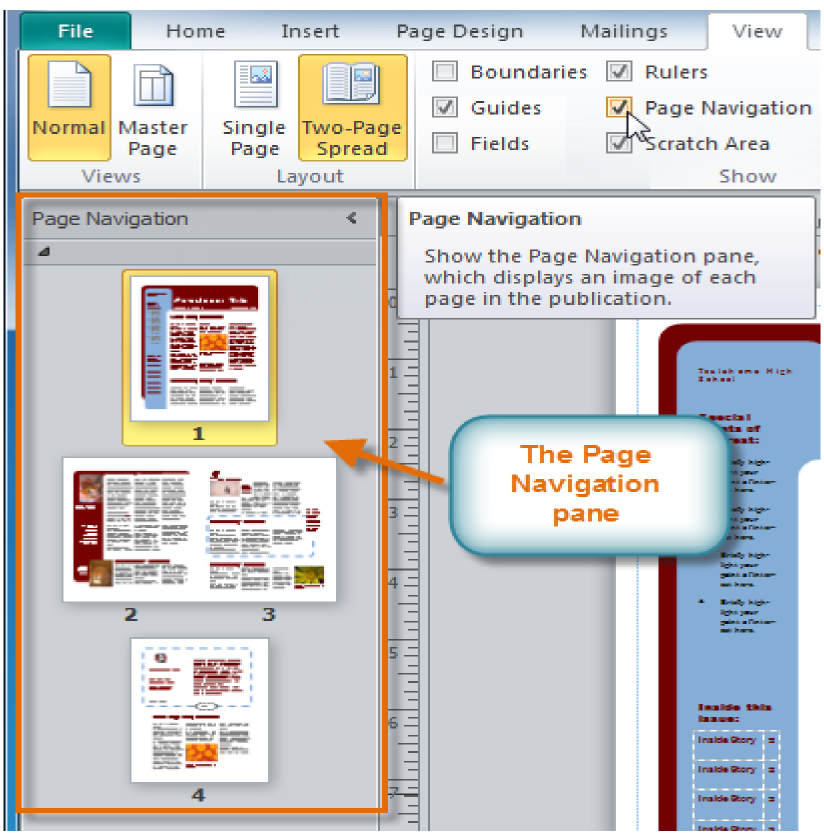
8. The new template will be applied to your publication.



Adding, rearranging, and deleting pages

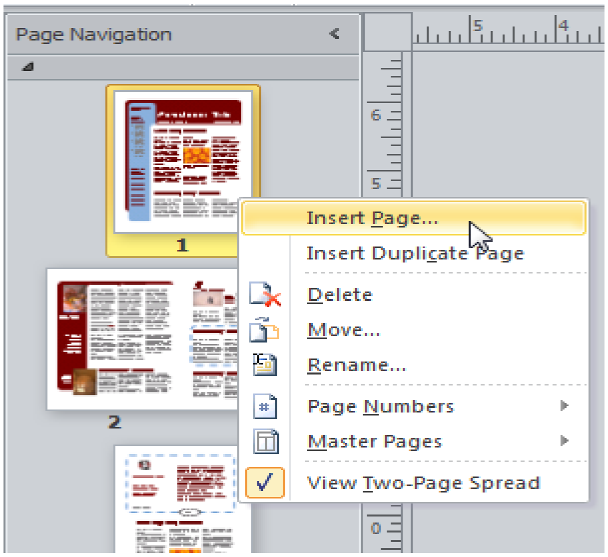
- If you're creating a **newsletter** or another type of publication with **multiple pages**, you might find the **Page Navigation** pane useful.
- The Page Navigation pane gives you a way to view and scroll through the pages in your publication. It also includes features that let you **add**, **move**, and **delete** pages.

- To open the **Page Navigation** pane, click the **View** tab on the Ribbon, then locate the **Show** group. Select the **Page Navigation check box**.

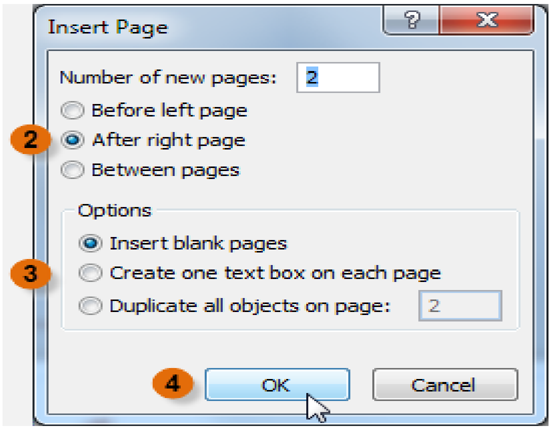


To add a new page:

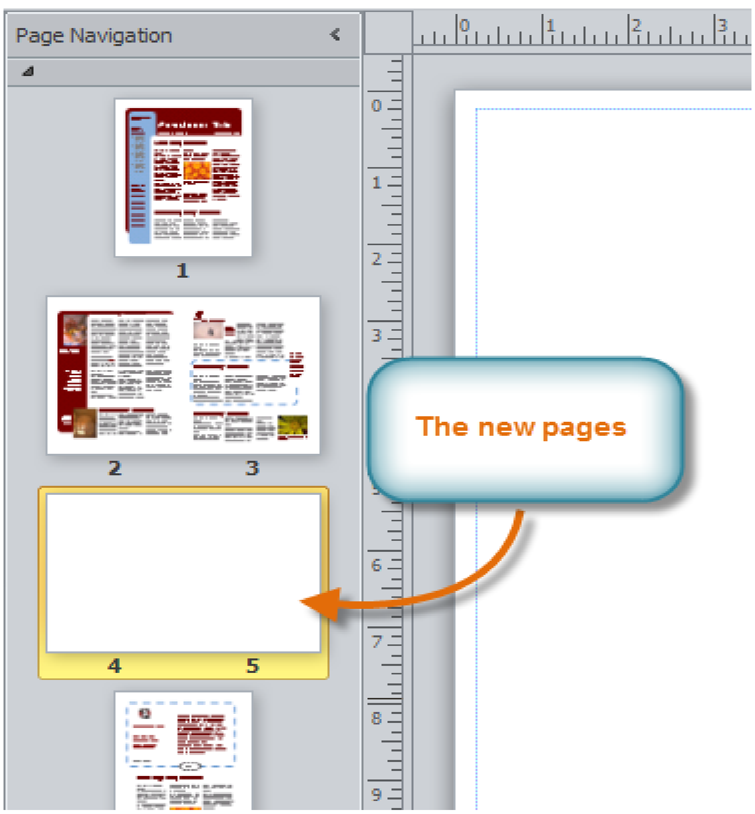
1. In the Page Navigation pane, right-click any page, then select **Insert Page**.



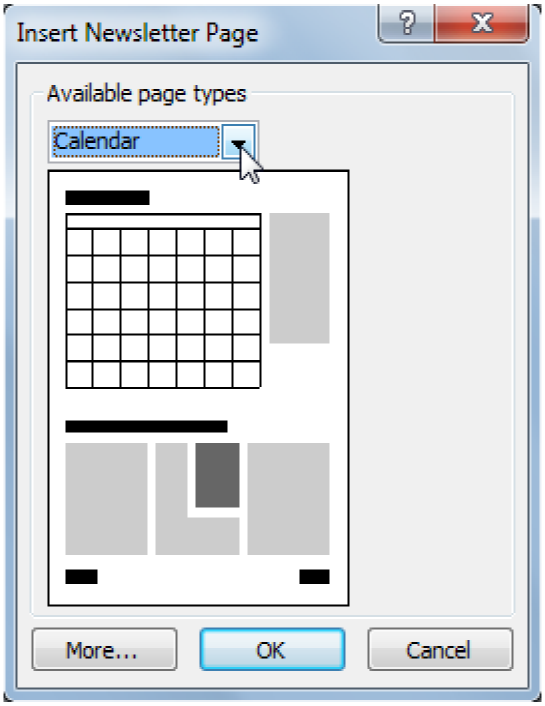
2. The **Insert Page** dialog box will appear. Specify the **number of pages to insert** and the location where you wish to **insert** them.
3. Choose what will appear on the new pages. By default, the pages will be **blank**, but you can also choose to create pages that include **one text box** or pages that are **duplicates** of an existing page.
4. Click **OK**.



5. The new page or pages will be inserted.

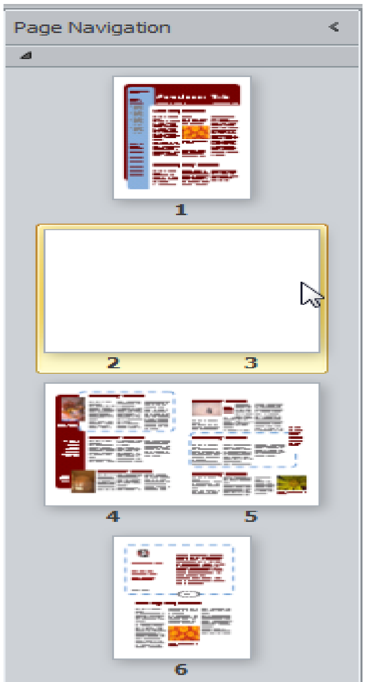


Depending on the template you're using, when you add a new page you may see a dialog box with page layout options.

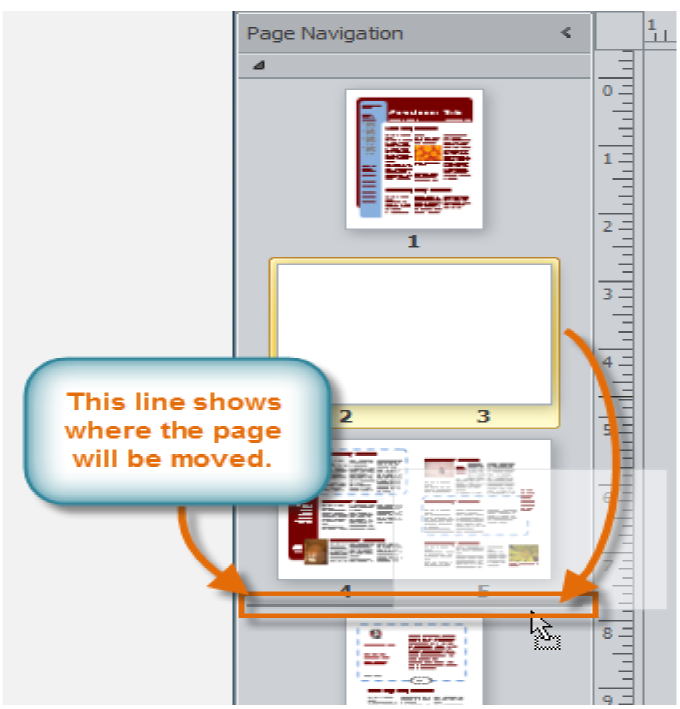


To move a page:

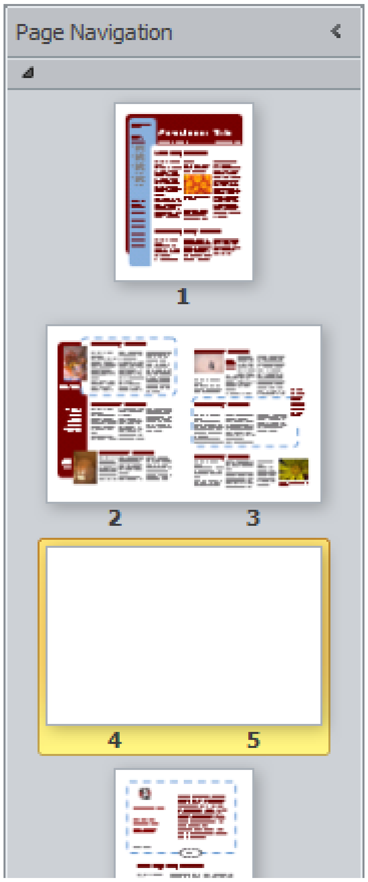
1. In the Page Navigation pane, locate the page you wish to move.



2. **Click** and **drag** the page to its new location, then **release** the mouse.

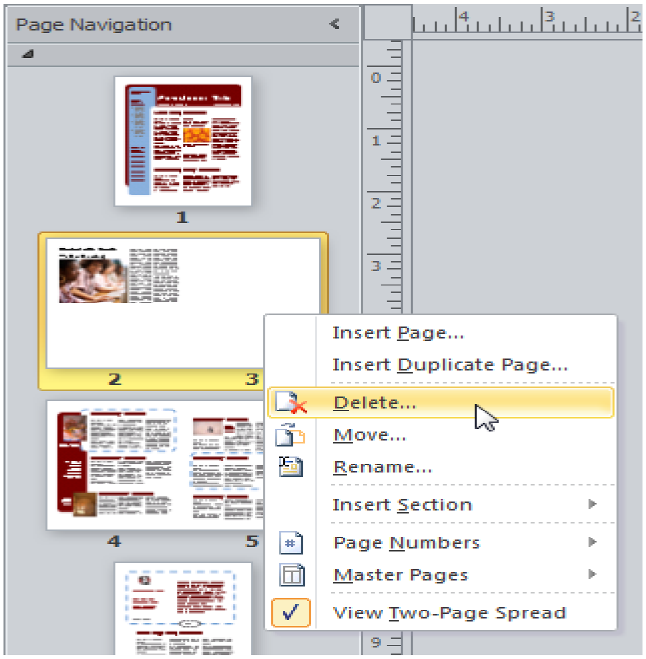


3. The new page order will be applied.

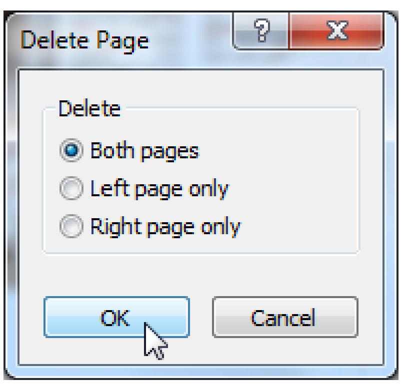


To delete a page

1. In the Page Navigation pane, right-click the page you wish to delete, then select **Delete** in the list that appears.



2. If the page is part of a **two-page** spread, Publisher will ask if you wish to delete one or both pages. Make your selection, then click **OK**.



3. The page will be deleted.

Challenge!

1. Open Publisher and **create** a new publication from a **template**. Be sure to review the **template options**.
2. Modify the page **margins** to make them **wider**.
3. Change the **page orientation** to see how it affects the layout of your publication.
4. Add a **new page** to your publication.
5. **Move** the page you just added so it is the **first** page in your publication.
6. Close the publication without saving it.

Formatting publication

Using text in Publisher

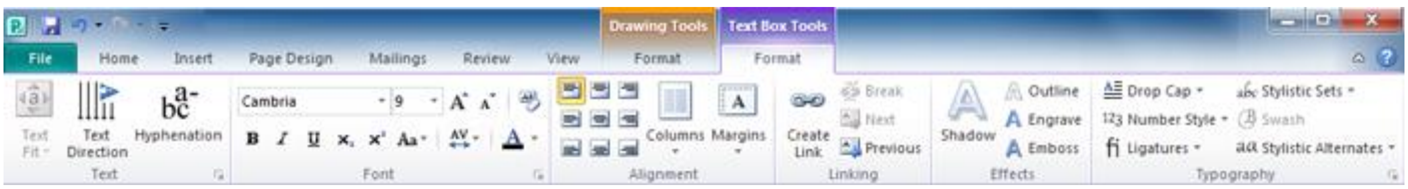
- As you enter text in Publisher, you'll need to adjust it to make it fit your publication. While most of Publisher's text tools are the same as those in other Office programs, a few are specifically designed to handle Publisher's unique publication tasks.

Text basics

- In order to use Publisher 2010 you should already feel comfortable using Microsoft Word to insert and edit text.

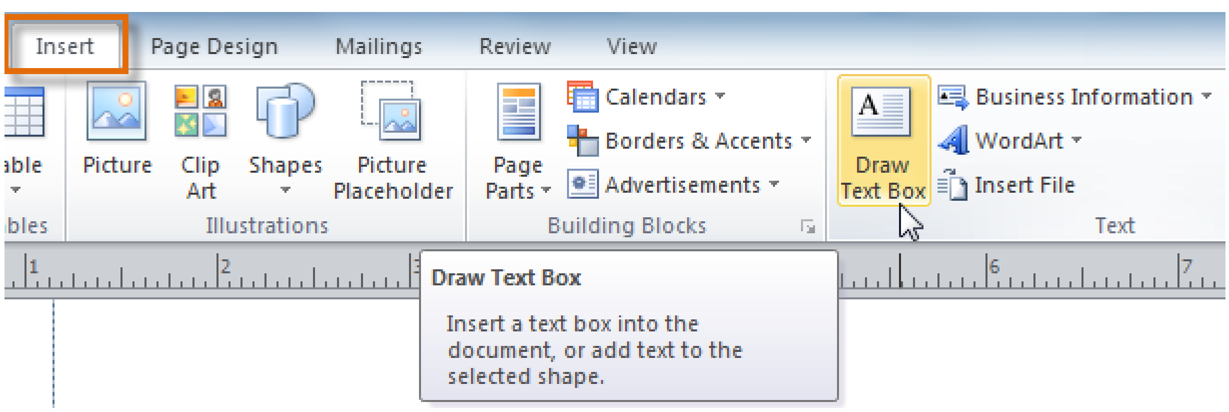
Working with text boxes

- In Publisher, text is contained in **text boxes**, which are blocks of text that you can place on the page.
- When you create or select a text box, the **Text Box Tools** tab will appear on the Ribbon. On this tab are commands that let you adjust and format your text box and the text it contains.

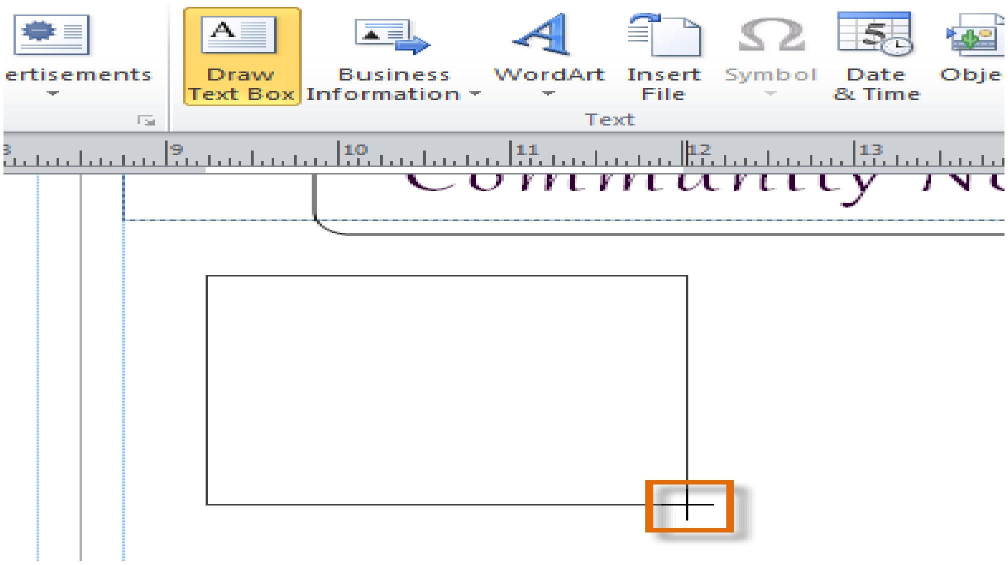


To insert a text box:

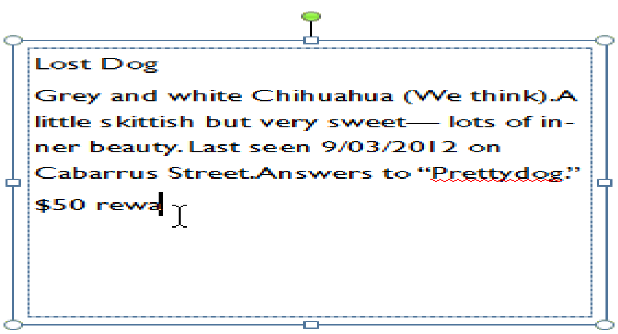
1. On the Ribbon, select the **Insert** tab, then locate the **Text** group.
2. Click the **Draw Text Box** command.



3. The cursor will turn into crosshairs +. Click anywhere onyour publication and **drag** your mouse to create the text box.

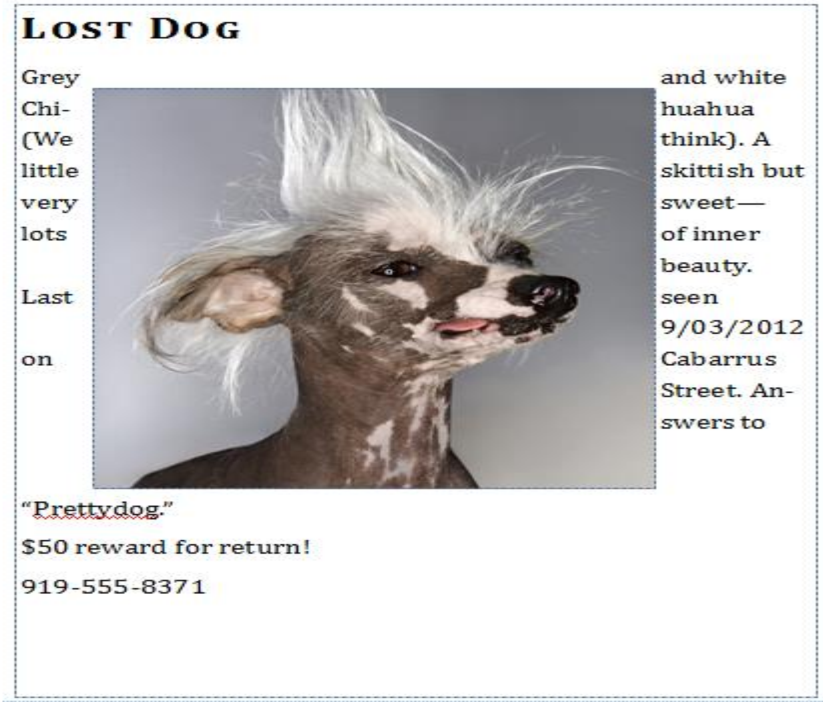


4. You can now start typing inside the text box.



Wrapping text

- If you place a text box near an **image** or another **object**, you may notice that the text is overlapping with the object or doesn't appear exactly where you want.
- To fix this problem, you'll need to change the object's **text wrapping settings**.



Lost Dog

Grey and white Chi-huahua (We think). A little skittish but very sweet— lots of inner beauty.

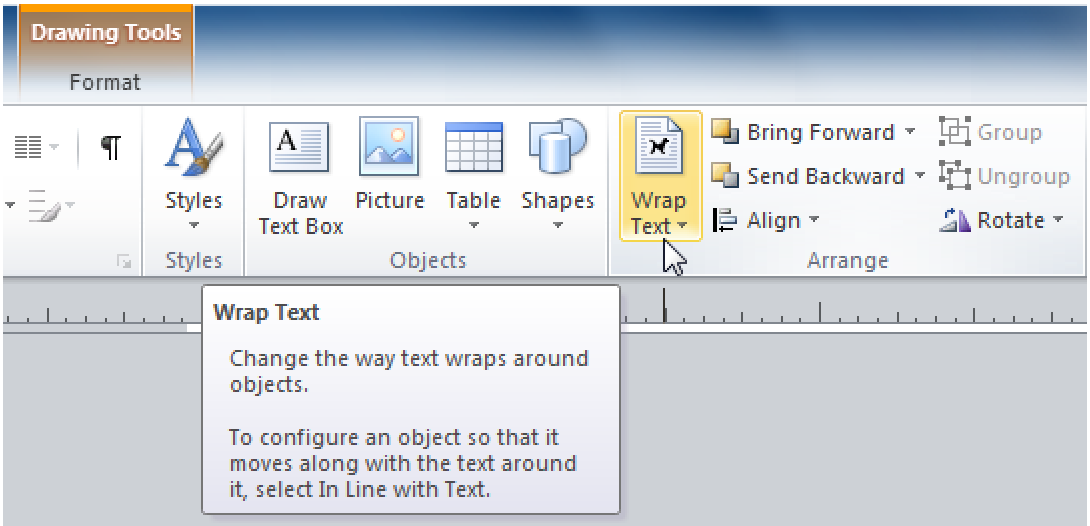
Last seen 9/03/2012 on Cabarrus Street. Answers to "Prettydog."

$50 reward for return!

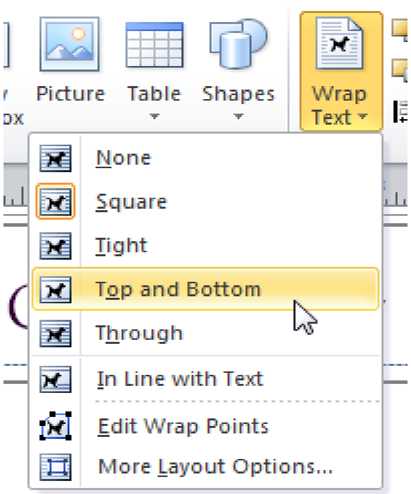
919-555-8371

To wrap text around an object:

1. Select the object, then click the **Format** tab that appears on the Ribbon.
2. Locate the **Arrange** group, then click the **Wrap Text** drop-down command.

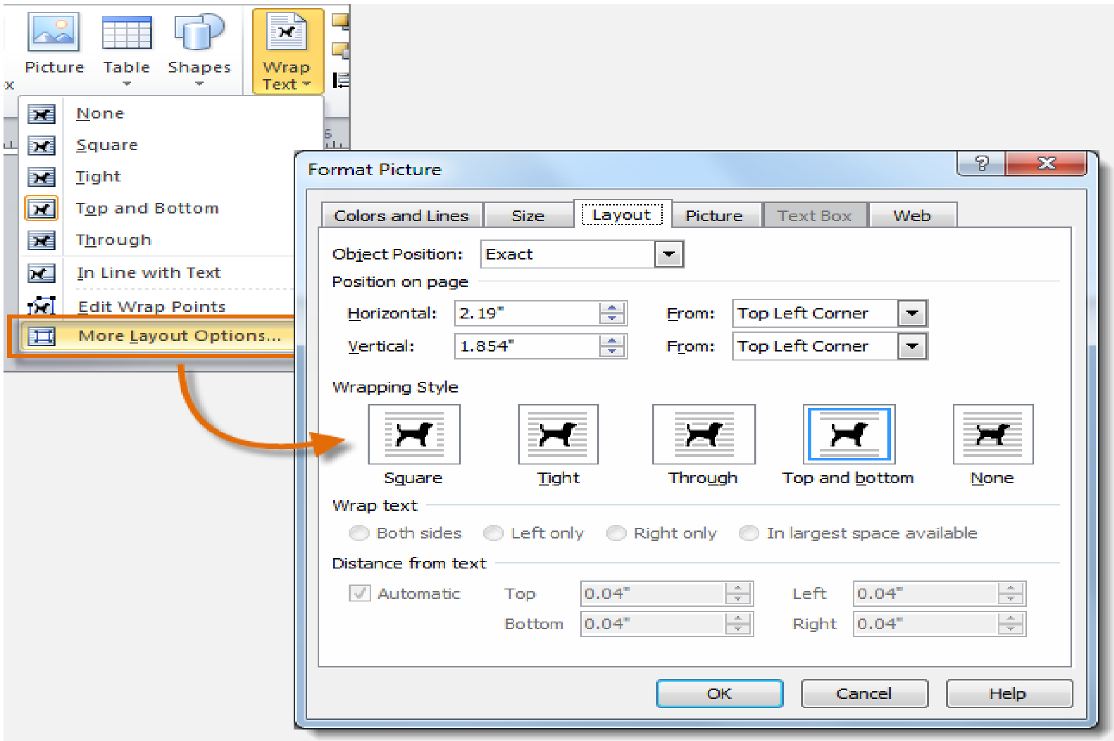


3. Select the desired wrap option. The text will adjust based on the option you have selected.



4. If necessary, reposition the object until the text wraps correctly.

- If you can't get your text to wrap the way you wish, click the **Wrap Text** command and select **More Layout Options** from the menu.
- You can make more precise changes in the Advanced Layout dialog box that appears.

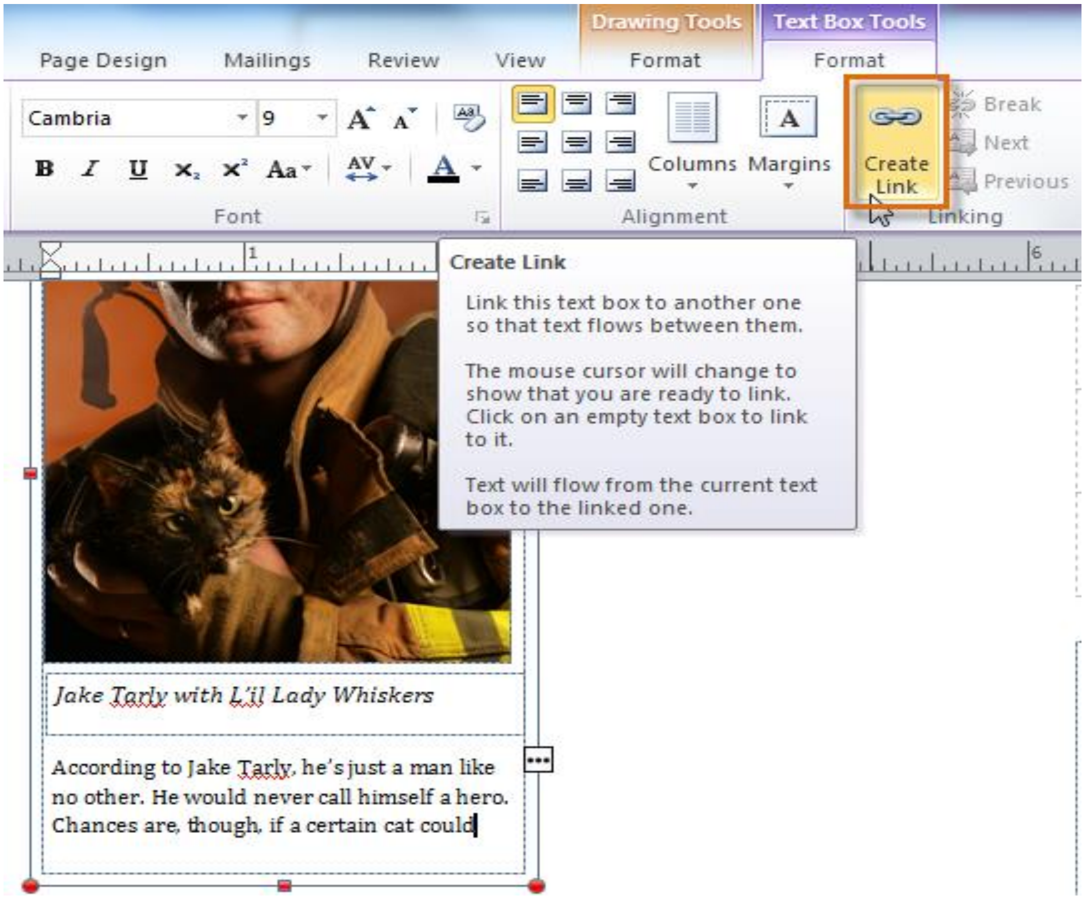


Connecting text boxes

- As you work with text boxes, you might find that a text box isn't large enough to contain all of the text you want to include.
- When you run out of room for text, you can use the **Link** command to **connect** text boxes.
- Once two or more text boxes are connected, text will overflow or continue from one text box to the next.

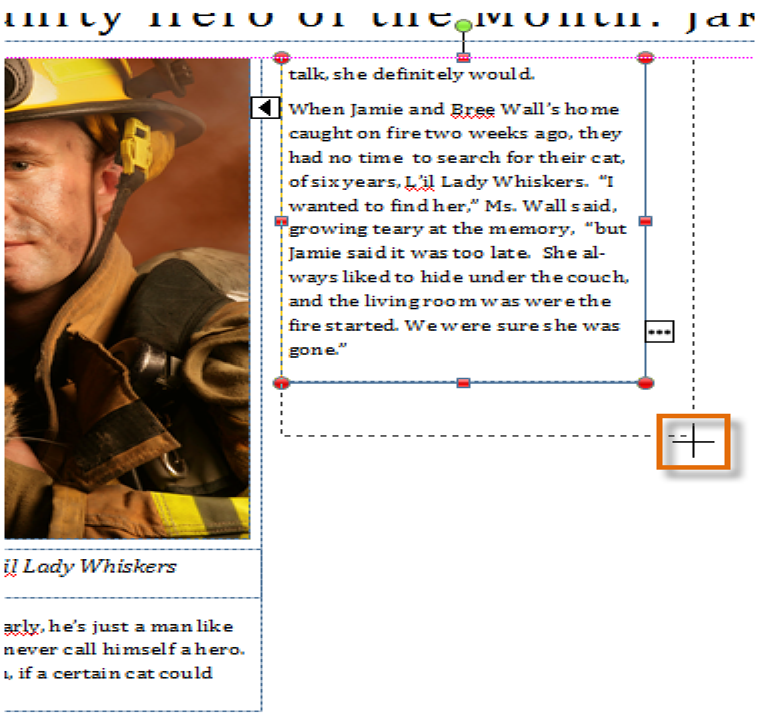
To connect to a new text box:

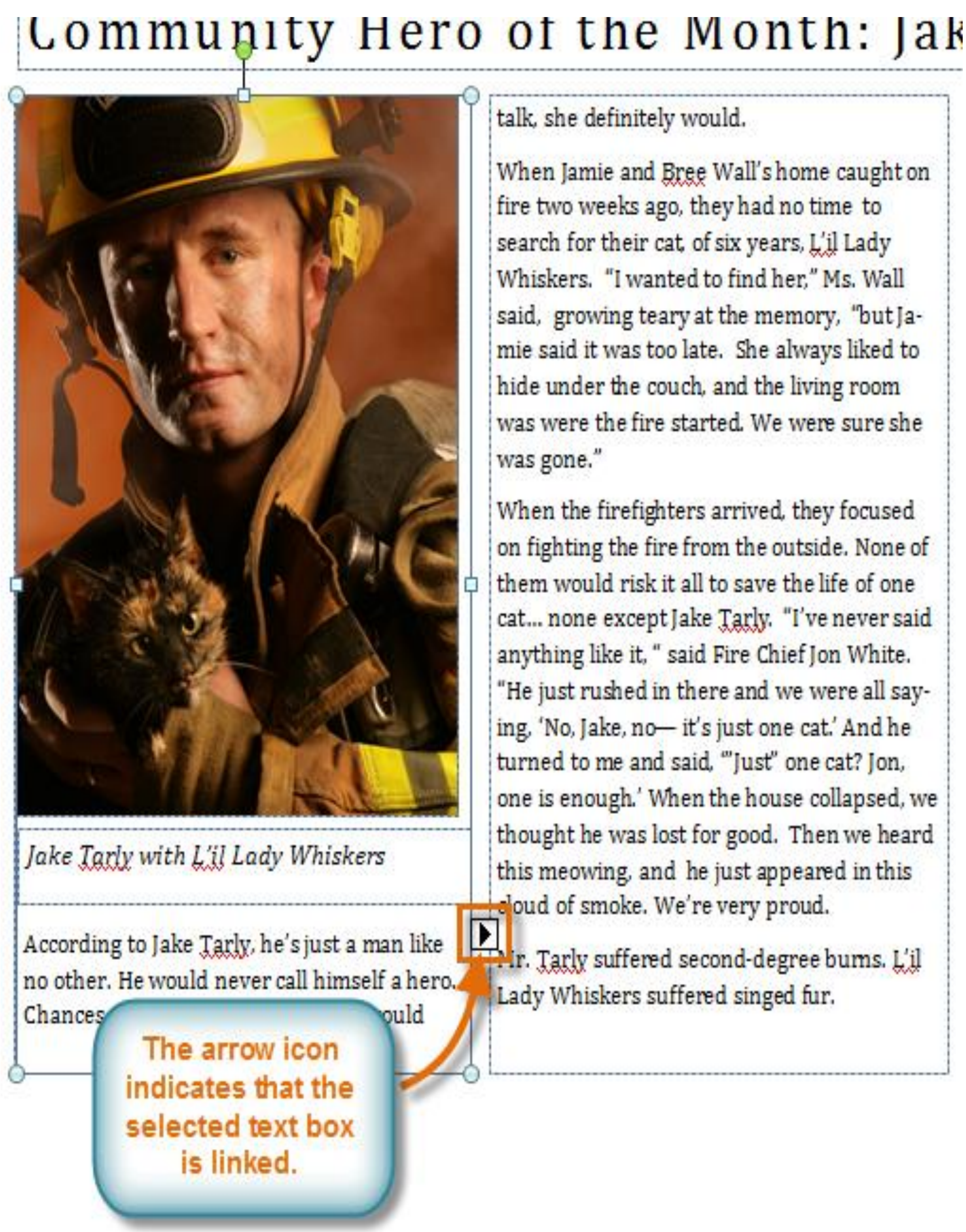
1. Select your text box.
2. Click the **Text Box Tools Format** tab, then locate the **Linking** group.
3. Click the **Create Link** command.



4. The **Link** icon will appear in place of your cursor. Click the spot on your publication where you would like to add the linked text box.

5. The text box will be added. Resize it as necessary.


talk, she definitely would.
When Jamie and Bree Wall's home caught on fire two weeks ago, they had no time to search for their cat, of six years, L'il Lady Whiskers. "I wanted to find her," Ms. Wall said, growing teary at the memory, "but Jamie said it was too late. She always liked to hide under the couch, and the living room was were the fire started. We were sure she was gone."
Lady Whiskers
he's just a man like
never call himself a hero.
if a certain cat could



6. Continue typing your text. Any text that overflows from the original text box will now appear in the connected box.

Modifying text boxes

Text Fit

- The **Text Fit** options allow you to format text boxes that **automatically adjust** font or text box size to get a good fit.
- There are four text fit options you can apply to any text box:

i. **Best Fit**,

- which makes the text larger or smaller to fit the text box

ii. **Shrink Text on Overflow**,

- which automatically shrinks the font size when the text box has no room for additional text

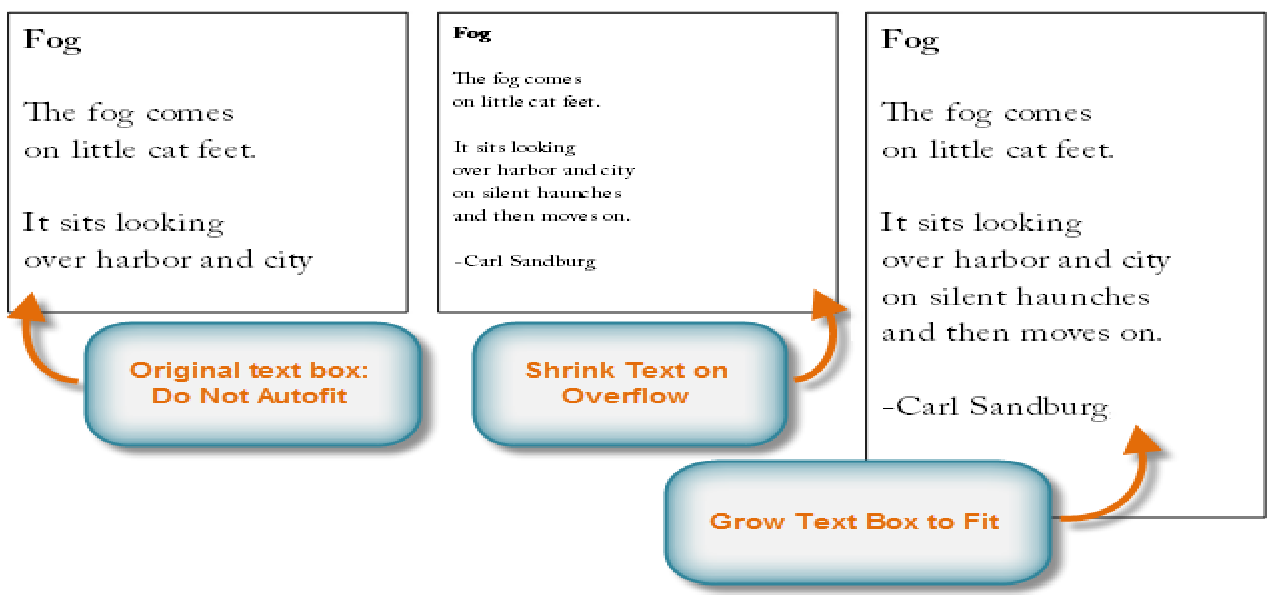
iii. **Grow Text Box to Fit**,

- which automatically enlarges the text box based on text size and length

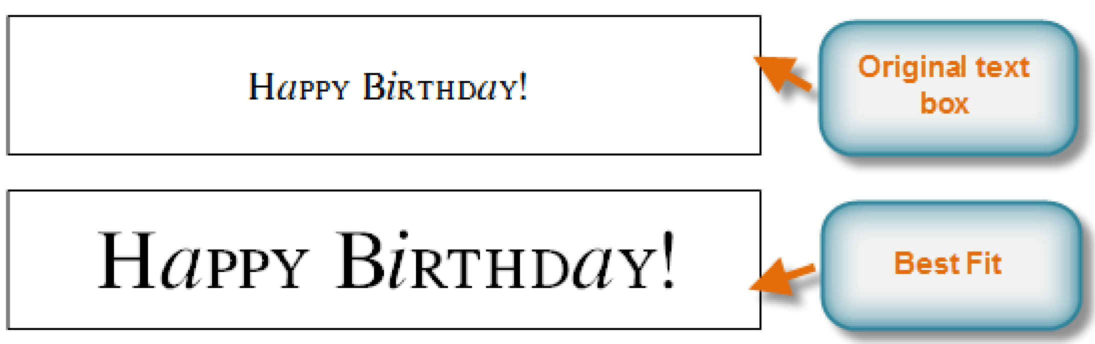
iv. **Do not Autofit**,

- which makes no automatic changes to the text or text box size; this is the default option

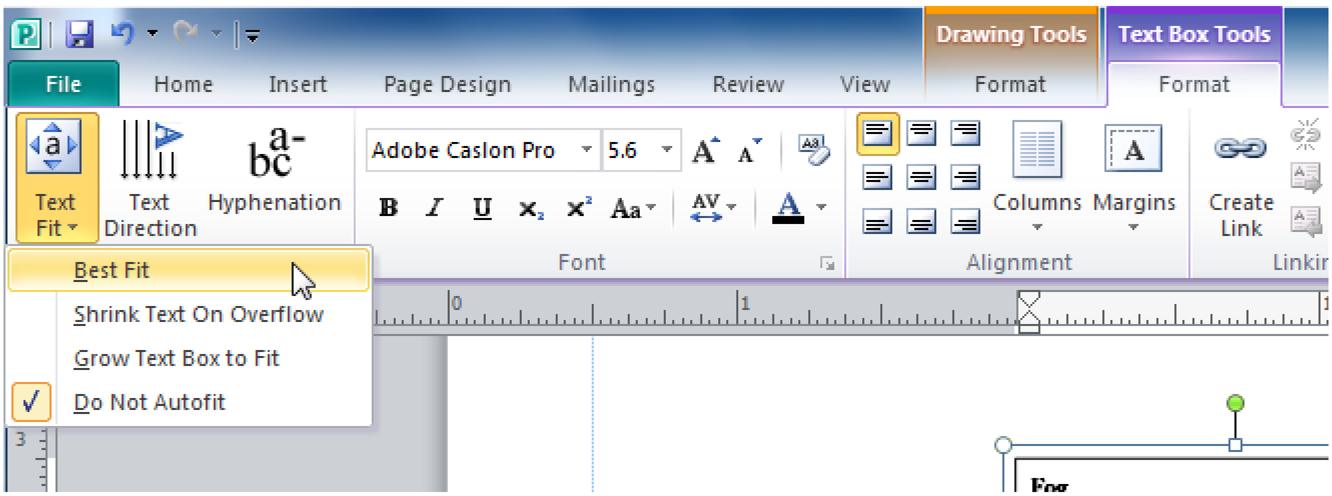
For instance, if your text box was too **small** for your text you might apply **Shrink Text on Overflow** or **Grow Text Box to Fit.**



- On the other hand, if you have a certain amount of space for your text box and want your text to fill the entire area, you might select **Best Fit**.



- To modify text fit, select the text box, then click the **Text Fit** drop-down command in the **Text** group of the **Text Box Tools** tab. Select the desired option.

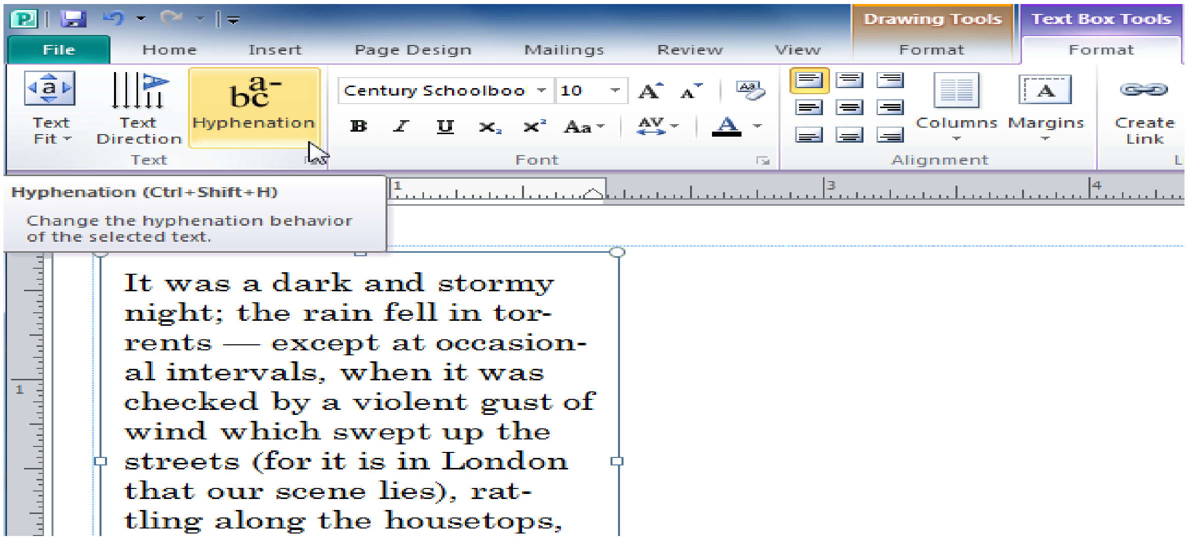


Hyphenation

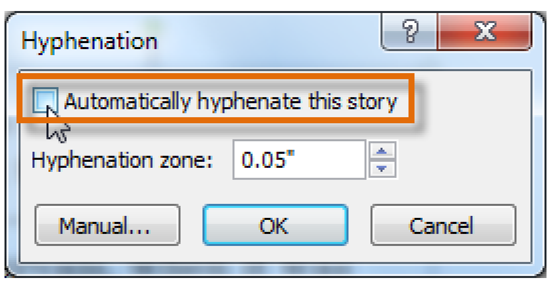
Publisher automatically **hyphenates** words at the ends of lines in order to improve text fit. You can control if and how your words are hyphenated by modifying your **hyphenation settings**.

To modify hyphenation settings:

1. Select a text box, then click the **Text Box Tools Format** tab on the Ribbon and locate the **Text** group.
2. Click the **Hyphenation** command.

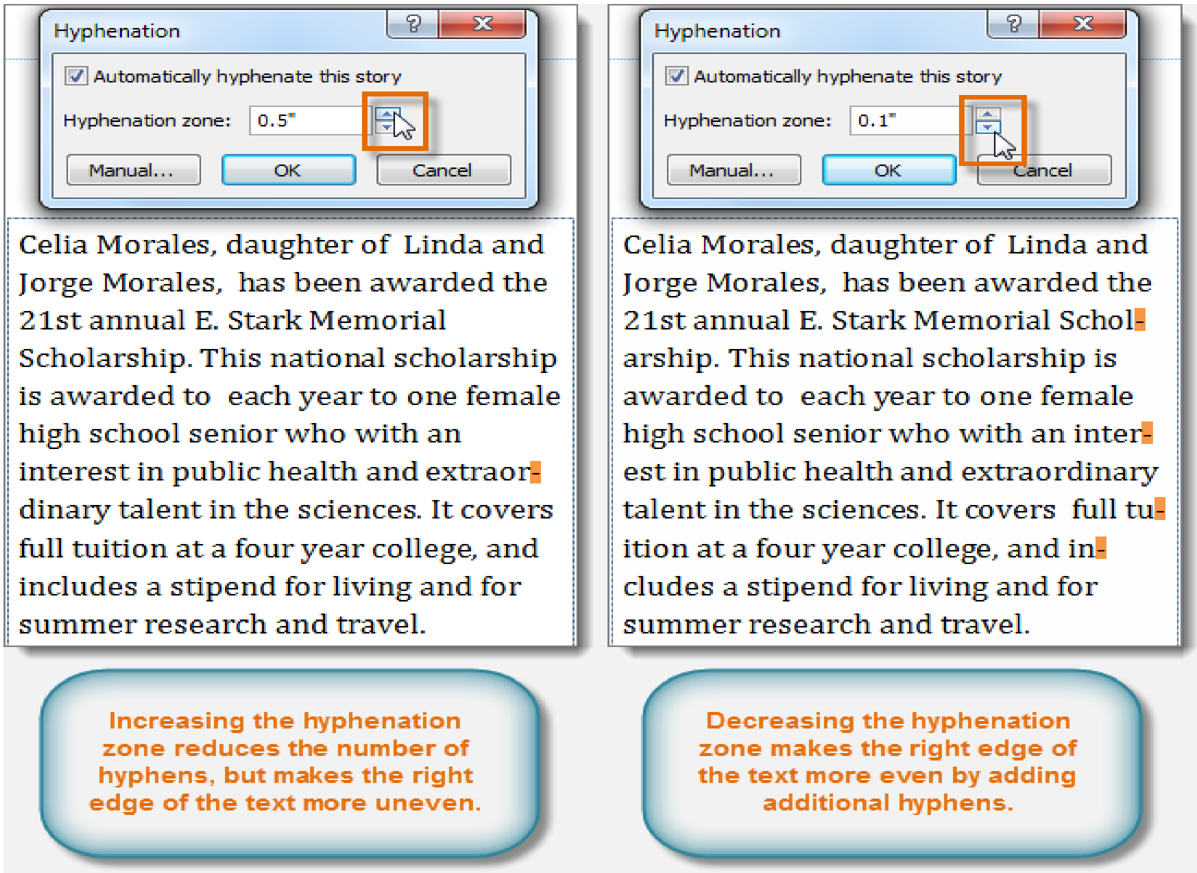


3. The **Hyphenation** dialog box will appear. Modify your hyphenation settings as desired.
   - To **remove** all hyphenation, uncheck the **Automatically hyphenate this story** box.

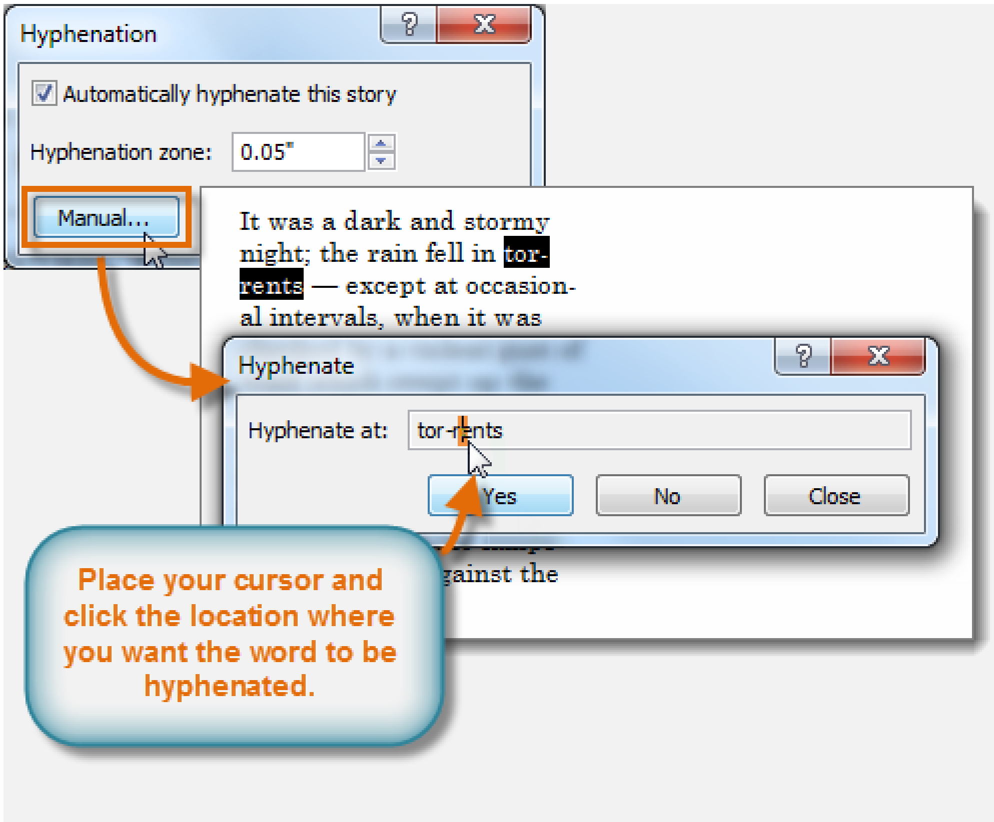


   - To change how frequently Publisher hyphenates words, use the **up and down arrows** to adjust the size of the **hyphenation zone**.
   - If you **increase** the size of the hyphenation zone, your publication will have fewer hyphens. If you **decrease** it, the

right edge of the text will appear more even, but your text will contain more hyphens.



- To specify exactly where each word should be hyphenated, click **Manual...** The **Hyphenate** dialog box will appear, containing one hyphenated word from your text box. To change where the hyphen appears in that word, simply **click** the place where you want the hyphen to appear, then click **Yes**.

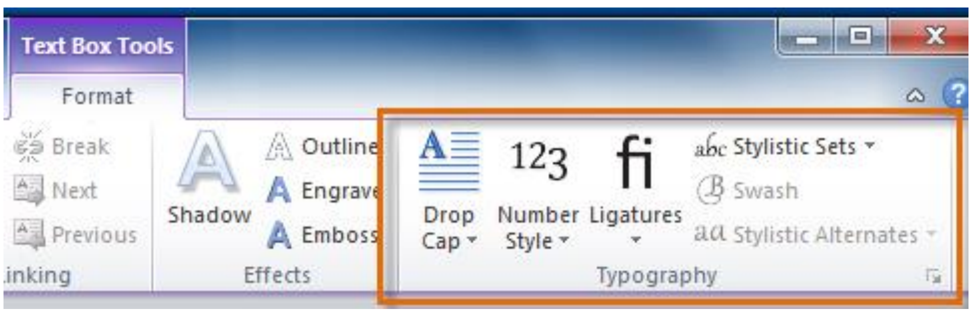


4. Click **OK**. The hyphenation will be adjusted.

Formatting text

- Publisher 2010 includes various **typography commands** that are designed to help you **embellish** your text.
- Although Publisher's developers have touted this as a significant feature, it's important to note that **many of these effects only work with a small number of fonts**, such as **Calibri**, **Cambria**, and **Gabriola**.

- Still, if you're using these fonts the typography commands can enhance the appearance of your text.
- Typography commands can be found in the **Text Box Tools** tab.
- To apply any command, simply select your text, then click the desired command. Certain commands, like **Stylistic Sets**, will include a drop-down list of choices.



There are six Publisher typography commands:

i. **Drop Cap**,

- which enlarges the first letter of the selected text

Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived in Liberty, and dedicated to the proposi-tion that all men are created equal.	Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived in Liberty, and dedicated to the proposition that all men are created equal.

ii. **Number Style**,

- which lets you choose between four different styles for number spacing and alignment

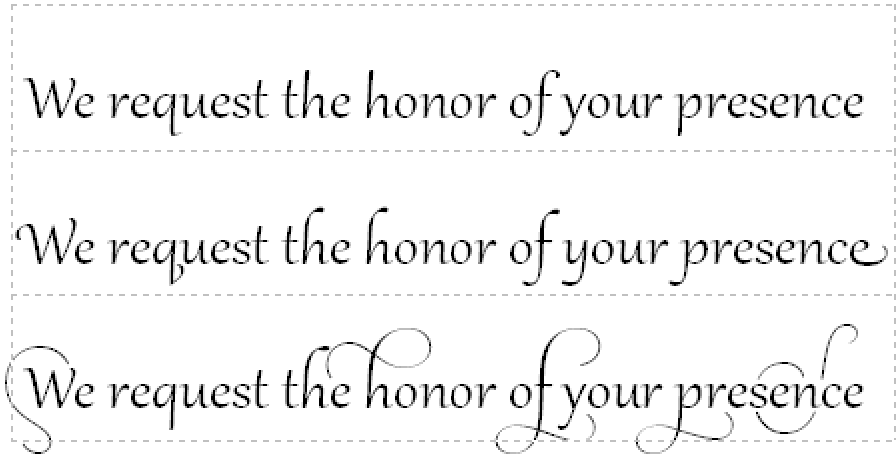
$\pi \approx 3.14159265$
$\pi \approx 3.14159265$
$\pi \approx 3.14159265$
$\pi \approx 3.14159265$

iii. **Ligatures**,

- which connect certain combinations of letters to make them easier to read

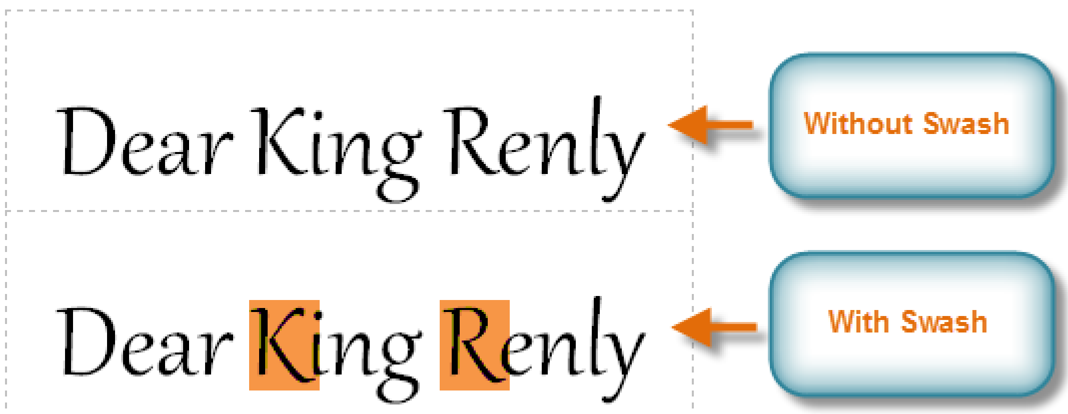
iv. Sets,

- which let you choose between various **embellishments** for your fonts, usually in the form of exaggerated **serifs** or **flourishes**



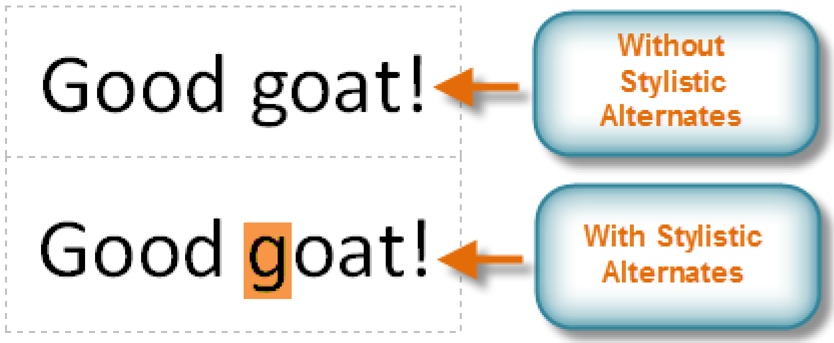
v. **Swash**,

- which embellishes **capital letters**



vii. **Stylistic Alternates**,

- which offer alternate versions of specific letters such as **g**

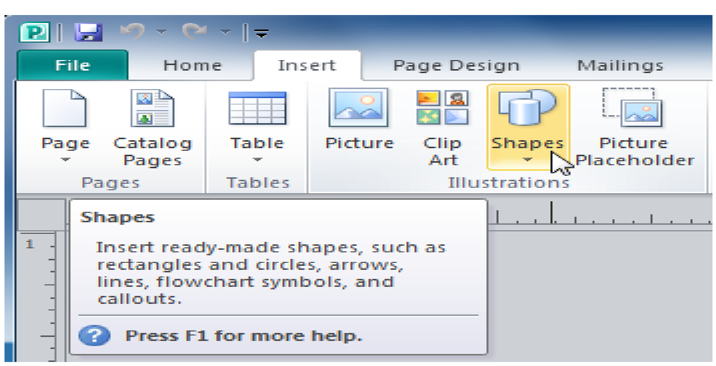


Working with shapes

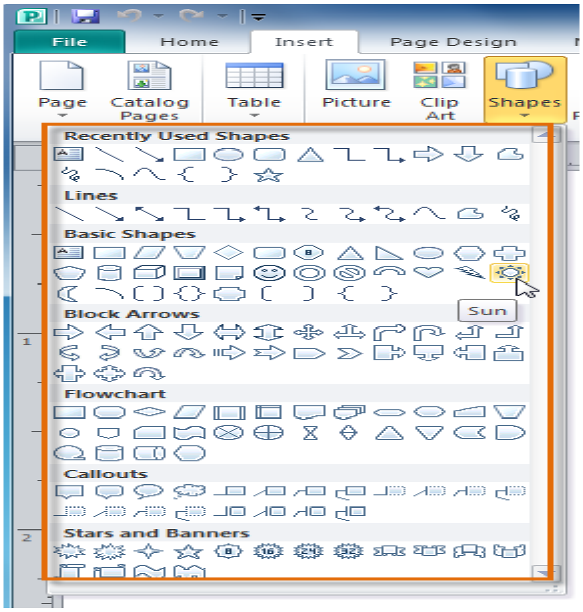
- Using shapes and objects is an easy way add graphic design elements to your publication. While you may not need shapes in every publication you create, they can add visual appeal.

To insert a shape:

1. Select the **Insert** tab, then locate the **Illustrations** group.
2. Click the **Shapes** drop-down command.

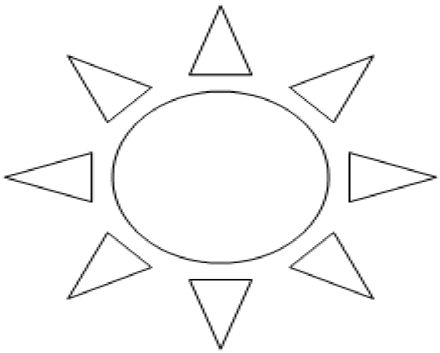


3. Select a shape from the drop-down menu.



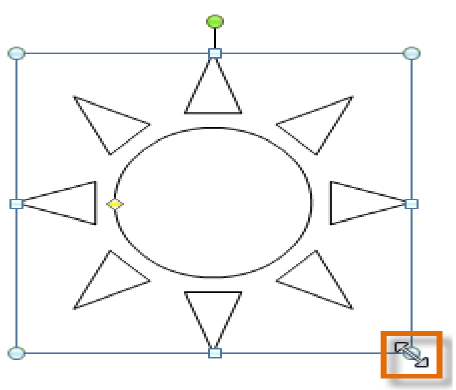
4. Click and drag the mouse until the shape is the desired size.

5. Release the mouse button. The shape will be added to your publication.

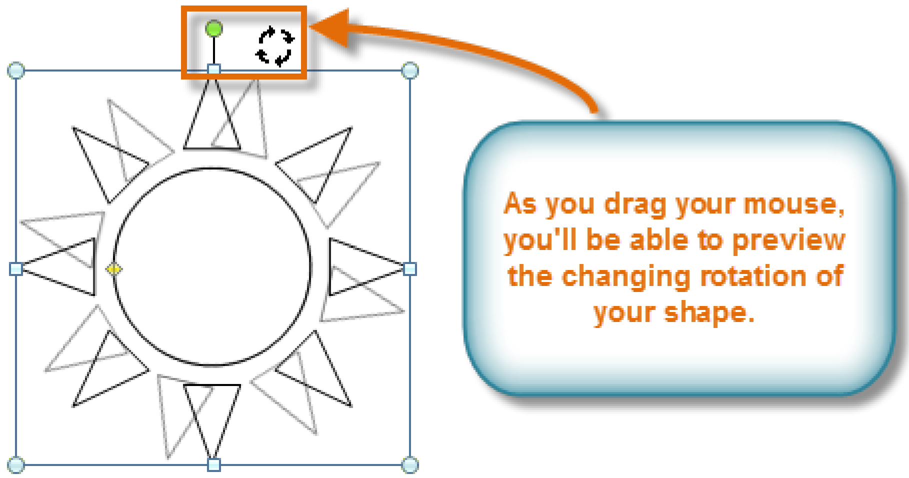


To resize a shape:

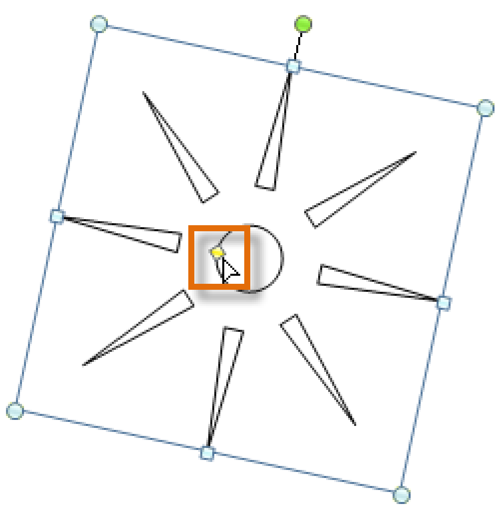
1. Select the shape.
2. Click and drag one of the **sizing handles** on the corners and sides of the text box until it is the desired size. You can:
    - Drag the top or bottom sizing handles to modify shape **height**
    - Drag the side handles to modify shape **width**
    - Drag the **corner handles** to modify **height** and **width** at the same time



3. To rotate the shape, click and drag the **green handle**.

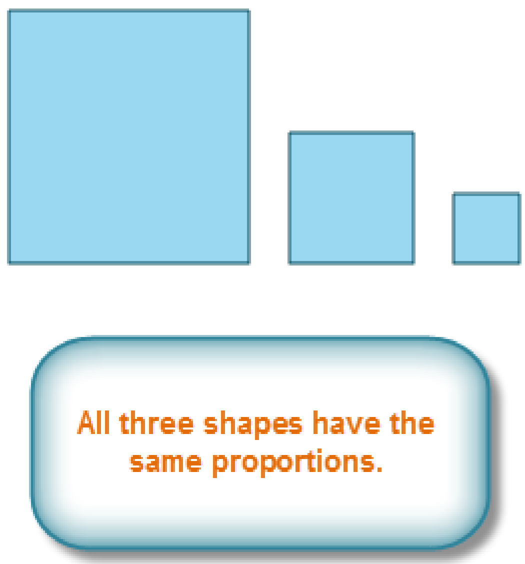


4. Some shapes also have one or more **yellow handles** that can be used to modify the shape. For example, with this sun shape you can adjust the diameter of the center circle and the length of the points.



- If you hold down the **shift** key while resizing a shape, the shape will keep its **proportions** instead of getting stretched out.

- For instance, if you hold down the shift key while you resize a **square**, the final shape will remain a perfect square with four equal sides.



Modifying shapes

To change to a different shape:

1. Select the shape, then click the **Format** tab and locate the **Shape Styles** group.
2. Click the **Change Shape** drop-down command.

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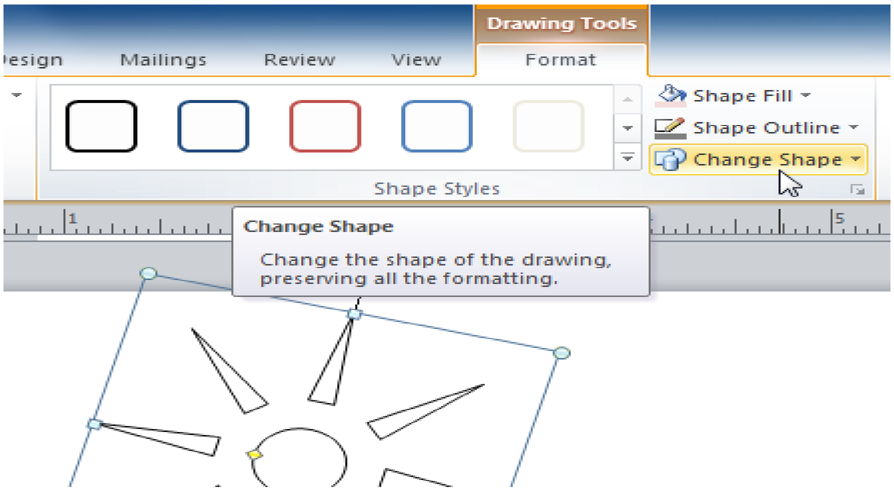
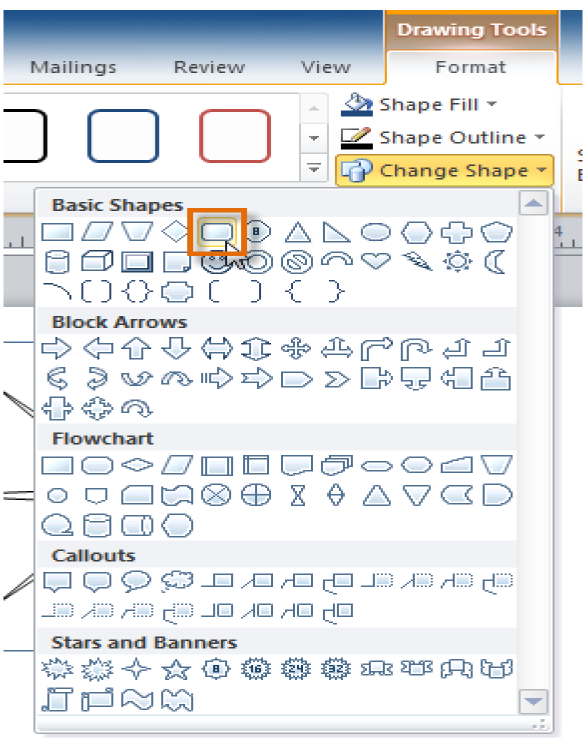
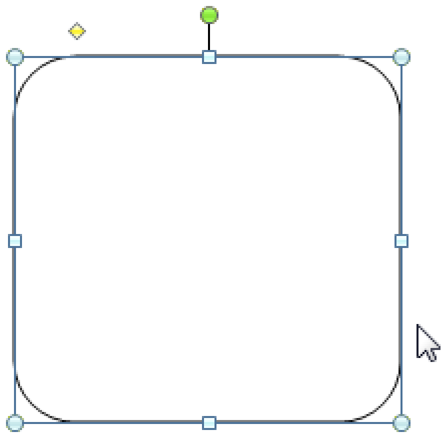


3. A drop-down list will appear. Select the desired shape.

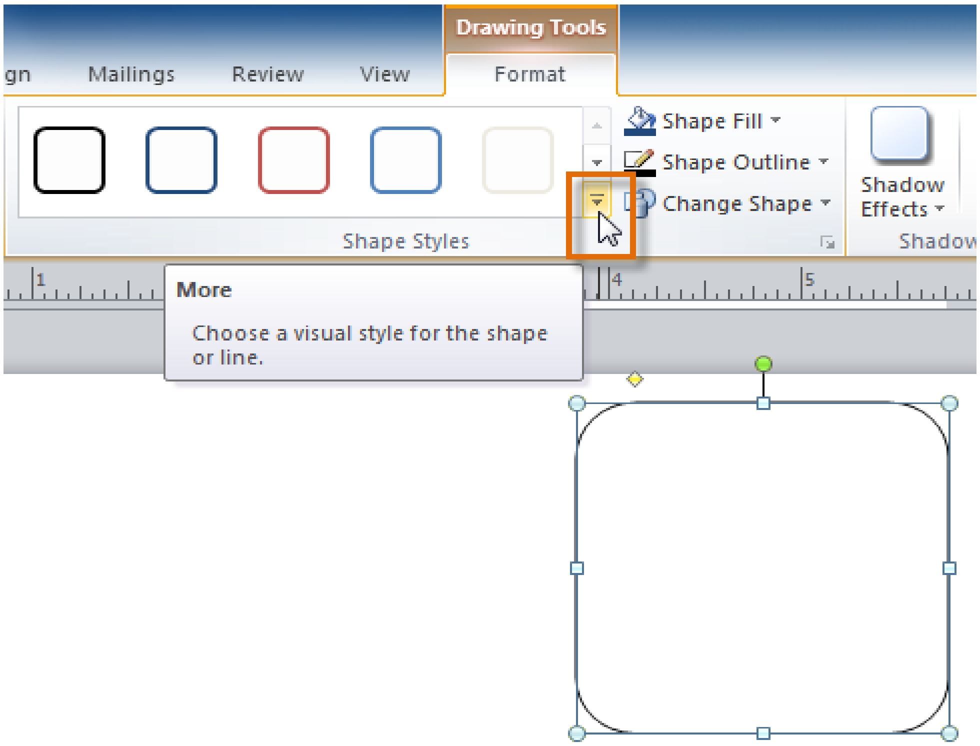


4. The shape will be changed.

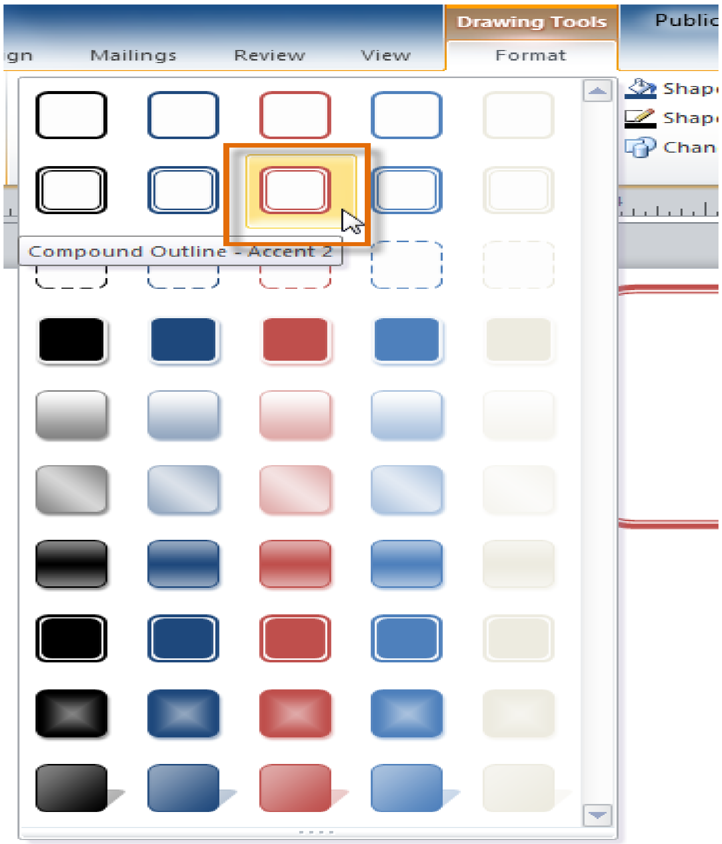


To change shape style:

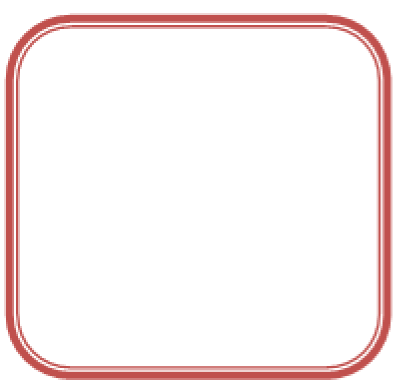
1. Select the shape, then click the **Format** tab and locate the **Shape Styles** group.
2. Click the **More Shape Styles** drop-down arrow.



3. A drop-down list of styles will appear. Move your cursor over the styles to see a live **preview** of the style in your publication, then select the desired style.

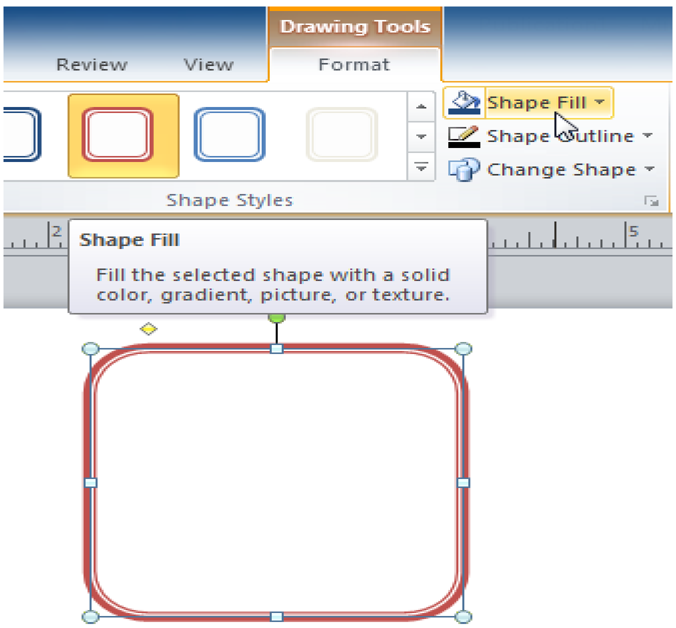


4. The style will be applied to the shape.

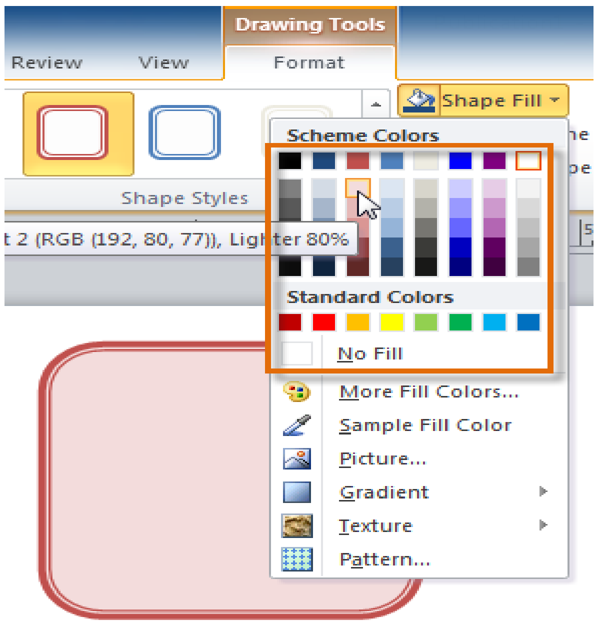


To change the shape fill color:

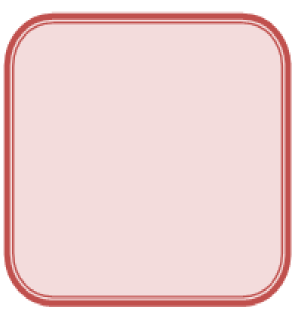
1. Select the shape, then click the **Format** tab and locate the **Shape Styles** group.
2. Click the **Shape Fill** drop-down command.



3. A drop-down list of colors will appear. Select the desired fill color from the list. You can also choose **No Fill** to remove the fill from your shape or **More Fill Colors** to select a custom color.

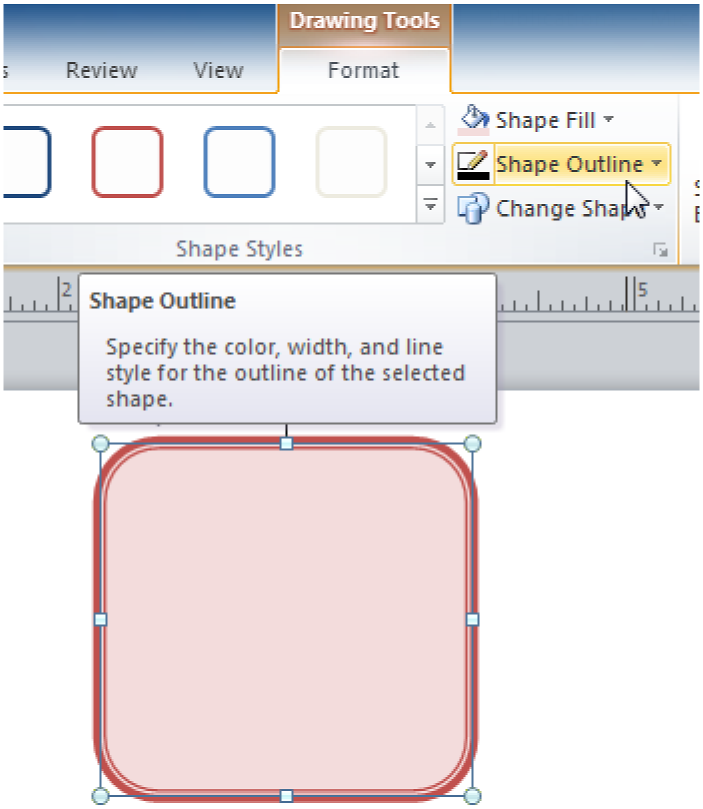


4. The new fill color will be applied.

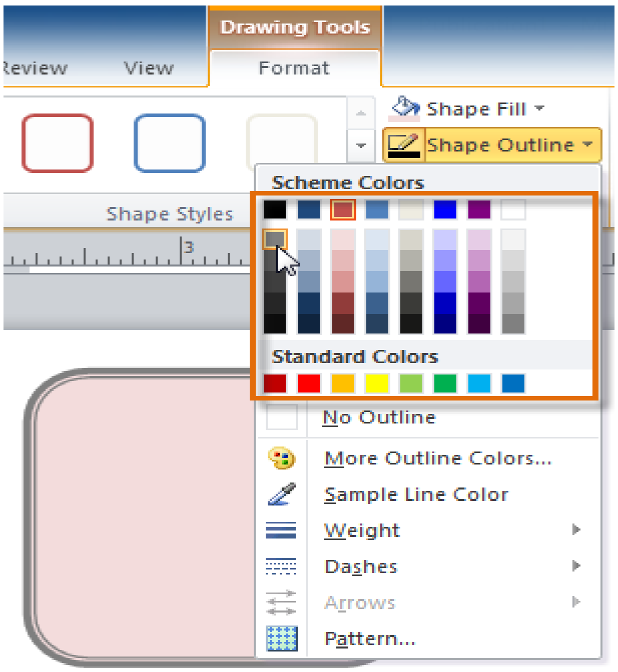


To change the shape outline:

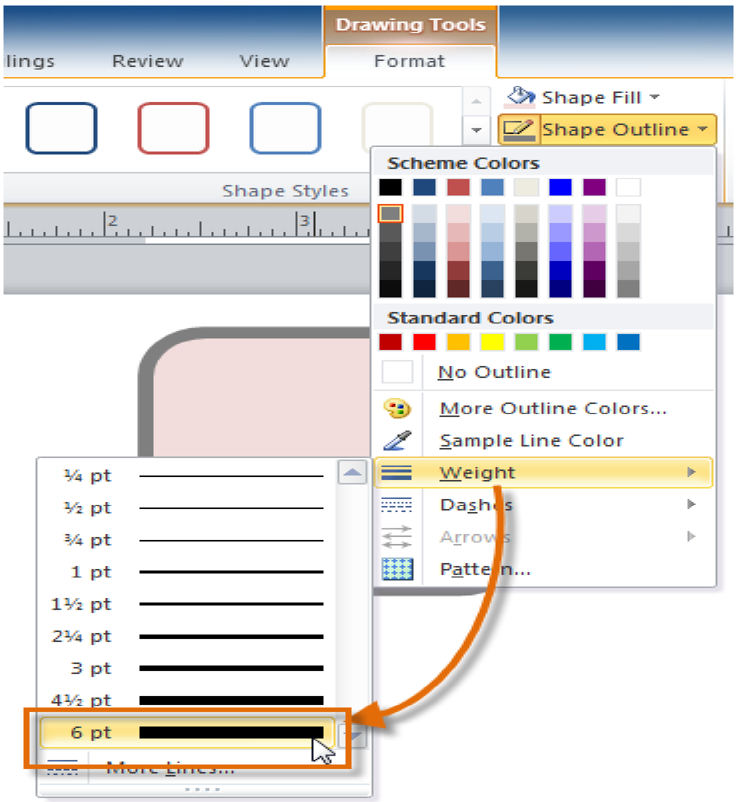
1. Select the shape, then click the **Format** tab and locate the **Shape Styles** group.
2. Click the **Shape Outline** drop-down command.



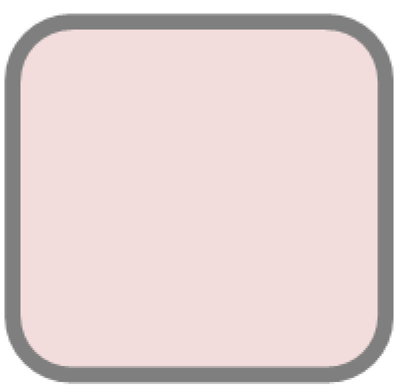
3. A drop-down list of options will appear. Select the desired outline color from the list. You can also choose **No Outline** to remove the outline from your shape or **More Outline Colors** to select a custom color.



4. If desired, further modify your shape outline by changing the outline's **weight** (thickness) and whether or not it is a **dashed** line.

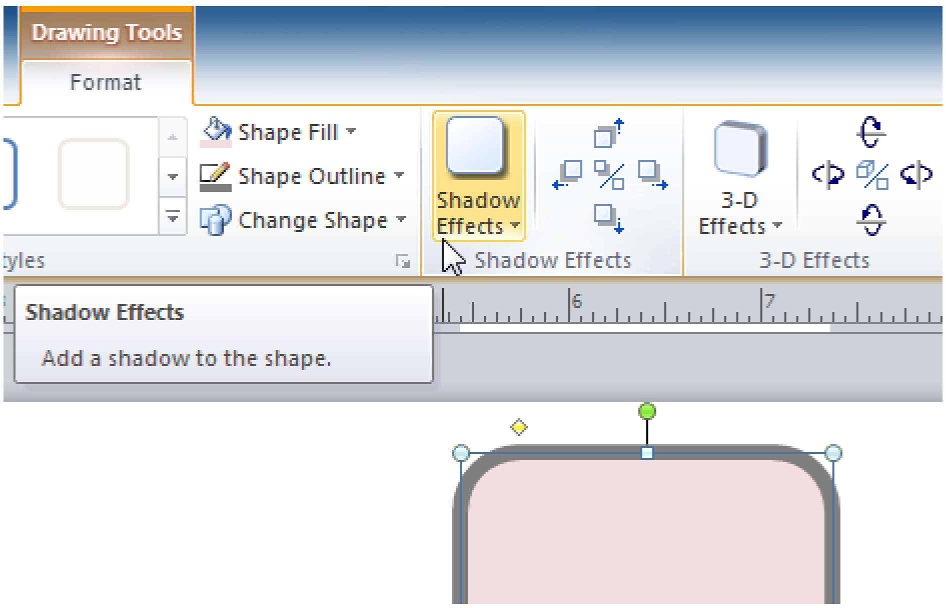


5. The shape outline will be modified.

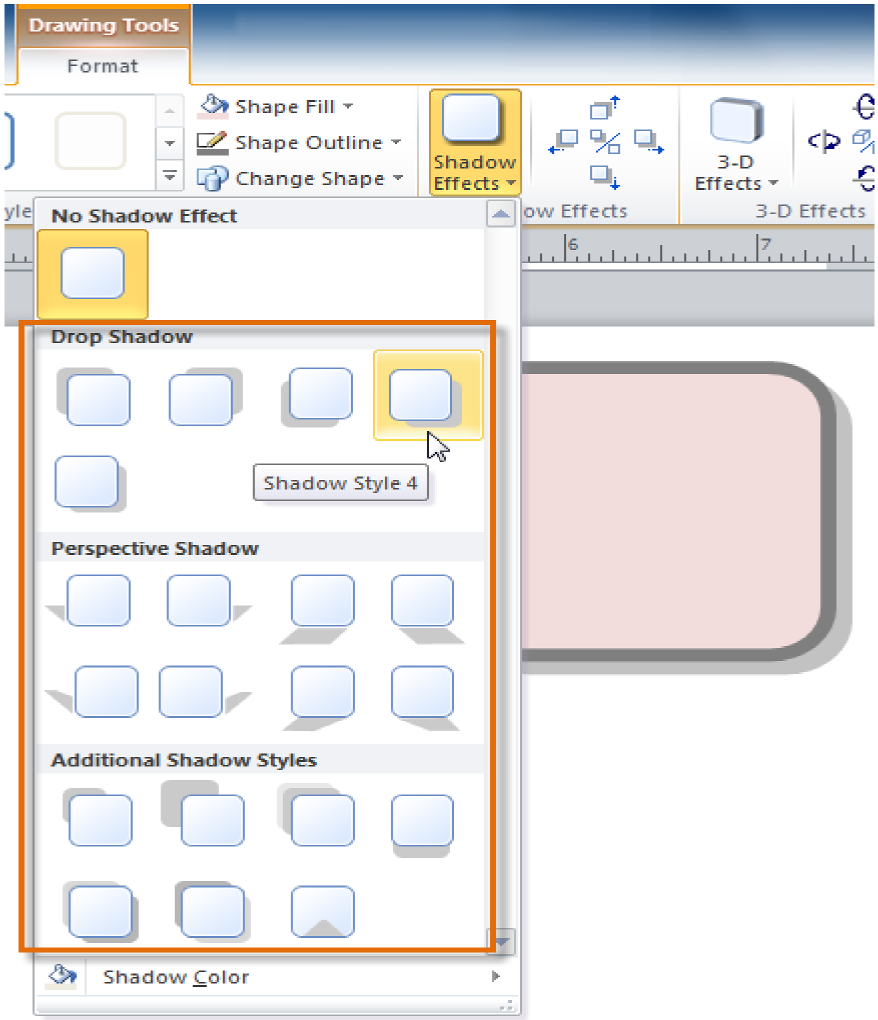


To add a shadow:

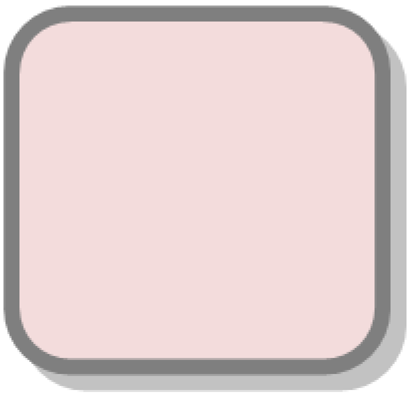
1. Select the shape, then click the **Format** tab and locate the **Shadow Effects** group.
2. Click the **Shadow Effects** drop-down command.



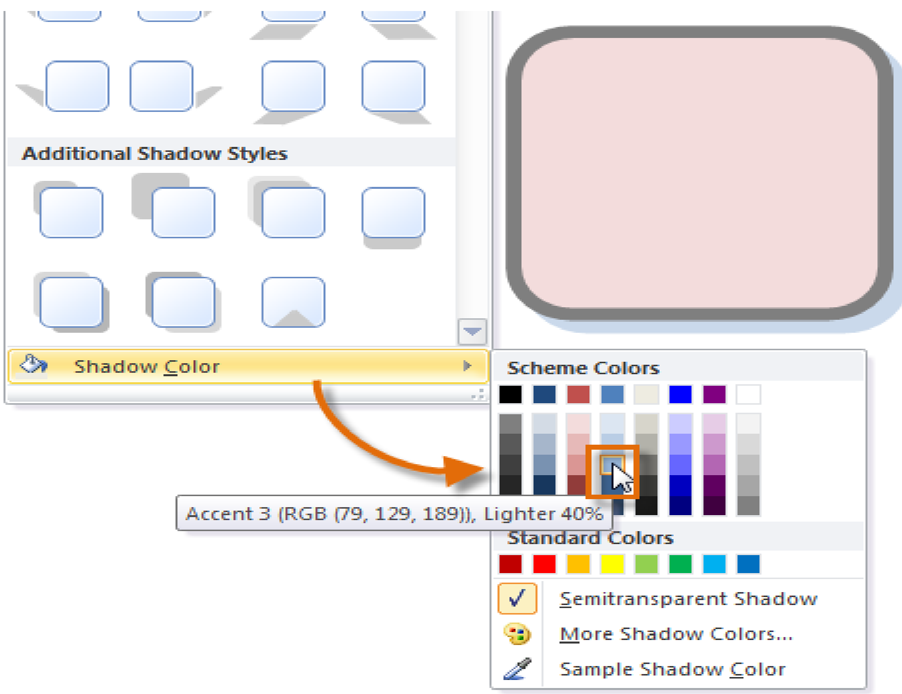
3. A drop-down menu with a list of shadow choices will appear. Move your mouse over a shadow effect to see a live preview of it in your publication.



4. Click the desired shadow effect to apply it to your shape.



- You can select **Shadow Options** from the drop-down menu and click the **Color** button to select a different shadow color for your shape.



Working with Building Blocks

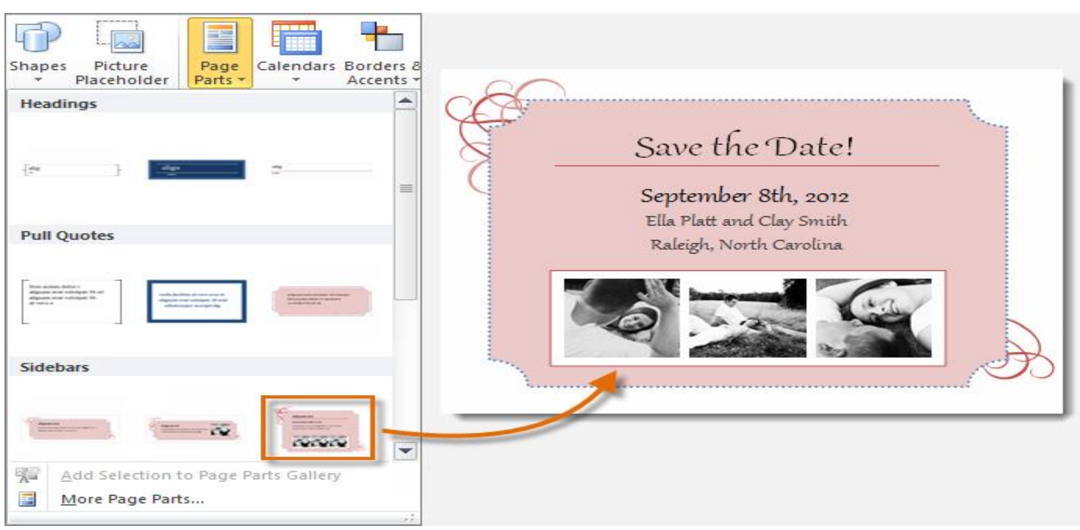
- **Building Blocks** are another type of object in Publisher.
- They usually contain some combination of **text**, **shapes**, and **images**, and they're meant to enhance the appearance of your publication.
- Once you insert a Building Block, you can modify it to suit your needs.

Types of Building Blocks:

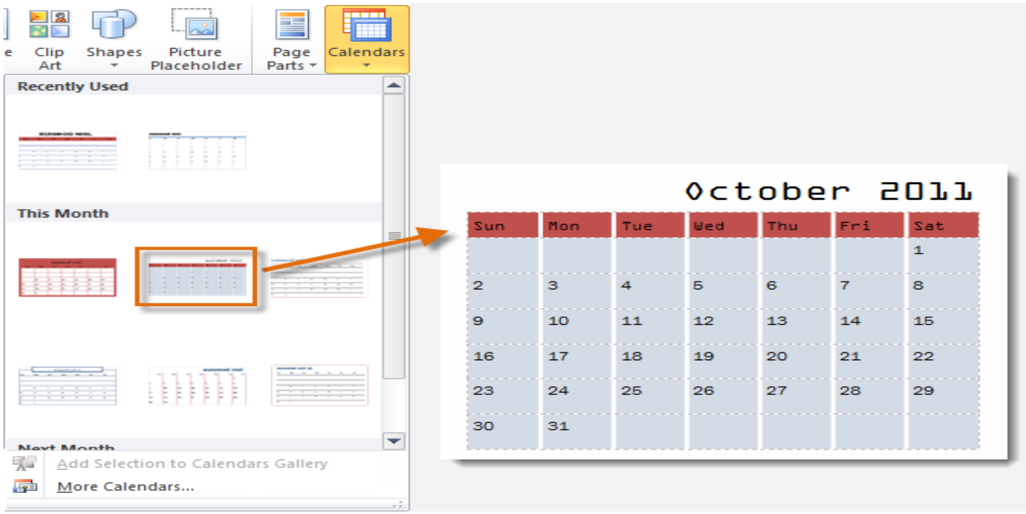
There are four types of Building Blocks:

i. **Page parts**,

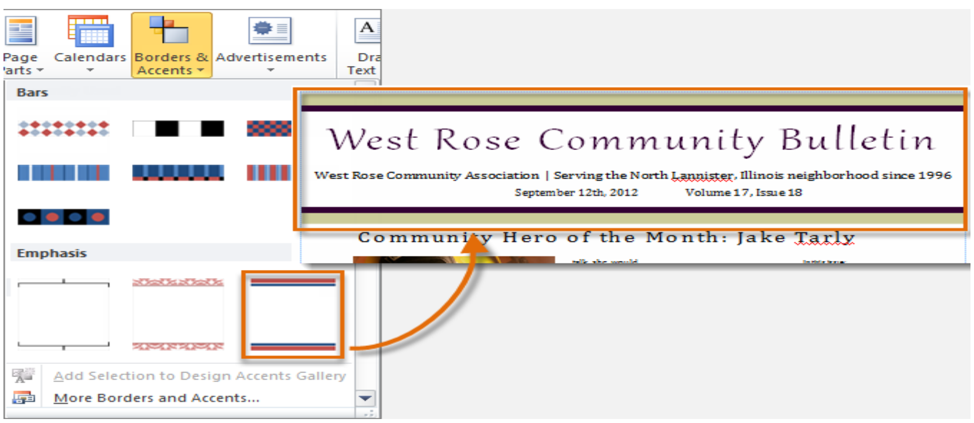
- which are stylized placeholders for your images and text



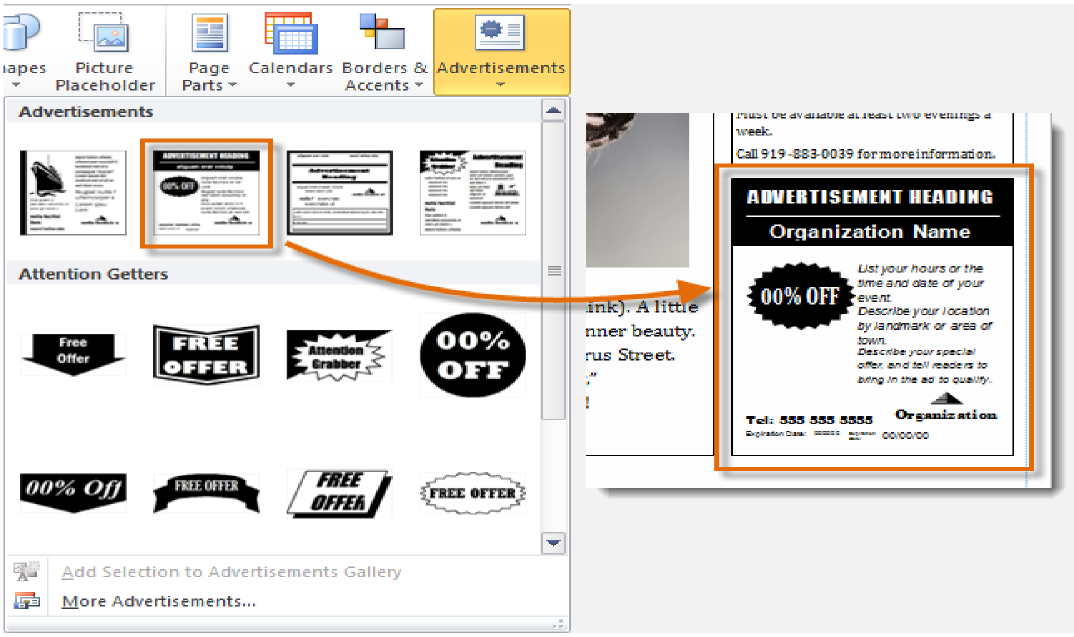
ii. **Calendars**



iii. Borders & Accents

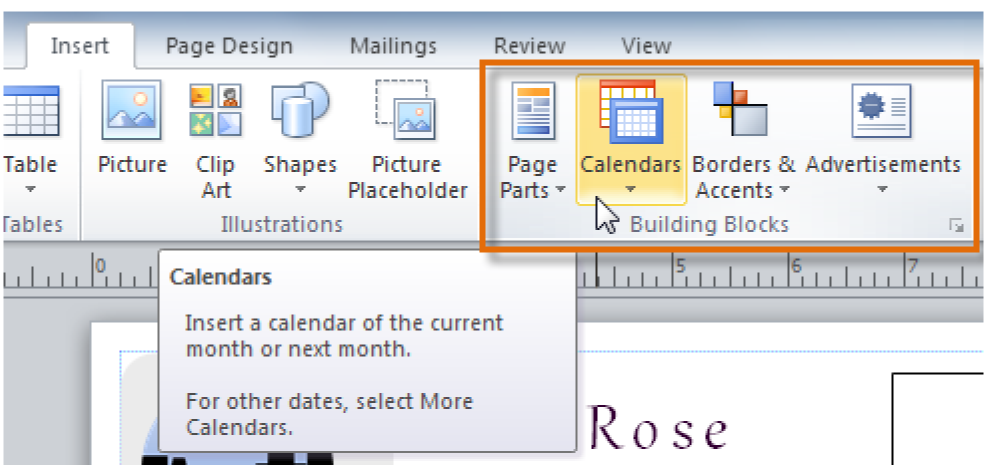


iv. Advertisements

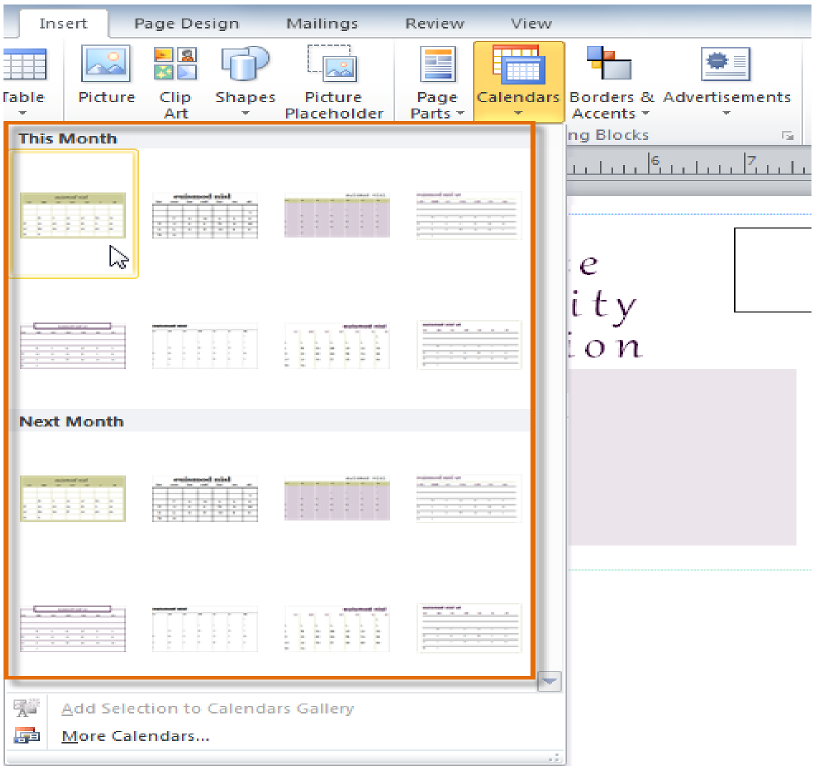


To insert a Building Block:

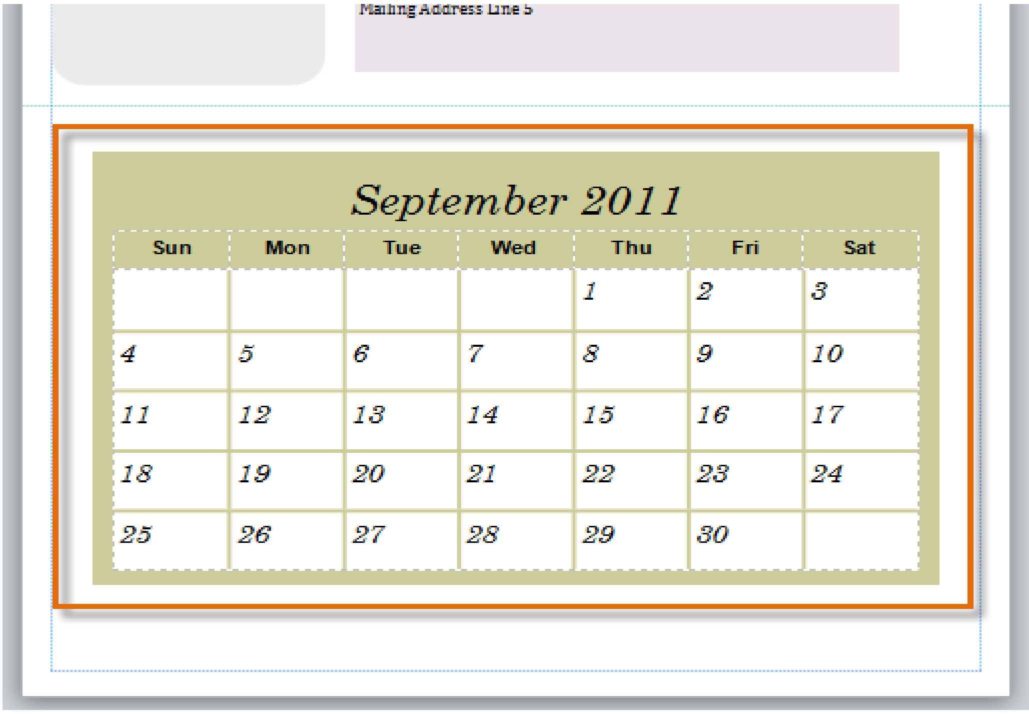
1. Select the **Insert** tab, then locate the **Building Blocks** group.
2. Click one of the four **Building Block drop-down commands**.



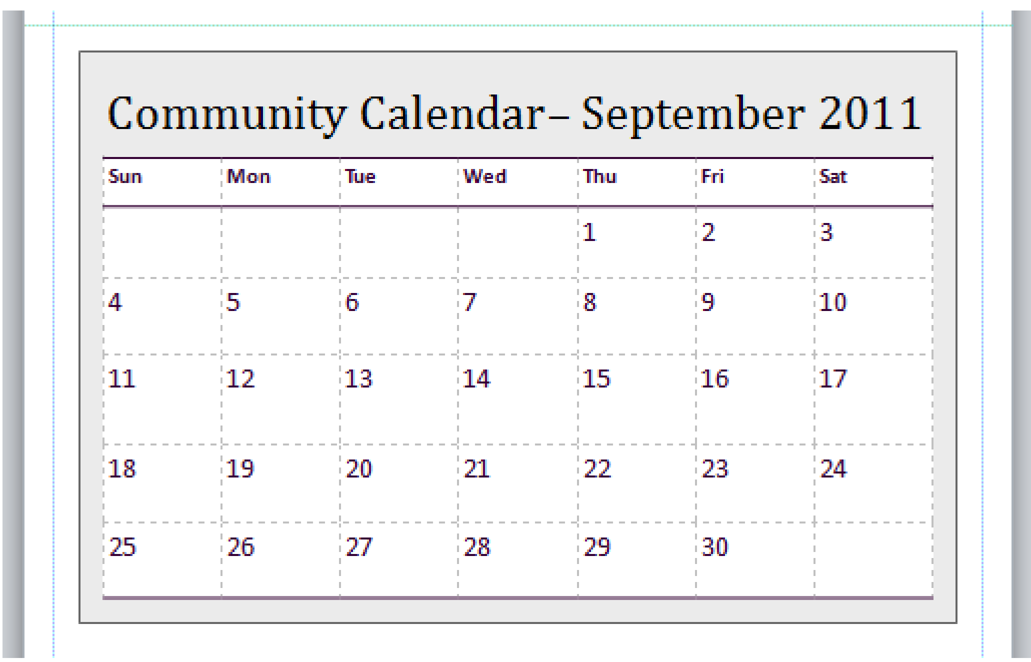
3. A drop-down menu will appear with Building Block styles and options. Select the desired Building Block.



4. The Building Block will be inserted.



5. If desired, modify the Building Block's text and formatting until you are satisfied with its appearance.



Arranging objects

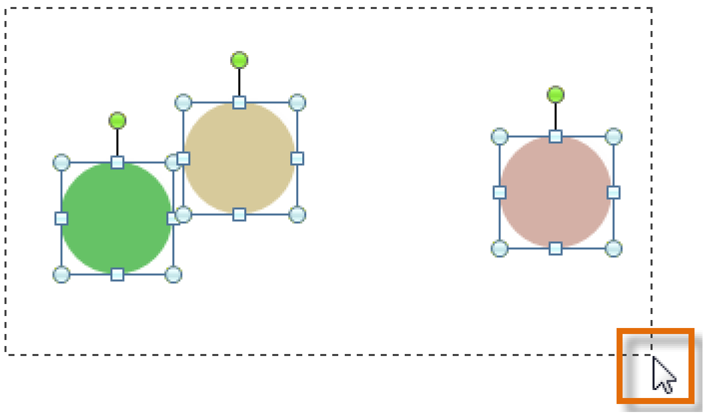
- Publisher offers a number of tools to help you **arrange** and **order** your objects.
- These tools work for any object, and they can help you lay out your pages quickly and precisely.

Aligning

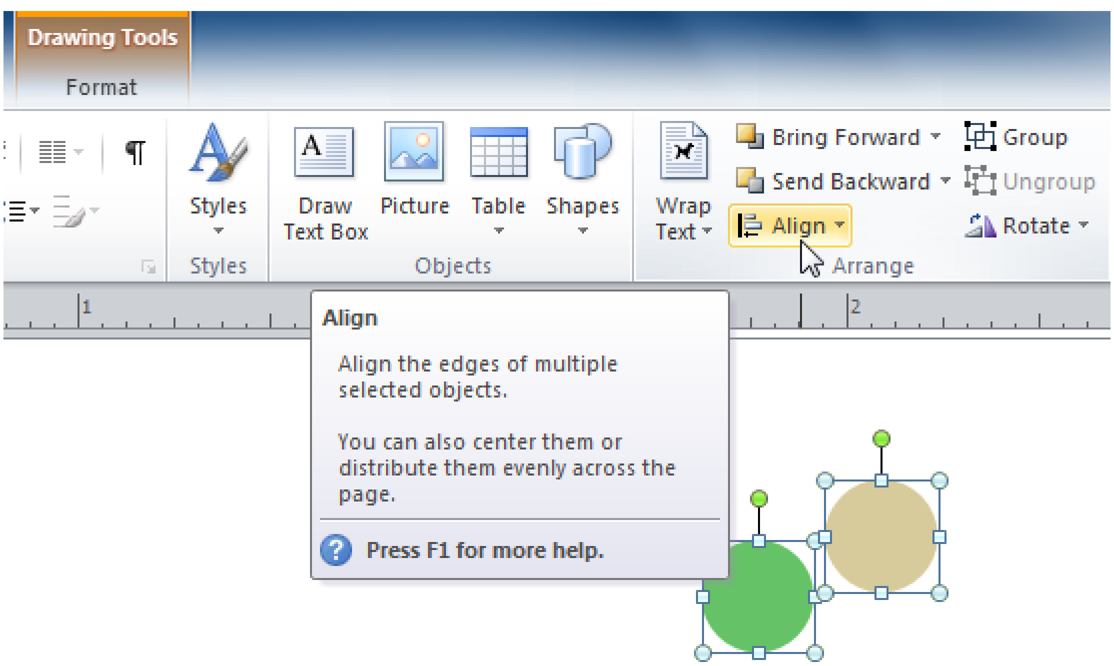
- You can **click** and **drag objects** to align them manually, but this can be difficult and time consuming.
- Publisher includes several commands that allow you to align your objects quickly and precisely. Objects can be aligned to **each other** or to the **page**.

To align two or more objects:

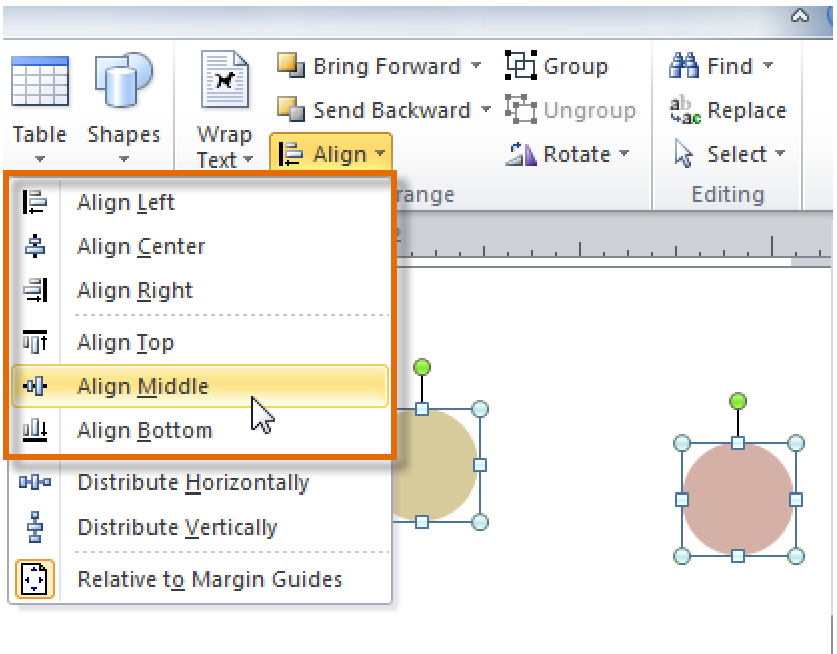
1. Click and drag your mouse to form a **selection box** around the objects you want to align. All of the objects will now have **sizing handles** to show that they are selected.



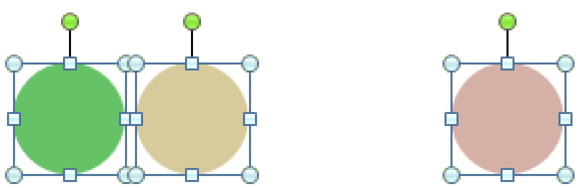
2. Click the **Format** tab, then locate the **Arrange** group.
3. Click the **Align** drop-down command.



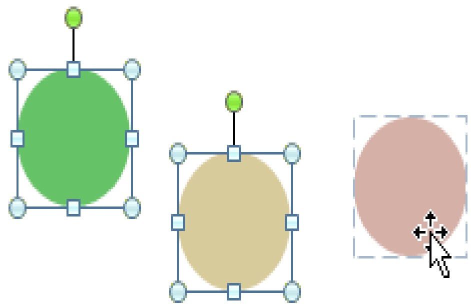
4. Select one of the six **alignment options**.



5. The objects will align to each other based on the option you have selected.

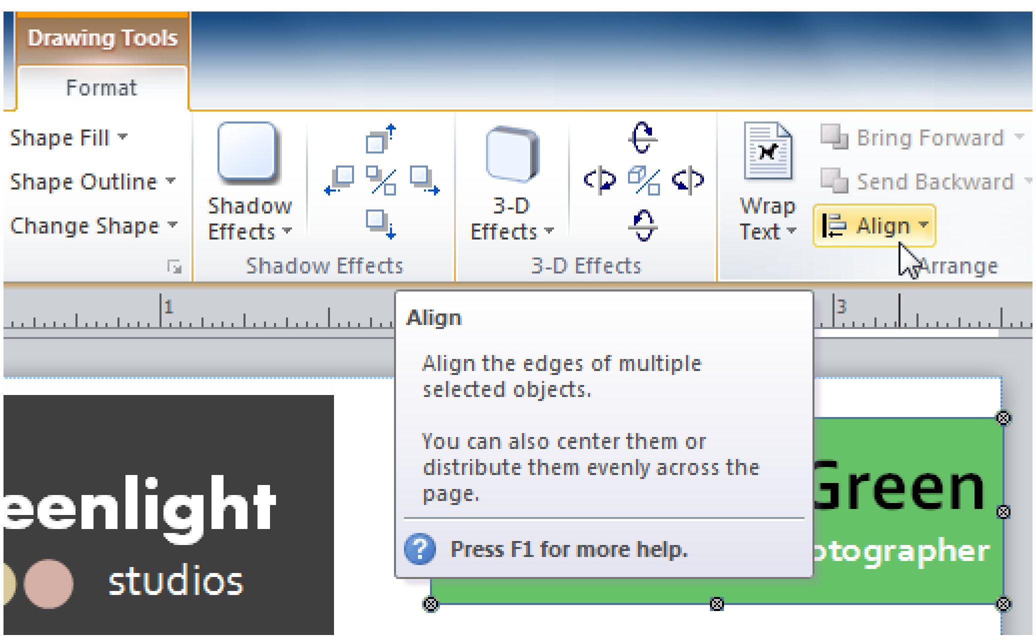


- Another way to select multiple objects at once is to simply hold down the **shift** key and click each object you wish to select.

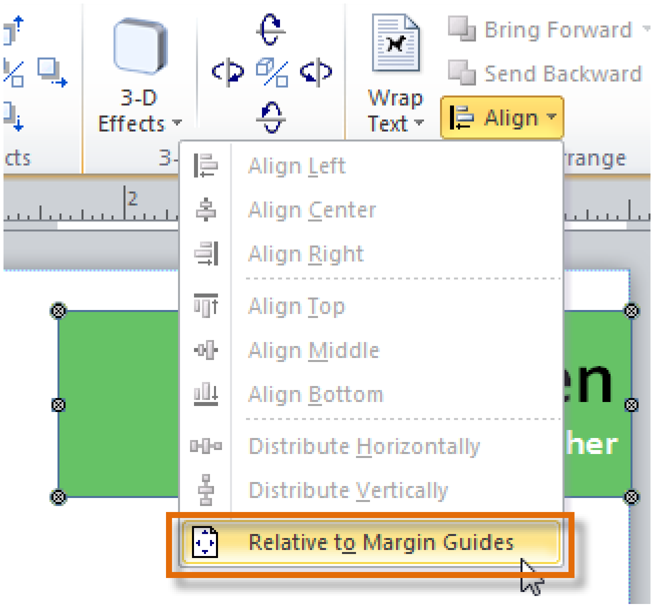


To align objects to the page:

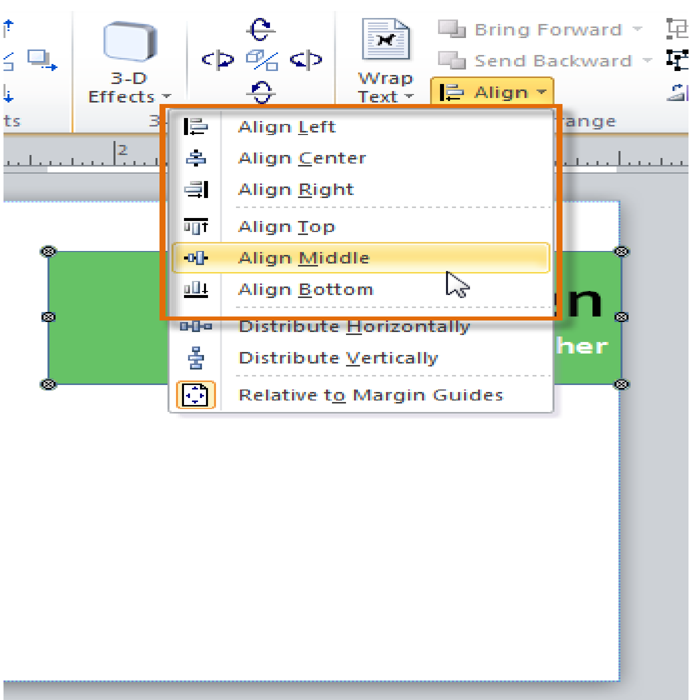
1. Select the object or objects you want to align.
2. Click the **Format** tab, then locate the **Arrange** group.
3. Click the **Align** drop-down command.



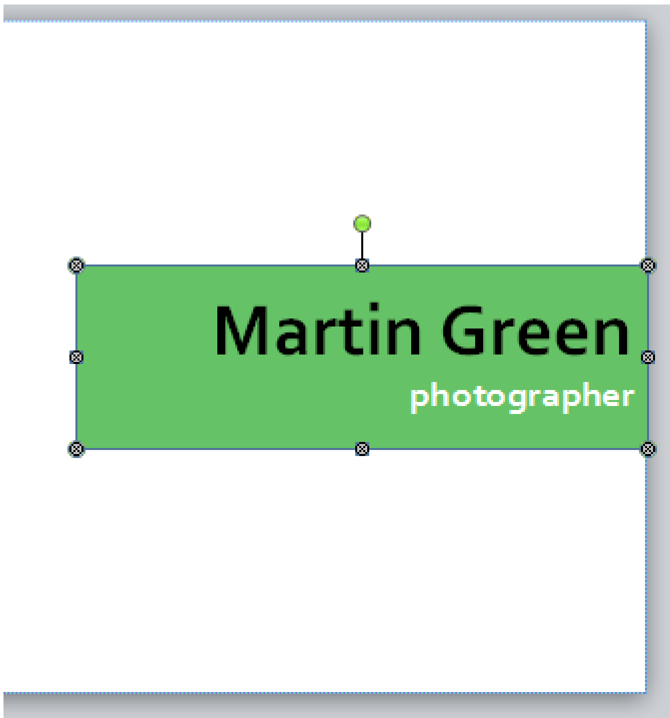
4. From the drop-down list that appears, select **Relative to Margin Guides**.



5. Select one of the six **alignment options**.



6. The objects will align to the page based on the option you have selected.



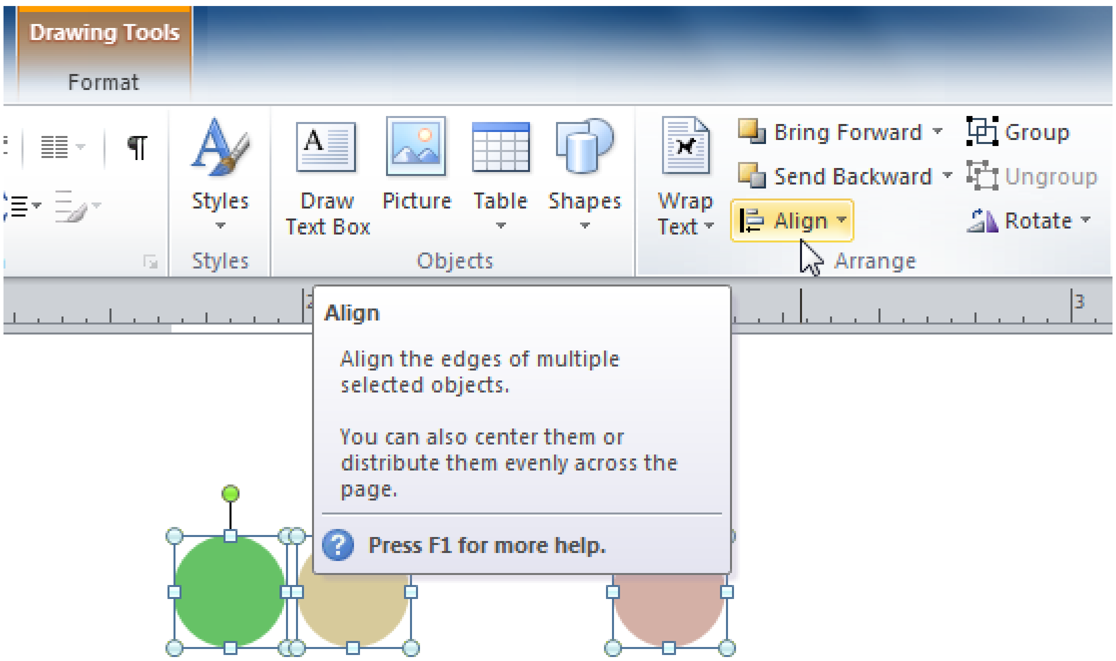
Distributing objects evenly

- If you have arranged objects in a row or column, you may want them to be an **equal distance** from one another for a neater appearance.
- You can do this by **distributing the objects** horizontally or vertically.

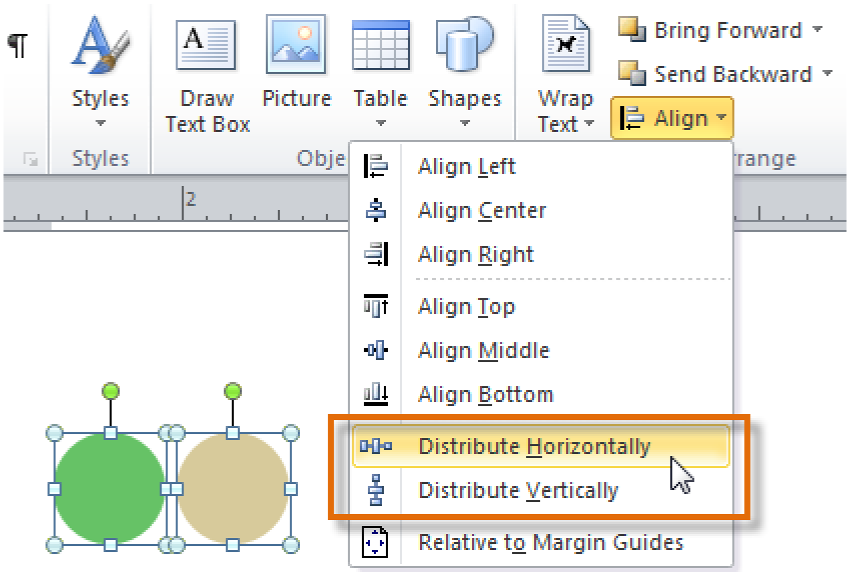
To distribute objects:

1. Select the objects you want to align.
2. Click the **Format** tab, then locate the **Arrange** group.

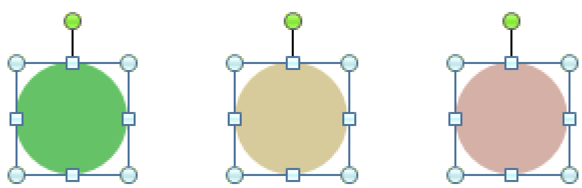
3. Click the **Align** drop-down command.



4. From the drop-down menu that appears, select **Distribute Horizontally** or **Distribute Vertically**.



5. The objects will be distributed evenly.



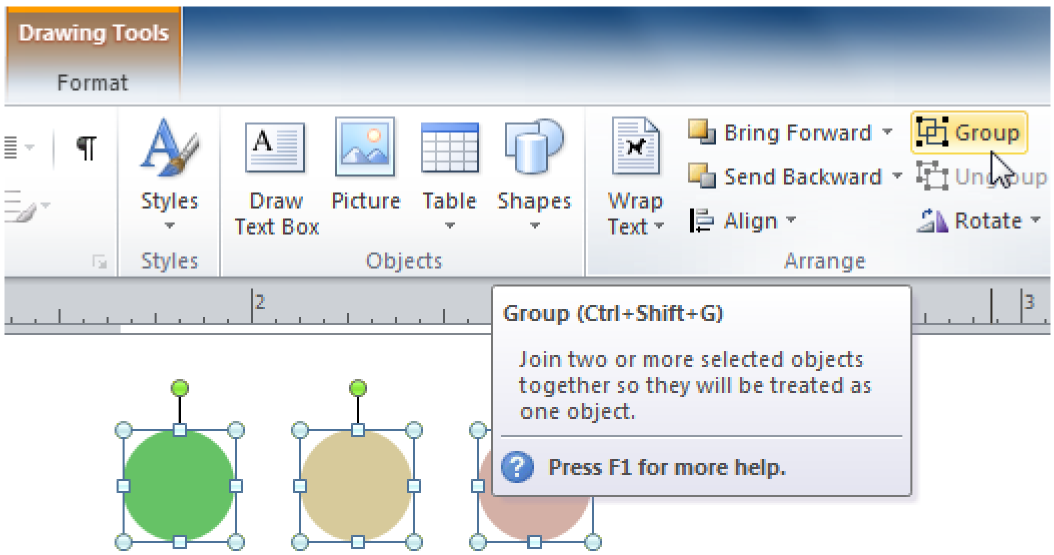
Grouping

- At times, you may want to **group** multiple objects into **one object** so they will stay together if they're moved.
- This can be easier than selecting all of the objects each time you want to move them.

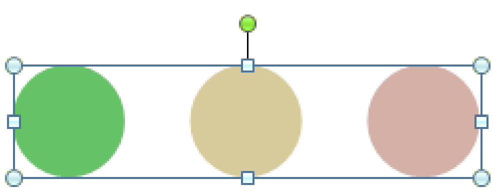
To group objects:

1. Select the objects you wish to group.

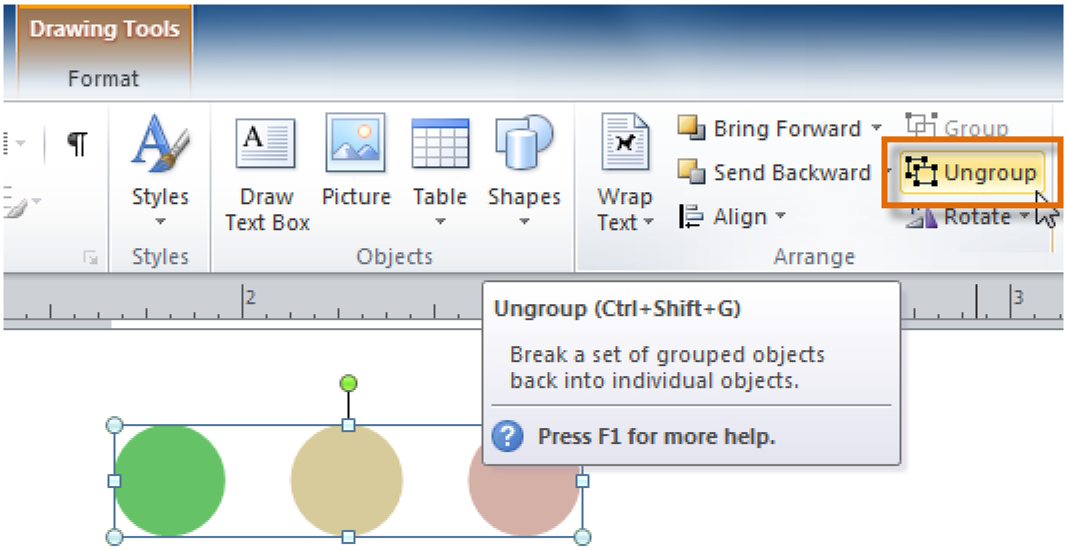
2. Click the **Format** tab, then locate the **Arrange** group.
3. Click the **Group** command.



4. The selected objects will now be grouped. There will be a **single box with sizing handles** around the entire group to show that they're one object.



- You can **ungroup** grouped objects at any time. Simply select the group, then click the **Ungroup** command.

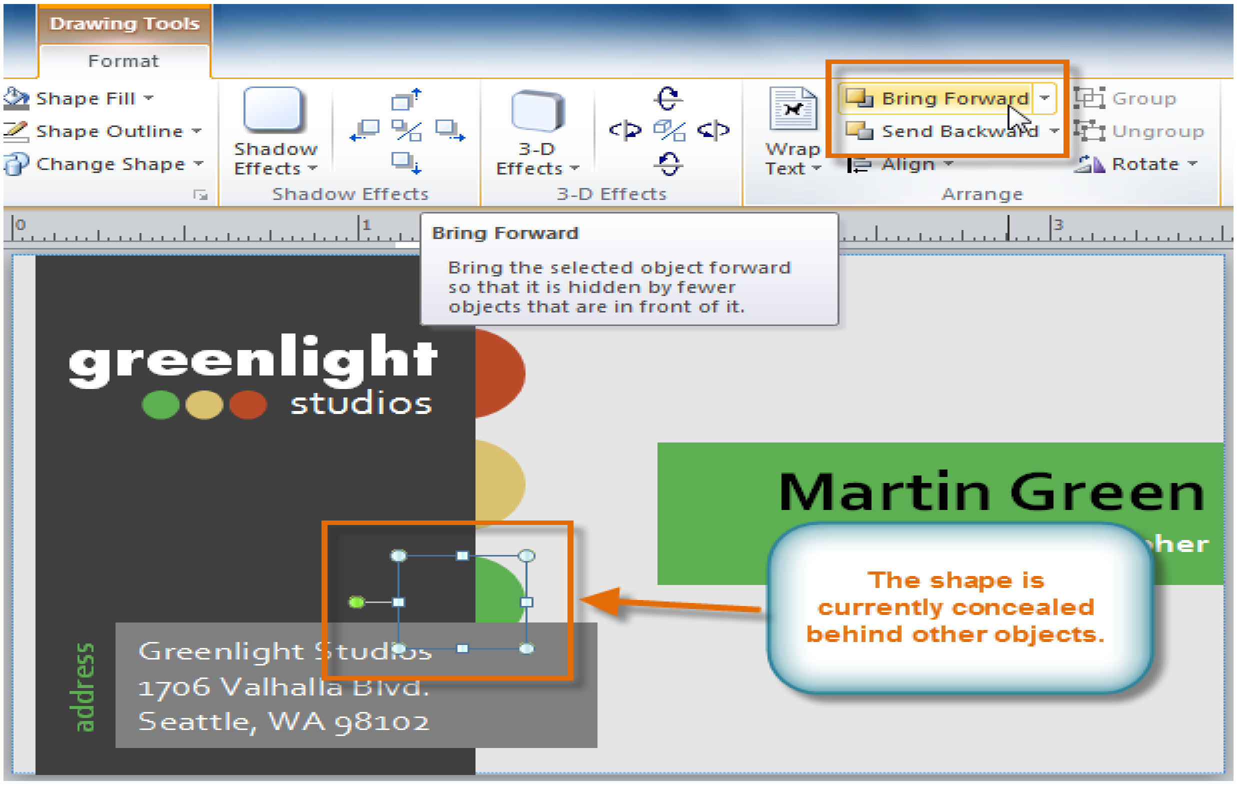


Moving objects backward and forward

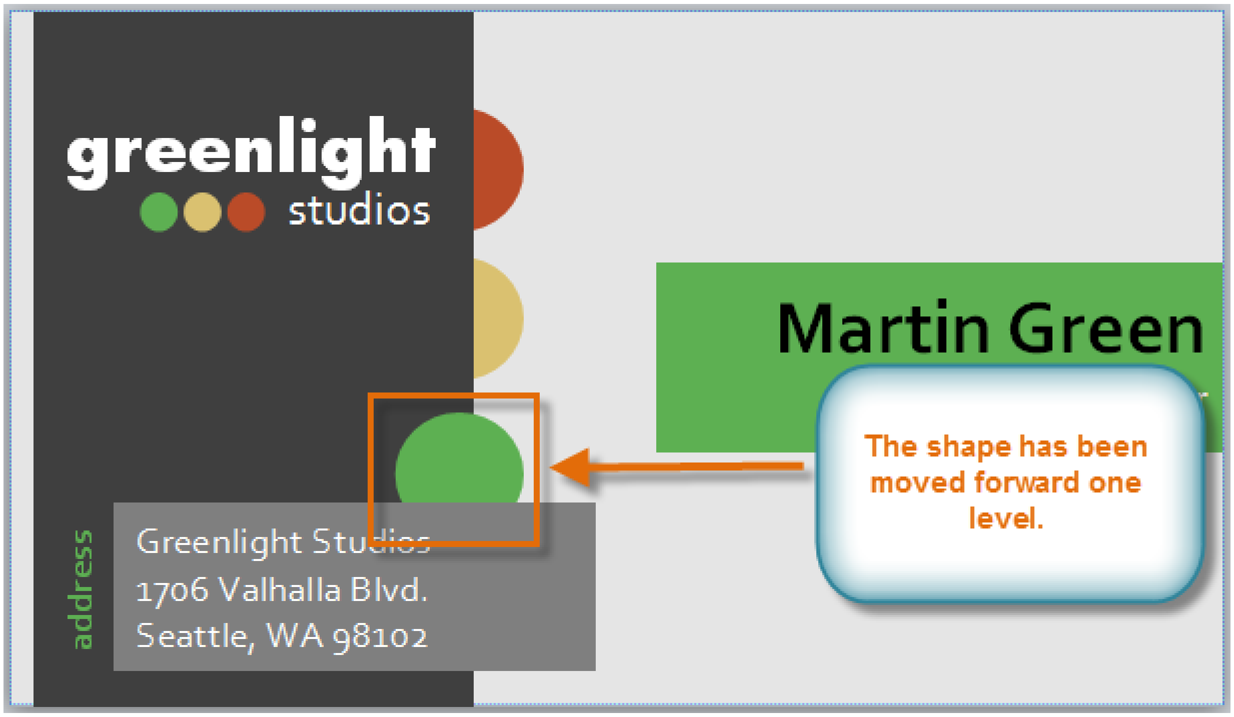
- In addition to aligning and grouping objects, Publisher gives you the ability to **arrange objects** in a **specific order**.
- Ordering is important when two or more objects **overlap**, as it will determine which objects are in the **front** or the **back**.

To change the ordering by one level:

1. Select the object you wish to move.
2. Click the **Format** tab, then locate the **Arrange** group.
3. Click the **Bring Forward** or **Send Backward** command to change the object's ordering by **one level**. If the object overlaps with more than one other object, you may need to click the command **several times** to achieve the desired ordering.



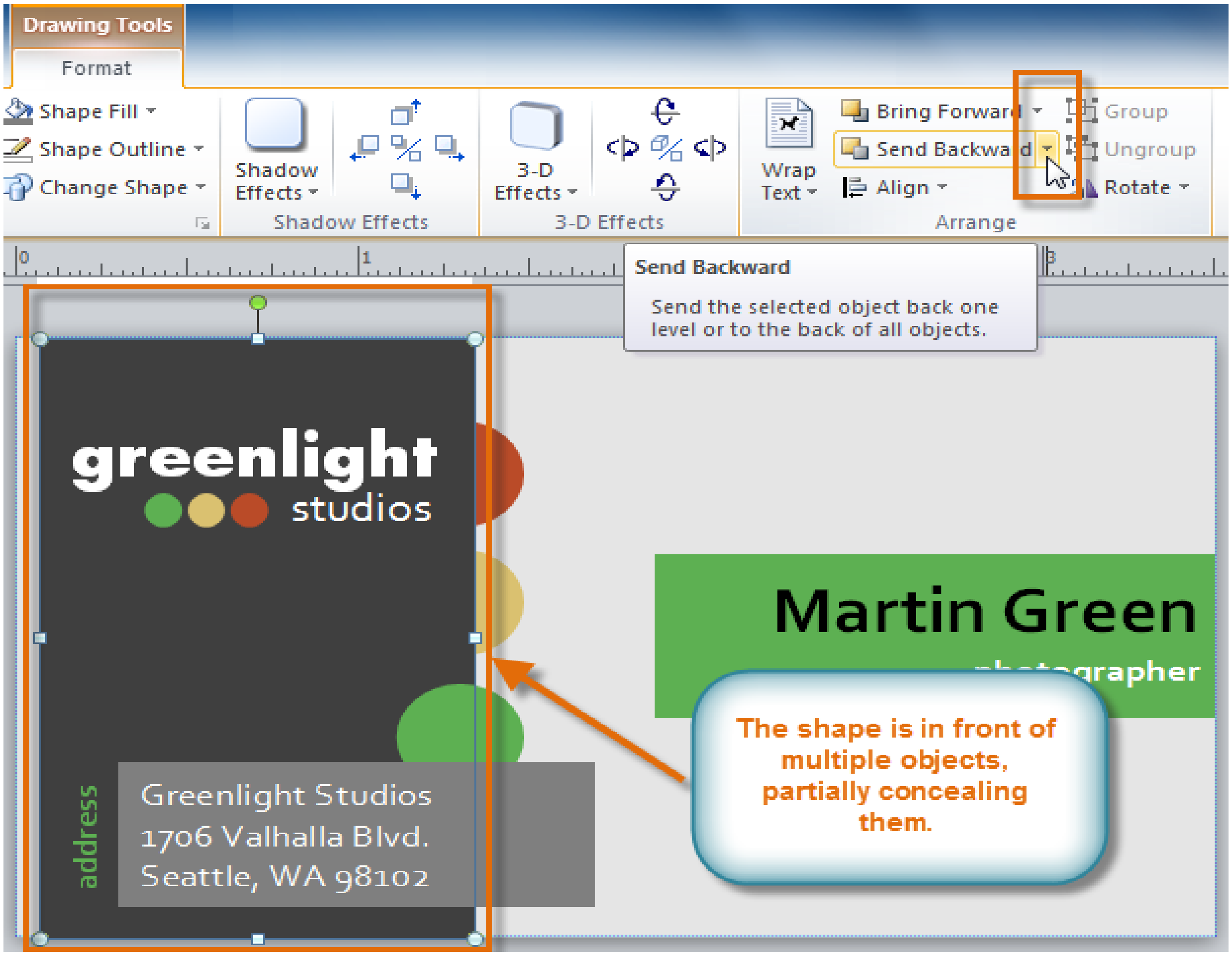
4. The objects will reorder themselves.



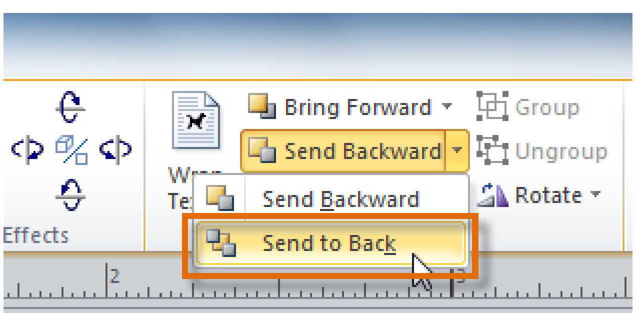
To bring an object to the front or back:

If you want to move an object behind or in front of several objects, it's usually faster to **bring it to front** or **send it to back** rather than clicking the ordering commands multiple times.

1. Select the object you wish to move.
2. Click the **Format** tab, then locate the **Arrange** group.
3. Click the **Bring Forward** or **Send Backward** drop-down command



4. From the drop-down menu, select **Bring to Front** or **Send to Back**.



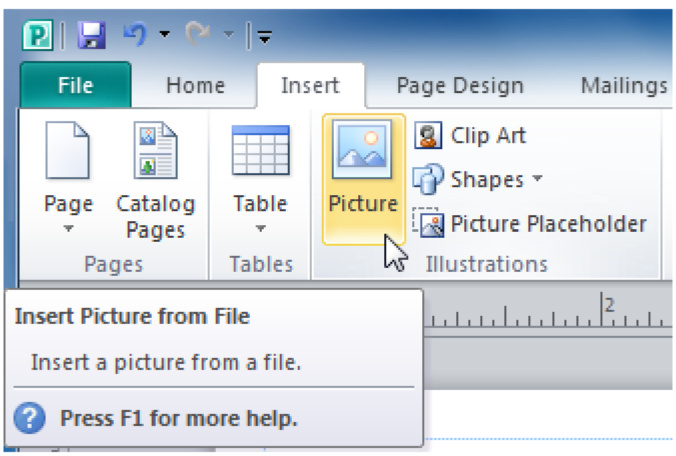
5. The objects will reorder themselves.

Adding pictures

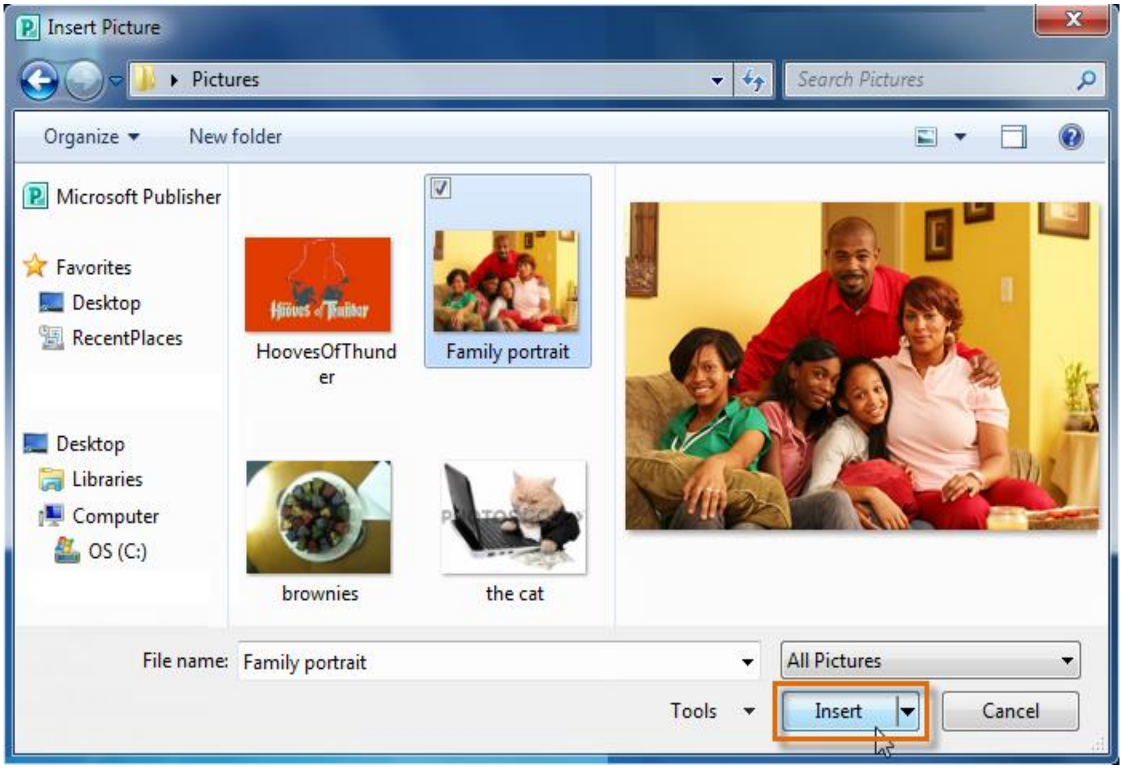
- To add a picture to your publication, you can either insert an image you have saved on your computer or choose one from Publisher's large selection of **Clip Art**.
- Once you've added images, you can then **edit** them as you wish.

To insert a picture from a file:

1. Select the **Insert** tab, then locate the **Illustrations** group.
2. Click the **Picture** command.



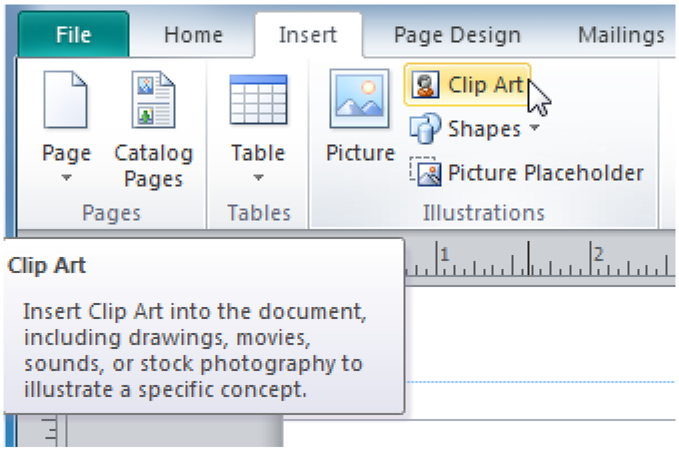
3. The **Insert Picture** dialog box will appear. Locate and select the picture you would like to insert, then click **Insert**.



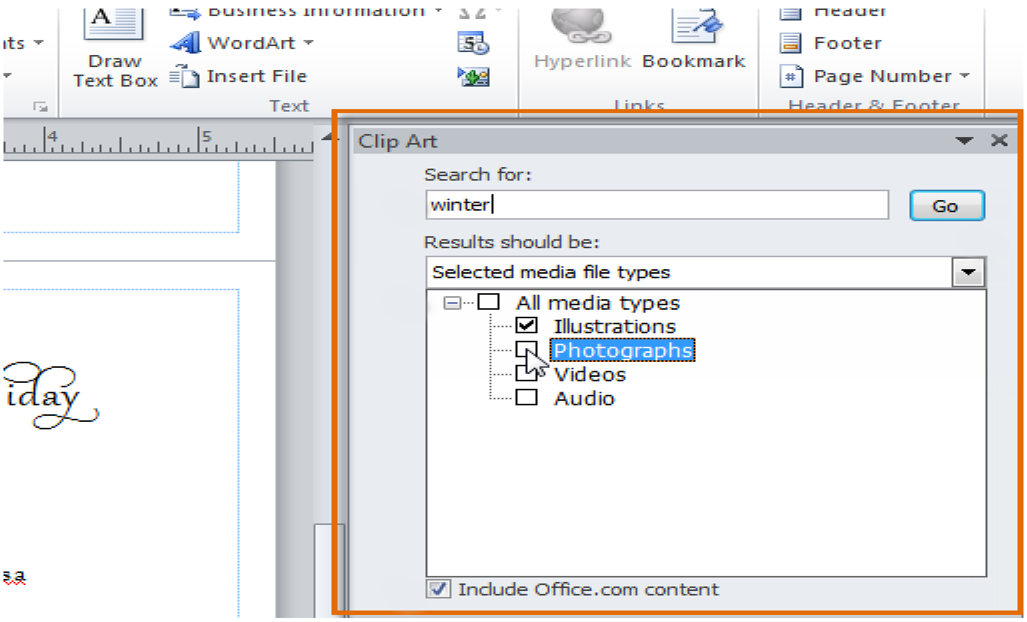
4. The picture will be added to your publication.

To Insert Clip Art:

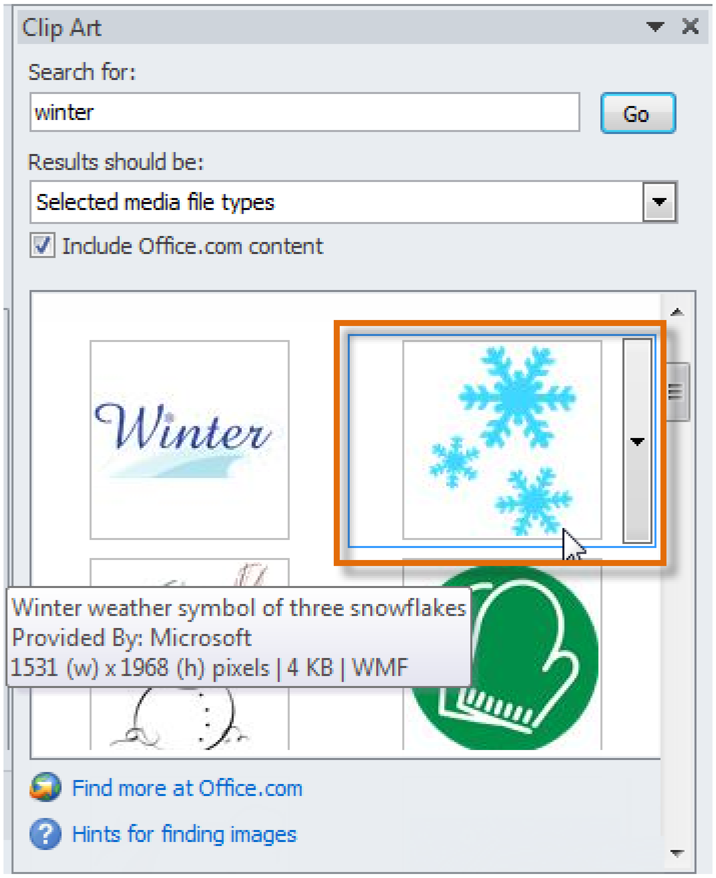
1. Select the **Insert** tab, then locate the **Illustrations** group.
2. Click the **Clip Art** command.



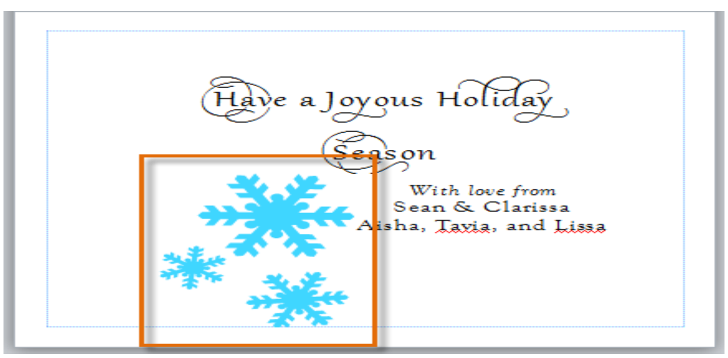
3. The **Clip Art** pane will appear on the right. Use the search tools to search for a suitable image.
   - Enter keywords in the **Search for:** field that are related to the image you wish to find.
   - Click the drop-down arrow in the **Results should be:** field, then **deselect** any types of media you do not wish to see.
   - If you would like to also search for Clip Art on Office.com, place a check mark next to **Include Office.com content**. Otherwise, it will just search for Clip Art on your computer.



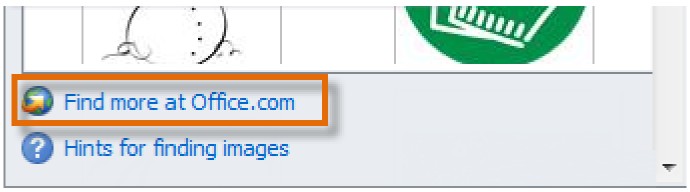
4. Click **Go** to begin your search.
5. Publisher will display pictures that meet your search terms. When you've found a picture you wish to use, **click** it.



6. The Clip Art will be added to your publication.

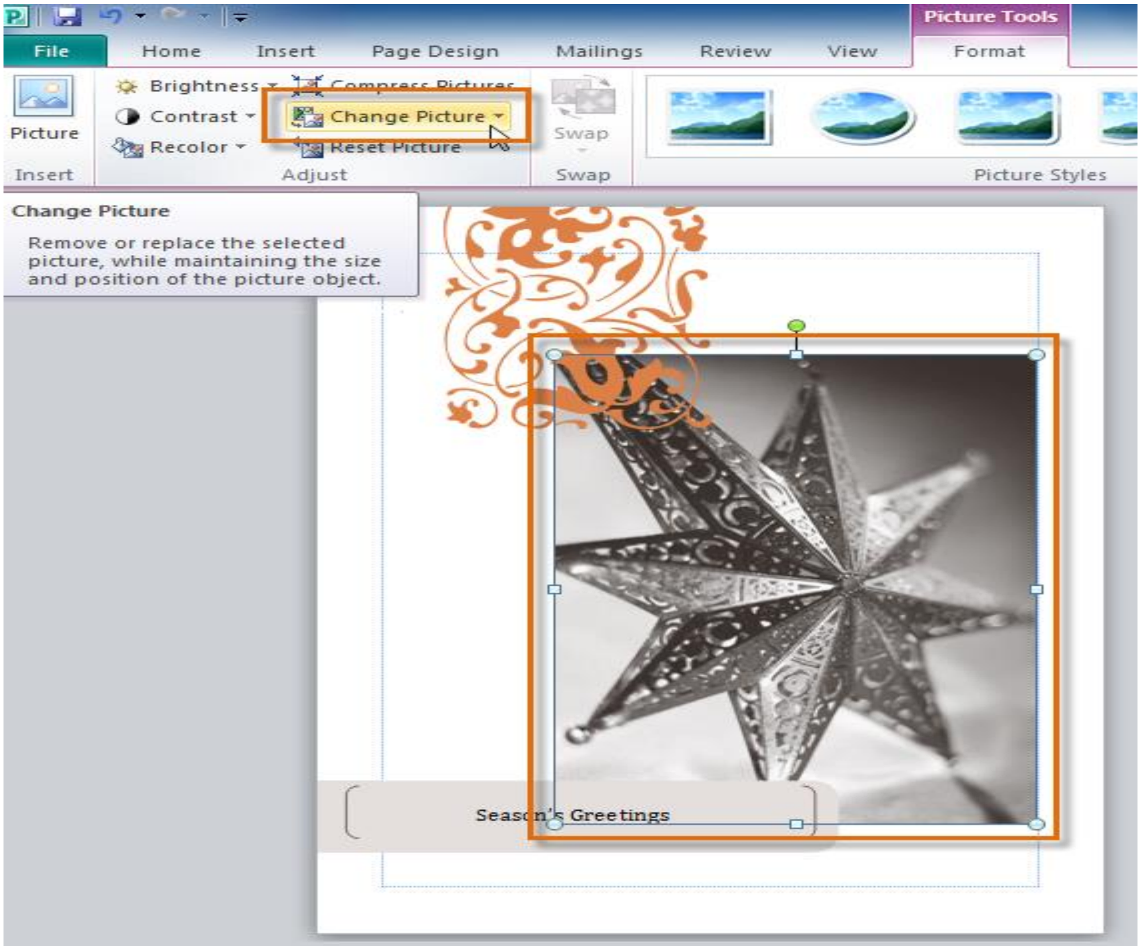


If you can't find Clip Art that suits your needs, you can also search on the Microsoft Office website by clicking the **Find more at Office.com** link at the bottom of the Clip Art pane.



Replacing existing pictures

- If you started your publication from a **template**, you're likely to want to replace some of the template's pictures with your own.
- The **Change Picture** command lets you insert new pictures in place of existing ones. When you use this command, the new picture will appear in the location of the original one, with the original picture's formatting applied.



Fitting pictures in your publication

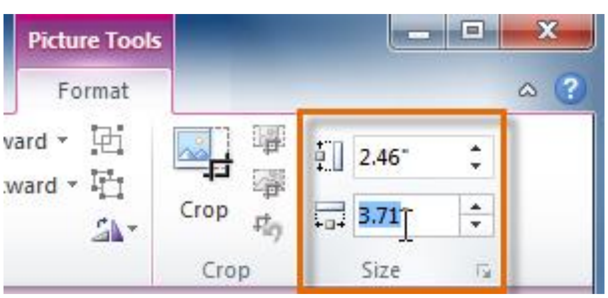
To get your picture to fit well on the page, you may have to adjust it by **resizing**, **cropping**, and **rearranging** it.

To resize a picture:

1. Select the picture.
2. **Click** one of the corner sizing handles and **drag** your mouse until the picture is the desired size.

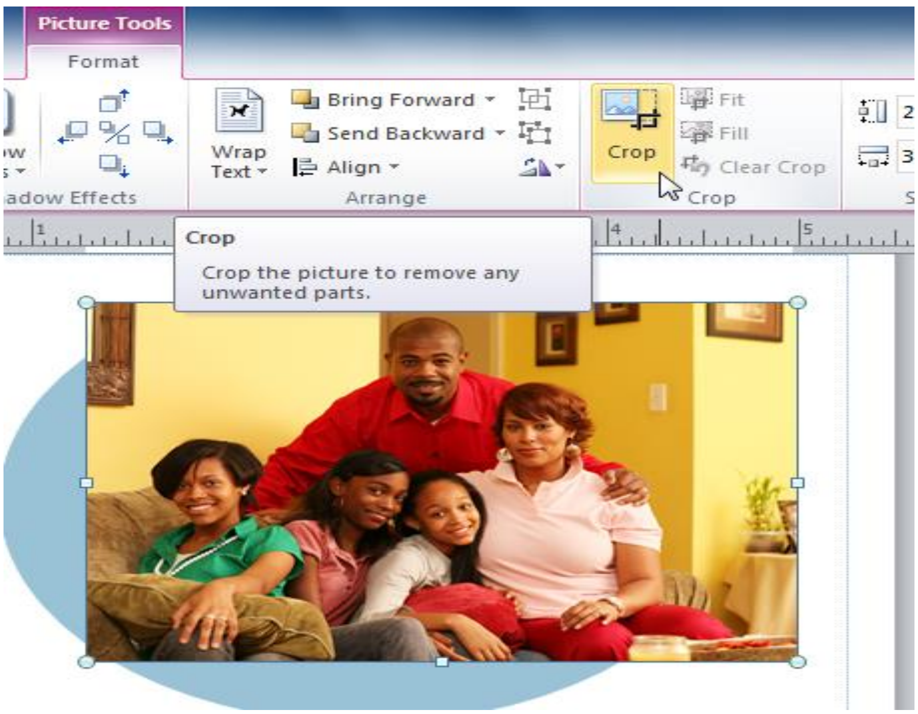
3. Release your mouse. The picture will be resized.

- If you know exactly how many inches tall and wide you want your picture to be, you can resize it to those specifications.
- Click the **Picture Tools Format** tab, then locate the **Size** group. Enter the desired **height** of your picture in the top box and the desired **width** in the bottom box.

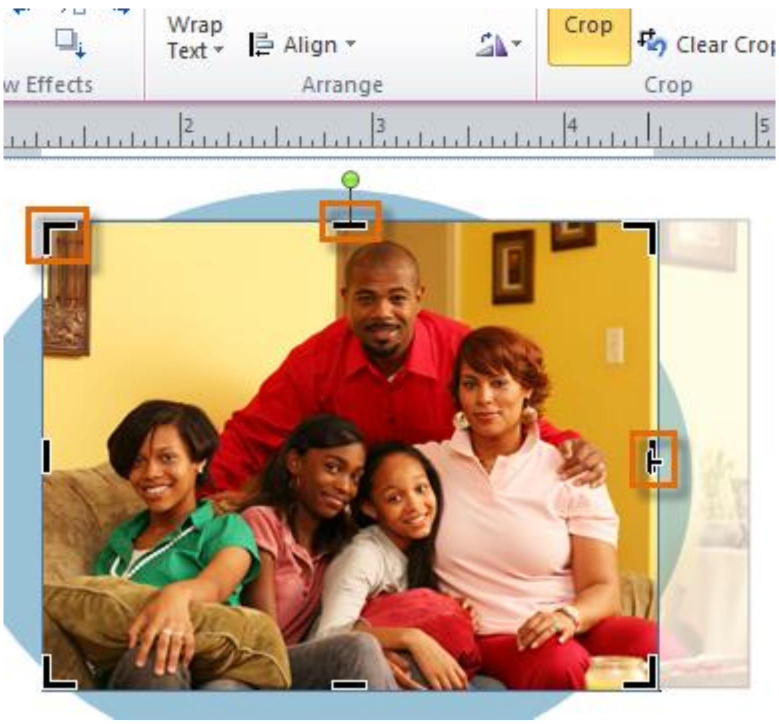


To crop a picture:

1. Select the picture, then click the **Picture Tools Format** tab and locate the **Crop** group.
2. Click the **Crop** command.



3. The black **cropping handles** will appear. Click and drag a **handle** to crop the picture. The areas that will be cropped will appear to be semi-transparent.



4. When you are satisfied with the appearance of your picture, click the **Crop** command again.
5. The picture will be cropped.

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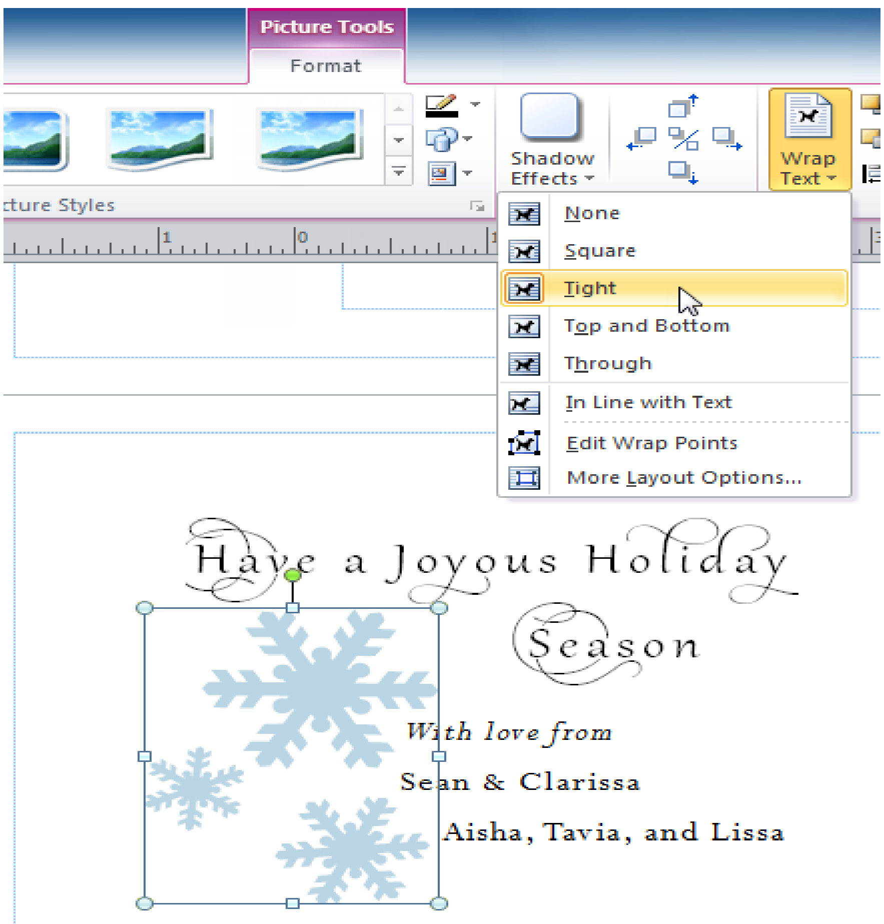
Arranging pictures

- To get your pictures to fit properly with text and other objects, you may have to **align** them and adjust their **text wrap settings**.
- The procedures for doing these things are identical to the procedures for working with **shapes** and other **objects**.



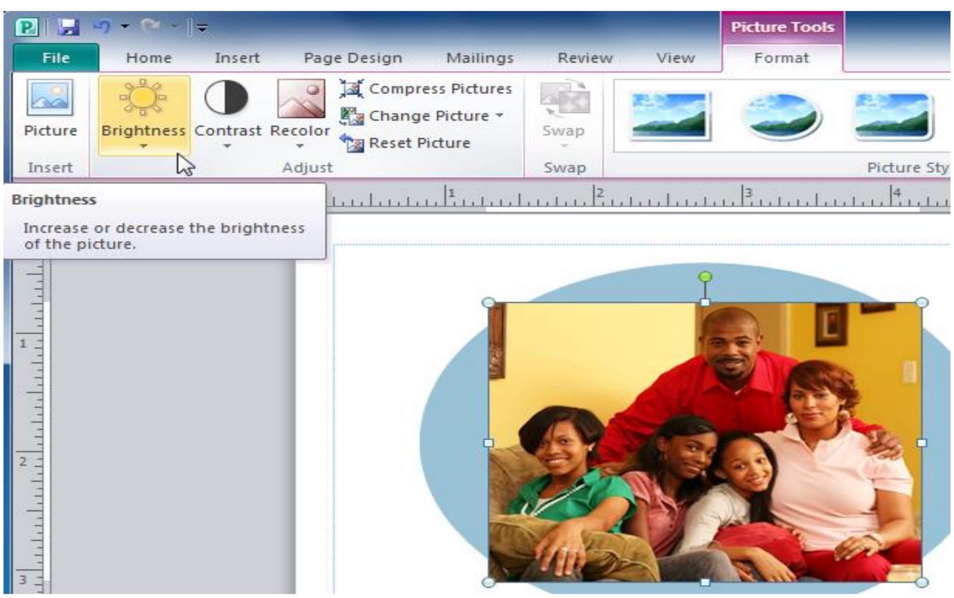
Modifying pictures

Brightness and contrast

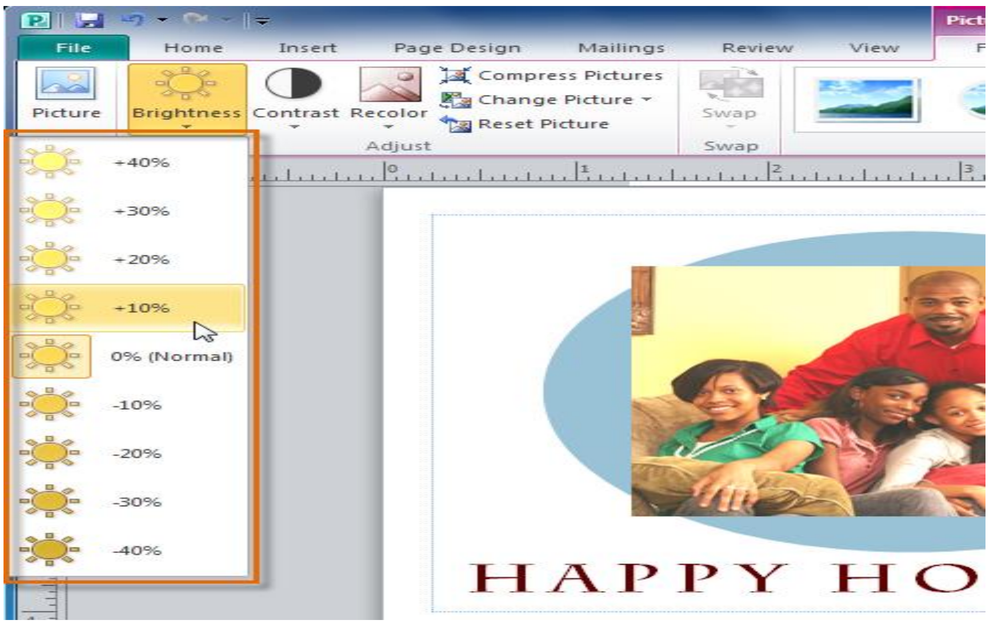
- One of the most basic edits you can make to a picture is modifying its **brightness** and **contrast**.
- Although these tools are separated into two commands in Publisher, they are most effective when used together.

To adjust brightness and contrast:

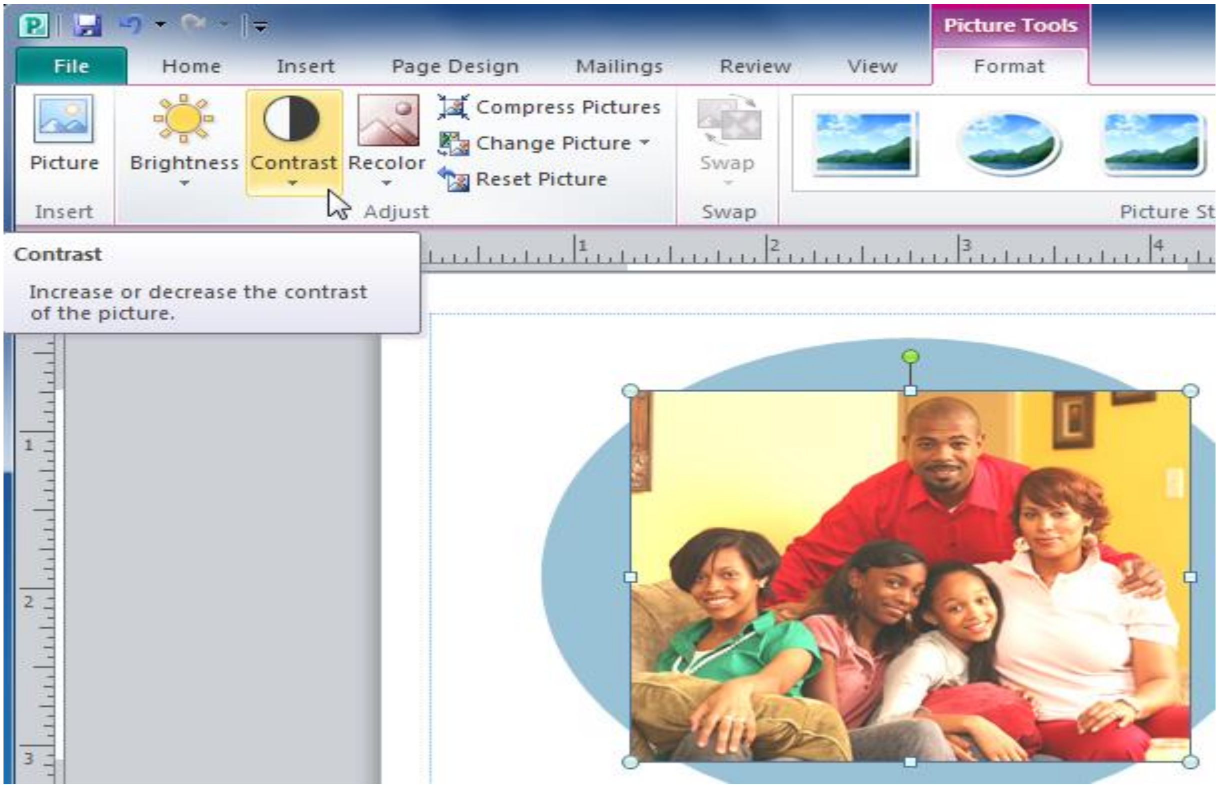
1. Select the picture you wish to adjust, then select the **Picture Tools Format** tab and locate the **Adjust** group.
2. Click the **Brightness** drop-down command.



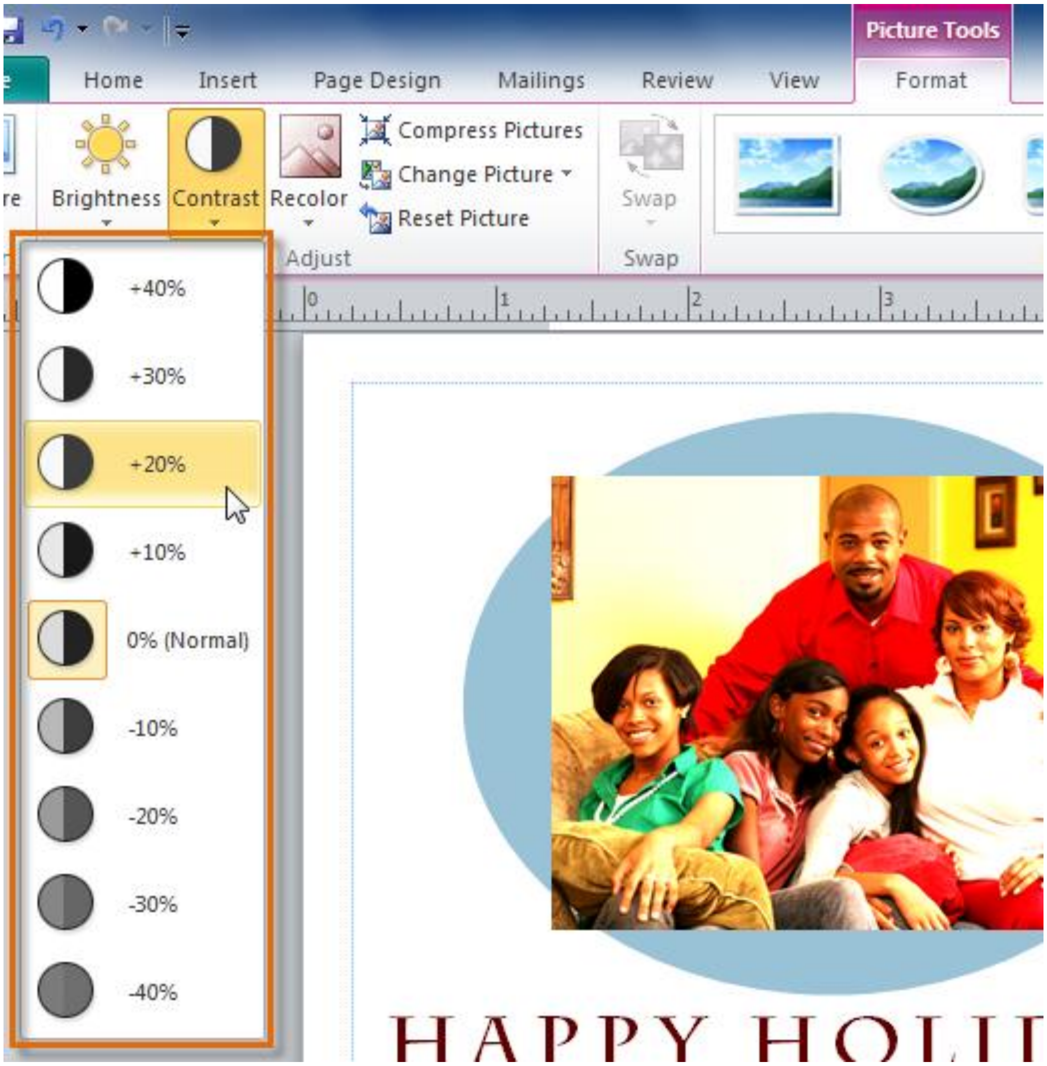
3. From the drop-down menu that appears, select the desired brightness. **Positive numbers (+)** will make the picture **brighter**, while **negative numbers (-)** will make the picture **darker**.



4. Click the **Contrast** drop-down command.

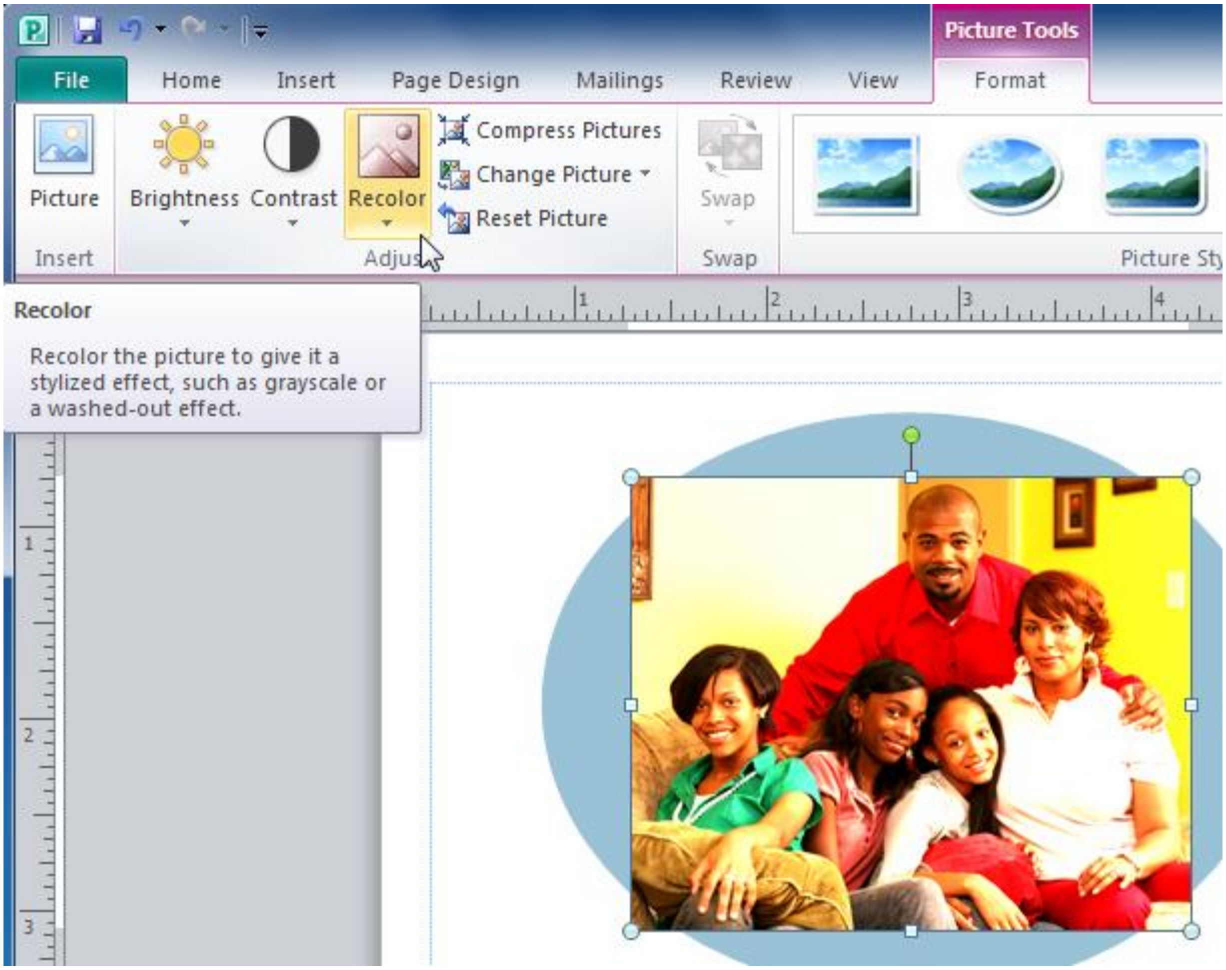


5. Select the desired **contrast** level. **Positive numbers (+)** will create a **greater contrast** between the light and dark areas of the picture, while **negative numbers (-)** will **reduce the contrast**.

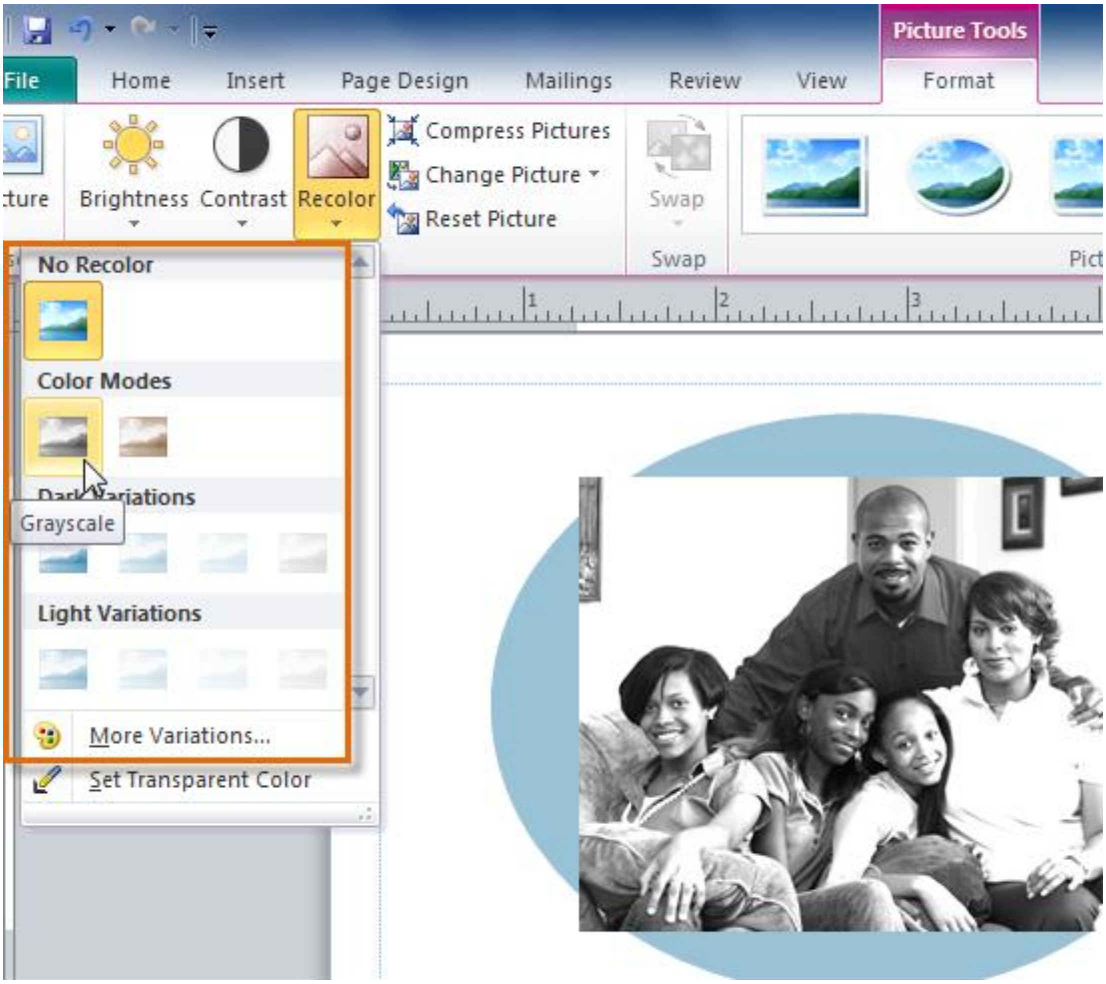


To recolor the picture:

1. Select the picture you wish to recolor, then select the **Picture Tools Format** tab and locate the **Adjust** group.
2. Click the **Recolor** drop-down command.



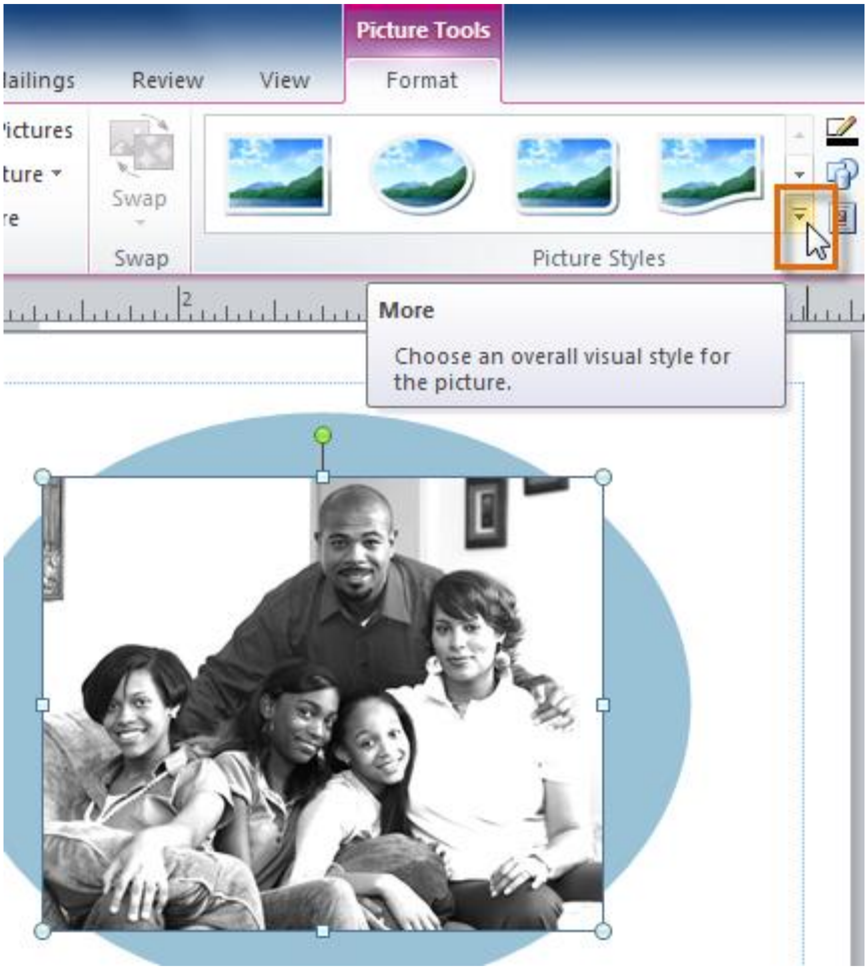
3. From the drop-down menu that appears, select a **recoloring option** or select **More Variations** to see additional color choices.



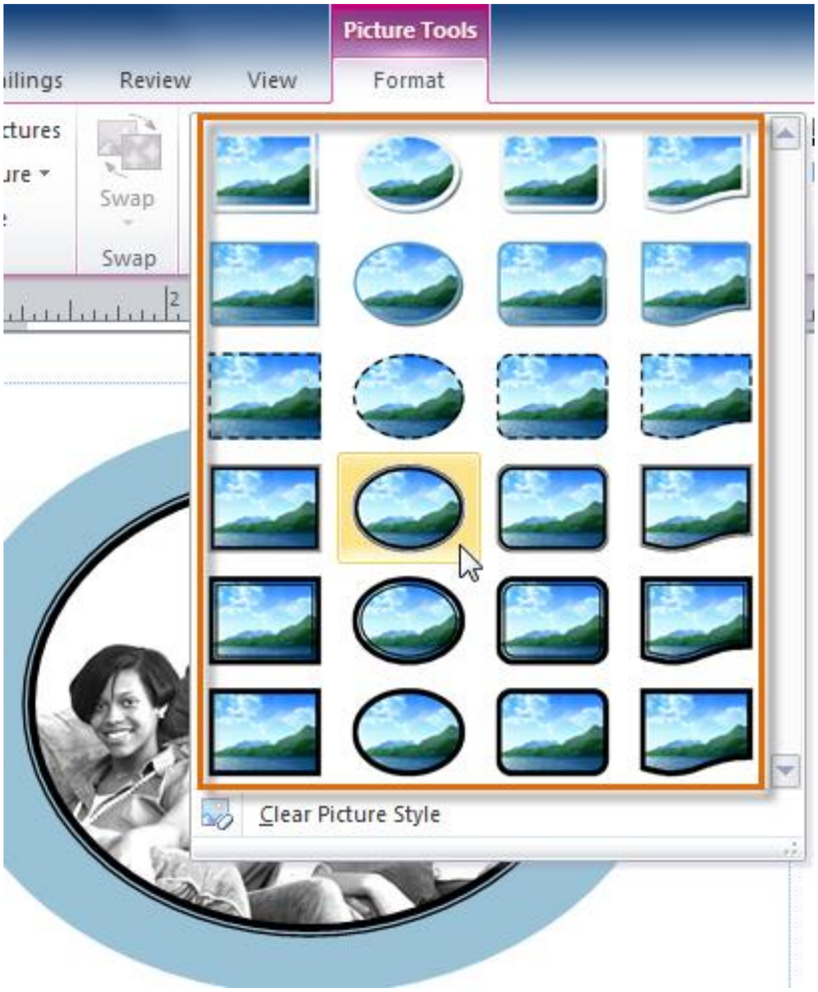
4. The picture will be recolored.

To apply a picture style:

1. Select the picture, then click the **Picture Tools Format** tab and locate the **Picture Styles** group.
2. Click the **More Picture Styles** drop-down arrow.



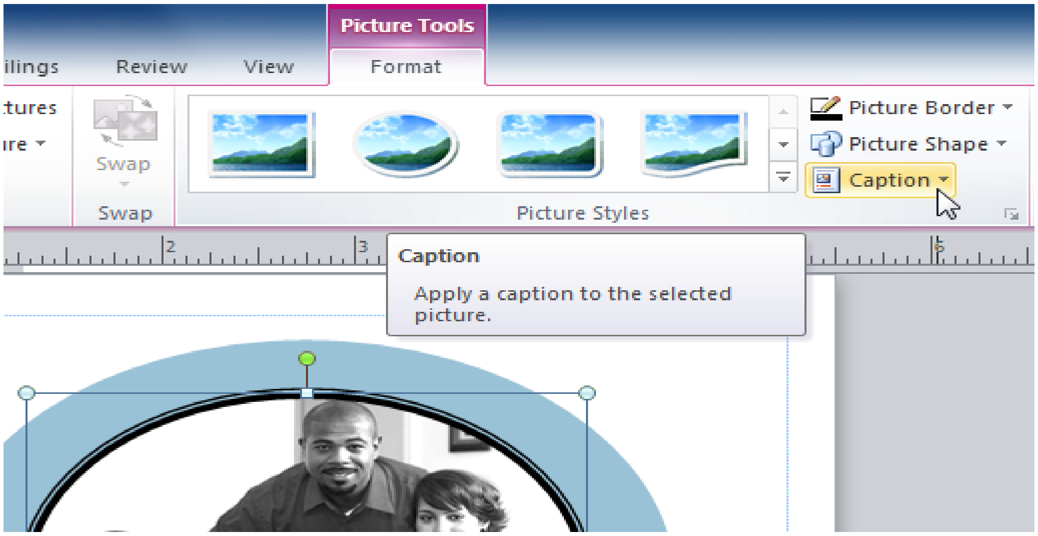
3. A drop-down list of styles will appear. Move your cursor over the styles to see a live **preview** of each style in your publication, then select the desired style.



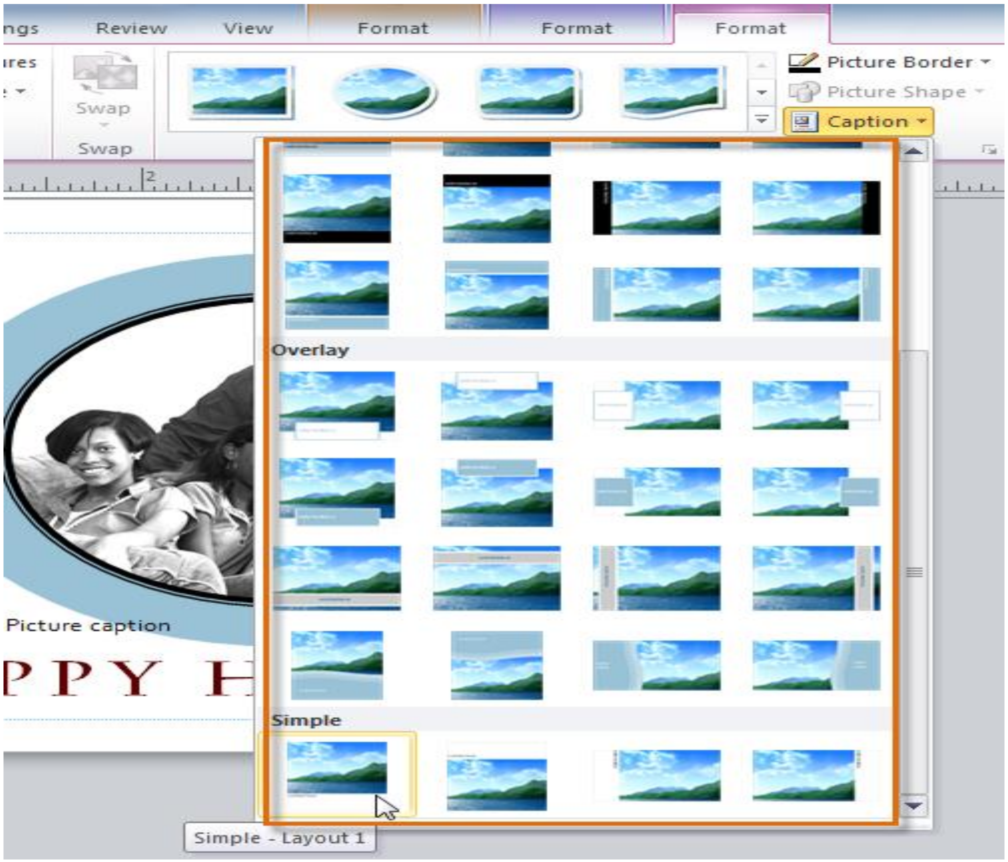
4. The style will be applied to the picture.

To add a caption:

1. Select the picture, then click the **Picture Tools Format** tab and locate the **Picture Styles** group.
2. Click the **Caption** drop-down command.



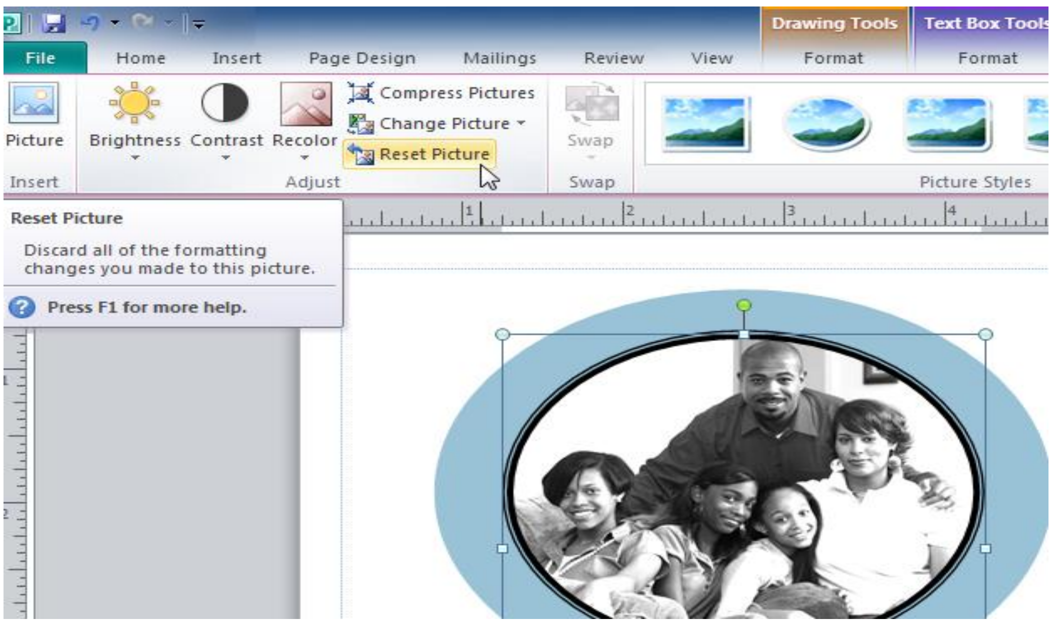
3. A drop-down list of **caption styles** will appear. Move your cursor over the caption styles to see a live **preview** of the captions with your picture, then select the desired caption style.



4. Click the caption **text box** and type your caption text.

To restore a picture to its original appearance:

1. Select the picture, then click the **Picture Tools Format** tab and locate the **Adjust** group.
2. Click the **Reset Picture** command.



3. The picture will be restored to its original appearance.

Preparing your pictures for publication

Compressing pictures

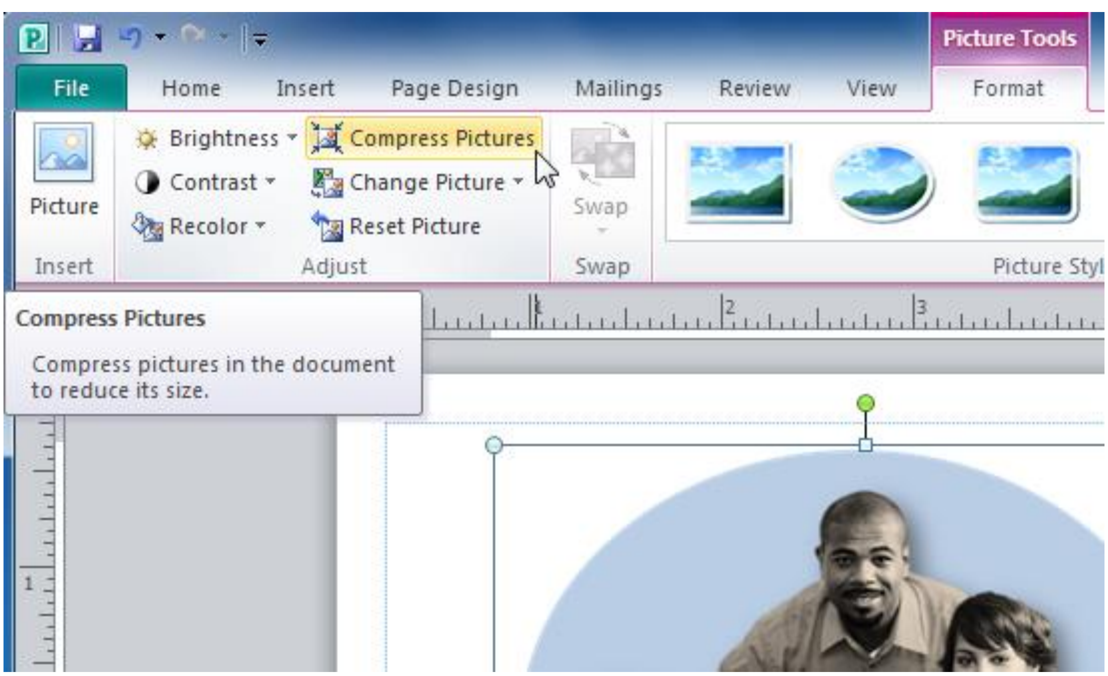
- You'll need to monitor the **file size** of publications that include pictures, especially if you send them via email.
- Large, high-resolution pictures can quickly cause your publication to become too large, which may make it difficult or impossible to attach to an email.
- In addition, **cropped areas** of pictures are saved with the publication by default, which can add to the file size.

- Publisher can reduce the file size by **compressing** pictures, lowering their **resolution**, and **deleting cropped areas**.

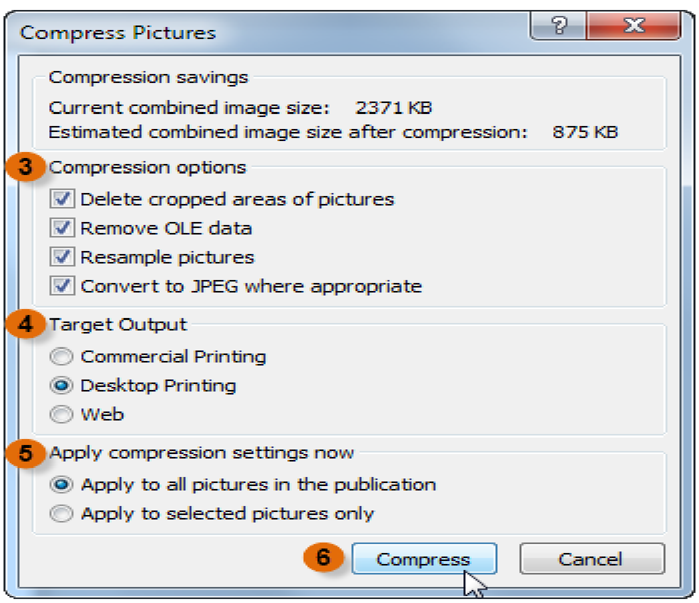
NB: Only compress pictures **after** you have edited and resized them. Attempting to **enlarge** or otherwise edit a compressed picture may result in a **blurry** or otherwise **low-quality image**.

To compress pictures:

1. Select a picture, then click the **Picture Tools Format** tab and locate the **Adjust** group.
2. Click the **Compress Pictures** command.



3. The **Compress Pictures** dialog box will appear. Review the settings in the **Compression Options** section. All four boxes should be checked.
4. In the **Target Output** section, select your planned method of publication.
5. Choose whether to compress **every picture** in the publication or **the selected picture only**.
6. When you are satisfied with the publication settings, click **OK**. The picture or pictures will be compressed.



Test printing

- Depending on the types of edits you make to your pictures, they may not print exactly as they appear onscreen.

- While this could signal a problem with your printer or ink, these print errors are often caused by problems in Publisher itself.
- If the pictures are especially blurry, grainy, oddly colored, or otherwise distorted from their onscreen appearance, consider **saving your publication as a PDF**, then opening and printing the PDF version.

CHAPTER 3: SOFTWARE INSTALLATION

Objectives

By the end of this chapter, the student should be able to:

Contents

- Install an operating system, device drivers, and application software
    - discussing procedure for installing software
    - installing windows operating system
    - installing *ubuntu* operating system
    - installing printer drivers, scanner drivers and USB drivers
    - installing MS office/open office
- Upgrade an operating stem, device drivers and application software
    - upgrading an operating system, device drivers and application software

Introduction

This section discusses basic skills on how to install computer software starting with Operating system, utility software and application software.

Definitions

Software installation refers to copying of computer program into a hard disk of computer in a form that it can be executed by the CPU.

- Software installation is also called program setup

- Today most of software installations platforms are GUI-based wizards that requires minimal input from the user.

Software Uninstallation: refers to an act of removing the program from the computer.

- This involves more than erasing the program folder since registry files and other code in the system need to be modified or deleted for complete uninstallation.

Factors to consider before software Installation

Before installation of any software you must consider the following:

1. System configurations compatibility

- Type and speed of processor
- RAM size
- Hard disk space

2. Reading the installation manual

- To get details on the minimal system requirements, warnings and user license agreement

3. User needs

- Software to be installed should be the one to meet the specific user needs
- This will ensure that no unnecessary software is installed which could end up taking valuable storage space.

Installation of Operating system

Operating systems exist in various categories such as

i. Windows Operating system

ii. Linux Operating system

iii. Mac-OS

Installing Windows 7 Operating system

Installation of any windows operating system involves some common steps.

This section will concentrate on windows 7 installation

Two methods can be used

1. **Upgrading** – changing a version of windows from lower to upper version for example upgrading from windows XP to windows 7
2. **Fresh installation** – installing a new copy of operating system

windows 7 installation Steps

1. Check for Minimum requirements for Windows 7 Installation

- The following system requirements must be satisfied for windows 7 to get installed

Part	Minimum Requirement
Processor (CPU)	1 GHz or faster CPU
Memory (RAM)	1 GB RAM (32-bit) or 2 GB RAM (64-bit)
Hard drive free space	16 GB available disk space (32-bit) or 20 GB (64-bit)
Disc drive	DVD drive
Graphics memory	128 MB or more

2. Get windows 7 set up into a bootable media

- Operating system boot by

a. Flash Drive (Bootable Pen Drive)

b. Bootable CD/DVD

c. Hard Disk

NB: If you purchase a DVD or CD of windows then it is already in bootable form

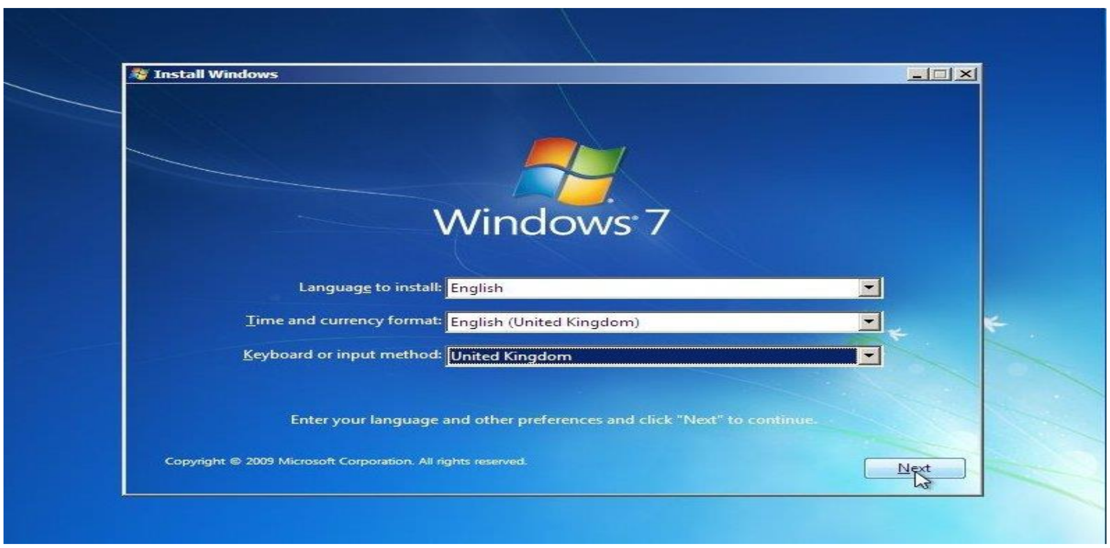
3. Insert Windows 7 DVD into optical drive or removable bootable flash containing windows 7 operating system and start the computer

- Then your computer will display the following interface

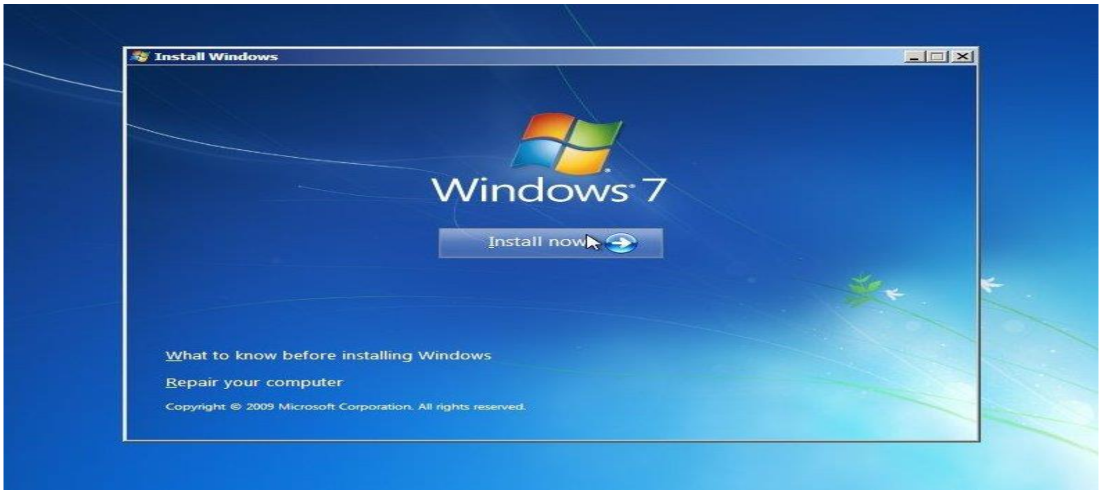
- After pressing any key, the windows files start loading as follows

- Followed by the following display

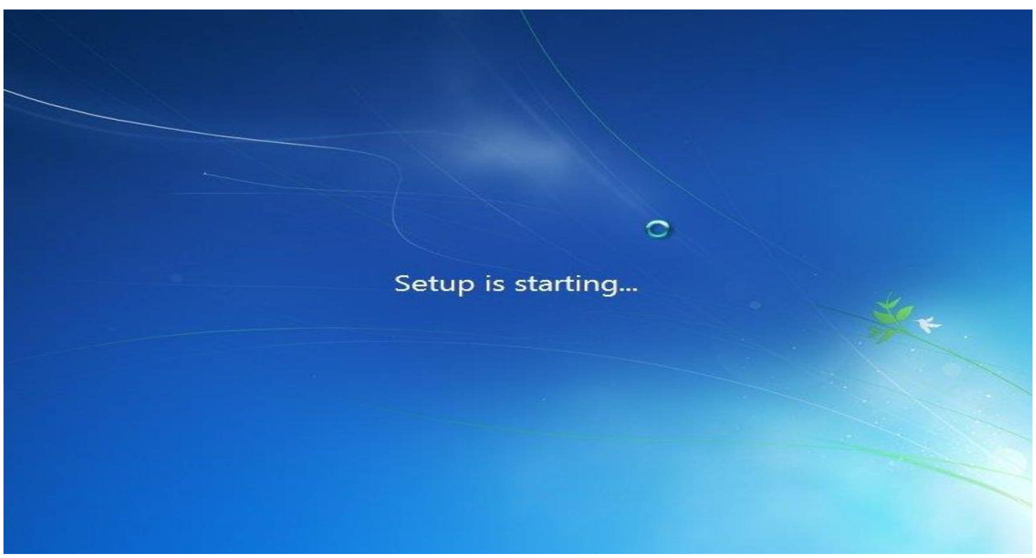
4. Select your language, time & currency format, keyboard or input method and click Next as displayed below



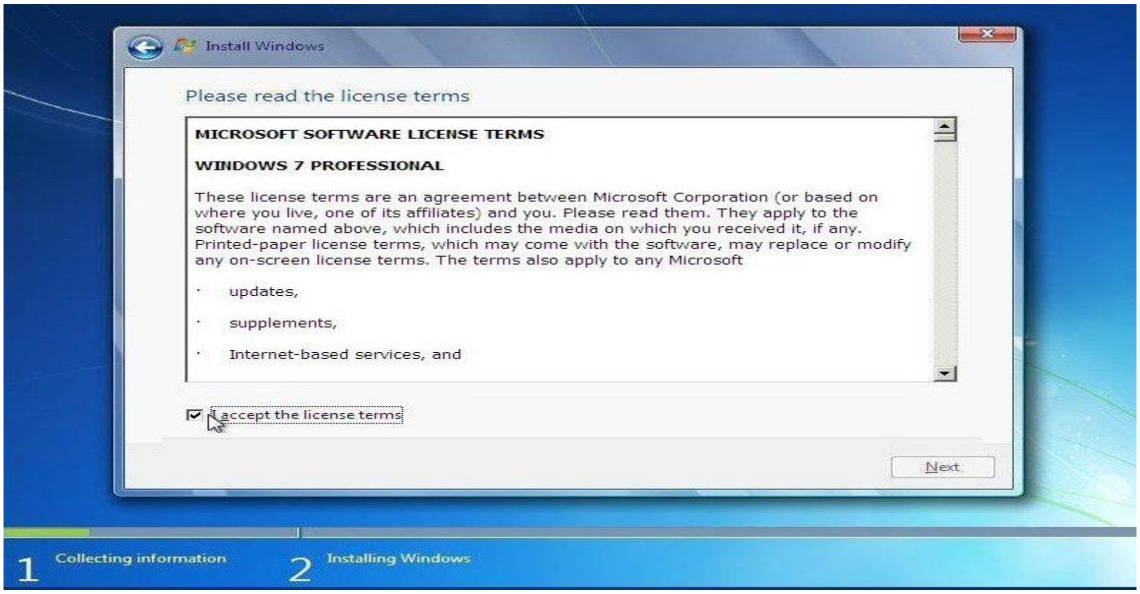
5. Click install now



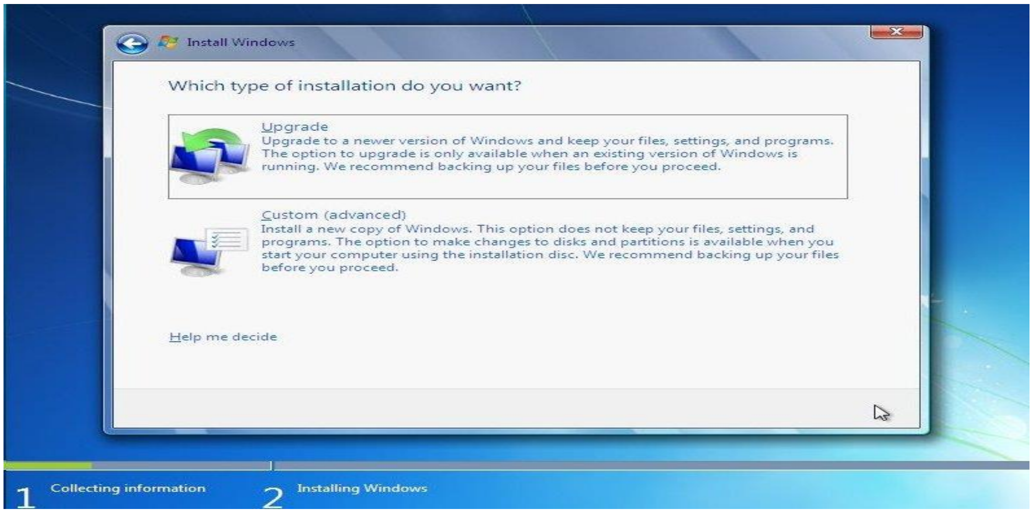
6. The following screen will displayed



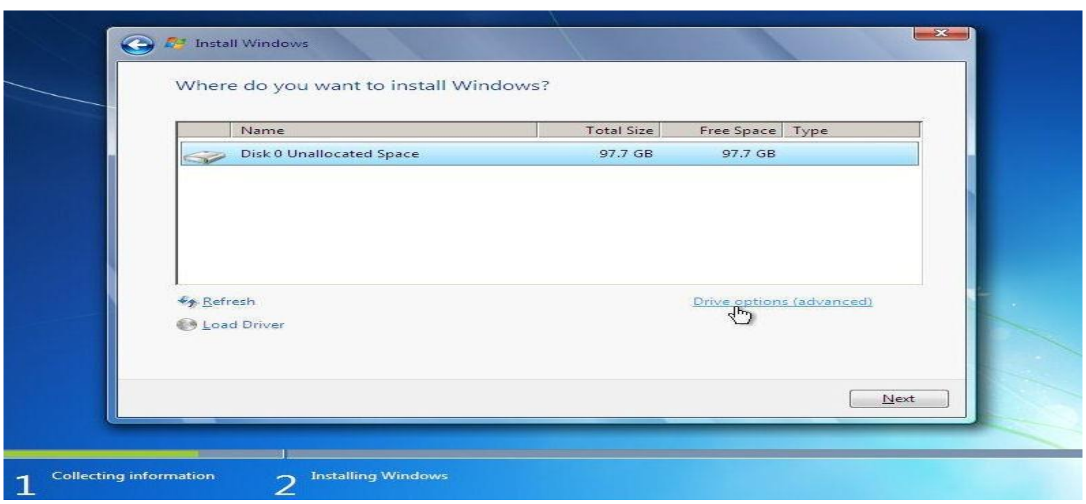
7. Check **I accept the license terms** and click **Next**.



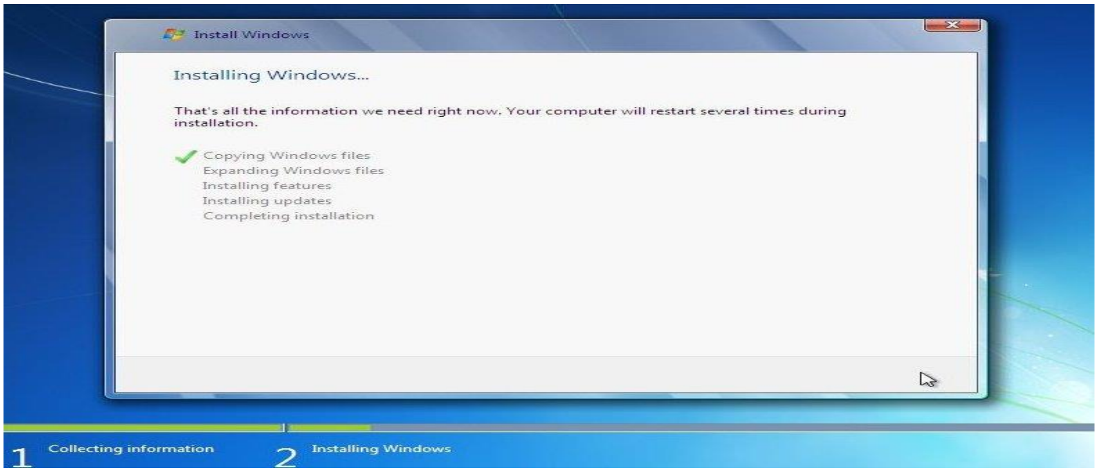
8. Click **Upgrade** if you already have a previous Windows version or **Custom (advanced)** if you don't have a previous Windows version or want to install a fresh copy of **Windows 7**.



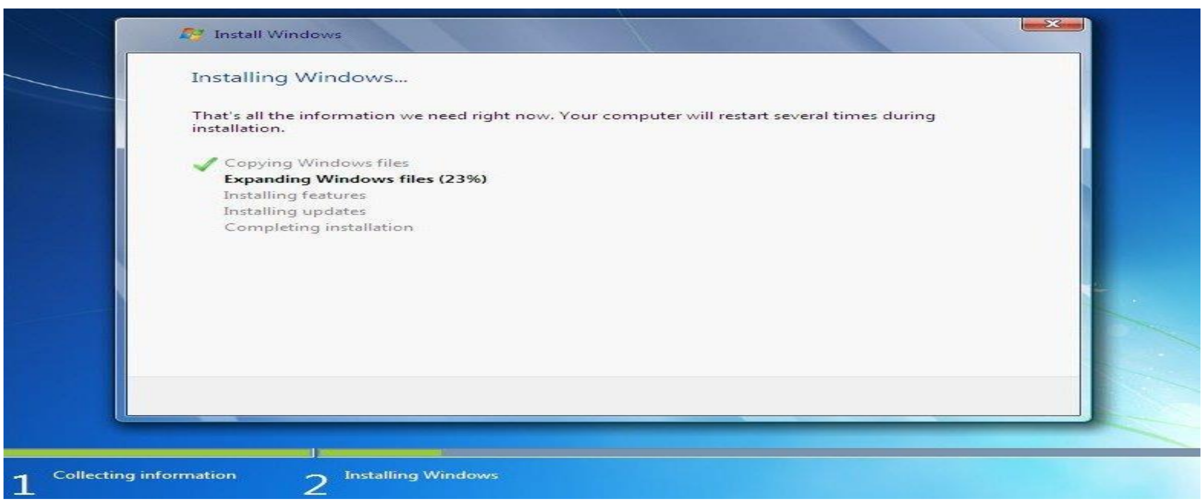
9. Select the drive where you want to install **Windows 7** and click **Next**. If you want to make any partitions, click **Drive options (advanced)**, make the partitions and then click **Next**.

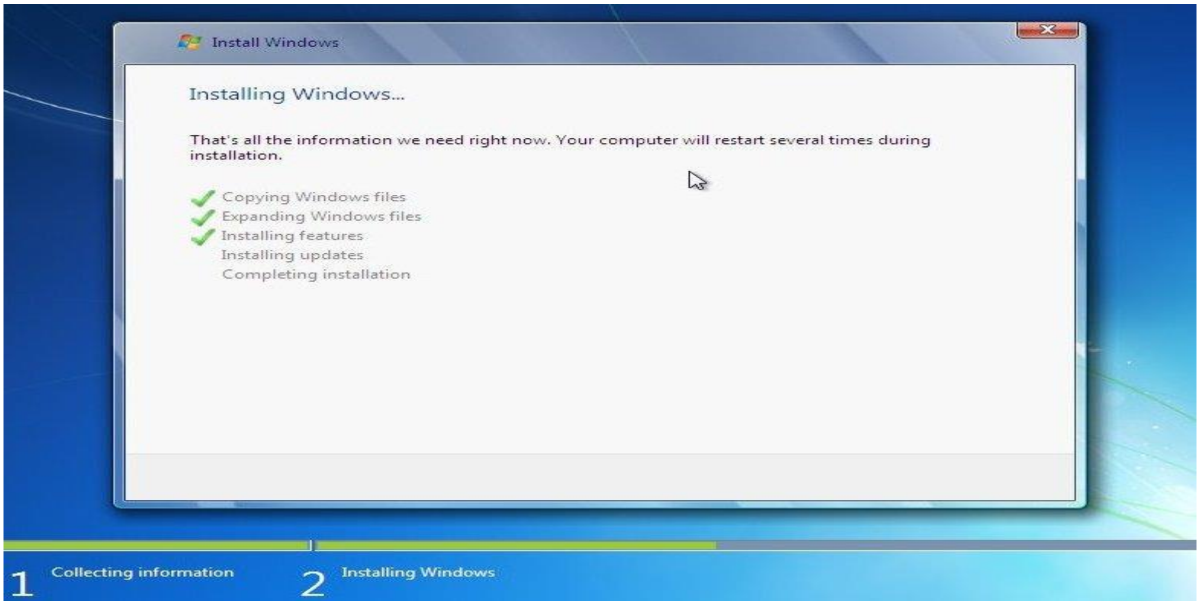


10. It will now start installing **Windows 7**. The first step, (i.e. **Copying Windows files**) was already done when you booted the **Windows 7 DVD** so it will complete instantly.

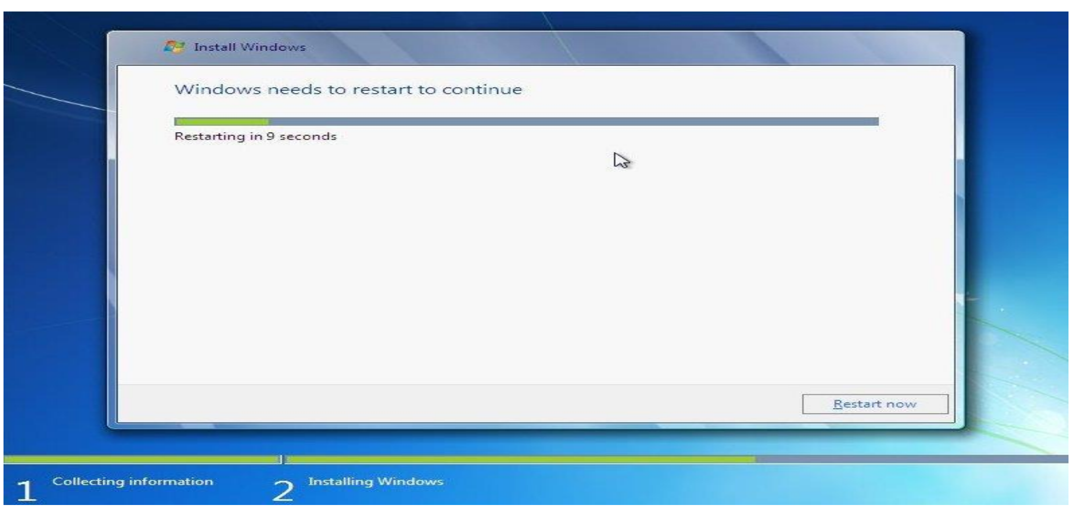


11. After completing the first step, it will expand (decompress) the files that it had copied.

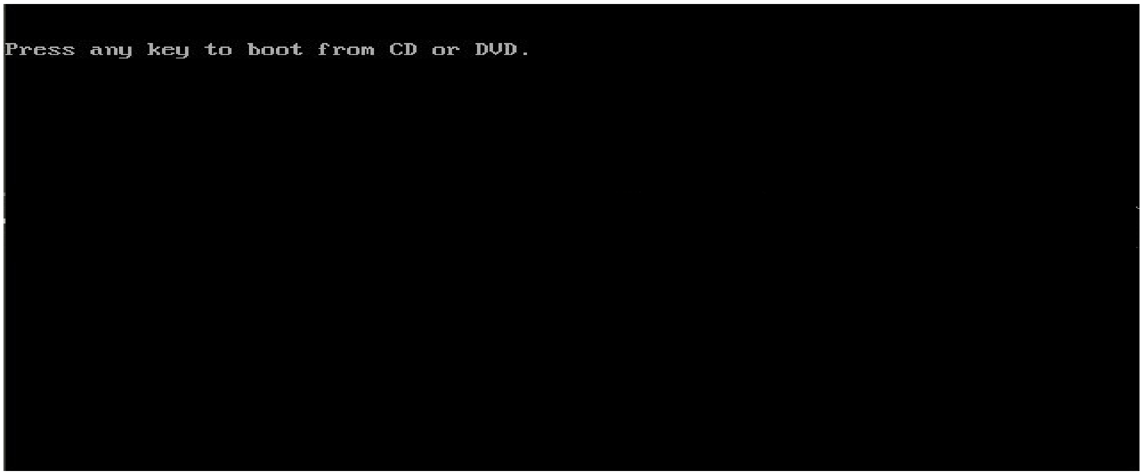




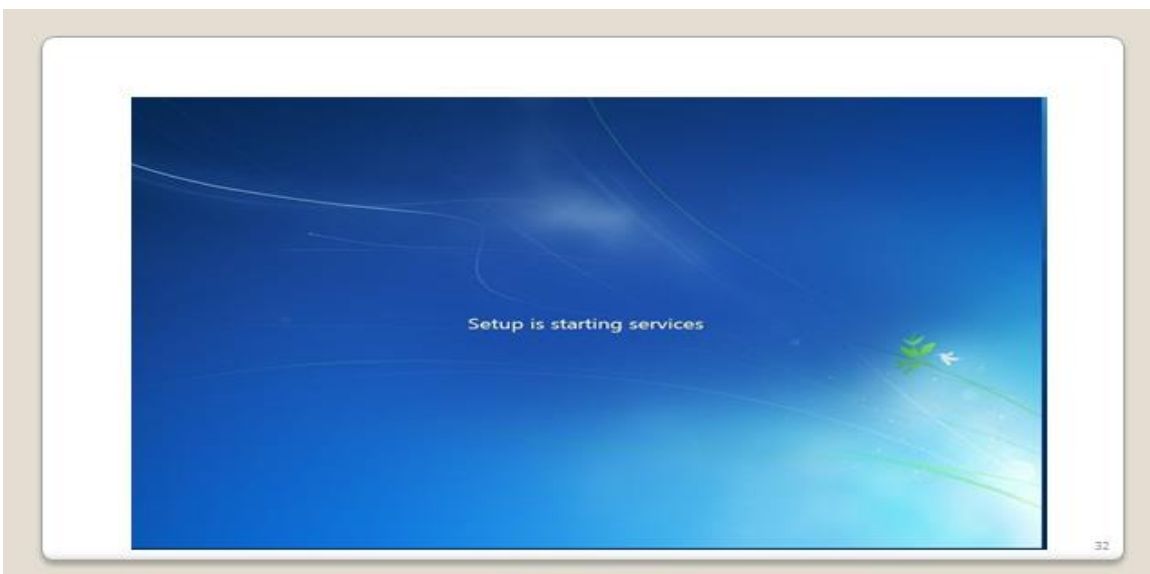
12. After that it will automatically restart after 15 seconds and continue the setup. You can also click **Restart now** to restart without any delays.



13. After installation, Please do not press key this time


Press any key to boot from CD or DVD.


Starting Windows
© Microsoft Corporation



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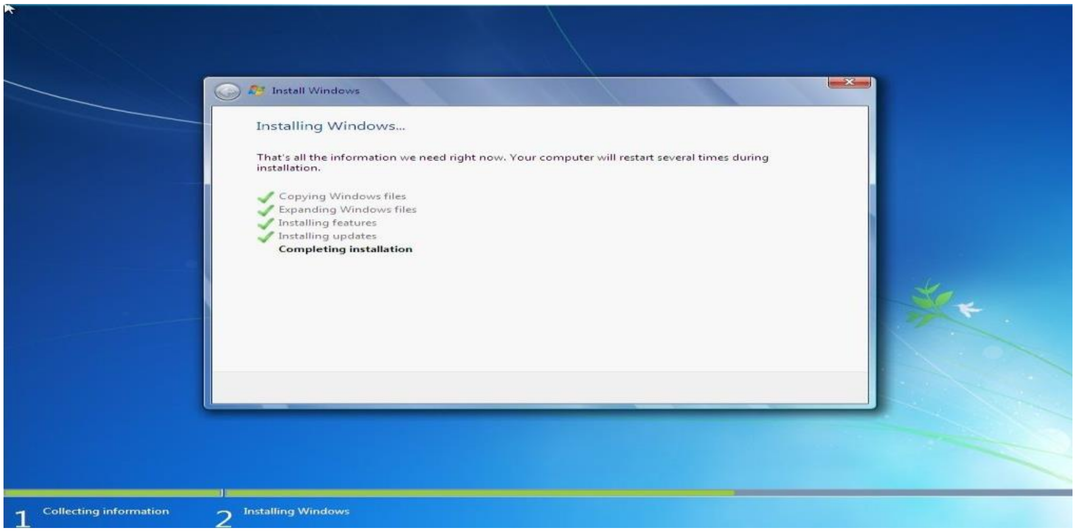
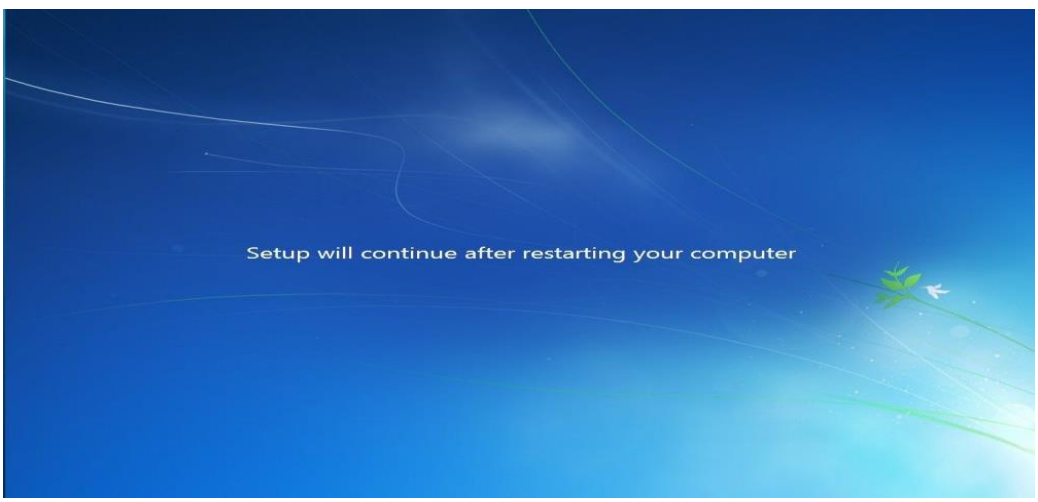
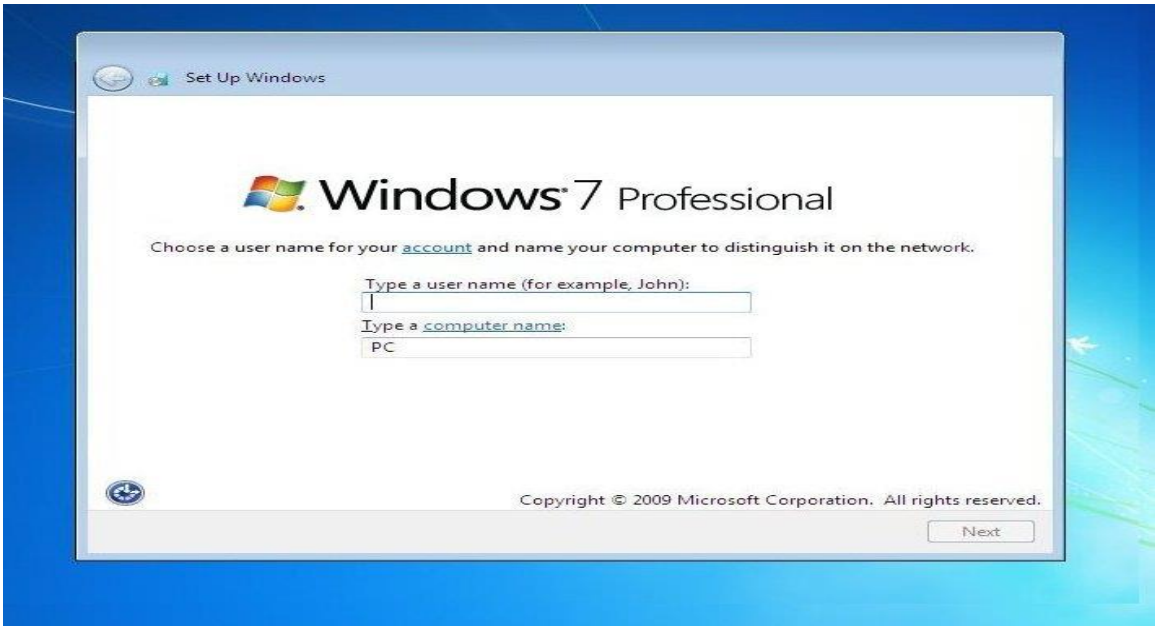
14. After restarting for the first time, it will continue the setup. This is the last step so it will take the most time than the previous steps.



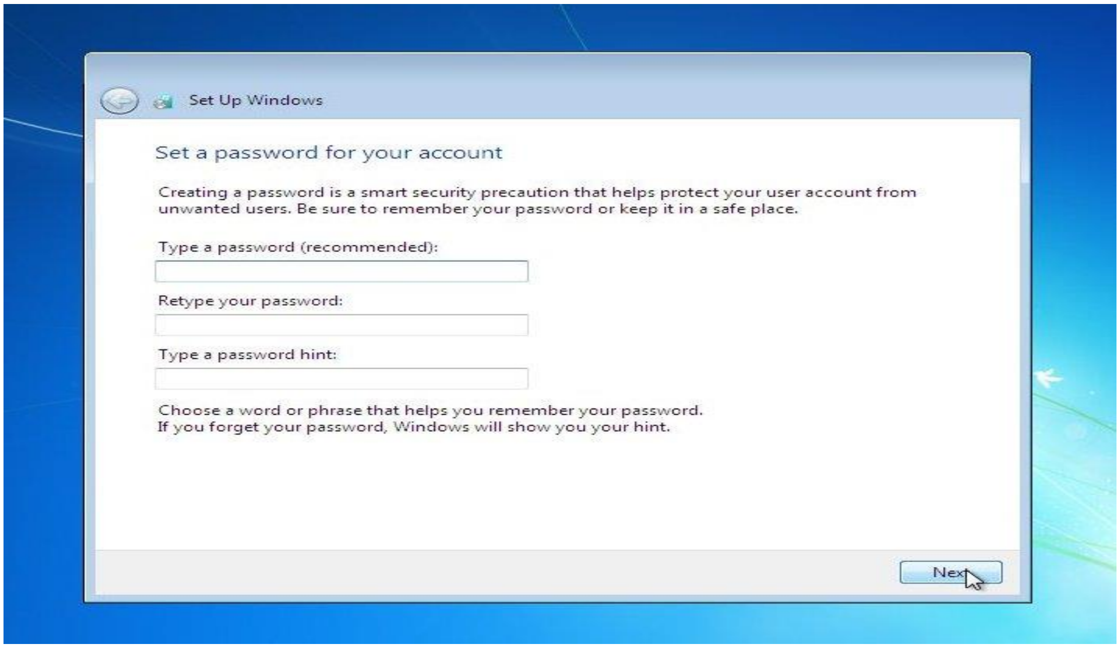


15. Type your desired user name in the text-box and click **Next**. It will automatically fill up the computer name.

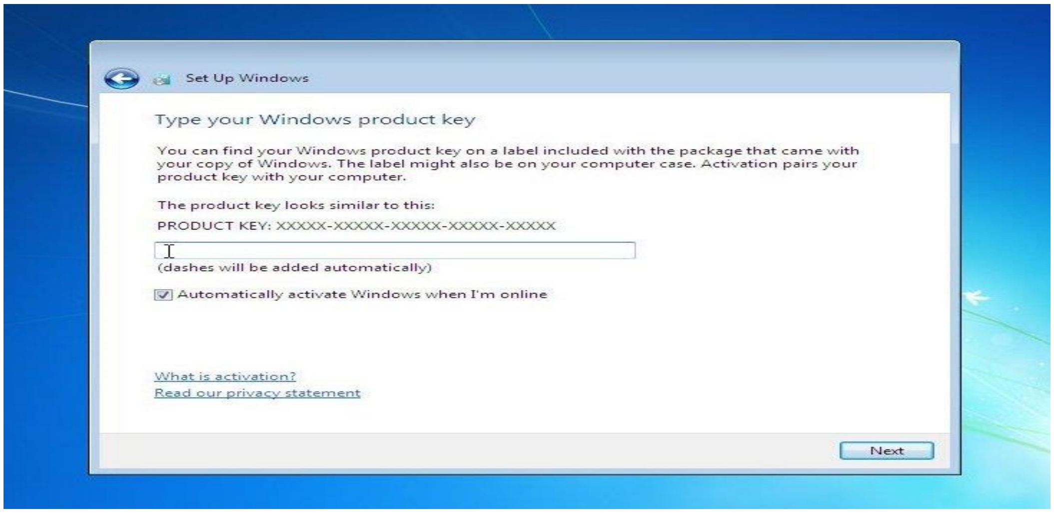


16. I

f you want to set a password, type it in the text-boxes and click **Next**.

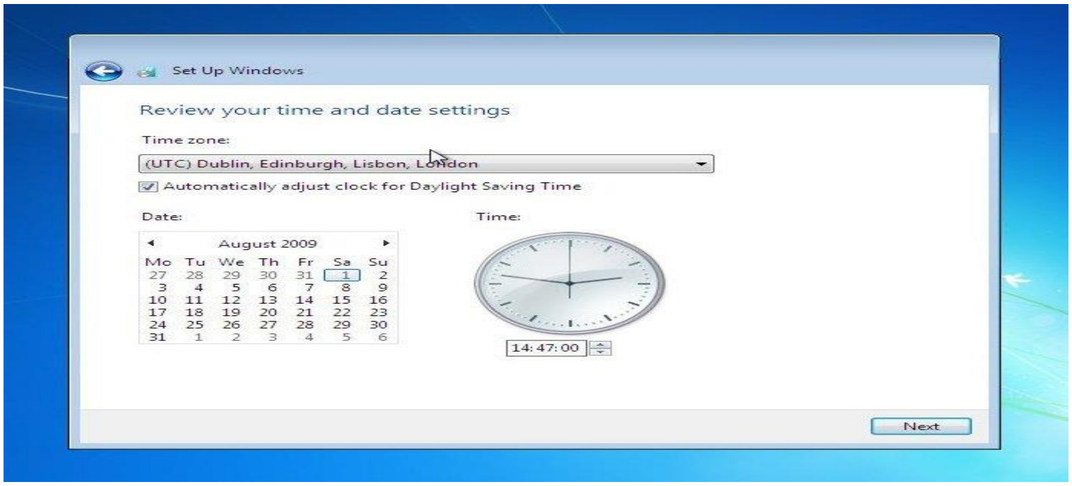


17. Type your product key in the text-box and click **Next**. You can also skip this step and simply click **Next** if you want to type the product key later. Windows will run only for 30 days if you do that.

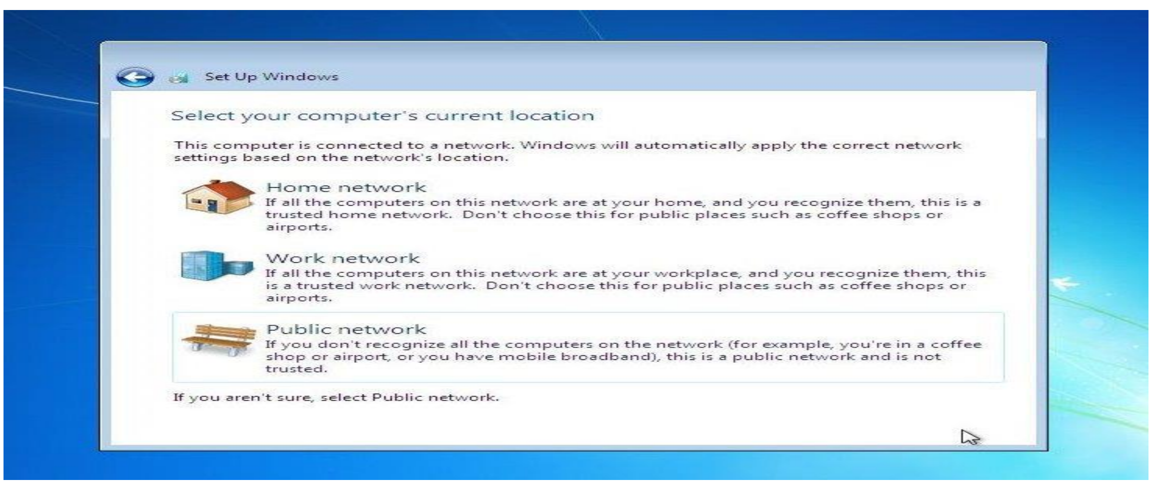


18. Select your desired option for **Windows Updates**.

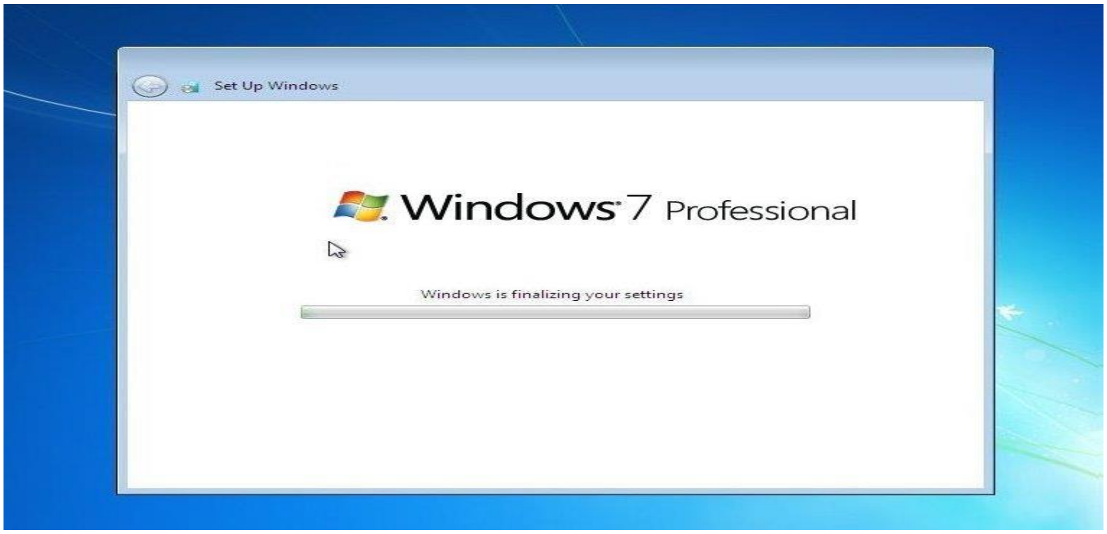
19. Select your time and click **Next**.



20. If you are connected to any network, it will ask you to set the network's location.

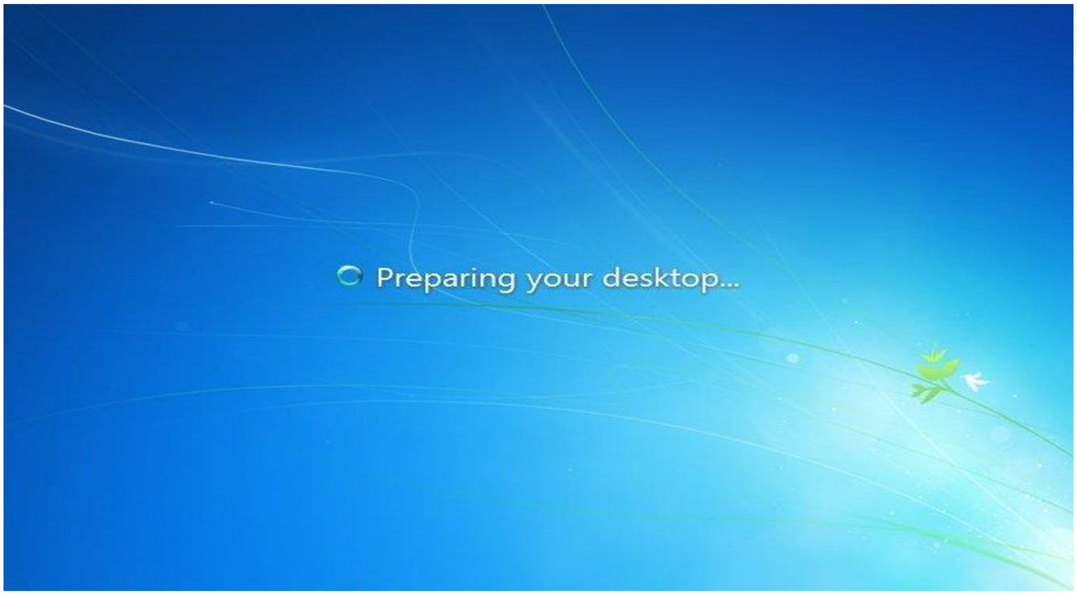


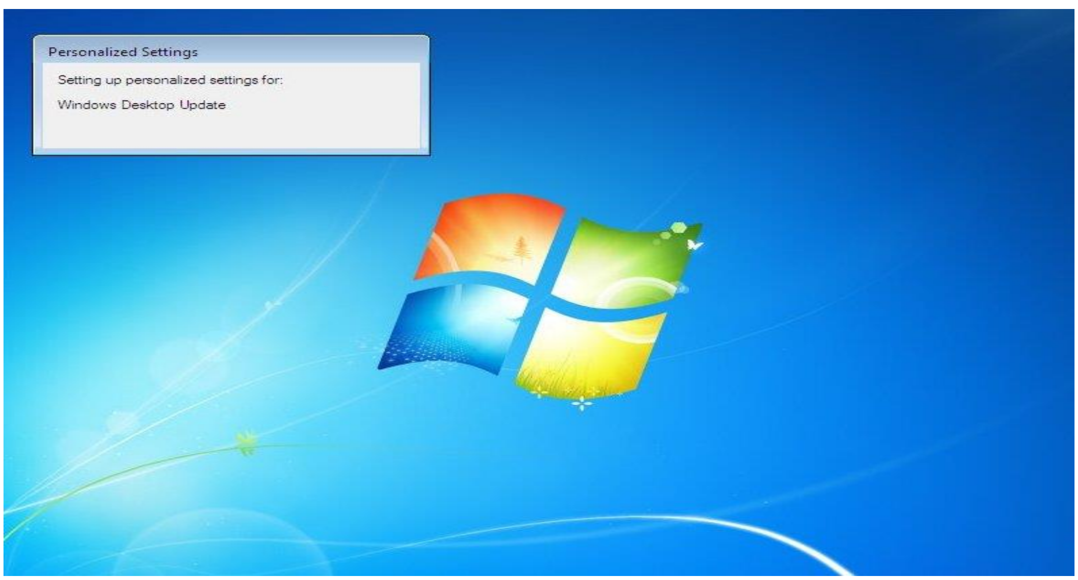
21. Window is finalizing your settings



Welcome screen

Preparing Desktop





And there you have a fresh copy of Windows 7 installed!

Installing Linux Ubuntu operating system

Linux Ubuntu can get installed through the following steps

1. Requirements

You'll need to consider the following before starting the installation:

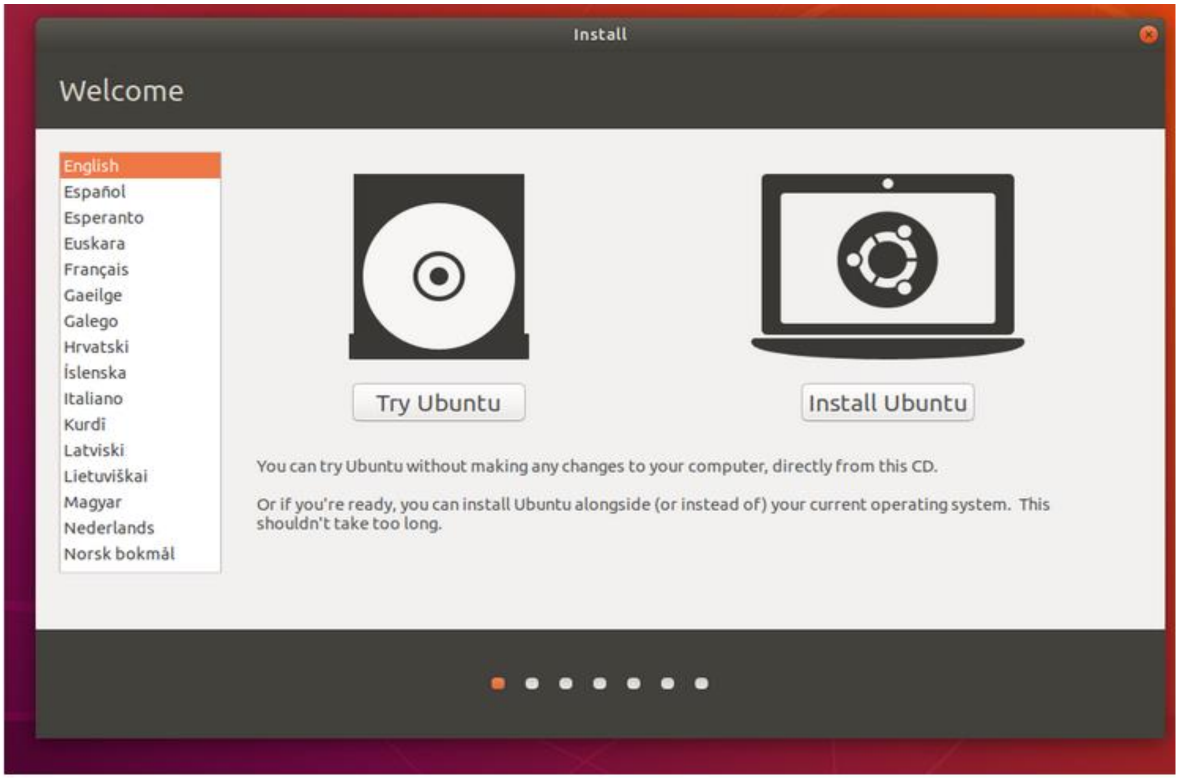
- Connect your laptop to a power source.
- Ensure you have at least 25GB of free storage space, or 5GB for a minimal installation.
- Have access to either a DVD or a USB flash drive containing the version of Ubuntu you want to install.
- Make sure you have a recent backup of your data. While it's unlikely that anything will go wrong, you can never be too prepared.

2. Boot from DVD

It's easy to install Ubuntu from a DVD. Here's what you need to do:

- Put the Ubuntu DVD into your optical/DVD drive.
- Restart your computer.

As soon as your computer boots you'll see the welcome window.



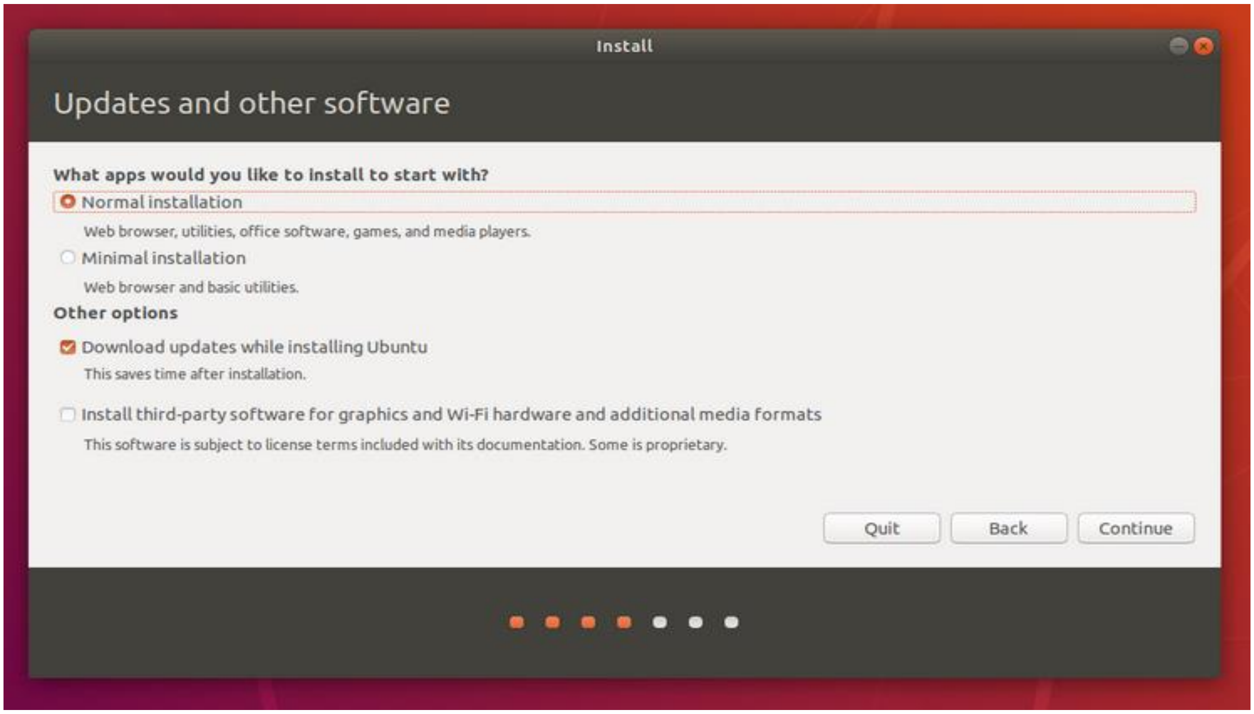
3. Boot from USB flash drive

- Most computers will boot from USB automatically.
- Simply insert the USB flash drive and either power on your computer or restart it.
- You should see the same welcome window we saw in the previous 'Install from DVD' step, prompting you to choose your language and either install or try the Ubuntu desktop.

NB : If your computer doesn't automatically boot from USB, try holding F12 when your computer first starts. With most machines, this will allow you to select the USB device from a system-specific boot menu.

4. Prepare to install Ubuntu

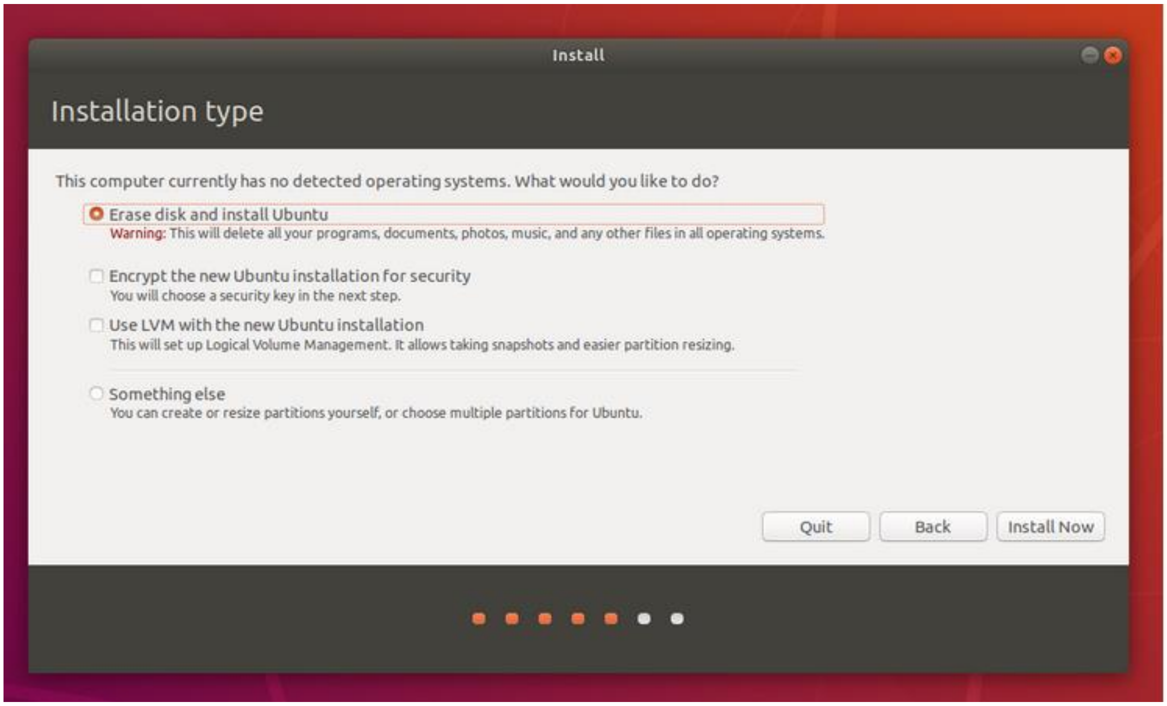
- You will first be asked to select your keyboard layout. If the installer doesn't guess the default layout correctly, use the 'Detect Keyboard Layout' button to run through a brief configuration procedure.
- After selecting *Continue* you will be asked *What apps would you like to install to start with?* The two options are '**Normal installation'** and '**Minimal installation'**. The first is the equivalent to the old default bundle of utilities, applications, games and media players - a great launchpad for any Linux installation. The second takes considerably less storage space and allows you to install only what you need.
- Beneath the installation-type question are two checkboxes; one to enable updates while installing and another to enable third-party software.
- It is recommended to enable both Download updates and Install third-party software.
- Stay connected to the internet so you can get the latest updates while you install Ubuntu.
- If you are not connected to the internet, you will be asked to select a wireless network, if available. We advise you to connect during the installation so we can ensure your machine is up to date



5. Allocate drive space

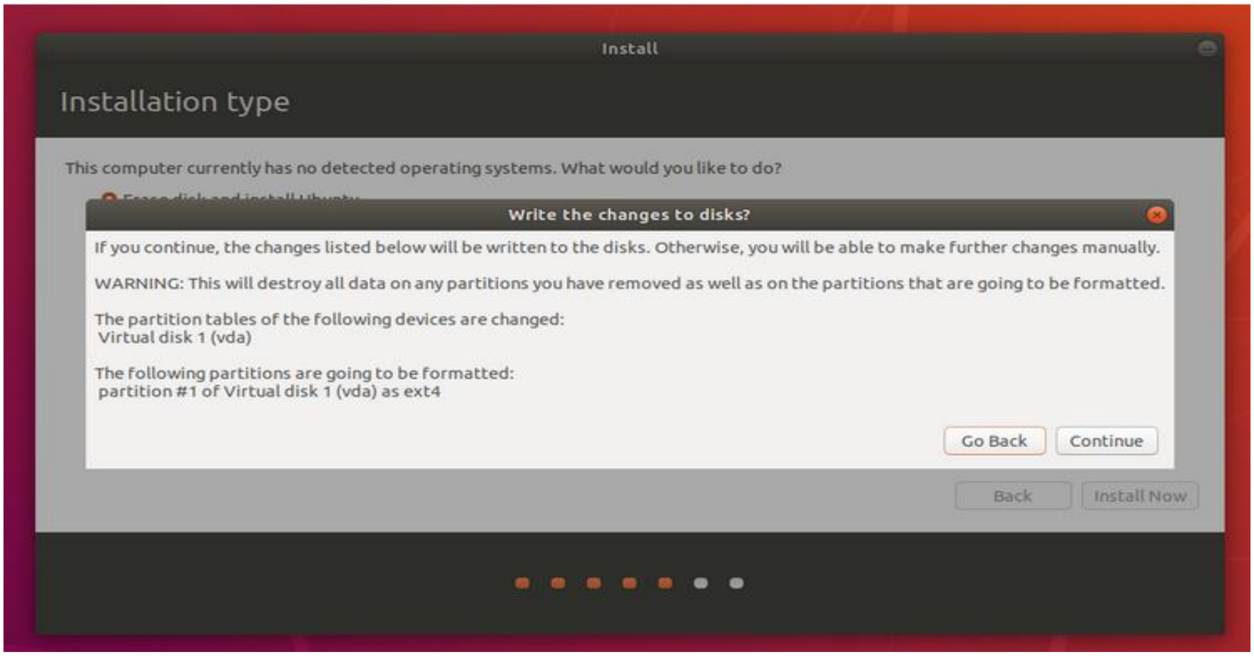
- Use the checkboxes to choose whether you'd like to install Ubuntu alongside another operating system, delete your existing operating system and replace it with Ubuntu, or — if you're an advanced user — choose the '**Something else**' option.

learnmalawi



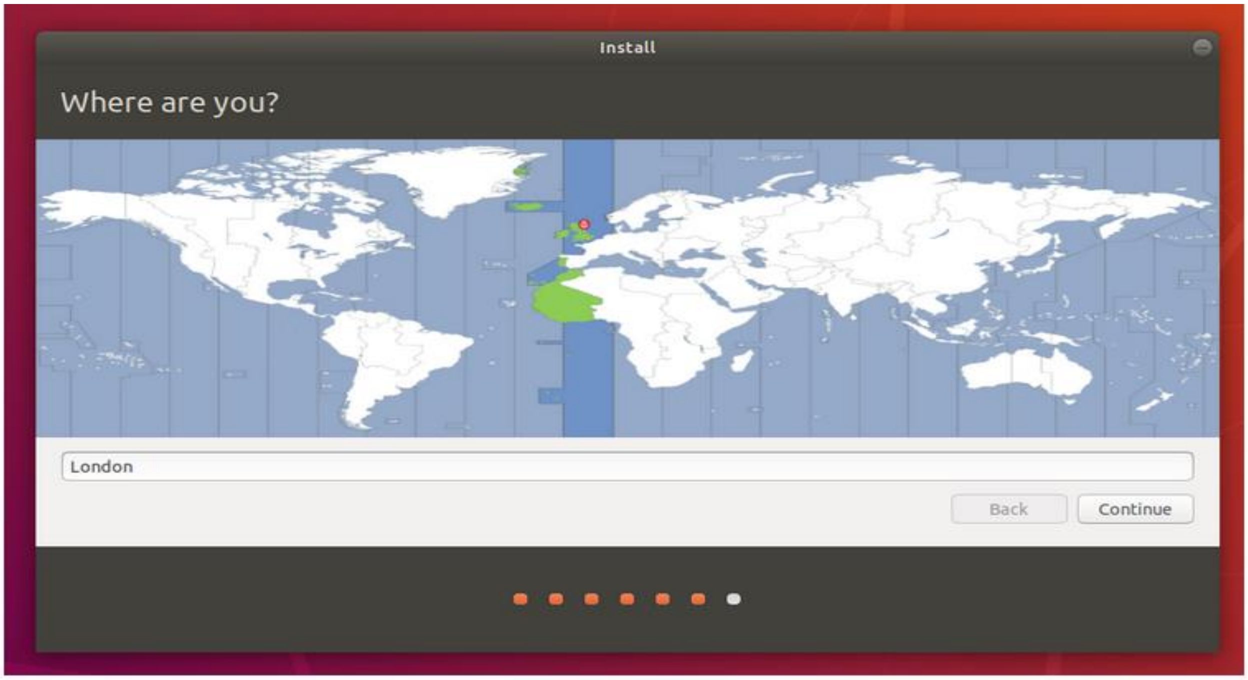
6. Begin installation

- After configuring storage, click on the 'Install Now' button. A small pane will appear with an overview of the storage options you've chosen, with the chance to go back if the details are incorrect.
- Click Continue to fix those changes in place and start the installation process.



7. Select your location

- If you are connected to the internet, your location will be detected automatically. Check your location is correct and click 'Forward' to proceed.
- If you're unsure of your time zone, type the name of a local town or city or use the map to select your location.



8. Login details

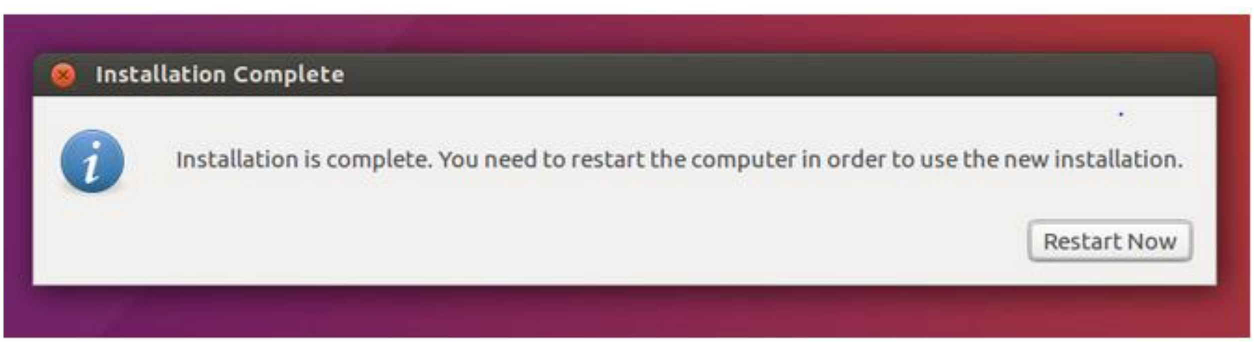
- Enter your name and the installer will automatically suggest a computer name and username. These can easily be changed if you prefer. The computer name is how your computer will appear on the network, while your username will be your login and account name.
- Next, enter a strong password. The installer will let you know if it's too weak.
- You can also choose to enable automatic login and home folder encryption. If your machine is portable, we recommend keeping automatic login disabled and enabling encryption. This should stop people accessing your personal files if the machine is lost or stolen.

9. Background installation

- The installer will now complete in the background while the installation window teaches you a little about how awesome Ubuntu is.
- Depending on the speed of your machine and network connection, installation should only take a few minutes.

10. Installation complete

- After everything has been installed and configured, a small window will appear asking you to restart your machine.
- Click on Restart Now and remove either the DVD or USB flash drive when prompted. If you initiated the installation while testing the desktop, you also get the option to continue testing.

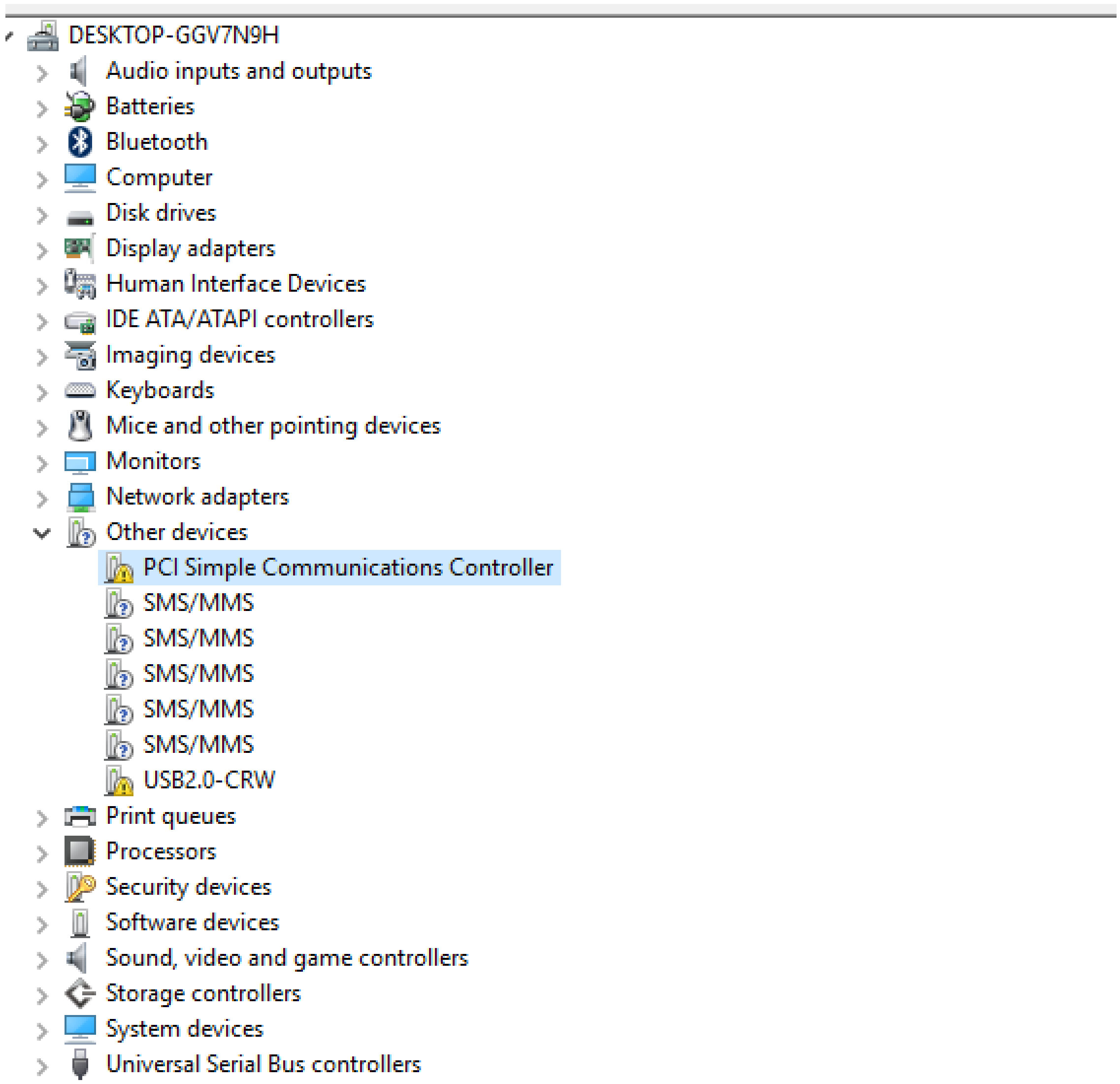


Installing Device drivers

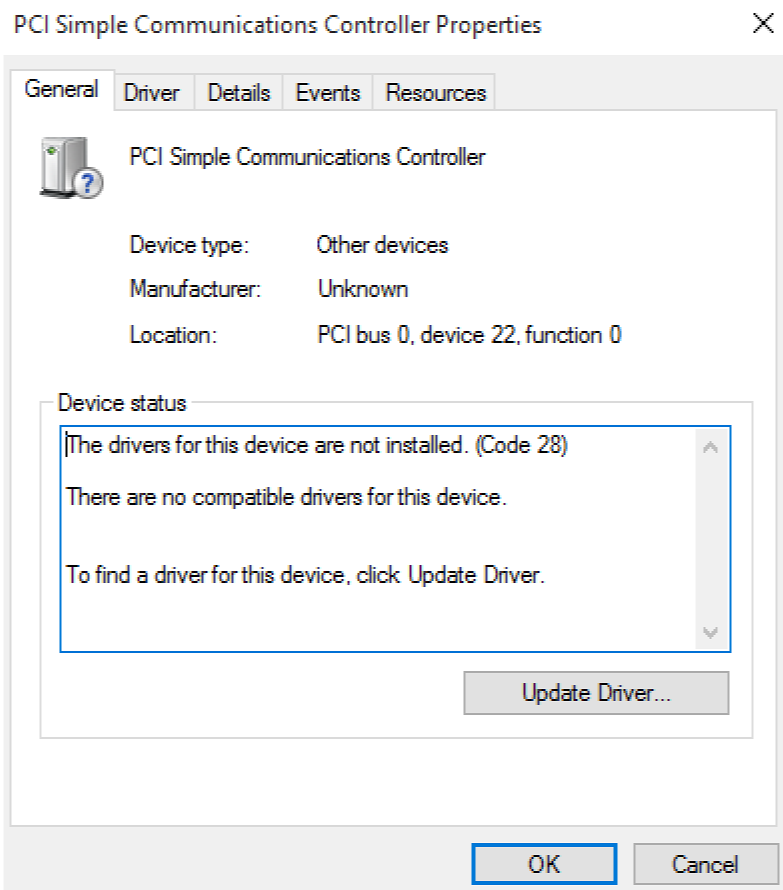
- A **device driver** is a computer program that controls a particular device that is connected to your computer.
- Typical **devices** are keyboards, printers, scanners, digital cameras and external storage **devices**. Each of these needs a **driver** in order to work properly

Installing device drivers in windows 7

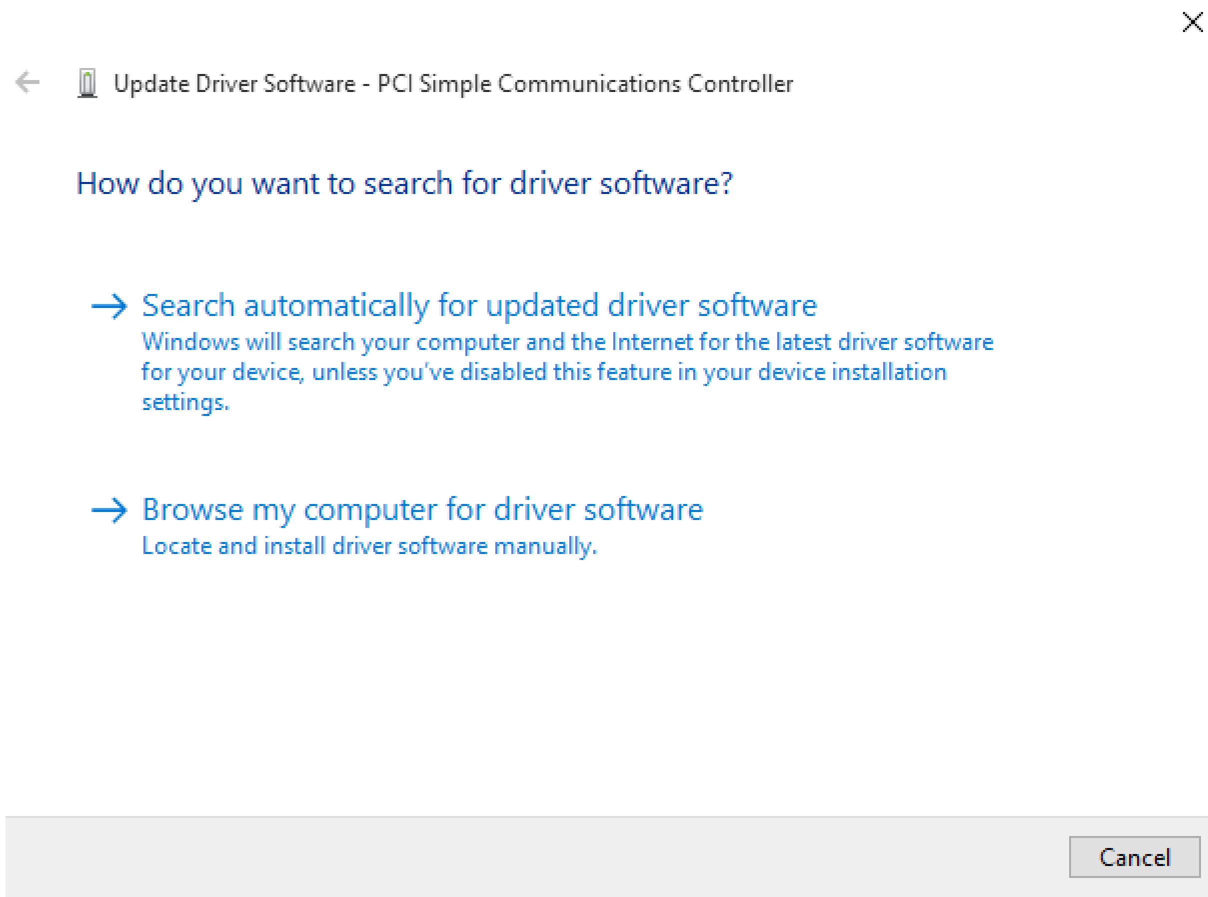
- Hardware devices need drivers to function properly. Most of these drivers are installed automatically by Windows 7 but if this is not the case these drivers need to be installed by the user.
- Opening the **Device Manager** (available in the Control Panel) will tell the user which hardware drivers need to be installed by marking them with a yellow exclamation mark.



- To install a driver
- Double click on a driver which you want to install especial the one with yellow exclamation mark and you will get to this box

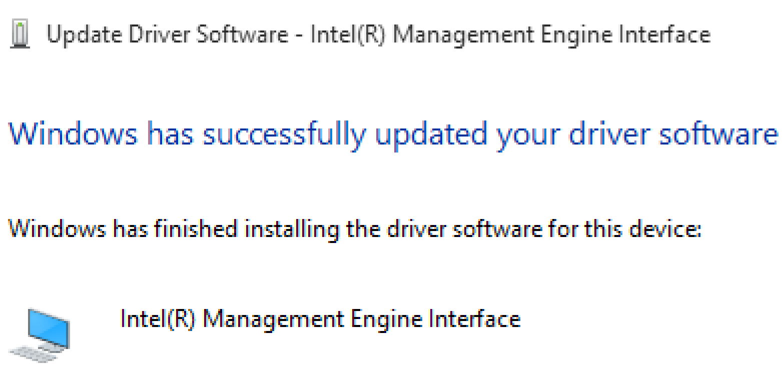


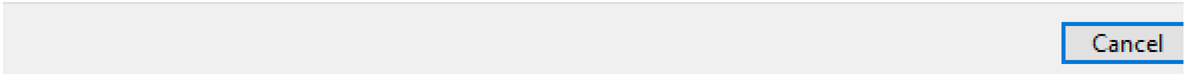
- Click on Update a drive which will give you two options
  a. To search a driver from internet
  b. To browse it from your local computer



NB you need internet connection for your computer to search a driver from internet.

- Your computer starts searching for a driver and installed automatically





Installing application software

Installation of application software is only limited to

- Hardware configuration
- Operating system
- License agreement

The following are examples of application software that can be installed

- Word processor
- Spreadsheet
- Database management system

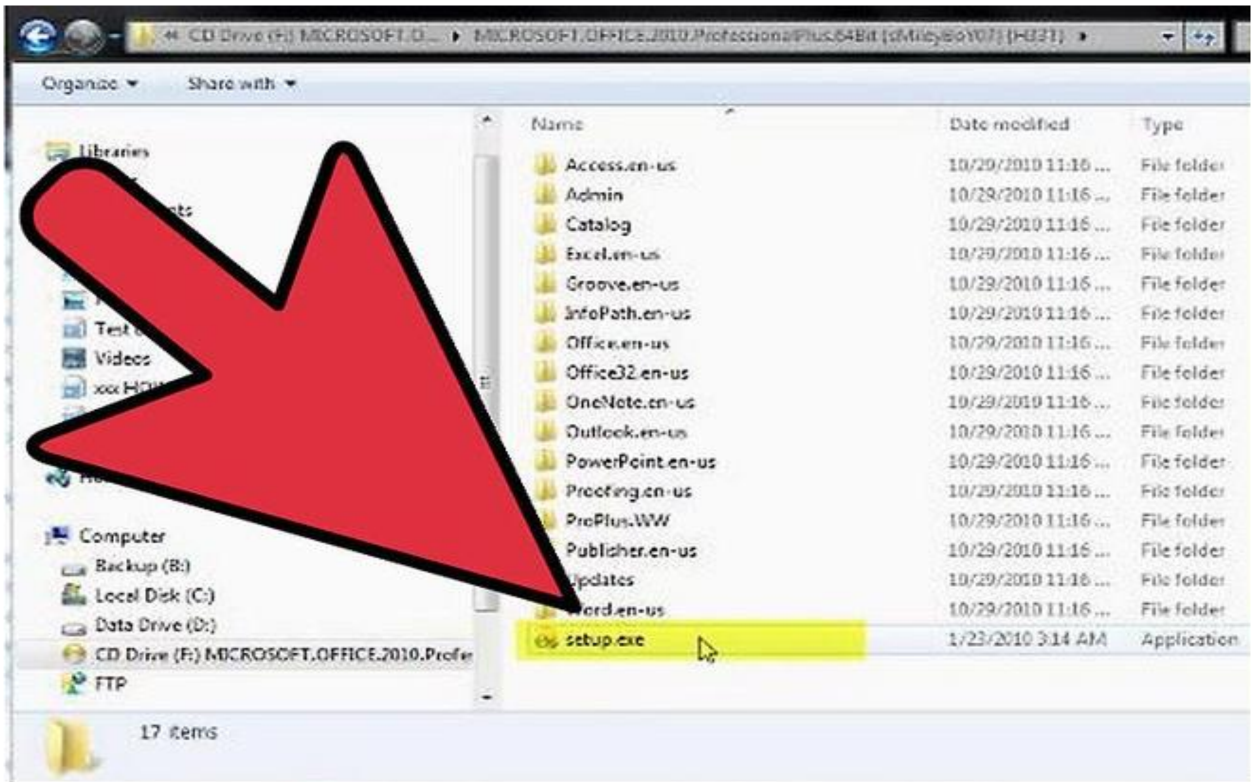
NB Before you install any application software, select only those application software relevant to a user of a particular computer ie whether general purpose or special purpose software

Installing Microsoft office 2010

- **Microsoft Office** is a suite of desktop productivity applications that is designed specifically to be used for **office** or business use.
- It mainly consists of Word, Excel, PowerPoint, Access, OneNote, Outlook and Publisher applications

To install Microsoft Office 2010, follow the following steps

1. **Insert your Office 2010 DVD.** Alternatively, open the downloaded Setup file that you received when you purchased Office 2010 online. Either method will follow the same steps.

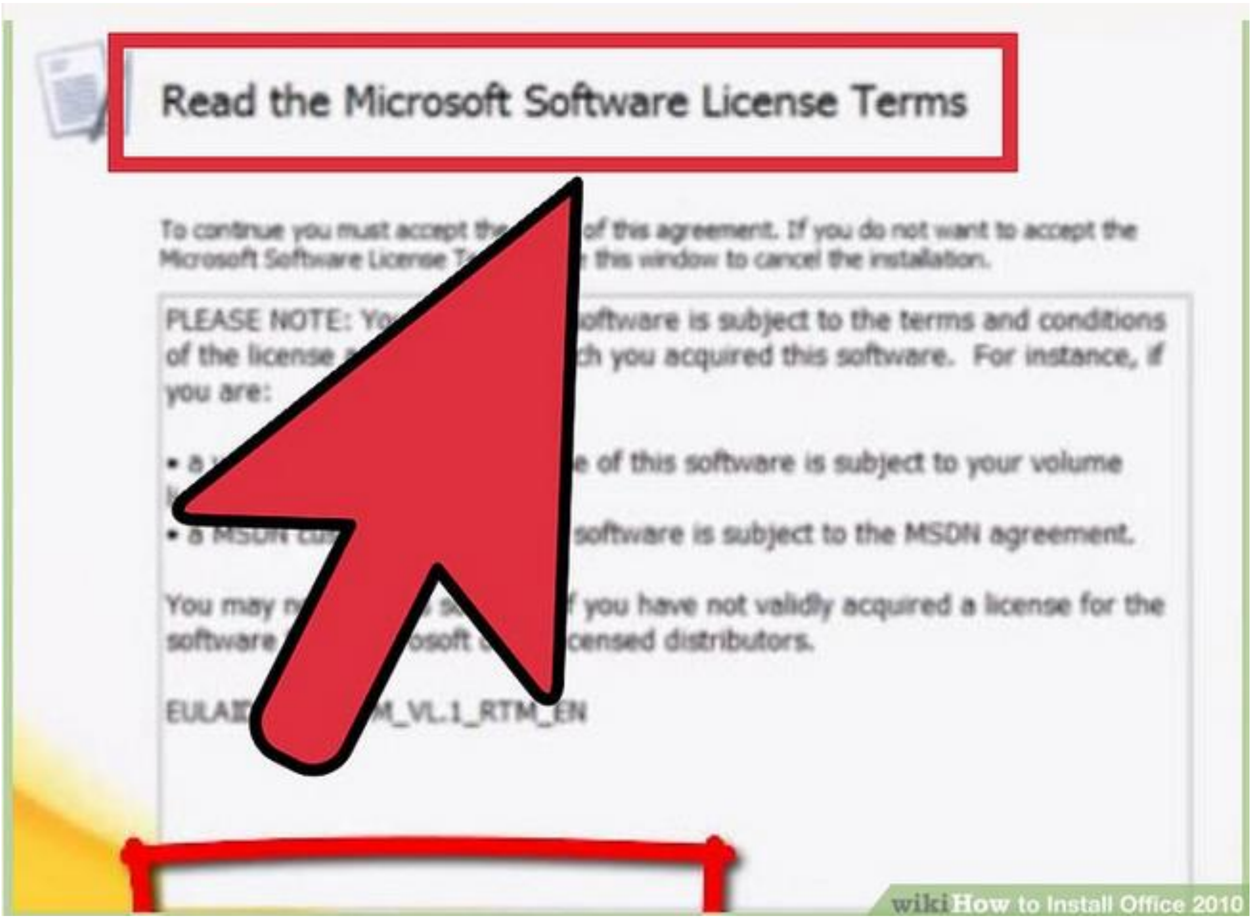


2. **Enter the Product Key.**

- This is the 25-character key found on the packaging that your Office 2010 came in.
- If you purchased online, the key will be displayed in the order confirmation window.
- You do not need to enter the dashes in between groups of characters

3. **Accept the License Terms.**

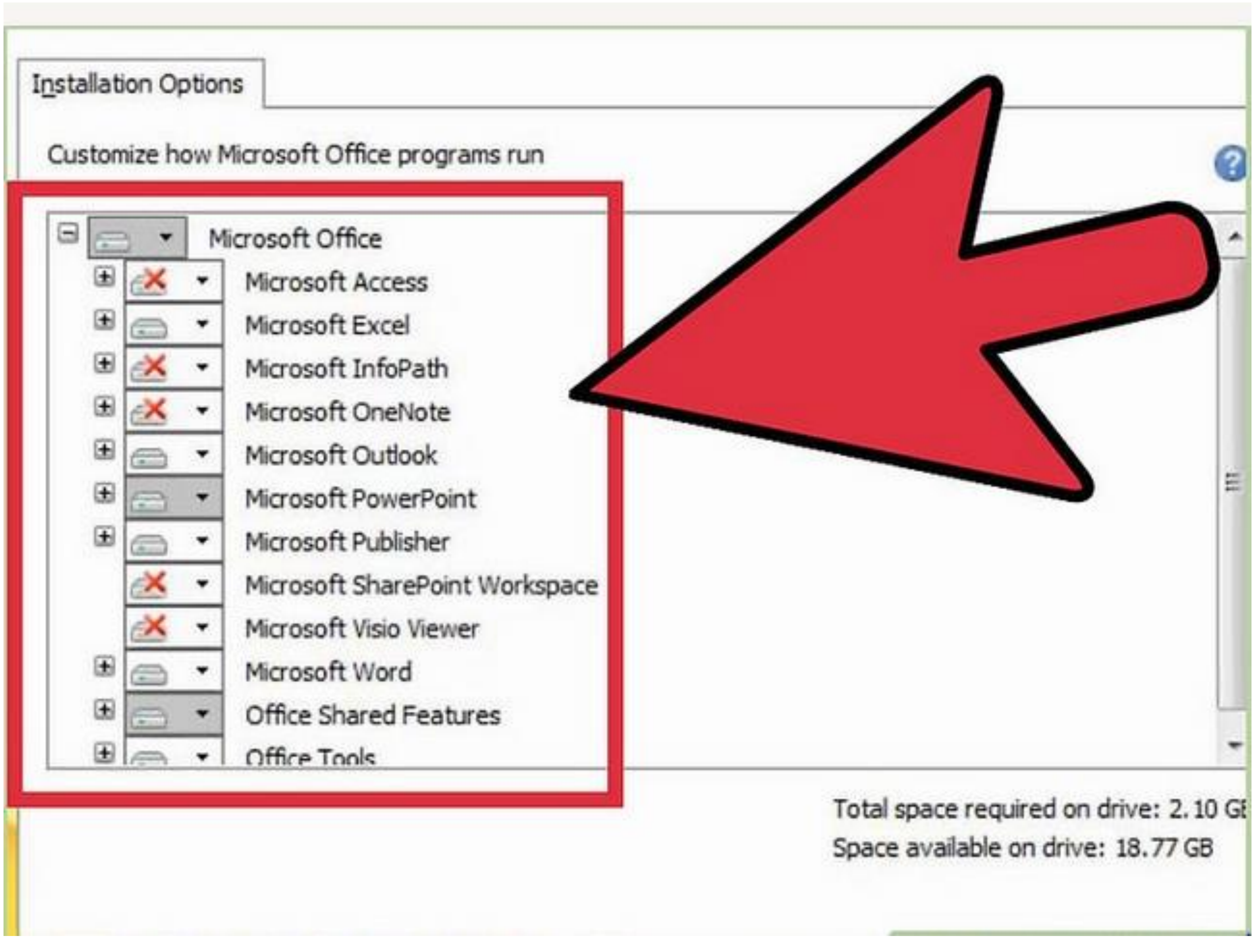
- In order to proceed with the installation, you need to check the box indicating that you have read and agree to Microsoft's terms of use.



4. **Choose your installation.**

- Clicking Install Now will install all of the Office products included in the version that you purchased. Office will be installed to your default hard drive (the same that Windows is installed on).
- Choose Customize to specify which products you want to install. For example, if you never use Excel and just need Word, use Customize to disable the Excel installation.

- You can also use the Customize option to install Office to a different location on your computer.



5. **Wait for installation to complete.**

- Once you have chosen your installation options, Office will be automatically installed. The amount of time this takes will vary depending on the version you are installing and the speed of your computer.

- Once Office is finished installing, you can access each of the individual Office programs from the Start menu.

Application software uninstallation

- If a software is no longer needed in a computer system, must be removed from a system
- This process is called software uninstallation

To remove programs and software components in Windows 7 from your computer hard disk drive, follow these steps:

1. Click Start, and then click **Control Panel**.
2. Under Programs, click Uninstall a program.

   The Uninstall or change a program window opens.

3. Select the program you want to remove.
4. Click Uninstall or Uninstall/Change at the top of the program list then windows will uninstall the selected software successfully

CHAPTER 4: TROUBLESHOOTING OF COMPUTERS

Objectives

By the end of this chapter, the student should be able to:

Contents

- Tracing hardware problems (Bios, boot sequence, sound)
- Tracing software problems
- Identifying sources of software problems
- Solutions to software problems
  - Repair software
  - Uninstall software
  - System restores
  - System update
  - Restart processes

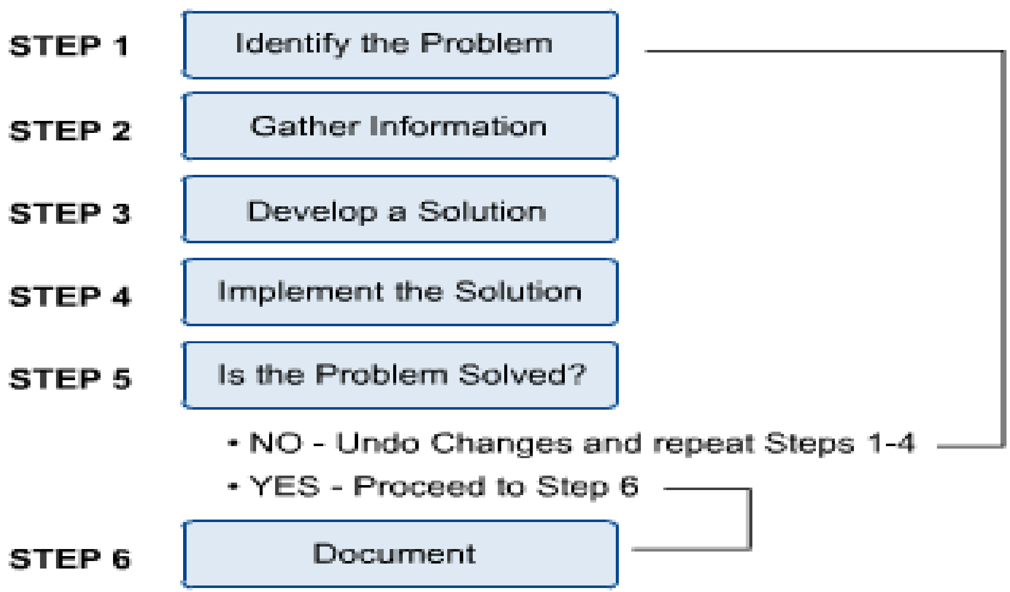
Meaning of Trouble Shooting

- Computer troubleshooting is the process of identifying and solving hardware and software related problems
- Effective troubleshooting uses techniques to diagnose and then fix computer problems.
- Rarely will simply guessing potential solutions for a problem work.

Troubleshooting is a cycle

- Refers to series of logical steps that are followed during the troubleshooting process.

The following flow chart summarises the steps that must be taken during troubleshooting cycle



Identify a problem

- This step should provide a clear problem statement that defines the problem as a set of symptoms and associated causes.
- This is done by identifying the general symptoms and then determining the possible causes that could result in these symptoms.
- The outcome of this step should be a written set of ideas and possibilities.

Identify the Problem

- Identify symptoms and associated causes
- List ideas and possibilities (brainstorm)
- Create a clear and concise problem statement

Gathering Information

- Gather information from customer
- Gather information using diagnostic tools and software
- Reproduce problem (if possible)
- Determine if the problem is hardware or software related
- Document your observations and results

Develop a Solution

- Compile gathered information
- Assess the information
- Consider the most likely cause
- Develop a solution based on your experience, and the information you gathered.

Implement the Solution

- Backup critical data
- Start with the simple things first
- Make changes one at a time
- Reverse any changes that caused any adverse effects

Is the Problem Solved?

- Perform test, visually, audibly inspect the computer
- Verify the computer problem is resolved
  - If the problem has been resolved
    - Show the user the problem is fixed
    - Document results
  - If the problem is still existing
    - Undo changes
    - Return to step one

Document

- Document all changes made to system
- Document the initial problem and subsequent resolution
- Document when the problem was resolved

- The following are some of computer problems and how they can be solved

1. The Computer Won't Start

- A computer that suddenly shuts off or has difficulty starting up could have a failing power supply.
- Check that the computer is plugged into the power point properly and, if that doesn't work, test the power point with another working device to confirm whether or not there is adequate power.

2. The Screen is Blank

- If the computer is on but the screen is blank, there may be an issue with the connection between the computer and the screen.
- First, check to see if the monitor is plugged into a power point and that the connection between the monitor and computer hard drive is secure.
- If the problem is on a laptop, then you may need to get a professional to fix it as some of the internal wires may be worn.

3. Abnormally Functioning Operating System or Software

- If the operating system or other software is either unresponsive or is acting up, then try restarting your computer and run a virus scan.

- To avoid having this happen, install reliable anti-virus software.

4. Windows Won't Boot

- If you are having troubles booting Windows, then you may have to reinstall it with the Windows recovery disk.

5. The Screen is Frozen

- When you computer freezes, you may have no other option than to reboot and risk losing any unsaved work.
- Freezes can be a ***sign of***

  i. insufficient ram,
  ii. registry conflicts,
  iii. corrupt or missing files, or spyware.

  - Press and hold the power button until the computer turns off, then restart it and get to work cleaning up the system so that it doesn't freeze again.

  6. Computer is Slow

  - If your computer is slower than normal, you can often fix the problem simply by cleaning the hard disk of unwanted files.
  - You can also ***install a firewall***, ***anti-virus*** and ***anti-spyware tools***, and ***schedule regular registry scans***.

- External hard drives are great storage solutions for overtaxed CPU's, and will help your computer run faster.

7.Strange Noises

- A lot of noise coming from your computer is generally a sign of either hardware malfunction or a noisy fan.
- Hard drives often make noise just before they fail, so you may want to back up information just in case, and fans are very easy to replace.

Slow Internet

- To improve your Internet browser performance, you need to clear cookies and Internet temporary files frequently.
- In the Windows search bar, type 'x' and hit enter to open the temporary files folder.

9. Overheating

- If a computer case lacks a sufficient cooling system, then the computer's components may start to generate excess heat during operation.
- To avoid your computer burning itself out, turn it off and let it rest if it's getting hot.

- Additionally, you can check the fan to make sure it's working properly.

10 **Dropped Internet Connections**

- Dropped Internet connections can be very frustrating.
- Often the problem is simple and may be caused by a bad cable or phone line, which is easy to fix.
- More serious problems include viruses, a bad network card or modem, or a problem with the driver.

11. PC Fans not working

- If you notice one or more fans in your PC aren't working, then it could be due to the dirt inside.
- You will have to open up the PC and **use a compressed air can or a leaf blower** to clean up the fans and other components.

12. PC crashes before loading the OS

- If your **PC only shows manufacturer logo and then crashes** right before it was supposed to load the operating system, then it's a **problem with RAM or hard disk**.

- As the OS is unable to load, then either the RAM is corrupted and can't hold the boot loader or the **hard drive is damaged** and can't load data inside it.
- If you have multiple RAM slots, then **taking out each one of them one by one and starting the PC** will help find the culprit.
- In the end, you will have to replace the corrupted RAM or the hard disk, whichever has the issue.

13. Blue Screen of Death

- The dreaded Blue Screen of Death (BSOD) can occur **due to both software and hardware problems**, but usually, it's a hardware problem.
- Whatever the cause, BSOD requires immediate attention as it's a **sign of a big problem**.

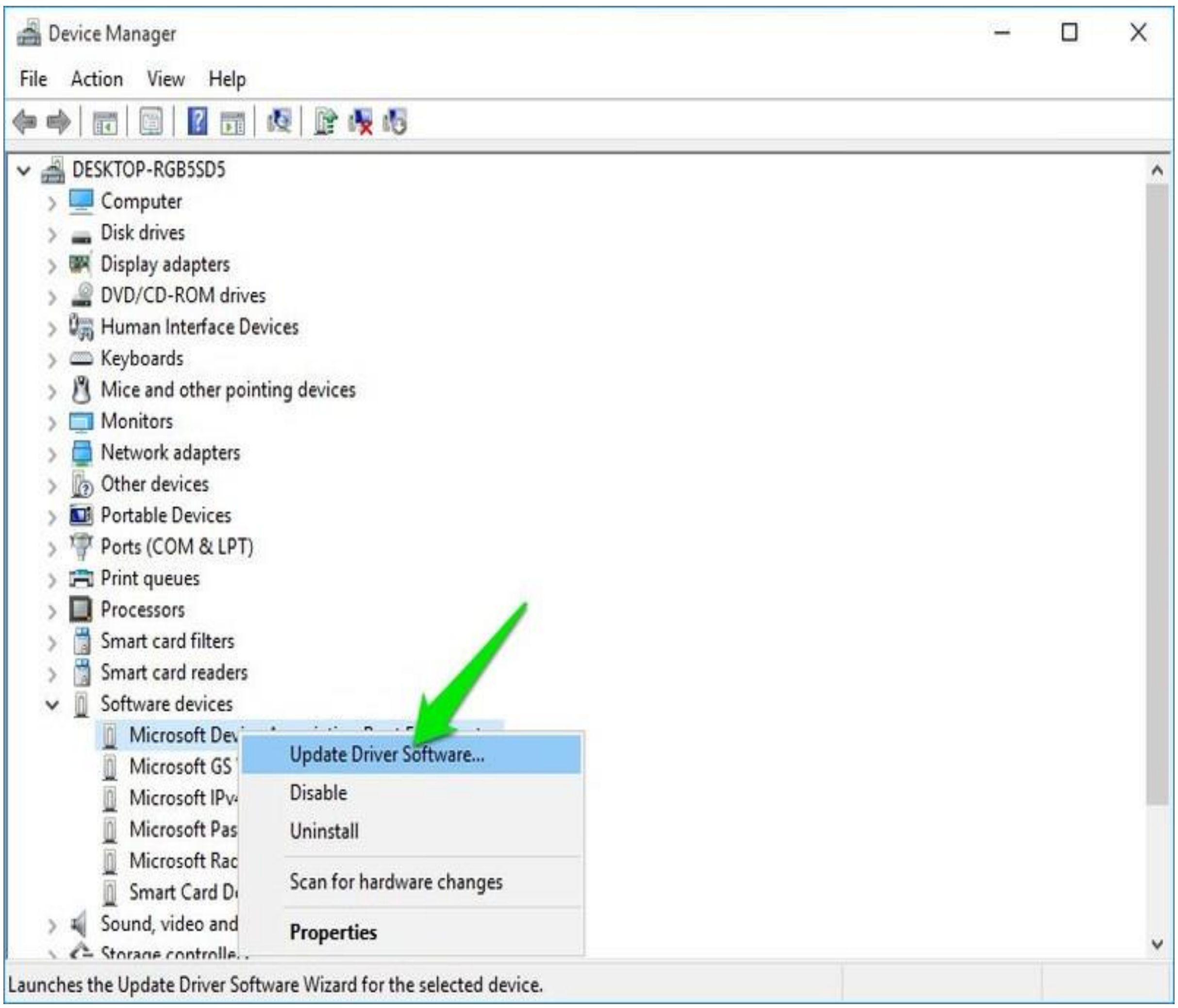
- Blue Screen View is a great Nirsoft utility that will show important information if you have recently suffered a Blue Screen of Death.
- You should be able to identify and solve the problem using this information. Below are some common reasons for BSOD and their solutions.

i) Corrupted drivers

- A corrupt driver may be the cause of BSOD. To find that out, use the following steps:

- Open **Device Manager** by typing **devmgmt.msc** in the *Run*.

- Here expand each menu and look for a yellow triangle icon next to each driver.
- If you find any, right-click on it and select **Update Driver Software** to update its driver.



- You can also use a third-party app like IObit Driver Booster to automatically find and fix driver problems.

ii) Too much pressure on the RAM

- If you **open too many programs that RAM can't handle**, then it may freeze the system and show BSOD. For that, you should either stop opening too many programs or upgrade the RAM.

iii) Faulty hard disk

- BSOD is also a **sign of a dying hard disk**, use the instructions in problem #ii above to identify hard disk problems.

iv) Heating PC

- Heating PC also leads to BSOD if **too much pressure is put on the components**. Use the instruction in problem #i) to solve it.

14 CPUs related problems

- Symptoms of a processor error can include slow performance, POST beep errors, or a system that is not operating properly.
- These errors usually indicate an internal error has occurred.
- Most CPUs will have an onboard fan. •
- his provides cooling directly to the CPU.
- CPUs must be set to receive the correct voltages to run properly.

- Motherboards that use Socket 5, Socket 7 or Super Socket 7 chips need to use voltage regulators. •
- Typically, the voltage regulators are built into the board.
- They must be set at the proper voltage, or the CPU can be damaged.

CHAPTER 5: INTRODUCTION TO COMMUNICATION NETWORKS

Objectives

By the end of this chapter, the student should be able to:

Contents

- Basic terminologies
  - communication
  - data communication
  - telecommunication
  - network
- History of communication technologies
  - Telephone
  - Cellular
  - internet
- Uses of networks
  - business applications
    - e-banking
    - mobile banking
    - auto teller machines
    - point of sale
  - home applications
  - remote access
  - mobile communication
- Elements of communication systems
- Network devices
  - Modems
  - Routers
  - Transmitters
  - Receivers
  - Switches
  - Repeaters
  - Hubs

  - Nics
  - Bridges
- Transmission signals (signals; analogue, digital, modulation)
  - data signals
  - Transmission of data signals
  - analogue signals
  - digital signals
  - modulation
- Transmission media
  - Physical
  - copper cables
  - optic fibre cables
  - wireless
- Transmission techniques
  - Parallel and serial data transmission techniques
- Transmission modes
  - Simplex
  - Half duplex
  - Full duplex

BASIC TERMINOLOGIES

Communication

- Process of using sounds, words, symbols, signs, pictures or signals to pass a message or information from one person to the other
- Message origin is called source or sender
- Message target recipient is called receiver
- The message is usually targeted for sending to the receiver

Data Communication

- Data communication refers to the process of transmitting data signals from one point to the other through communication channel

Telecommunication

- Telecommunication the use of technology to exchange messages in form of data and information over wired or wireless communication media.
- Telecommunication technologies include telegraph, telephone, Radio, television and computers

Telecommunication network

Telecommunication network is interconnection of telecommunication equipment like telephone, mobiles, radios, televisions and computers using transmission media or links

The network enables the flow of data or information from the source to destination

Computer network

A computer network is an interconnection of computers using transmission media and networking devices to enable exchange of data and information.

Information and Communication Technology (ICT)

ICT means the convergence of computer networks with telecommunication networks like telephone, mobiles, radios and televisions to provide a communication platform through which people can share information.

History of communication technologies

- Evolution has not spared communication technologies in the following aspects

  i) telephone

  ii) cellular

  iii) internet

  history of Telephone

- Telephone was inverted after telegraph by Alexander Graham who was trying to improve telegraph in 1876
- Telegraph used dots and dashes (beeps) coded as electric signals to transmit text characters over long distances
- The following were the generalised categories of telephone evolution

  i) Rotary dial telephone
  ii) Touch tone dial telephone
  iii) Mobile telephone
  iv) Smart phones

Rotary Dialing

- The first rotary dial was invented in 1896.
- Prior to that, telephone owners would have to push a button on their telephone the required number of pulses by tapping in order to call a number.
- Understandably, the rotary dial was seen as a superior alternative to this system. By 1943, the last button tapping telephone had been phased out.
- Rotary dials worked by generating pulses in a certain frequency range based on where the rotary dial turned.

Touch Tone Phones

- The first touch tone phone was invented in 1941.
- These phones used tones in the voice frequency range – much different from the pulses generated by rotary dials.
- You pressed the buttons on the phone to make a phone call.

Cordless Phones

- Cordless phones started to hit the market in the 1970s. In 1986, the FTC had released the frequency range between 47 and 49 MHz for use by cordless phones.
- This wider frequency range meant phones could work wirelessly with less interference and less power required in order to run.
- As cordless phones became more and more popular, the FTC would eventually grant more and more frequency range to cordless phones over the years.

The First Cell Phones

- Cell phones have obviously exploded with growth over the past 20-odd years.
- But the first cell phone dates back to post-World War II America.
- In 1947, researchers began theorizing that a mobile telephone was possible.

Smart phones

- The first smartphone, created by IBM, was invented in 1992 and released for purchase in 1994. It was called the Simon Personal Communicator (SPC).
- While not very compact and sleek, the device still featured several elements that became staples to every smartphone that followed.

The following are some of the other key features of a smartphone:

- Internet connectivity.
- A **mobile** browser.
- The ability to sync more than one email account to a device.
- Embedded memory.
- A hardware or software-based QWERTY keyboard.
- Wireless synchronization with other devices, such as laptop or desktop computers.

Uses of networks

Computer networks have become invaluable to organizations as well as individuals. Some of its main uses are as follows −

i. **Information and Resource Sharing**

- Computer networks allow organizations having units which are placed apart from each other, to share information in a very effective manner.
- Programs and software in any computer can be accessed by other computers linked to the network. It also allows sharing of hardware equipment, like printers and scanners among varied users.

ii. **Retrieving Remote Information**

- Through computer networks, users can retrieve remote information on a variety of topics.

- The information is stored in remote databases to which the user gains access through information systems like the World Wide Web.

iii. **Speedy Interpersonal Communication**

- Computer networks have increased the speed and volume of communication like never before. Electronic Mail (email) is extensively used for sending texts, documents, images, and videos across the globe.
- Online communications have increased by manifold times through social networking services.

iv. **E-Commerce**

- Computer networks have paved way for a variety of business and commercial transactions online, popularly called e-commerce.
- Users and organizations can pool funds, buy or sell items, pay bills, manage bank accounts, pay taxes, transfer funds and handle investments electronically.

v. **Highly Reliable Systems**

- Computer networks allow systems to be distributed in nature, by the virtue of which data is stored in multiple sources.
- This makes the system highly reliable. If a failure occurs in one source, then the system will still continue to function and data will still be available from the other sources.

vi. **Cost–Effective Systems**

- Computer networks have reduced the cost of establishment of computer systems in organizations.
- Previously, it was imperative for organizations to set up expensive mainframes for computation and storage.
- With the advent of networks, it is sufficient to set up interconnected personal computers (PCs) for the same purpose.

vii. **VoIP**

- VoIP or Voice over Internet protocol has revolutionized telecommunication systems
- Through this, telephone calls are made digitally using Internet Protocols instead of the regular analog phone lines.

Elements of communication systems

- Several components are involves in any communication system
- These include sender, transmitter, Channel, Receiver, Message user, protocol.

Sender

- This is a pwrson who creaytes the message

Transmitter

- This is a device that sends a message

Channel

- This is method how a message is transmitted
- Can also be a medium through which a message passes

Receiver

- This is a device that receives the message

  Message user

- This is a person who uses received message

  Protocol

- These are set of rules and procedures that govern communication on the network

Network devices

- These are Hardware devices that are used to connect computers, printers, fax machines and other electronic devices to a network .
- These devices transfer data in a fast, secure and correct way over same or different networks.
- Network devices may be **inter-network** or **intra-network.**
- Some devices are installed on the device, like NIC card or RJ45 connector, whereas some are part of the network, like router, switch, etc. Let us explore some of these devices in greater detail.

Modem

- Modem is a device that enables a computer to send or receive data over telephone or cable lines.
- The data stored on the computer is digital whereas a telephone line or cable wire can transmit only analog data.

Digital Data Waveform

Analog Data Waveform

- The main function of the modem is to convert digital signal into analog and vice versa.
- Modem is a combination of two devices − **modulator** and **demodulator**.
- The **modulator** converts digital data into analog data when the data is being sent by the computer.
- The **demodulator** converts analog data signals into digital data when it is being received by the computer.

Types of Modem

- Modem can be categorized in several ways like direction in which it can transmit data, type of connection to the transmission line, transmission mode, etc.
- Depending on direction of data transmission, modem can be of these types −

i. **Simplex** − A simplex modem can transfer data in only one direction, from digital device to network (modulator) or network to digital device (demodulator).

ii. **Half duplex** − A half-duplex modem has the capacity to transfer data in both the directions but only one at a time.

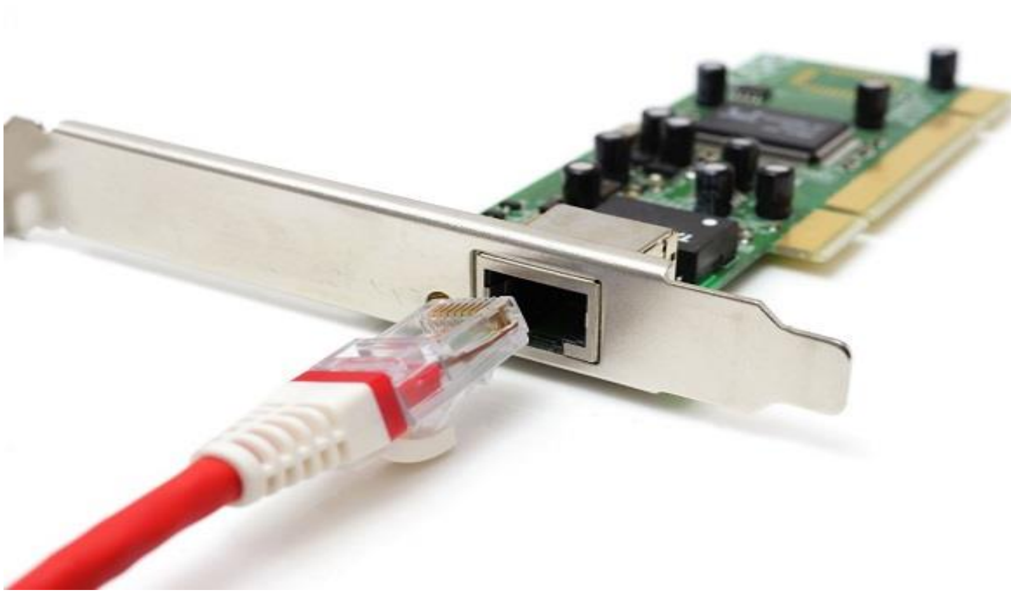
iii. **Full duplex** − A full duplex modem can transmit data in both the directions simultaneously.

RJ45 Connector

- RJ45 is the acronym for **Registered Jack 45. RJ45 connector** is an 8-pin jack used by devices to physically connect to **Ethernet** based **local area networks (LANs)**.
- **Ethernet** is a technology that defines protocols for establishing a LAN.
- The cable used for Ethernet LANs are twisted pair ones and have **RJ45 connector pins** at both ends.
- These pins go into the corresponding socket on devices and connect the device to the network.

Ethernet Card

- **Ethernet card**, also known as **network interface card (NIC)**, is a hardware component used by computers to connect to **Ethernet LAN** and communicate with other devices on the LAN.
- The earliest **Ethernet cards** were external to the system and needed to be installed manually.
- In modern computer systems, it is an internal hardware component.
- The NIC has **RJ45 socket** where network cable is physically plugged in.



- **Ethernet card speeds** may vary depending upon the protocols it supports.
- Old Ethernet cards had maximum speed of **10 Mbps**.
- However, modern cards support fast Ethernets up to a speed of **100 Mbps**.
- Some cards even have capacity of **1 Gbps**.

Router

- A **router** is a **network layer** hardware device that transmits data from one LAN to another if both networks support the same set of protocols.
- So a **router** is typically connected to at least two LANs and the **internet service provider** (ISP).
- It receives its data in the form of **packets**, which are **data frames** with their **destination address** added.
- Router also strengthens the signals before transmitting them. That is why it is also called **repeater**.

Routing Table

- A router reads its routing table to decide the best available route the packet can take to reach its destination quickly and accurately.
- The routing table may be of these two types

i. **Static**

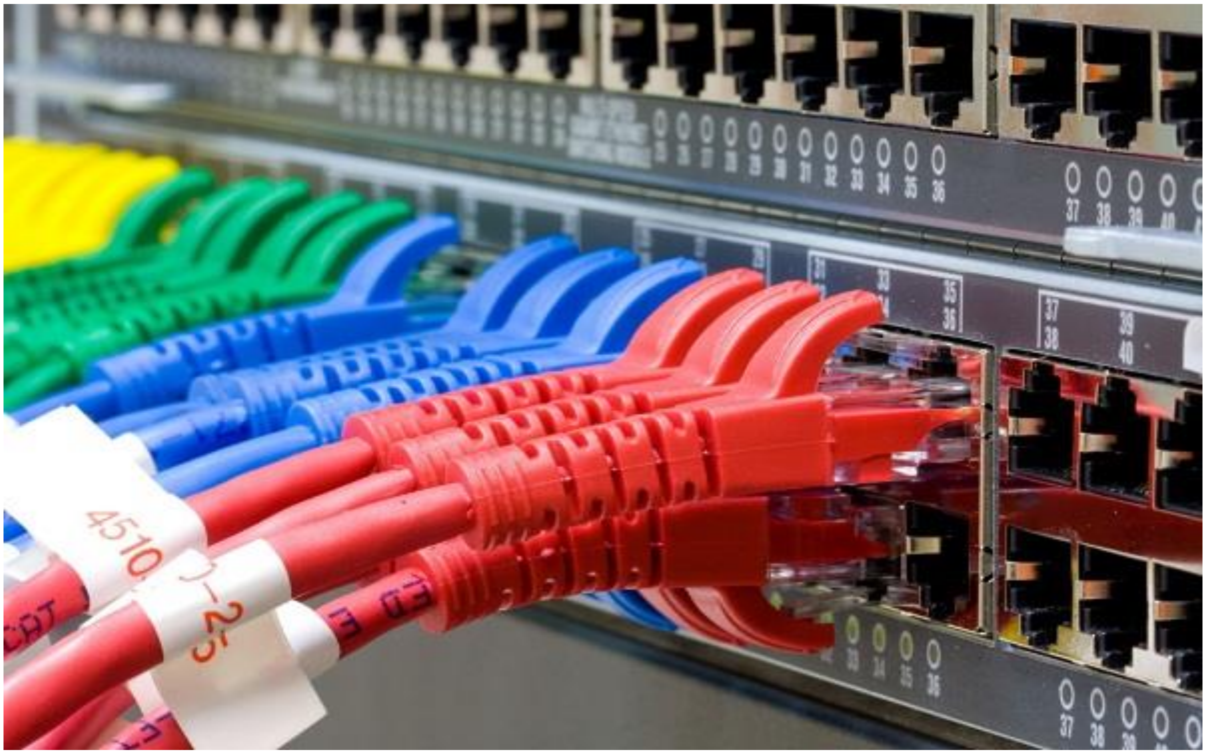
- In a static routing table the routes are fed manually. So it is suitable only for very small networks that have maximum two to three routers.

ii. **Dynamic**

- In a dynamic routing table, the router communicates with other routers through protocols to determine which routes are free.
- This is suited for larger networks where manual feeding may not be feasible due to large number of routers.

Switch

- **Switch** is a network device that connects other devices to **Ethernet** networks through **twisted pair** cables.
- It uses **packet switching** technique to **receive, store** and **forward data packets** on the network.
- The switch maintains a list of network addresses of all the devices connected to it.
- On receiving a packet, it checks the destination address and transmits the packet to the correct port.
- Before forwarding, the packets are checked for collision and other network errors. The data is transmitted in full duplex mode



- Data transmission speed in switches can be double that of other network devices like hubs used for networking.
- This is because switch shares its maximum speed with all the devices connected to it. This helps in maintaining network speed even during high traffic. In fact, higher data speeds are achieved on networks through use of multiple switches.

Gateway

- **Gateway** is a network device used to connect two or more dissimilar networks.
- In networking parlance, networks that use different protocols are **dissimilar networks**.

- A gateway usually is a computer with multiple **NICs** connected to different networks.
- A gateway can also be configured completely using software. As networks connect to a different network through gateways, these gateways are usually hosts or end points of the network.

- **Gateway** uses **packet switching** technique to transmit data from one network to another. In this way it is similar to a **router**, the only difference being router can transmit data only over networks that use same protocols.

Wi-Fi Card

- **Wi-Fi** is the acronym for **wireless fidelity.**
- **Wi-Fi technology** is used to achieve **wireless connection** to any network.
- **Wi-Fi card** is a **card** used to connect any device to the local network wirelessly.

- The physical area of the network which provides internet access through Wi-Fi is called **Wi-Fi hotspot**.
- Hotspots can be set up at home, office or any public space. Hotspots themselves are connected to the network through wires.

- A **Wi-Fi card** is used to add capabilities like **teleconferencing, downloading** digital camera images, **video chat**, etc. to old devices. Modern devices come with their in-built **wireless network adapter**.

Repeater

- A repeater operates at the physical layer.

- Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted so as to extend the length to which the signal can be transmitted over the same network.
- important point to be noted about repeaters is that they do not amplify the signal.
- When the signal becomes weak, they copy the signal bit by bit and regenerate it at the original strength.
- It is a 2 port device.

Hub

- A hub is basically a multiport repeater.
- A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations.
- Hubs cannot filter data, so data packets are sent to all connected devices. In other words, collision domain of all hosts connected through Hub remains one. Also, they do not have intelligence to find out best path for data packets which leads to inefficiencies and wastage.

Types of Hub

***i. Active Hub*:**

- These are the hubs which have their own power supply and can clean, boost and relay the signal along with the network.
- It serves both as a repeater as well as wiring centre.
- These are used to extend the maximum distance between nodes.

***ii. Passive Hub* :**

- These are the hubs which collect wiring from nodes and power supply from active hub
- These hubs relay signals onto the network without cleaning and boosting them and can’t be used to extend the distance between nodes.

Bridge

- A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of source and destination.
- It is also used for interconnecting two LANs working on the same protocol.
- It has a single input and single output port, thus making it a 2 port device.

TRANSMISSION SIGNALS

- Data Signal refers to voltage level that represent flow of data or information from one point to another in a network
- When data is sent over physical medium, it needs to be first converted into electromagnetic signals.
- Signals can either be Digital or analog

Digital signals

- These are discrete in nature and represent sequence of voltage pulses.
- Digital signals are used within the circuitry of a computer system.

Analog Signals

- Analog signals are in continuous wave form in nature and represented by continuous electromagnetic waves.

NB :

- Data itself can be analog such as human voice, or digital such as file on the disk.
- Both analog and digital data can be represented in digital or analog signals.

Transmission Impairment

- When signals travel through the medium, they tend to deteriorate.
- The following are causes of signal impairment

i. Attenuation

- For the receiver to interpret the data accurately, the signal must be sufficiently strong. When the signal passes through the medium, it tends to get weaker.
- As it covers distance, it loses strength.

ii. Dispersion

- As signal travels through the media, it tends to spread and overlaps.
- The amount of dispersion depends upon the frequency used.

iii. Delay distortion

- Signals are sent over media with pre-defined speed and frequency.
- If the signal speed and frequency do not match, there are possibilities that signal reaches destination in arbitrary fashion.
- In digital media, this is very critical that some bits reach earlier than the previously sent ones.

iv. Noise

- Random disturbance or fluctuation in analog or digital signal is said to be Noise in signal, which may distort the actual information being carried.
- Noise can be characterized in one of the following class

Thermal Noise

- Heat agitates the electronic conductors of a medium which may introduce noise in the media. Up to a certain level, thermal noise is unavoidable.

Intermodulation

- When multiple frequencies share a medium, their interference can cause noise in the medium.
- Intermodulation noise occurs if two different frequencies are sharing a medium and one of them has excessive strength or the component itself is not functioning properly, then the resultant frequency may not be delivered as expected.

v. Crosstalk

- This sort of noise happens when a foreign signal enters into the media.
- This is because signal in one medium affects the signal of second medium.

vi. Impulse

- This noise is introduced because of irregular disturbances such as lightening, electricity, short-circuit, or faulty components.
- Digital data is mostly affected by this sort of noise.

TRANSMISSION MEDIA: PHYSICAL ;COPPER CABLES, OPTIC FIBRE CABLES; WIRELESS)

- Transmission media which is also called **transmission medium** is the means through which we send our data from one place to another.
- Transmission media is categorized into two main types

  a. Wired (bounded) /Guided media
  b. Wireless (unbounded)/ Unguided Media

Wired (Bounded) /Guided midea

- Guided media, which are those that provide a conduit from one device to another, include i. **Twisted-Pair Cable**,

  ii **Coaxial Cable**, and

  iii. **Fibre-Optic Cable**.

- A signal travelling along any of these media is directed and contained by the physical limits of the medium.
- Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current.
- **Optical fibre** is a cable that accepts and transports signals in the form of light.

Twisted Pair Cable

- This cable is the most commonly used and is cheaper than others.

- It is lightweight, cheap, can be installed easily, and they support many different types of network.

NB Some important points :

- Its frequency range is 0 to 3.5 kHz.
- Typical attenuation is 0.2 dB/Km @ 1kHz.
- Typical delay is 50 µs/km.
- Repeater spacing is 2km.

- A twisted pair consists of two conductors(normally copper), each with its own plastic insulation, twisted together.
- One of these wires is used to carry signals to the receiver, and the other is used only as ground reference.

  The receiver uses the difference between the two.

- In addition to the signal sent by the sender on one of the wires, interference(noise) and crosstalk may affect both wires and create unwanted signals.
- If the two wires are parallel, the effect of these unwanted signals is not the same in both wires because they are at different locations relative to the noise or crosstalk sources. This results in a difference at the receiver.

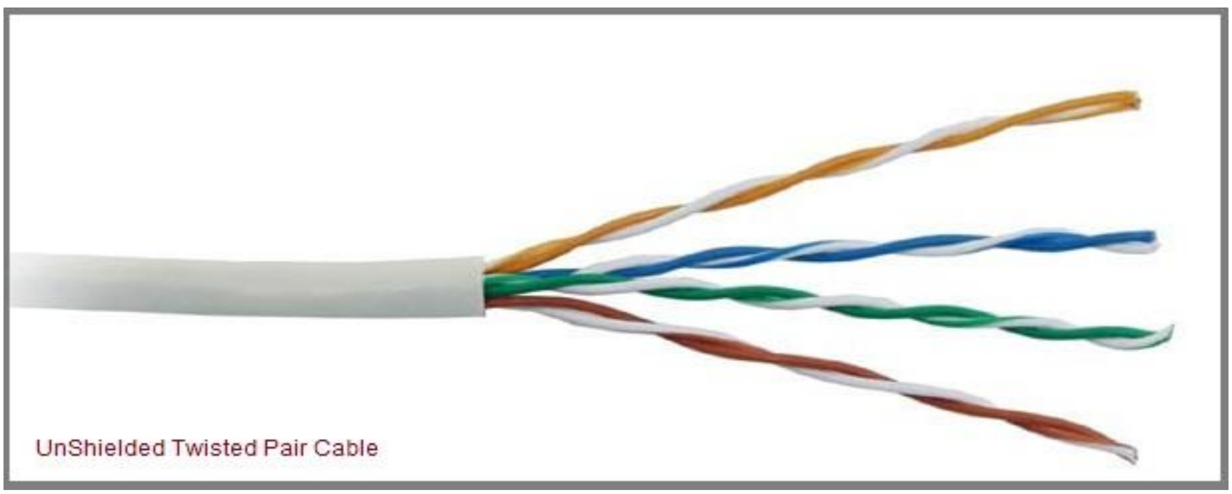
Twisted Pair is of two types:

i. Unshielded Twisted Pair (UTP)

ii. Shielded Twisted Pair (STP)

Unshielded Twisted Pair Cable (UTP)

- It is the most common type of telecommunication when compared with Shielded Twisted
- Pair Cable which consists of two conductors usually copper, each with its own colour plastic insulator.
- Identification is the reason behind coloured plastic insulation.
- UTP cables consist of 2 or 4 pairs of twisted cable.
- Cable with 2 pair use **RJ-11** connector and 4 pair cable use **RJ-45** connector.



Advantages of Unshielded Twisted Pair Cable

- Installation is easy
- Flexible
- Cheap

- It has high speed capacity,
- 100 meter limit
- Higher grades of UTP are used in LAN technologies like Ethernet.
- It consists of two insulating copper wires (1mm thick). The wires are twisted together in a helical form to reduce electrical interference from similar pair.

Disadvantages of Unshielded Twisted Pair Cable

- Bandwidth is low when compared with Coaxial Cable
- Provides less protection from interference.

Shielded Twisted Pair Cable

- This cable has a metal foil or braided-mesh covering which encases each pair of insulated conductors.
- Electromagnetic noise penetration is prevented by metal casing.
- Shielding also eliminates crosstalk .
- It has same attenuation as unshielded twisted pair.
- It is faster the unshielded and coaxial cable. It is more expensive than coaxial and unshielded twisted pair.

Advantages of Shielded Twisted Pair Cable

- Easy to install
- Performance is adequate
- Can be used for Analog or Digital transmission
- Increases the signalling rate
- Higher capacity than unshielded twisted pair
- Eliminates crosstalk

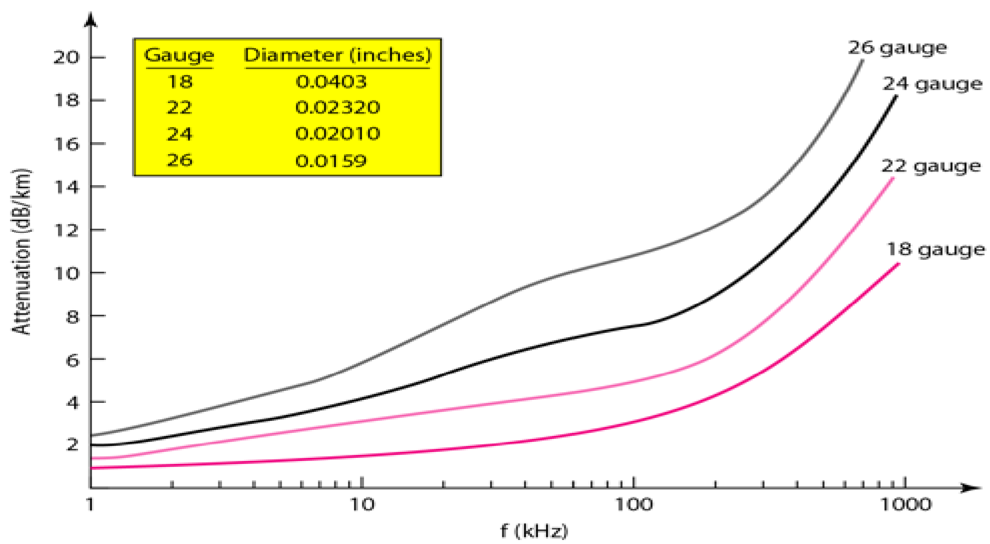
Disadvantages of Shielded Twisted Pair Cable

- Difficult to manufacture
- Heavy

Performance of Shielded Twisted Pair Cable

- One way to measure the performance of twisted-pair cable is to compare attenuation versus frequency and distance.

- As shown in the below figure, a twisted-pair cable can pass a wide range of frequencies. However, with increasing frequency, the attenuation, measured in decibels per kilometre (dB/km), sharply increases with frequencies above 100kHz. Note that gauge is a measure of the thickness of the wire.



Applications of Shielded Twisted Pair Cable

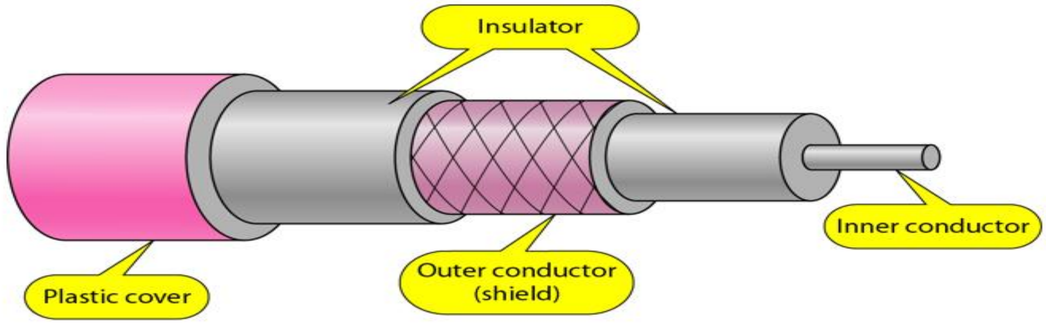
- In telephone lines to provide voice and data channels. The DSL lines that are used by the telephone companies to provide high-data-rate connections also use the high-bandwidth capability of unshielded twisted-pair cables.
- Local Area Network, such as 10Base-T and 100Base-T, also use twisted-pair cables.

Coaxial Cable

- Coaxial is called by this name because it contains two conductors that are parallel to each other.
- Copper is used in this as centre conductor which can be a solid wire or a standard one.
- It is surrounded by PVC installation, a sheath which is encased in an outer conductor of metal foil, barid or both.
- Outer metallic wrapping is used as a shield against noise and as the second conductor which completes the circuit.
- The outer conductor is also encased in an insulating sheath. The outermost part is the plastic cover which protects the whole cable.

Here the most common coaxial standards.

- 50-Ohm RG-7 or RG-11 : used with thick Ethernet.
- 50-Ohm RG-58 : used with thin Ethernet
- 75-Ohm RG-59 : used with cable television
- 93-Ohm RG-62 : used with ARCNET.



Coaxial Cable Standards

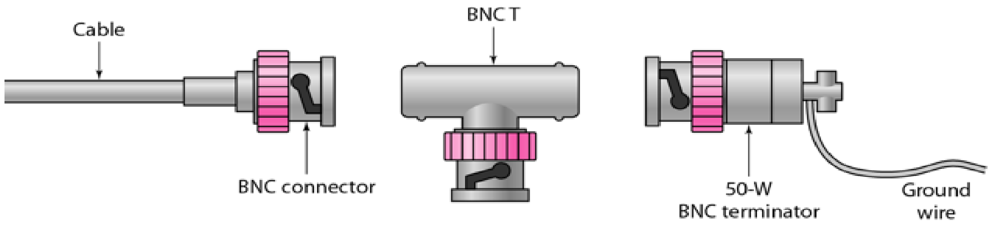
- Coaxial cables are categorized by their **Radio Government**(RG) ratings.
- Each RG number denotes a unique set of physical specifications, including the wire gauge of the inner conductor, the thickness and the type of the inner insulator, the construction of the shield, and the size and type of the outer casing.
- Each cable defined by an RG rating is adapted for a specialized function, as shown in the table below:

Category	*Impedance*	*Use*
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Coaxial Cable Connectors

- To connect coaxial cable to devices, we need coaxial connectors.
- The most common type of connector used today is the Bayonet Neill-Concelman (BNC) connector.
- The below figure shows 3 popular types of these connectors:

  i. The BNC Connector,
  ii. The BNC T connector and
  iii. The BNC terminator.



- The **BNC connector** is used to connect the end of the cable to the device, such as a TV set.
- The **BNC T connector** is used in Ethernet networks to branch out to a connection to a computer or other device.
- The **BNC terminator** is used at the end of the cable to prevent the reflection of the signal.

There are two types of Coaxial cables: **Baseband** and **broadband**

1. BaseBand

- This is a 50 ohm (Ω) coaxial cable which is used for digital transmission.
- It is mostly used for LAN's.
- Baseband transmits a single signal at a time with very high speed.
- The major drawback is that it needs amplification after every 1000 feet.

2. BroadBand

- This uses analog transmission on standard cable television cabling.
- It transmits several simultaneous signal using different frequencies.
- It covers large area when compared with Baseband Coaxial Cable.

Advantages of Coaxial Cable

- Bandwidth is high
- Used in long distance telephone lines.
- Transmits digital signals at a very high rate of 10Mbps.
- Much higher noise immunity
- Data transmission without distortion.
- The can span to longer distance at higher speeds as they have better shielding when compared to twisted pair cable

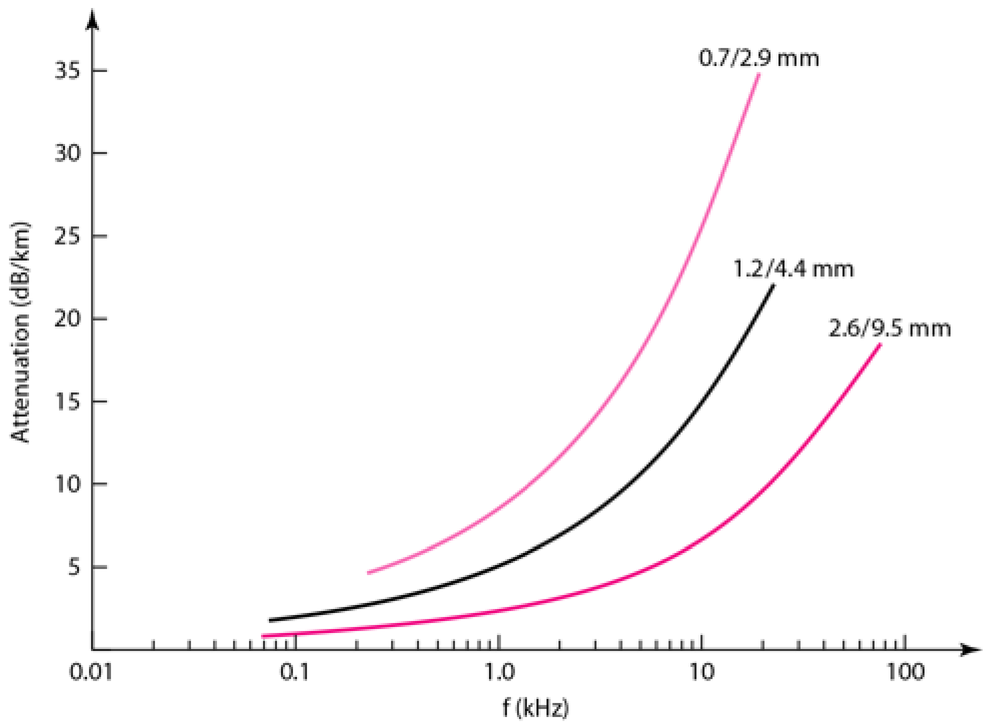
Disadvantages of Coaxial Cable

- Single cable failure can fail the entire network.
- Difficult to install and expensive when compared with twisted pair.
- If the shield is imperfect, it can lead to grounded loop.

Performance of Coaxial Cable

- We can measure the performance of a coaxial cable in same way as that of Twisted Pair Cables.

- From the below figure, it can be seen that the attenuation is much higher in coaxial cable than in twisted-pair cable.
- In other words, although coaxial cable has a much higher bandwidth, the signal weakens rapidly and requires the frequent use of repeaters.



Applications of Coaxial Cable

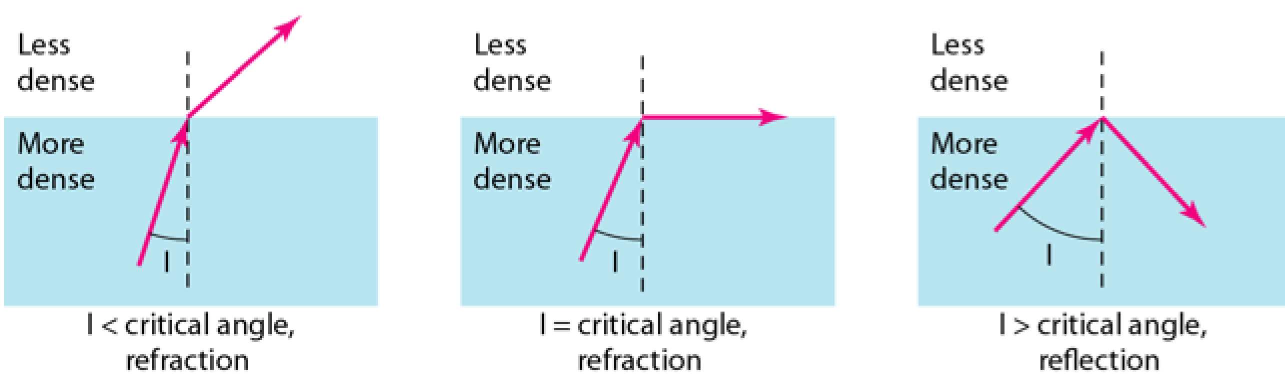
- Coaxial cable was widely used in analog telephone networks, where a single coaxial network could carry 10,000 voice signals.
- Cable TV networks also use coaxial cables. In the traditional cable TV network, the entire network used coaxial cable. Cable TV uses RG-59 coaxial cable.
- In traditional Ethernet LANs. Because of it high bandwidth, and consequence high data rate, coaxial cable was chosen for digital transmission in early Ethernet LANs. The 10Base-2, or Thin Ethernet, uses

RG-58 coaxial cable with BNC connectors to transmit data at 10Mbps with a range of 185 m.

FIBER OPTIC CABLE

- A fibre-optic cable is made of glass or plastic and transmits signals in the form of light.
- For better understanding we first need to explore several aspects of the **nature of light**.
- Light travels in a straight line as long as it is mobbing through a single uniform substance.
- If ray of light travelling through one substance suddenly enters another substance (of a different density), the ray changes direction.

The below figure shows how a ray of light changes direction when going from a more dense to a less dense substance.



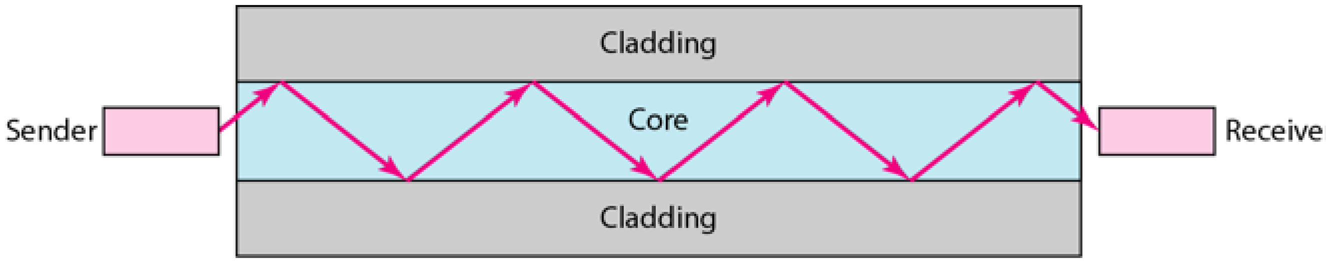
Bending of a light ray

As the figure shows:

If the **angle of incidence I** (the angle the ray makes with the line perpendicular to the interface between the two substances) is **less** than the **critical angle**, the ray **refracts** and moves closer to the surface.
If the angle of incidence is **greater** than the critical angle, the ray **reflects**(makes a turn) and travels again in the denser substance

If the angle of incidence is **equal** to the critical angle, the ray refracts and **moves parallel** to the surface as shown
Note: *The critical angle is a property of the substance, and its value differs from one substance to another.*

- Optical fibres use reflection to guide light through a channel. A glass or plastic core is surrounded by a cladding of less dense glass or plastic. The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it.

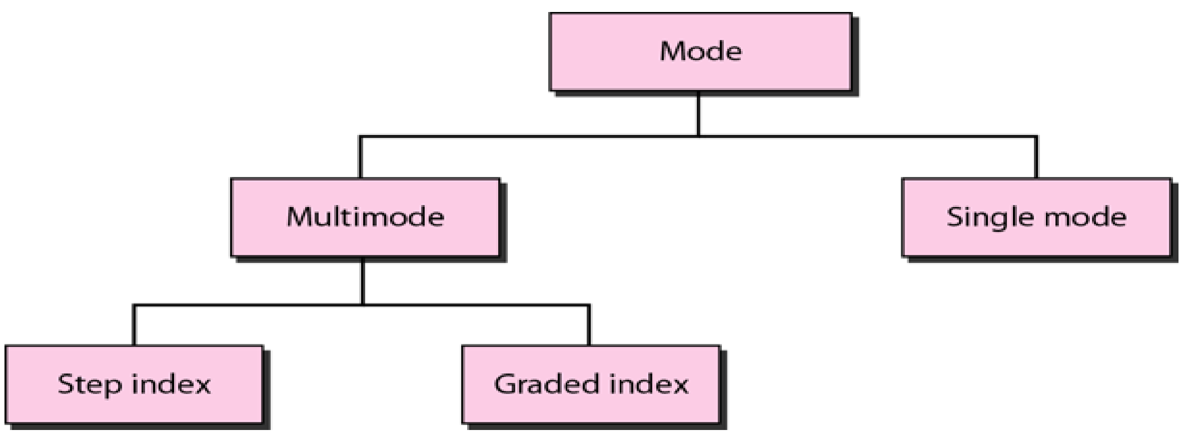


Propagation Modes of Fiber Optic Cable

- Current technology supports two modes

    i. Multimode and
    ii. Single mode

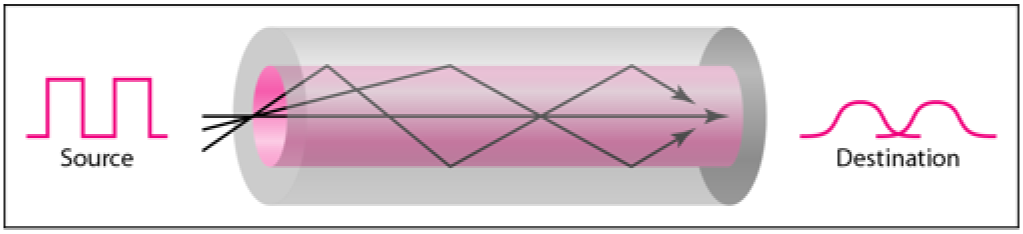
    Multimode can be implemented in two forms: **Step-index** and **Graded-index**.



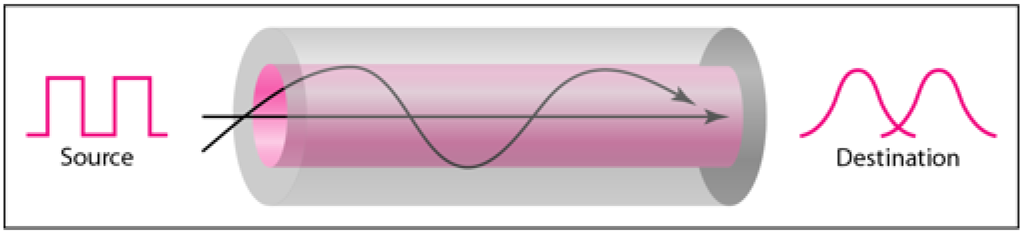
Multimode Propagation Mode

vii. Multimode is so named because multiple beams from a light source move through the core in different paths.

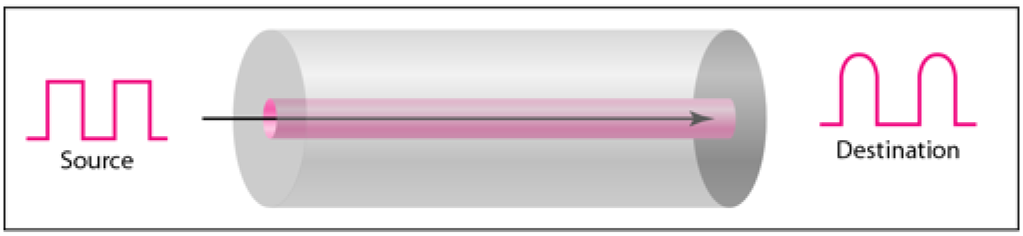
viii. How these beams move within the cable depends on the structure of the core as shown in the figure below



a. Multimode, step index



b. Multimode, graded index



c. Single mode

In **multimode step-index fibre**,

- The density of the core remains constant from the centre to the edges.
- A beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding.
- The term **step-index** refers to the suddenness of this change, which contributes to the distortion of the signal as it passes through the fibre.

• In **multimode graded-index fibre**,

- This distortion gets decreases through the cable.
- The word **index** here refers to the index of refraction.
- This index of refraction is related to the density.

- A graded-index fibre, therefore, is one with varying densities.
- Density is highest at the centre of the core and decreases gradually to its lowest at the edge.

Single Mode

- **Single mode** uses **step-index fibre** and a highly focused source of light that limits beams to a small range of angles, all close to the horizontal.
- The single-mode fibre itself is manufactured with a much smaller diameter than that of multimode fibre, and with substantially lower density.
- The decrease in density results in a critical angle that is close enough to 90 degree to make the propagation of beams almost horizontal.

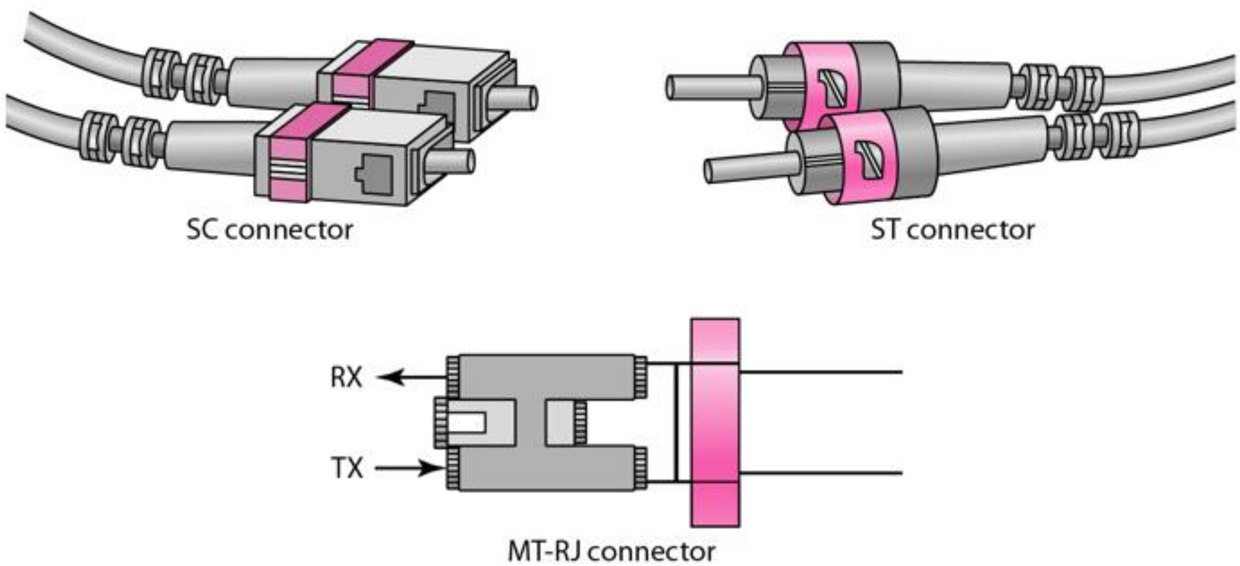
Fibre Sizes for Fiber Optic Cable

- Optical fibres are defined by the ratio of the diameter or their core to the diameter of their cladding, both expressed in micrometers.
- The common sizes are shown in the figure below

Type	*Core (μm)*	*Cladding (μm)*	*Mode*
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

Fibre Optic Cable Connectors

There are three types of connectors for fibre-optic cables, as shown in the figure below.



- The **Subscriber Channel(SC)** connector is used for cable TV. It uses push/pull locking system.
- The **Straight-Tip(ST)** connector is used for connecting cable to the networking devices.
- The **MT-RJ** is a connector that is the same size as RJ45.

Advantages of Fibre Optic Cable

Fibre optic has several advantages over metallic cable:

- Higher bandwidth
- Less signal attenuation
- Immunity to electromagnetic interference

- Resistance to corrosive materials
- Light weight
- Greater immunity to tapping

Disadvantages of Fibre Optic Cable

There are some disadvantages in the use of optical fibre:

- Installation and maintenance
- Unidirectional light propagation
- High Cost

Performance of Fibre Optic Cable

- Attenuation is flatter than in the case of twisted-pair cable and coaxial cable.
- The performance is such that we need fewer(actually one tenth as many) repeaters when we use the fibre-optic cable.

Applications of Fibre Optic Cable

- Often found in backbone networks because its wide bandwidth is cost-effective.
- Some cable TV companies use a combination of optical fibre and coaxial cable thus creating a hybrid network.
- Local-area Networks such as 100Base-FX network and 1000Base-X also use fibre-optic cable.

WIRELESS TRANSMISSION MEDIA

- Wireless transmission media which are also called **Unguided medium** transport electromagnetic waves without using a physical conductor.
- Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

We can divide wireless transmission into three broad groups:

1. Radio waves
2. Micro waves
3. Infrared waves

Radio Waves

- Electromagnetic waves ranging in frequencies between 3 KHz and 1 GHz are normally called radio waves.
- Radio waves are omnidirectional. When an antenna transmits radio waves, they are propagated in all directions.
- This means that the sending and receiving antennas do not have to be aligned.
- A sending antenna send waves that can be received by any receiving antenna.
- The **omnidirectional property** has disadvantage, too. The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signal suing the same frequency or band.
- Radio waves, particularly with those of low and medium frequencies, can **penetrate walls**. This characteristic can be both an advantage and a disadvantage. It is an advantage because, an AM radio can receive signals

inside a building. It is a disadvantage because we cannot isolate a communication to just inside or outside a building.

Omnidirectional Antenna for Radio Waves

Radio waves use omnidirectional antennas that send out signals in all directions.

Applications of Radio Waves

- The omnidirectional characteristics of radio waves make them useful for multicasting in which there is one sender but many receivers.
- AM and FM radio, television, maritime radio, cordless phones, and paging are examples of multicasting.

Micro Waves

- Electromagnetic waves having frequencies between 1 and 300 GHz are called micro waves.
- Micro waves are unidirectional.
- When an antenna transmits microwaves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned.
- The **unidirectional property** has an obvious advantage. A pair of antennas can be aligned without interfering with another pair of aligned antennas.

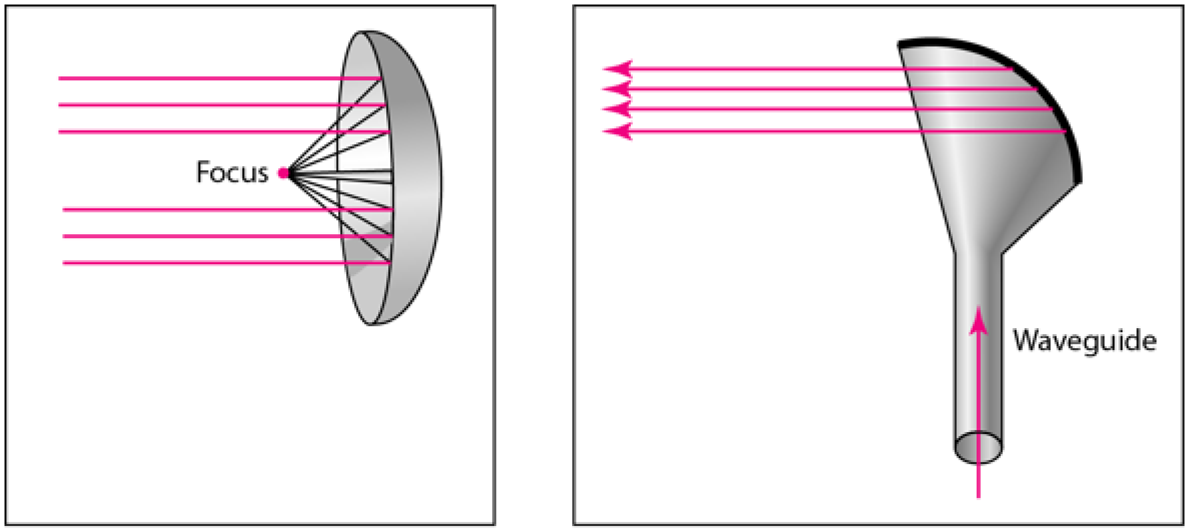
The following describes some characteristics of microwaves propagation:

- Microwave propagation is line-of-sight. Since the towers with the mounted antennas need to be in direct sight of each other, towers that are far apart need to be very tall.
- Very high-frequency microwaves cannot penetrate walls. This characteristic can be a disadvantage if receivers are inside the buildings.
- The microwave band is relatively wide, almost 299 GHz. Therefore, wider sub-bands can be assigned and a high date rate is possible.
- Use of certain portions of the band requires permission from authorities.

Unidirectional Antenna for Micro Waves

- Microwaves need unidirectional antennas that send out signals in one direction.
- Two types of antennas are used for microwave communications:

  i. **Parabolic Dish** and
  ii. **Horn**.



a. Dish antenna b. Horn antenna

- A **parabolic antenna** works as a funnel, catching a wide range of waves and directing them to a common point.
- In this way, more of the signal is recovered than would be possible with a single-point receiver.
- A **horn antenna** looks like a gigantic scoop.
- Outgoing transmissions are broadcast up a stem and deflected outward in a series of narrow parallel beams by the curved head.
- Received transmissions are collected by the scooped shape of the horn, in a manner similar to the parabolic dish, and are deflected down into the stem.

Applications of Micro Waves

Microwaves, due to their unidirectional properties, are very useful when unicast(one-to-one) communication is needed between the sender and the receiver. They are used in cellular phones, satellite networks and wireless LANs.

There are 2 types of Microwave Transmission :

1. Terrestrial Microwave
2. Satellite Microwave

Advantages of Microwave Transmission

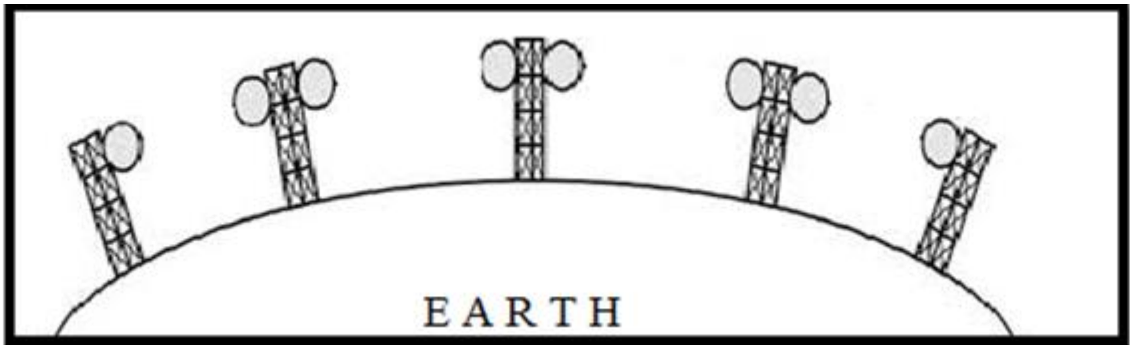
- Used for long distance telephone communication
- Carries 1000's of voice channels at the same time

Disadvantages of Microwave Transmission

- It is very costly

Terrestrial Microwave

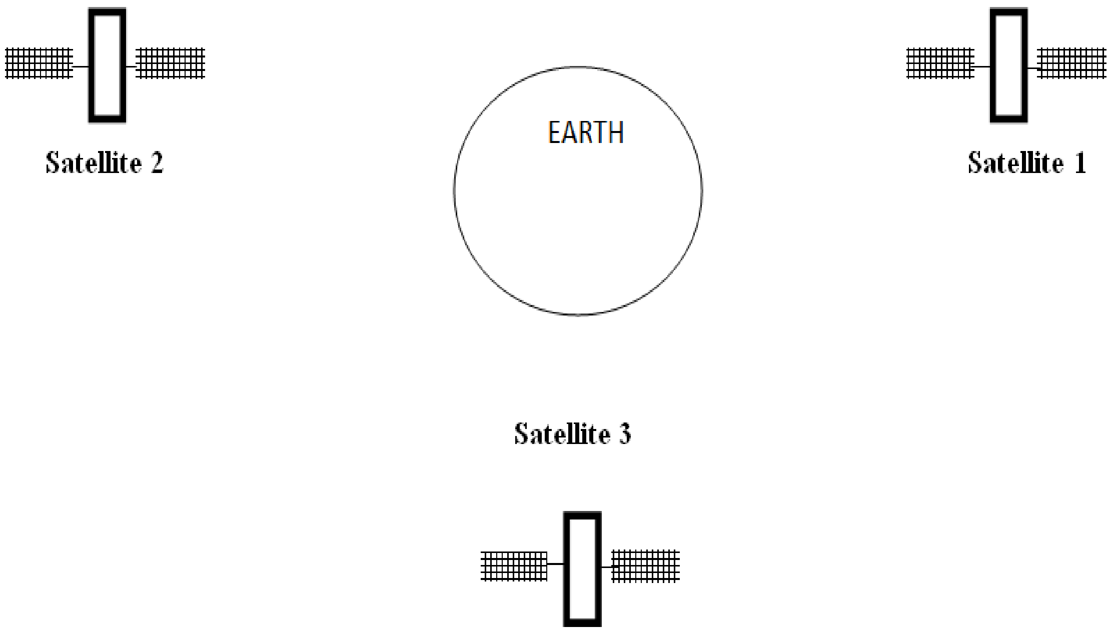
- For increasing the distance served by terrestrial microwave, repeaters can be installed with each antenna .
- The signal received by an antenna can be converted into transmittable form and relayed to next antenna as shown in below figure. It is an example of telephone systems all over the world



The microwave communication can either parabolic or horn type of antennas

Satellite Microwave

- This is a microwave relay station which is placed in outer space.
- The satellites are launched either by rockets or space shuttles carry them.
- These are positioned 36000 Km above the equator with an orbit speed that exactly matches the rotation speed of the earth.
- As the satellite is positioned in a geo-synchronous orbit, it is stationery relative to earth and always stays over the same point on the ground. This is usually done to allow ground stations to aim antenna at a fixed point in the sky.



Features of Satellite Microwave

- Bandwidth capacity depends on the frequency used.
- Satellite microwave deployment for orbiting satellite is difficult.

Advantages of Satellite Microwave

- Transmitting station can receive back its own transmission and check whether the satellite has transmitted information correctly.
- A single microwave relay station which is visible from any point.

Disadvantages of Satellite Microwave

- Satellite manufacturing cost is very high
- Cost of launching satellite is very expensive
- Transmission highly depends on whether conditions, it can go down in bad weather

Infrared Waves

- Infrared waves, with frequencies from 300 GHz to 400 THz, can be used for short-range communication.
- Infrared waves, having high frequencies, cannot penetrate walls. This advantageous characteristic prevents interference between one system and another, a short-range communication system in on room cannot be affected by another system in the next room.
- When we use infrared remote control, we do not interfere with the use of the remote by our neighbours. However, this same characteristic makes infrared signals useless for long-range communication.
- In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication.

Applications of Infrared Waves

- The infrared band, almost 400 THz, has an excellent potential for data transmission. Such a wide bandwidth can be used to transmit digital data with a very high data rate.
- The Infrared Data Association(IrDA), an association for sponsoring the use of infrared waves, has established standards for using these signals for communication between devices such as keyboards, mouse, PCs and printers.
- Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

Factors to be considered while selecting a Transmission Medium

When selecting the suitable transmission media, the following factors must be considered

1. Transmission Rate
2. Cost and Ease of Installation
3. Resistance to Environmental Conditions
4. Distances

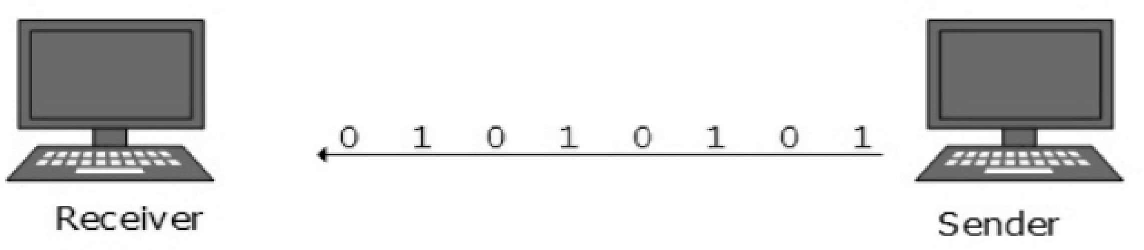
TRANSMISSION TECHNIQUES

Data signals can be transmitted using two main techniques

1. Serial data transmission
2. Parallel data transmission

Serial data transmission

- In serial data transmission, bits are sent sequentially (one after the other) down the same wire (channel)
- Using a single wire reduces costs but slows down the speed of transmission.
- Sending data sequentially is perfect for transmitting over longer distances as there are no synchronization issues.

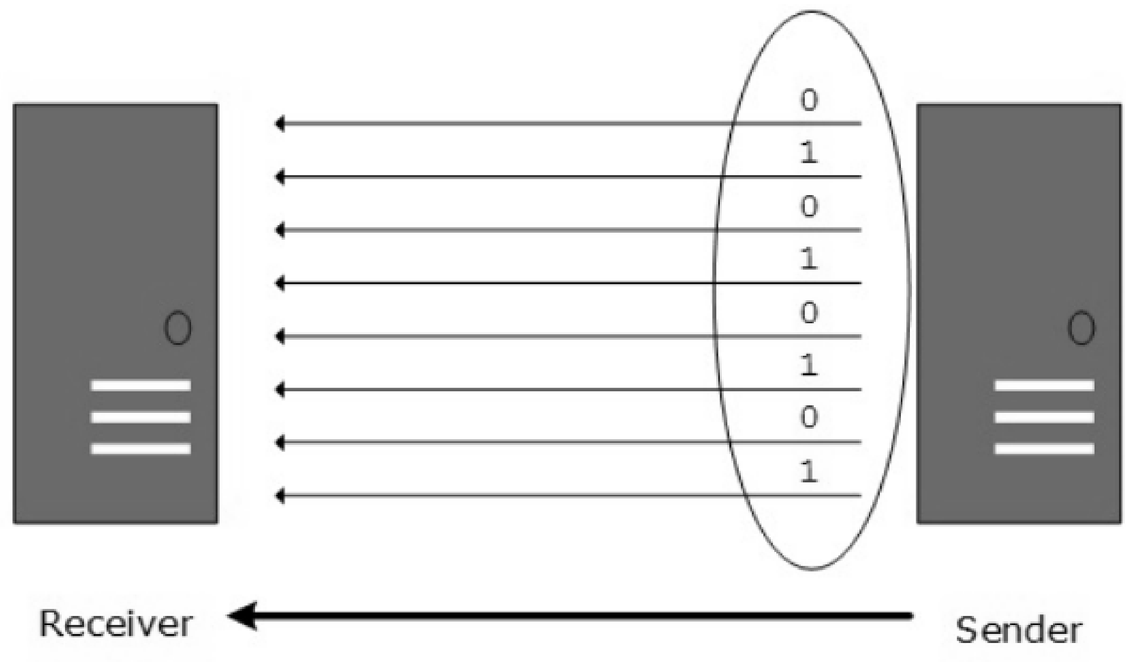


Uses of serial transmission

- Transmission to another computer or to external devices
- Medium to long distances
- Universal Serial Bus (USB)

Parallel data transmission

- In parallel data transmission, multiple bits are sent simultaneously down different wires (channels) within the same cable.
- Data is synchronized by a clock, however this becomes problematic over longer distances where synchronization errors may start to occur.
- Using parallel wires is more expensive but transmission is faster.



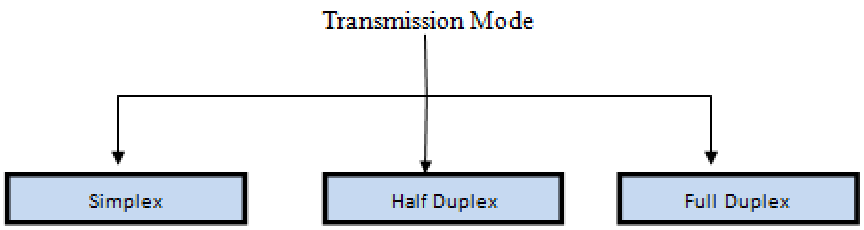
Uses of parallel transmission

- Fast transmission within a computer system
- Short distances
- Integrated Circuits (IC), Busses

TRANSMISSION MODES

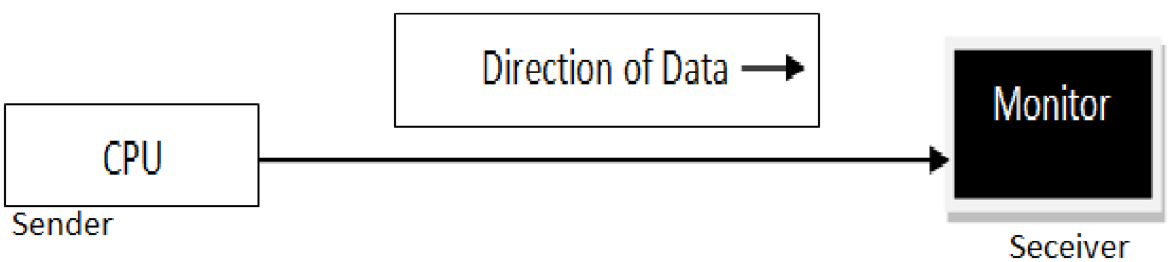
- Transmission mode refers to the mechanism of transferring of data between two devices connected over a network.
- It is also called **Communication Mode**.
- These modes direct the direction of flow of information.
- There are three types of transmission modes. They are:

1. Simplex Mode
2. Half duplex Mode
3. Full duplex Mode



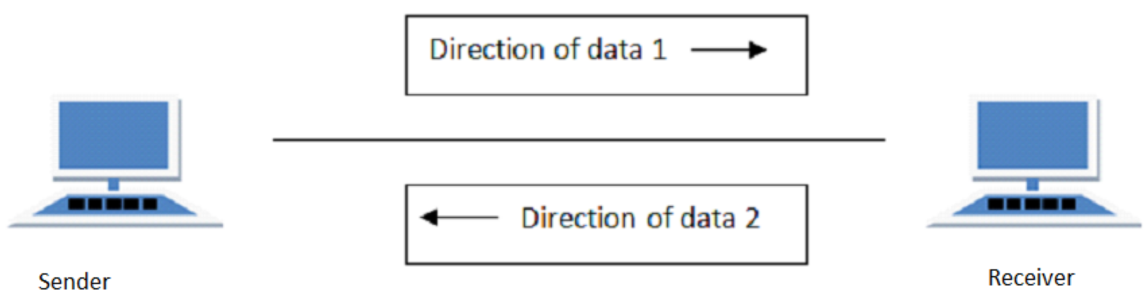
SIMPLEX Mode

- In this type of transmission mode, data can be sent only in one direction i.e. communication is unidirectional.
- We cannot send a message back to the sender.
- Unidirectional communication is done in Simplex Systems where we just need to send a command/signal, and do not expect any response back.
- Examples of simplex Mode are **loudspeakers**, **television broadcasting**, **television** and **remote**, **keyboard** and **monitor** etc.



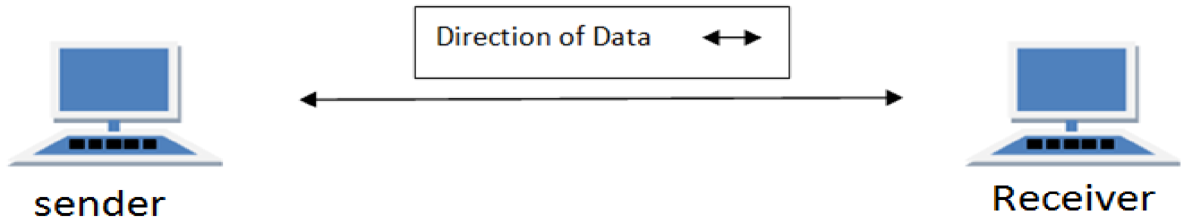
HALF DUPLEX Mode

- Half-duplex data transmission means that data can be transmitted in both directions on a signal carrier, but not at the same time.
- **For example**, on a local area network using a technology that has half-duplex transmission, one workstation can send data on the line and then immediately receive data on the line from the same direction in which data was just transmitted. Hence half-duplex transmission implies a bidirectional line (one that can carry data in both directions) but data can be sent in only one direction at a time.
- Example of half duplex is a walkie- talkie in which message is sent one at a time but messages are sent in both the directions.

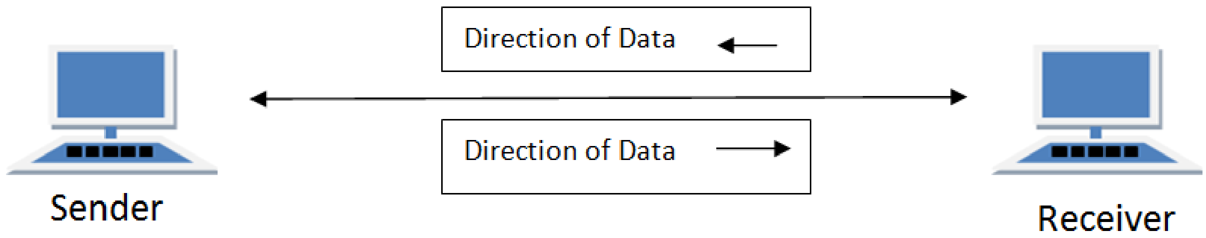


FULL DUPLEX Mode

- In full duplex system we can send data in both the directions as it is bidirectional at the same time in other words, data can be sent in both directions simultaneously.
- Example of Full Duplex is a Telephone Network in which there is communication between two persons by a telephone line, using which both can talk and listen at the same time.



In full duplex system there can be two lines one for sending the data and the other for receiving data.



CHAPTER 6: Programming Fundamentals

Objectives

By the end of this chapter, the student should be able to:

Contents

- definitions (computer program, programming, compiler, translator, assembler)
- history of programming languages (machine language, assembly language, high level languages)
- programming languages (examples: pascal, fortran, COBOL, C, C++, Java, HTML, visual basic)
- programming techniques
  - constructing a computer programme
    - programme-structure
    - compiling a computer programme
    - running a computer programme
  - program constructs
    - discussing data types
    - discussing data variables
    - modifying data variables
    - discussing how to access data types
    - key words (calculate, count)
    - input and output (read, display, print)
      - designing programs that Read from the keyboard using the Read command
      - writing to the terminal using the Write command
    - Operators (arithmetic: addition, subtraction, multiplication, division, logical: and, or, not comparison operators; equal to, not equal to, less than, greater than)
      - using arithmetic operators: + (plus), - (minus), // (forward slash, * (multiplication)

        - using Boolean and comparison operators in programming (arithmetic: Addition, Subtraction, Multiplication, Division, logical: and, or, not comparison operators; equal to, not equal to, less than, greater than)
    - Control structures (sequence, selection, loops)
        - designing programmes that use control structures
- Debugging programmes

Definitions of terms in programming

computer program

- Set of instructions written using a programming language to tell computer to perform a task
- A program is like a recipe containing a list of ingredients referred to as variables and list of instructions (statements) that instructs a computer what to do with the variables

Programming language

- A formal language that specifies syntax and semantic rules for writing a computer programs
- Examples of programming languages include Pascal, FORTRAN and COBOL

Computer programming

- This is a process of writing a computer program using programming language
- The person who writes programs is referred to as programmer

- Programmers are also called software developers or software engineers

Algorithm

- This is a design of a computer program prior to the implementation
- Algorithms are also referred to as program blue prints

Source code

- This is set of instructions written by a programmer that are not converted into machine readable form
- This is usually a text file written in programming languages like BASIC, Pascal or C++

Object code

- This is set of instructions that have been converted into machine readable form
- Assemblers, interpreters and compliers are used to convert a source code into an object code

Assembler

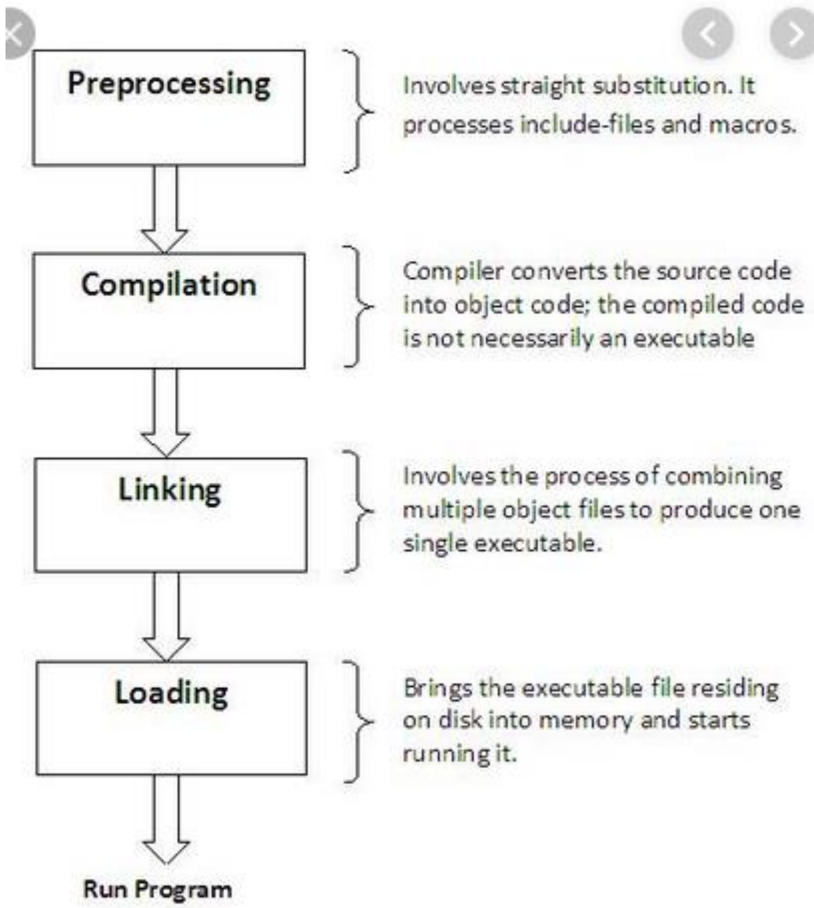
- Converts assembly language into machine readable form

Interpreters

- Converts source code into an object code statement by statement allowing CPU to execute one line at a time
- Interpreted line is not stored in the computer memory hence every time the program is needed for execution, each line has to be interpreted
- Interpreters were mostly used by early computers that did not have memory

Compiler

- Converts entire source code into object code
- The process of converting a source code into an object code by compilers is called compilation
- The following are the steps of compilation process



a. **Preprocess** – source code is prepared for the conversion
b. **Compile** – source code is changed into an object code
c. **Linking** – libraries are added to an object code to generated an executable file which is loaded into RAM and executed by CPU

Differences between interpreters and compilers

Interpreters	Compilers
1. Translate the source code to object code one statement at a time	1. Translate the entire source code at once before execution
2. Translate the program each time it is to run hence slower than compliers	3. Complied program (object code) can be saved on a storage media and run as required, hence they are faster than interpreters
Interpreted object code takes less memory compared to compiled program	Compiled programs require more me mory since object files are larger

A Brief History of Programming Languages

- Programming languages have been developed in stages called generations

1. The first generation programming languages (Machine Language

- The first generation program language is pure machine code, that is just ones and zeros, e.g.

- Programmers have to design their code by hand then transfer it to a computer by using a punch card, punch tape or flicking switches.
- There is no need to translate the code and it will run straight away. This may sound rather archaic,

Advantages of Machine language

i. Code can be fast and efficient

ii. Code can make use of specific processor features such as special registers

Drawbacks of Machine Language

i. Lack of portability

- Code cannot be ported to other systems

ii. Code is difficult to edit and update since very cumbersome to be understood by human beings

Second generation programming (Assembly Language)

- Second-generation programming languages are a way of describing Assembly code which you may have already met.
- Uses codes resembling English, programming becomes much easier.
- The usage of these mnemonic codes such as LDA for load and STA for store means the code is easier to read and write.

- To convert an assembly code program into object code to run on a computer requires an **Assembler** and each line of assembly can be replaced by the equivalent one line of object (machine) code

Assembly Code		Object Code
LDA A ADD #5 STA A JMP #3	-> Assembler ->	000100110100 001000000101 001100110100 010000000011

Advantages of second generation languages (Assembly Language)

i. Code can be fast and efficient
ii. Code can make use of specific processor features such as special registers
iii. As it is closer to plain English, it is easier to read and write when compared to machine code

Drawbacks of Assembly Language

Lack of portability

- Code cannot be ported to other systems i.e a code only runs on a machine where it was written

NB: first and second generation programming languages are referred to Low level Languages because computers can understand their code with minimal effort.

Third generation Programming Language (High Level Language)

- Also called structured programming languages
- Structured programming languages has ability to break down a program into smaller components called modules, each performing a particular task
- 3GLs are close to human language hence can be read and understood by people who are experts in programming.

  Examples of 3GLs include

  i. **Pascal**
     - Enable teaching and learning structuall programming to be easy

  ii. **FOTRAN** (**FOR**mula **TRAN**slator)
     - For writing programs that solve scientific, mathematics and engineering problems

  iii. **COBOL** : (Common Business – Oriented Language)
     - Suitable for developing programs that solve business problems

  iv. **BASIC** :(Beginners All-purpose Symbolic Instruction Code)
     - Suitable for developing business and education systems

v. **C** :

- suitable for developing operating systems

vi. Ada

- suitable for developing military, industrial and real time systems

NB Third generation (High Level Languages) codes are **imperative**.

- Imperative means that code is executed line by line, in sequence. For example:

```
dim x as integer
x = 3
dim y as integer
y = 5
x = x + y
console.writeline(x)
```

Would output: 8

Fourth generation Programming Language

- Fourth-generation languages are designed to reduce programming effort and the time it takes to develop software, resulting in a reduction in the cost of software development.
- They are not always successful in this task, sometimes resulting in inelegant and hard to maintain code.

- Examples of 4GLs includes
  i. Structured Query Language (**SQL),**
  ii. Oracle Reports
  iii. CSS

An example of code written using SQL is as follows

```
--an example of a Structured Query Language (SQL) to select criminal details from a
database
SELECT name, height, DoB FROM criminals WHERE numScars = 7;
```

Scripting Languages

- A scripting language is a programming language designed for integrating and communicating with other programming languages.
- Some of the most widely used scripting languages are
  i. JavaScript,
  ii. VBScript,
  iii. PHP,
  iv. Perl,
  v. Python,
  vi. Ruby,
  vii. ASP and
  viii. Tcl.

- Since a scripting language is normally used in conjunction with another programming language, they are often found alongside HTML, Java or C++.

HTML (HyperText Markup Language)

- is the standard markup language for creating Web pages.
- HTML stands for Hyper Text Markup Language
- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements tell the browser how to display the content
- HTML elements are represented by tags
- HTML tags label pieces of content such as "heading", "paragraph", "table", and so
- Browsers do not display the HTML tags, but use them to render the content of the page

Components of HTML code

Consider the following HTML code

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

Code Explanation

- The **<!DOCTYPE html**> declaration defines this document to be HTML5
- The **<html>** element is the root element of an HTML page
- The **<head>** element contains meta information about the document
- The **<title>** element specifies a title for the document

- The **<body>** element contains the visible page content
- The **<h1>** element defines a large heading
- The **<p>** element defines a paragraph

Advantages of High level language

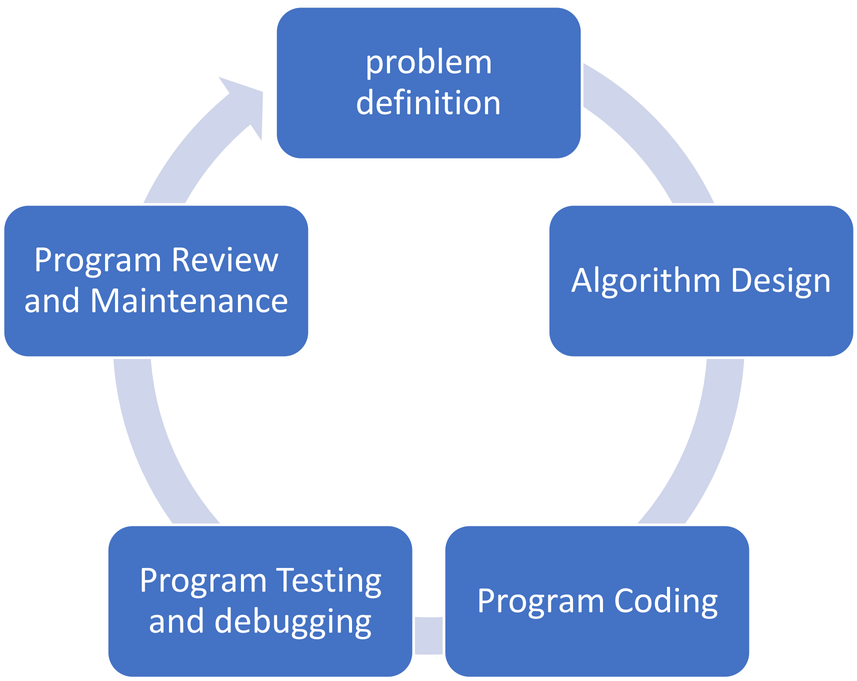
1. High level languages are programmer friendly. They are easy to write, debug and maintain.
2. It provides higher level of abstraction from machine languages.
3. It is machine independent language.
4. Easy to learn.
5. Less error prone, easy to find and debug errors.
6. High level programming results in better programming productivity.

Disadvantages of High level language

1. It takes additional translation times to translate the source to machine code.
2. High level programs are comparatively slower than low level programs.
3. Compared to low level programs, they are generally less memory efficient.
4. Cannot communicate directly with the hardware.

Program development process

- A program development process consists of various steps that are followed to develop a computer program.
- These steps are followed in a sequence in order to develop a successful and beneficial computer program.
- Following is the brief description about program development process.



- A programmer has to go through the following stages to develop a computer program:

1. Defining and Analyzing The Problem

2. Designing The Algorithm

3. Coding or Writing The Program
4. Program Testing Execution
5. Debugging
6. Program review and maintenance

Defining and Analyzing The Problem

- In this step, a programmer studies the problem.
- He decides the best way to solve these problems.
- Studying a problem is also necessary because it helps a programmer to decide about the following things:
    - The facts and figures which are necessary for developing the program.
    - way in which the program will be designed
    - Also, the language in which the program will be most suitable.
    - What is the desired output and in which form it is needed, etc

Designing an Algorithm

- An algorithm is a sequence of steps that must be carried out before a programmer starts preparing his program.
- The programmer designs an algorithm to help visual possible alternatives in a program also.

Coding or Writing the Program

- The next step after designing the algorithm is to write the program in a high-level language.
- This process is known as coding.

Program Testing

- The process of executing the program to find out **errors** or **bugs** is called test execution.
- It helps a programmer to check the logic of the program. It also ensures that the program is error-free and workable.
- The following are common types of errors that might be encountered

i. Syntax error

- Due to improper use of language
- Also called grammatical errors
- These errors can be detected by a compiler

ii. Logical error

- Due to improper use of Logic in a program
- Can not be detected by a compiler
- A program runs successfully but gives a wrong output

Tracing an error in a program

The following steps are used

a. Dry-run (desk check)

- Uses trace table to check whether an algorithm has errors
- Usually dry run checks for logical errors

b. Debugging

- Uses debug utility in an editor to check for syntax errors

c. Test data

- Carry out trial runs using test data to check for logical run-time errors
- Achieved by entering valid and invalid input to test whether the program produces desired result

Debugging

- Debugging is a process of detecting, locating and correcting the bugs in a program. It is performed by running the program again and again.

Program review and maintenance

- This is continuous updating and fixing of program errors after installation
- Maintenance of a program stops only when a program has been replace or if it becomes unusable

Final Documentation

- When the program is finalized, its documentation is prepared. Final documentation is provided to the user.
- It guides the user how to use the program in the most efficient way.
- Furthermore, another purpose of documentation is to allow other programmers to modify the code if necessary.
- Documentation should also be done in each step during the development of the program.

Characteristics of a good program

A good computer program should have following characteristics:

1. **Portability**:

- Portability refers to the ability of an application to run on different platforms (operating systems) with or without minimal changes.

2. **Readability**:

- The program should be written in such a way that it makes other programmers or users to follow the logic of the program without much effort.
- If a program is written structurally, it helps the programmers to understand their own program in a better way.

3. **Efficiency**:

- Every program requires certain processing time and memory to process the instructions and data.
- As the processing power and memory are the most precious resources of a computer, a program should be laid out in such a manner that it utilizes the least amount of memory and processing time.

4. **Structural**:

- To develop a program, the task must be broken down into a number of subtasks.
- These subtasks are developed independently, and each subtask is able to perform the assigned job without the help of any other subtask.
- If a program is developed structurally, it becomes more readable, and the testing and documentation process also gets easier.

5. **Flexibility**:

- A program should be flexible enough to handle most of the changes without having to rewrite the entire program.
- Most of the programs are developed for a certain period and they require modifications from time to time.
- For example, in case of payroll management, as the time progresses, some employees may leave the company while some others may join.

Hence, the payroll application should be flexible enough to incorporate all the changes without having to reconstruct the entire application.

6. **Generality**:

- Apart from flexibility, the program should also be general.
- Generality means that if a program is developed for a particular task, then it should also be used for all similar tasks of the same domain.
- For example, if a program is developed for a particular organization, then it should suit all the other similar organizations.

7. **Documentation**:

- Documentation is one of the most important components of an application development.
- Even if a program is developed following the best programming practices, it will be rendered useless if the end user is not able to fully utilize the functionality of the application.
- A well-documented application is also useful for other programmers because even in the absence of the author, they can understand it.

Programming in Visual Basic

- VB.Net is a simple, modern, object-oriented computer programming language developed by Microsoft to combine the power of .NET

Framework and the common language runtime with the productivity benefits that are the hallmark of Visual Basic.

- VB.NET is implemented by Microsoft's .NET framework. Therefore, it has full access to all the libraries in the .Net Framework. It's also possible to run VB.NET programs on Mono, the open-source alternative to .NET, not only under Windows, but even Linux or Mac OSX.

The following reasons make VB.Net a widely used professional language –

- Modern, general purpose.
- Object oriented.
- Component oriented.
- Easy to learn.
- Structured language.
- It produces efficient programs.
- It can be compiled on a variety of computer platforms.
- Part of .Net Framework.

Setting environment for VB.net programming

The following tools are required to set up environment for writing and running Visual basic programs

1. The .Net Framework

- The .Net framework is a revolutionary platform that helps you to write the following types of applications

i. Windows applications
ii. Web applications
iii. Web services

2. Integrated Development Environment (IDE) For VB.Net

- Microsoft provides the following development tools for VB.Net programming –

- Visual Studio 2010 (VS)
- Visual Basic 2010 Express (VBE)
- Visual Web Developer

NB : for you to start programming in VB, make sure one of the above IDE is installed in your computer

Structure of VB program

To understand structure of VB program, lets consider the following example of simple program written in VB.net

VB.Net Hello World Example

A VB.Net program basically consists of the following parts –

- Namespace declaration
- A class or module
- One or more procedures
- Variables
- The Main procedure
- Statements & Expressions
- Comments

Let us look at a simple code that would print the words "Hello World" –

```
Imports System
Module Module1
   'This program will display Hello World
   Sub Main()
      Console.WriteLine("Hello World")
      Console.ReadKey()
   End Sub
```

End Module

When the above code is compiled and executed, it produces the following result –

Hello, World!

Let us look various parts of the above program –

- The first line of the program **Imports System** is used to include the System namespace in the program.
- The next line has a **Module** declaration, the module *Module1*. VB.Net is completely object oriented, so every program must contain a module of a class that contains the data and procedures that your program uses.
- Classes or Modules generally would contain more than one procedure. Procedures contain the executable code, or in other words, they define the behavior of the class. A procedure could be any of the following –
    - Function
    - Sub
    - Operator
    - Get
    - Set

- AddHandler
- RemoveHandler
- RaiseEvent

- The next line('This program) will be ignored by the compiler and it has been put to add additional comments in the program.
- The next line defines the Main procedure, which is the entry point for all VB.Net programs. The Main procedure states what the module or class will do when executed.
- The Main procedure specifies its behavior with the statement

  Console.WriteLine("Hello World") *WriteLine* is a method of the *Console* class defined in the *System* namespace. This statement causes the message "Hello, World!" to be displayed on the screen.

- The last line **Console.ReadKey()** is for the VS.NET Users. This will prevent the screen from running and closing quickly when the program is launched from Visual Studio .NET.

Compile & Execute VB.Net Program

If you are using Visual Studio.Net IDE, take the following steps –

- Start Visual Studio.
- On the menu bar, choose File → New → Project.
- Choose Visual Basic from templates

- Choose Console Application.
- Specify a name and location for your project using the Browse button, and then choose the OK button.
- The new project appears in Solution Explorer.
- Write code in the Code Editor.
- Click the Run button or the F5 key to run the project. A Command Prompt window appears that contains the line Hello World.

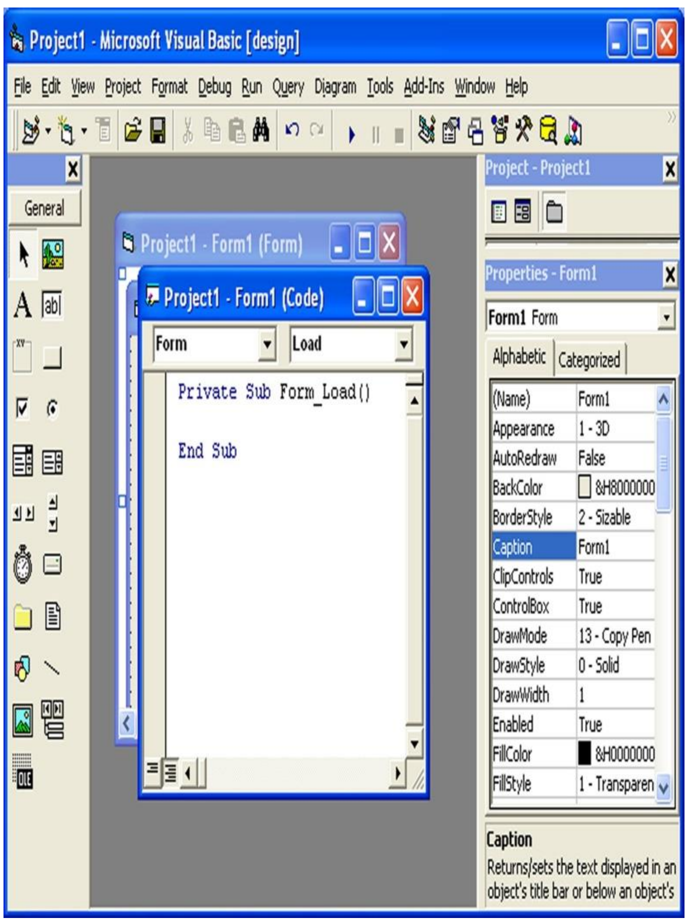
You can compile a VB.Net program by using the command line instead of the Visual Studio IDE –

- Open a text editor and add the above mentioned code.
- Save the file as **helloworld.vb**
- Open the command prompt tool and go to the directory where you saved the file.
- Type **vbc helloworld.vb** and press enter to compile your code.
- If there are no errors in your code the command prompt will take you to the next line and would generate **helloworld.exe** executable file.
- Next, type **helloworld** to execute your program.
- You will be able to see "Hello World" printed on the screen.

.1 Creating Your First Application

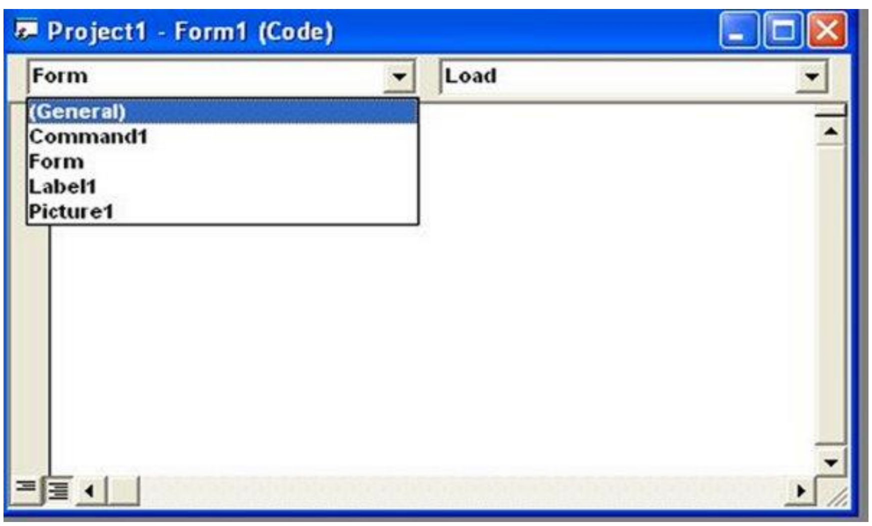
- First of all, launch Microsoft Visual Basic 6 compiler that you have installed earlier.

- In the New Project Dialog , choose Standard EXE to enter Visual Basic 6 integrated development environment.
- In the VB6 IDE, a default form with the name **Form1** will appear.
- Next, double click on Form1 to bring up the source code window for Form1, as shown in Figure below



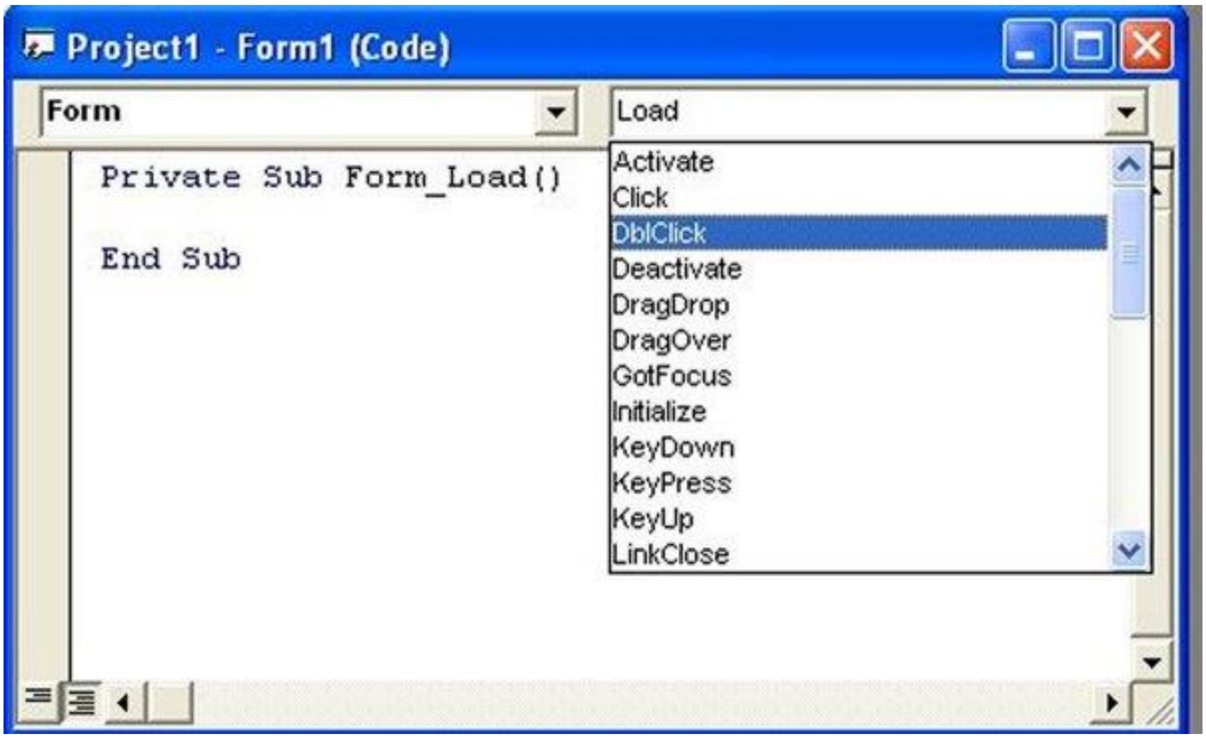
- The top of the source code window consists of a list of objects and their associated events or procedures.

- In the source code window, the object displayed is **Form1** and the associated procedure is **Load**.
- When you click on the object box, the drop-down list will display a list of objects you have inserted into your form.
- Here, you can see a form with the name Form1, a command button with the name Command1, a Label with the name Label1 and a Picture Box with the name Picture1



- Some of the procedures associated with the object Form1 are Activate, Click, DblClick (which means Double-Click) , DragDrop, keyPress and more.

- Each object has its own set of procedures.
- You can always select an object and write codes for any of its procedure in order to perform certain tasks.



- You do not have to worry about the beginning and the end statements

  (i.e. **Private Sub Form_Load…….End Sub**.);

- Just key in the lines in between the above two statements exactly as are shown here.

- When you press **F5** to run the program, you will be surprised that nothing showed up
- .In order to display the output of the program, you have to add the **Form1.show** statement or you can just use **Form_Activate ( )** event procedure
- The command **Print** does not mean printing using a printer but it means displaying the output on the computer screen.
- Now, press F5 or click on the run button to run the program and you will get the output as

Example 1

Private Sub Form_Load ( )

Form1.show

Print "Welcome to Visual Basic tutorial"

End Sub

Example 2

Private Sub Form_Activate ( )

Print 20 + 10

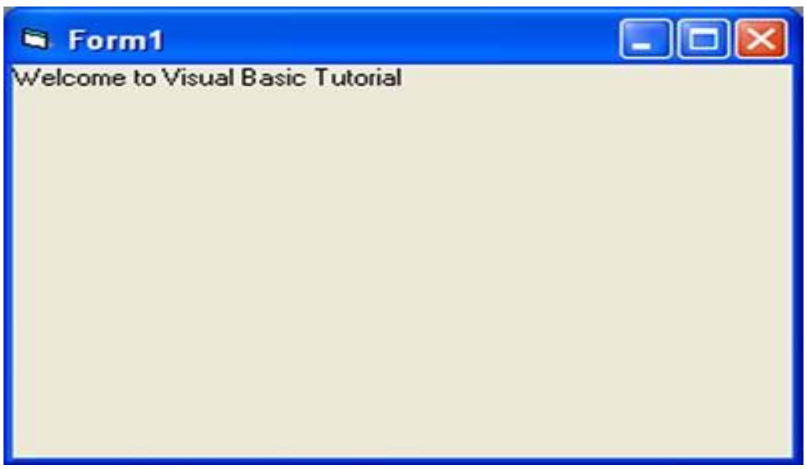
Print 20 - 10

Print 20 * 10

Print 20 / 10

End Sub

Output for example 1



Output for example 2

You can also use the **+** or the **&** operator to join two or more texts (string) together

Example 3

Private Sub

A = "Tom"

B = "likes"

C = "to"

D = "eat"

E = "burger"

Print A + B + C + D + E

End Sub

Example 4

Private Sub

A = "Tom"

B = "likes"

C = "to"

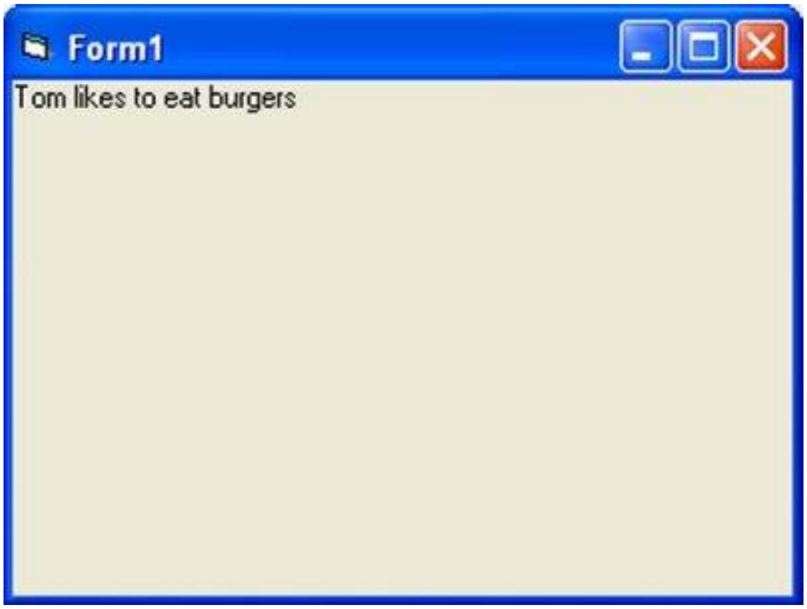
D = "eat"

E = "burger"

Print A & B & C & D & E

End Sub

OUTPUT for example 3 and 4



Steps in Building a Visual Basic Application

Step 1: Design the interface by adding controls to the form and set their properties

Step 2: Write code for the event procedures

Example 5 Changing Background and Foreground Color at Random

- In this example, we want to show you how to write code to change the background and the foreground color randomly.

- We will place two command buttons and a label on the form. One of the command buttons will be used to change the background color while the other one will be used to change the foreground color.
- The Label is for displaying the foreground color. There are two events here, **change background color** and **change foreground color**. Therefore, we need to write code for the two event procedures.

- To make the program more interesting, we will use the Rnd() function, the Int() function and the RGB codes to change the color randomly.

- The Rnd() function creates a random number between 0 and 1 and the RGB code uses a combination of three integers to form a certain color.
- The Int() is a function that converts a number into an integer by truncating its decimal part and the resulting integer is the largest integer that is smaller than the number. For example, Int(0.2)=0, Int(2.4)=2, Int(4.8)=4.
- Therefore, Int(Rnd()*256) returns the smallest integer 0 and the biggest integer 255.
- The format of RGB code is RGB(a,b,c), where a, b, c range from 0 to 255. For example, RGB(255,0,0) is red, RGB(255,255,255) is white and (0,0,0) is black. Do not worry about the jargons, you will learn them in later.

1. Now, rename the controls as follows:

- Form1-MyForm
- Label1-LblMessage
- Command1-cmd_bgColor
- Command2-cmd_fgColor

2. Next, change the caption of the Label to "Please Change My Color".

- In addition, change the caption of Command1 button to "Change Background Color" and change the caption of Command2 button to "Change Foreground Color"

3. Now, enter the following code

```
Private Sub cmd_bgColor_Click()
 Dim r, g, b As Integer
 r = Int(Rnd() * 256)
 g = Int(Rnd() * 256)
 b = Int(Rnd() * 256)
 MyForm.BackColor = RGB(r, g, b)
End Sub

Private Sub Cmd_fgColor_Click()
 Dim r, g, b As Integer
 r = Int(Rnd() * 256)
 g = Int(Rnd() * 256)
```

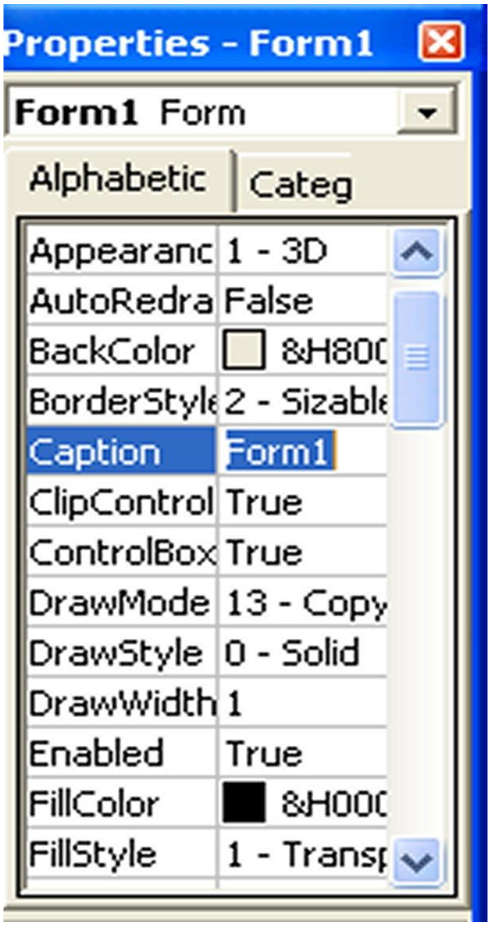
```
 b = Int(Rnd() * 256)
 Lbl_Msg.ForeColor = RGB(r, g, b)
End Sub
```

- When you run the program, each time you press on the 'Change Background Color' button, you will see different background color.
- Similarly, each time you press on the 'Change Foreground Color', you will see the message on the Label changes color. The output is shown below

The Control Properties

- Before writing an event procedure for the control to response to an event, you have to set certain properties for the control to determine its appearance and how will it work with the event procedure.

- You can set the properties of the controls in the properties window or at runtime.
- In the properties window, the item appears at the top part is the object currently selected.
- At the bottom part, the items listed in the left column represent the names of various properties associated with the selected object while the items listed in the right column represent the states of the properties.
- Properties can be set by highlighting the items in the right column then change them by typing or selecting the options available.



- For example, in order to change the caption, just highlight Form1 under the name Caption and change it to other names.
- You may also alter the appearance of the form by setting it to 3D or flat, change its foreground and background color, change the font type and font size, enable or disable, minimize and maximize buttons and more.
- You can also change the properties at runtime to give special effects such as change of color, shape, animation effect and so on.

Example 6 Program to change background color

This example changes the background colour of the form using the **BackColor property**.

```
Private Sub Form_Load()
Form1.Show
Form1.BackColor = &H000000FF&
End Sub
```

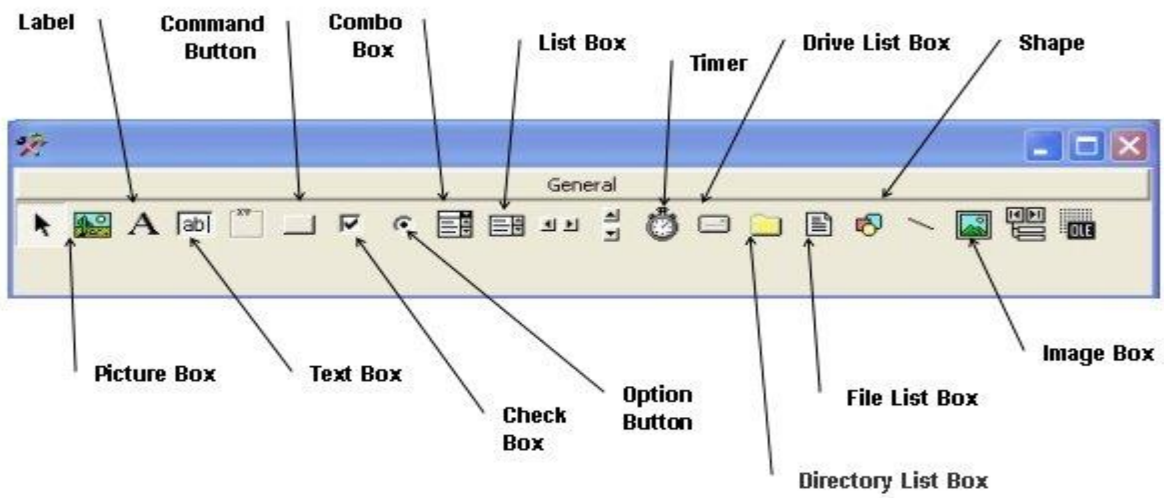
Example 7: Program to change shape

- This example is to change the control's Shape using the **Shape property**. This code will change the shape to a circle at runtime.

```
Private Sub  Form_Load()
Shape1.Shape = 3
End Sub
```

Handling some of the common Controls

Below is the VB6 toolbox that shows the basic controls.



The TextBox

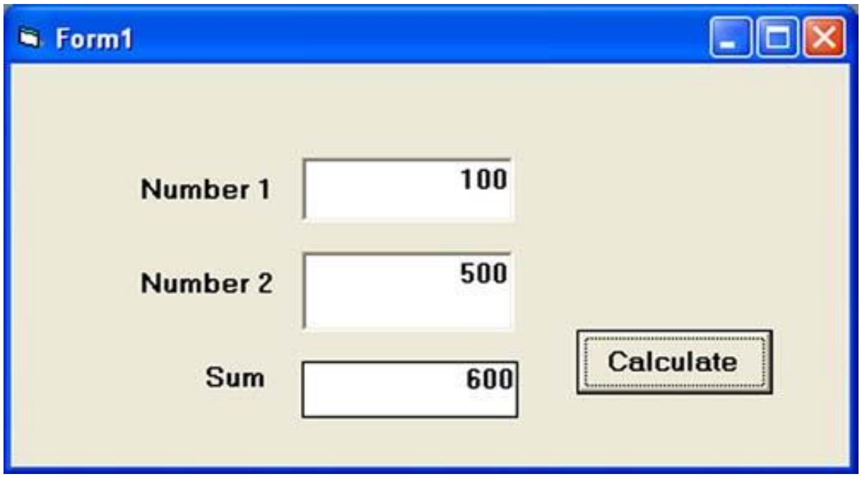
- The text box is the standard control for accepting input from the user as well as to display the output.
- It can handle string (text) and numeric data but not images or pictures.
- A string entered into a text box can be converted to a numeric data by using the function Val(text).
- The following example illustrates a simple program that processes the input from the user.

Example 8

- In this program, two text boxes are inserted into the form together with a few labels.

- The two text boxes are used to accept inputs from the user and one of the labels will be used to display the sum of two numbers that are entered into the two text boxes.
- Besides, a command button is also programmed to calculate the sum of the two numbers using the plus operator.
- The program use creates a variable sum to accept the summation of values from text box 1 and text box 2.
- The procedure to calculate and to display the output on the label is shown below.

```
 Private Sub Command1_Click()
'To add the values in TextBox1 and TextBox2
 Sum = Val(Text1.Text) +Val(Text2.Text)
'To display the answer on label 1
  Label1.Caption = Sum
End Sub
```



The Label

- The label is a very useful control for Visual Basic, as it is not only used to provide instructions and guides to the users, it can also be used to display outputs.
- One of its most important properties is Caption. Using the syntax Label.
- Caption, it can display text and numeric data. You can change its caption in the properties window and also at runtime.

The Command Button

- The command button is one of the most important controls as it is used to execute commands.
- It displays an illusion that the button is pressed when the user click on it.

- The most common event associated with the command button is the Click event, and the syntax for the procedure is

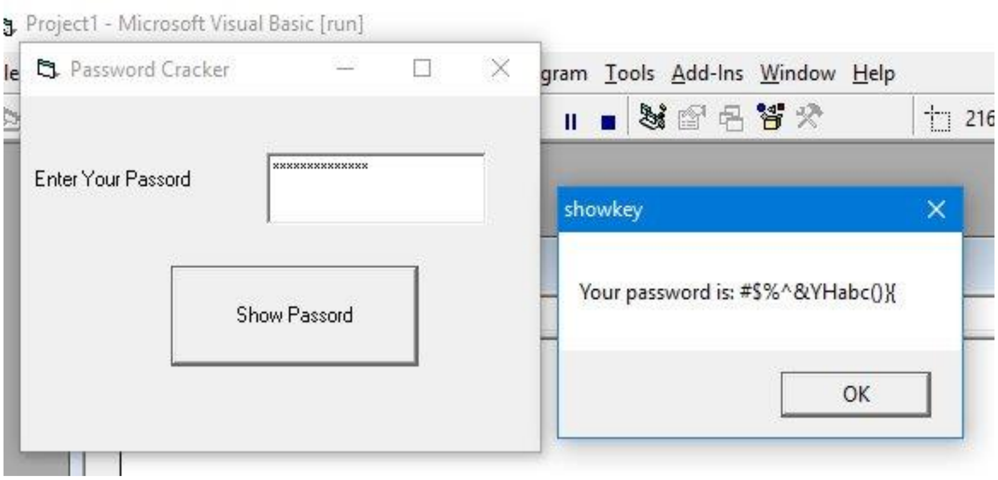
```
Private Sub Command1_Click ()
 Statements
End Sub
```

Example 9 A Simple Password Cracker

- In this program, we want to crack a secret passoword entered by the user.
- In the design phase, insert a command button and change its name to cmd_ShowPass.
- Next, insert a TextBox and rename it as TxtPassword and delete Text1 from the Text property.
- Besides that, set its PasswordChr to *. Now, enter the following code in the code window.

```
Private Sub cmd_ShowPass_Click()
 Dim yourpassword As String
 yourpassword = Txt_Password.Text
 MsgBox ("Your password is: " & yourpassword)
End Sub
```

Run the program and enter a password, then click on the Show Password button to reveal the password, as shown below.



- You can also reveal the password by setting the PasswordChr property back to normal mode, as follows:

```
Private Sub cmd_ShowPass_Click()
Dim yourpassword As String
Txt_Password.PasswordChar = ""
End Sub
```

The PictureBox

- The Picture Box is one of the controls that is used to handle graphics.
- You can load a picture at design phase by clicking on the picture item in the properties window and select the picture from the selected folder.

- You can also load the picture at runtime using the **LoadPicture** method.
- For example, the statement will load the picture grape.gif into the picture box.

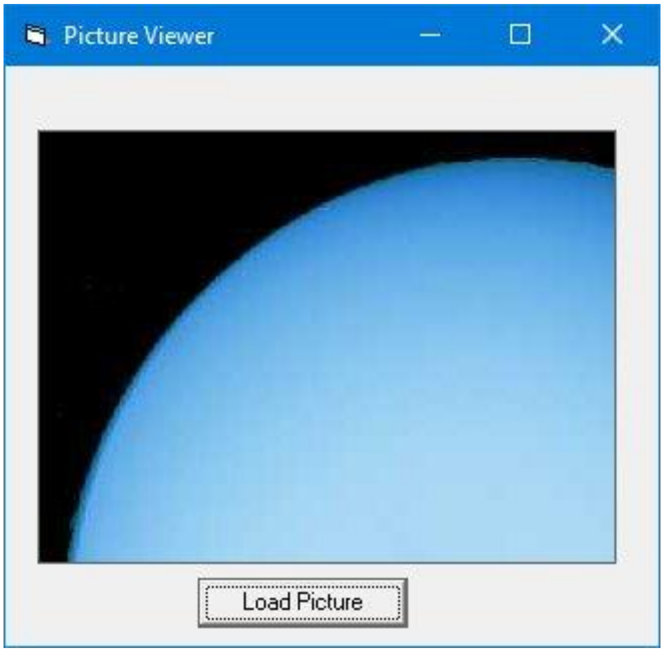
```
Picture1.Picture=LoadPicture ("C:\VBprogram\Images\grape.gif")
```

Example : Loading Picture

In this program, insert a command button and a picture box. Enter the following code:

```
Private Sub cmd_LoadPic_Click()
MyPicture.Picture = LoadPicture("C:\Users\admin.DESKTOP-
G1G4HEK\Documents\My Websites|vbtutor|vb6|images|uranus.jpg")
End Sub
```

* You must ensure the path to access the picture is correct. Besides that, the image in the picture box is not resizable. The output is shown below



The Image Control

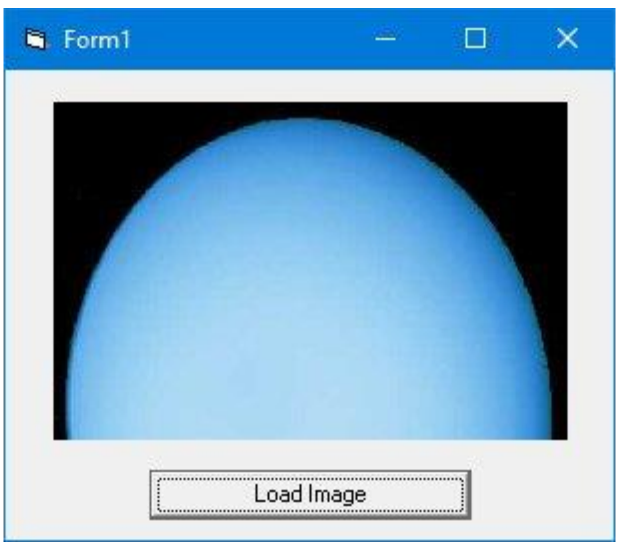
- The Image Control is another control that handles images and pictures.
- It functions almost identically to the pictureBox. However, there is one major difference, the image in an Image Box is stretchable, which means it can be resized.
- This feature is not available in the PictureBox. Similar to the Picture Box, it can also use the LoadPicture method to load the picture.
- For example, the statement loads the picture grape.gif into the image box.

Image1.Picture=LoadPicture ("C:\VBprogram\Images\grape.gif")

Example :: Loading Image

- In this program, we insert a command button and an image control into the form. Besides that, we set the image Strech property to true.
- Next, enter te following code:

```
Private Sub cmd_LoadImg_Click()
 MyImage.Picture = LoadPicture("C:\Users\admin.DESKTOP-
G1G4HEK\Documents\My Websites\vbtutor\vb6\images\uranus.jpg")
End Sub
```



* Note the difference between the image in Figure 3.5 and Figure 3.6.

The ListBox

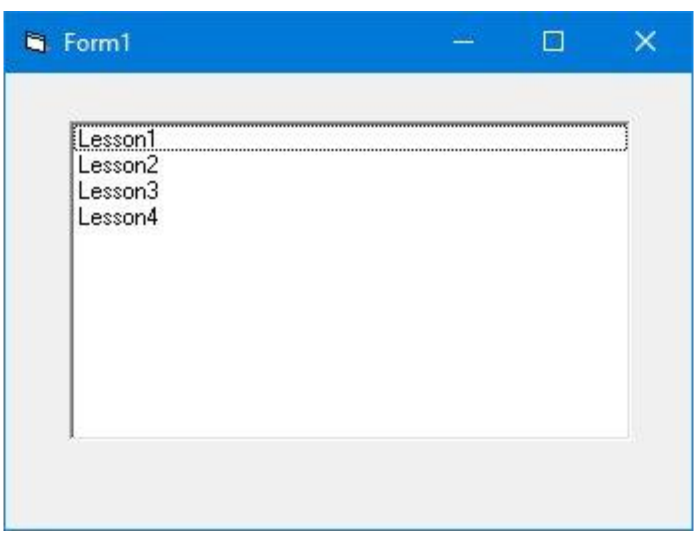
- The function of the ListBox is to present a list of items where the user can click and select the items from the list.

- In order to add items to the list, we can use the **AddItem method**. For example, if you wish to add a number of items to list box 1, you can key in the following statements

Example

```
Private Sub Form_Load ( )
 List1.AddItem "Lesson1"
 List1.AddItem "Lesson2"
 List1.AddItem "Lesson3"
 List1.AddItem "Lesson4"
End Sub
```

The Output



- The items in the list box can be identified by the **ListIndex** property, the value of the ListIndex for the first item is 0, the second item has a ListIndex 1, and the third item has a ListIndex 2 and so on

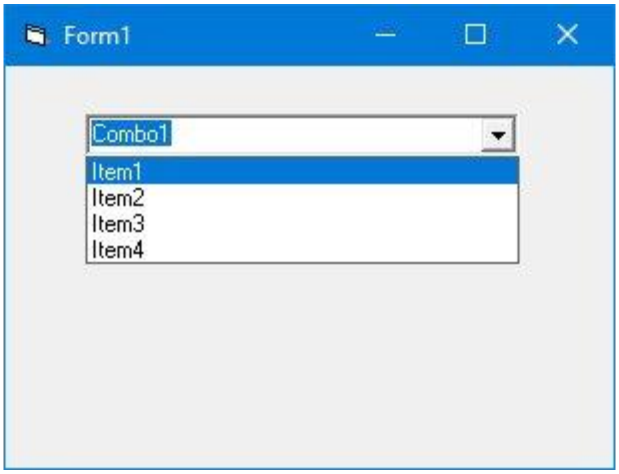
The ComboBox

- The function of the Combo Box is also to present a list of items where the user can click and select the items from the list.
- However, the user needs to click on the small arrowhead on the right of the combo box to see the items which are presented in a drop-down list.
- In order to add items to the list, you can also use the **AddItem method**.
- For example, if you wish to add a number of items to Combo box 1, you can key in the following statements

Example

```
Private Sub Form_Load ( )
 Combo1.AddItem "Item1"
 Combo1.AddItem "Item2"
 Combo1.AddItem "Item3"
 Combo1.AddItem "Item4"
End Sub
```

The Output



The CheckBox

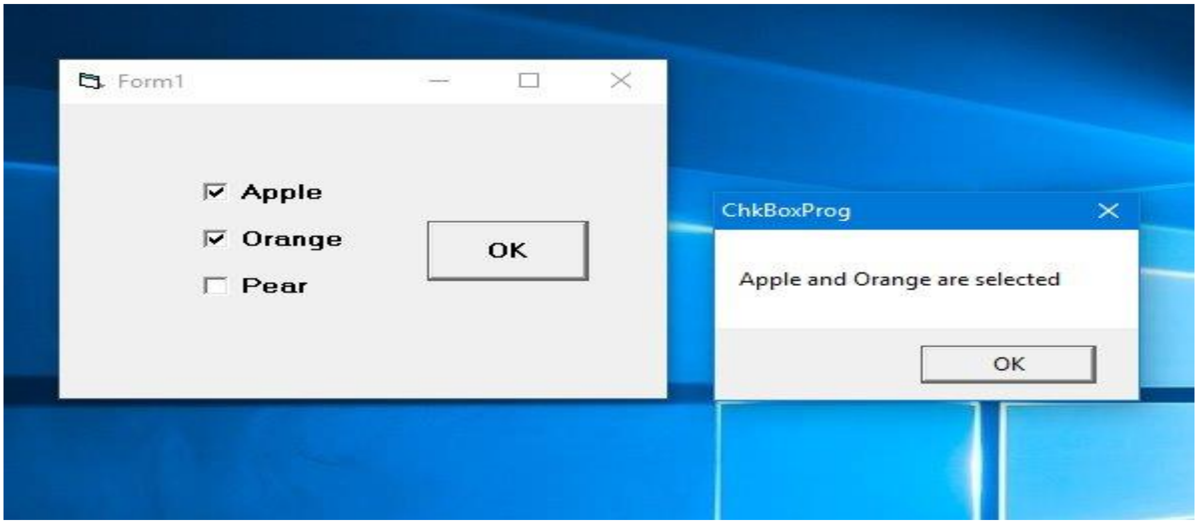
- The Check Box control lets the user selects or unselects an option.
- When the Check Box is checked, its value is set to 1 and when it is unchecked, the value is set to 0.
- You can include the statements Check1.Value=1 to mark the Check Box and Check1. Value=0 to unmark the Check Box, as well as use them to initiate certain actions.
- For example, the program in Example below will show which items are selected in a message box.

Example

```
Private Sub Cmd_OK_Click()
 If Check1.Value = 1 And Check2.Value = 0 And Check3.Value = 0 Then
  MsgBox "Apple is selected"
```

```
 ElseIf Check2.Value = 1 And Check1.Value = 0 And Check3.Value = 0 Then
  MsgBox "Orange is selected"
  ElseIf Check3.Value = 1 And Check1.Value = 0 And Check2.Value = 0
Then
  MsgBox "Orange is selected"
 ElseIf Check2.Value = 1 And Check1.Value = 1 And Check3.Value = 0 Then
  MsgBox "Apple and Orange are selected"
 ElseIf Check3.Value = 1 And Check1.Value = 1 And Check2.Value = 0 Then
  MsgBox "Apple and Pear are selected"
 ElseIf Check2.Value = 1 And Check3.Value = 1 And Check1.Value = 0 Then
  MsgBox "Orange and Pear are selected"
 Else
  MsgBox "All are selected"
 End If
End Sub
```

The Output



The OptionButton

- The OptionButton control also lets the user selects one of the choices. However, two or more Option buttons must work together because as one of the option buttons is selected, the other Option button will be unselected
- In fact, only one Option Box can be selected at one time.
- When an option box is selected, its value is set to “True” and when it is unselected; its value is set to “False”.

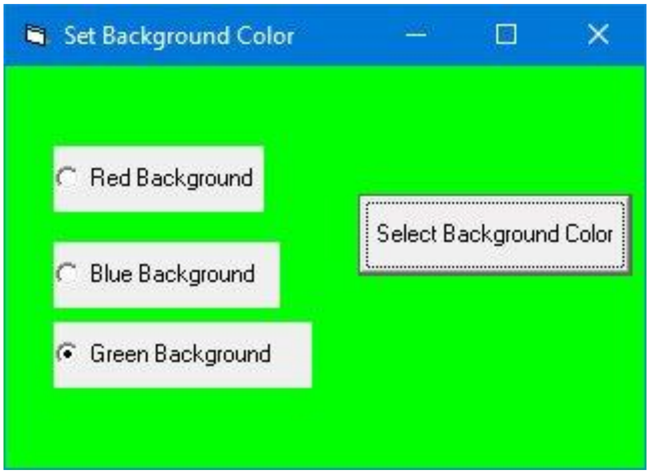
Example

- In this example, we want to change the background color of the form according to the selected option.

- We insert three option buttons and change their captions to "Red Background","Blue Background" and "Green Background" respectively.
- Next, insert a command button and change its name to cmd_SetColor and its caption to "Set Background Color". Now, click on the command button and enter the following code in the code window:

```
Private Sub cmd_SetColor_Click()
 If Option1.Value = True Then
 Form1.BackColor = vbRed
 ElseIf Option2.Value = True Then
 Form1.BackColor = vbBlue
 Else
 Form1.BackColor = vbGreen
 End If
End Sub
```

Run the program, select an option and click the "Set Background Color" produces the output, as shown below



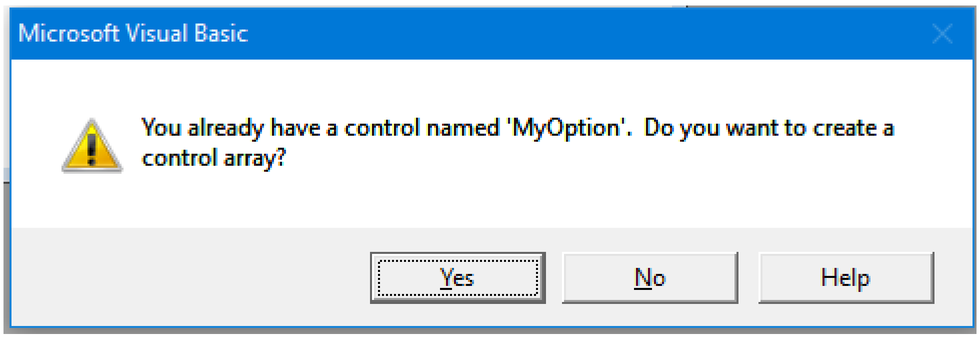
The Shape Control

- In the following example, the shape control is placed in the form together with six OptionButtons.
- To determine the shape of the shape control, we use the shape property.
- The property values of the shape control are 0, 1, and 2,3,4,5 which will make it appear as a rectangle, a square, an oval shape, a circle, a rounded rectangle and a rounded square respectively.

Example

- In this example, we insert six option buttons.
- It is better to make the option buttons into a control array as they perform similar action, i.e to change shape.
- In order to create a control array, click on the first option button, rename it as MyOption. Next, click on the option button and select copy then paste.

- clicking the paste button, a popup dialog will ask you whether you wish to create a control array, select yes.
- The control array can be accessed via its index value, MyOtion(Index. In addition, we also insert a shape control.



- Now, enter the code in the code window. We use the If..Then..Else program structure to determine which option button is selected by the user. Private Sub MyOption_Click(Index As Integer)

If Index = 0 Then

MyShape.Shape = 0

ElseIf Index = 1 Then

MyShape.Shape = 1

ElseIf Index = 2 Then

MyShape.Shape = 2

ElseIf Index = 3 Then

MyShape.Shape = 3

ElseIf Index = 4 Then

MyShape.Shape = 4

ElseIf Index = 5 Then

MyShape.Shape = 5

End If

End Sub

- Run the program and you can change the shape of the shape control by clicking one of the option buttons.
- The output is shown in Figure below.

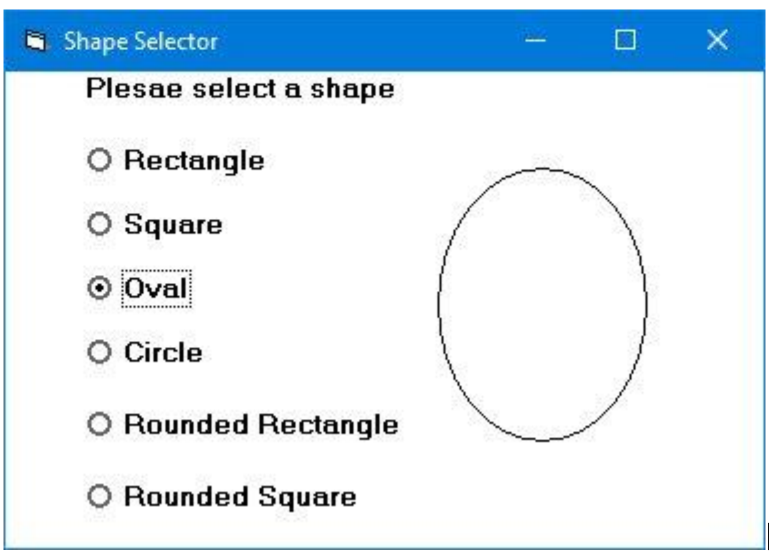
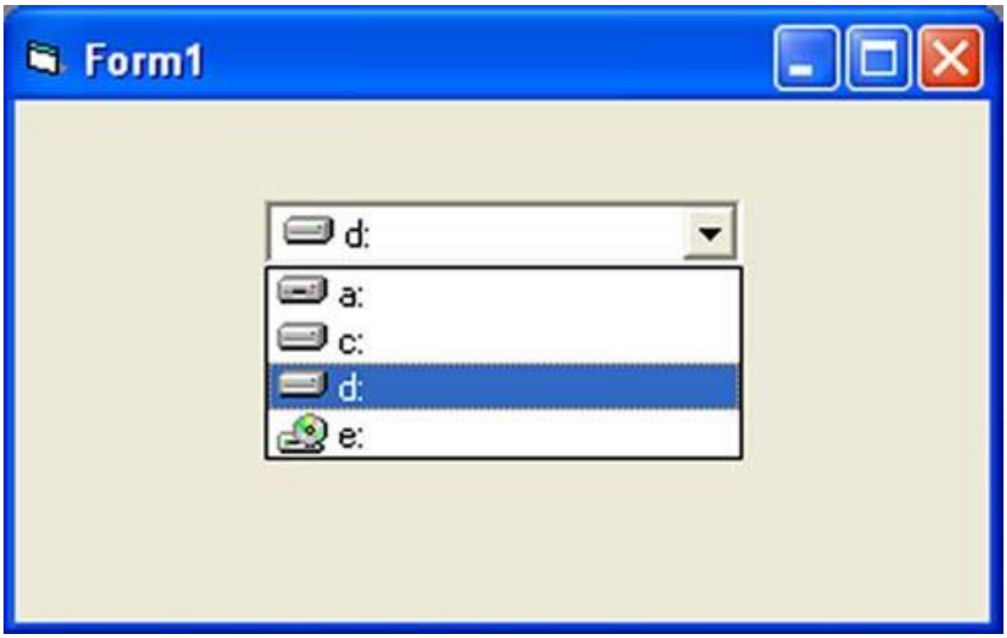


Figure 3.

The DriveListBox

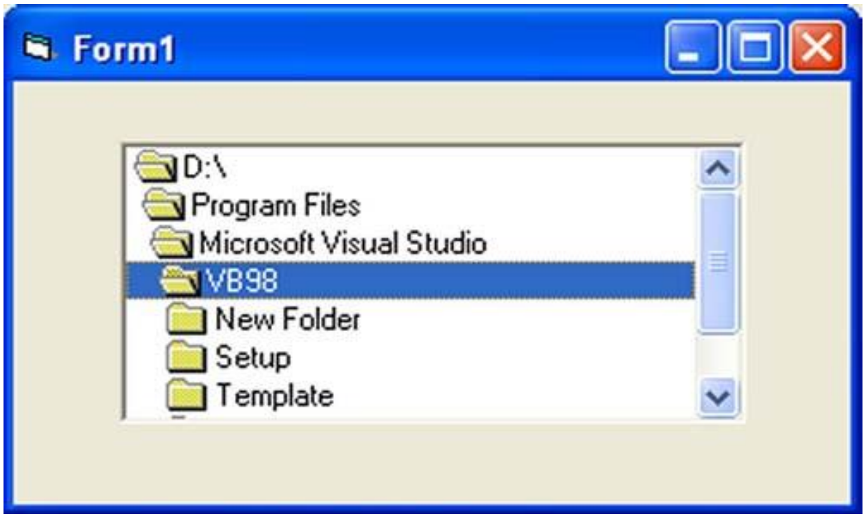
- The DriveListBox is for displaying a list of drives available in your computer.

- When you place this control into the form and run the program, you will be able to select different drives from your computer as shown below

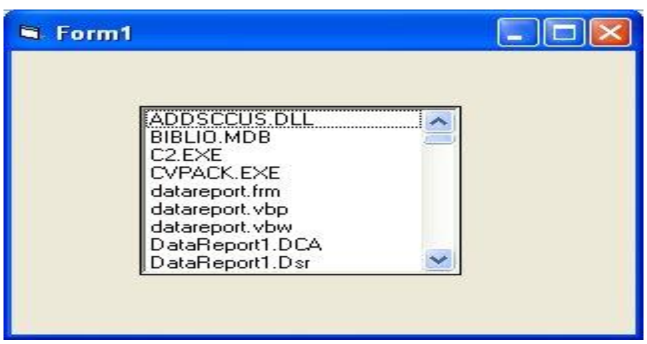


The DirListBox

- The DirListBox means the Directory List Box.
- It is for displaying a list of directories or folders in a selected drive.
- When you place this control into the form and run the program, you will be able to select different directories from a selected drive in your computer as shown below



DirListBox



The FileListBox

- You can coordinate the Drive List Box, the Directory List Box and the File List Box to search for the files you want.

Event Procedure

- For each event, you need to write an event procedure so that it can perform an action or a series of actions.
- To start writing code for an event procedure, you need to double-click an object to enter the VB code window.
- For example, if you want to write code for the event of clicking a command button, you double-click the command button and enter the codes in the event procedure that appears in the code window, as shown below

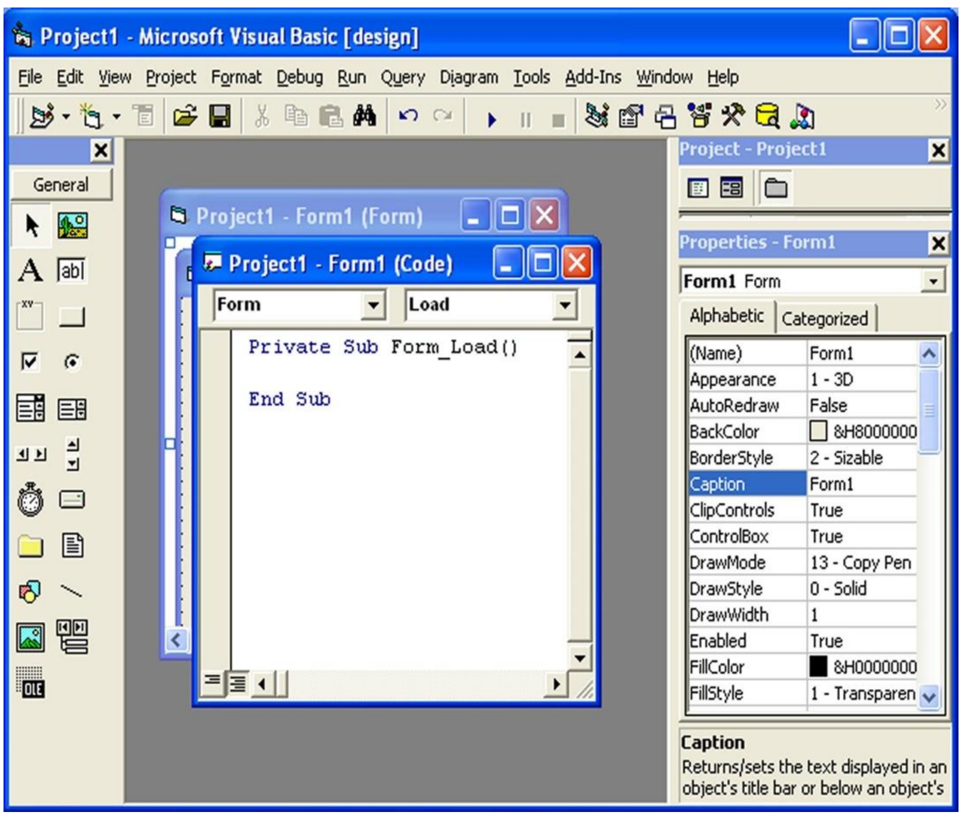


Figure 4.1

The structure of an event procedure is as follows:

Private Sub Command1_Click

VB Statements

End Sub

- You enter the codes in the space between **Private Sub Command1_Click............ End Sub**.
- The keyword **Sub** actually stands for a sub procedure that made up a part of all the procedures in a program or a module.
- The program code is made up of a number of VB statements that set certain properties or trigger some actions.
- The syntax of the Visual Basic's program code is almost like the normal English language, though not exactly the same, so it is fairly easy to learn.
- The syntax to set the property of an object or to pass a certain value to it is :

Object.Property

where **Object** and **Property** are separated by a period (or dot).

- For example, the statement **Form1.Show** means to show the form with the name Form1, **Iabel1.Visible=true** means label1 is set to be visible, **Text1.text="VB"** is to assign the text VB to the text box with the name Text1, Text2.text=100 is to pass a value of 100 to the text box with the name text2, **Timer1.Enabled=False** is to disable the timer with the name Timer1 and so on.

- Let's examine a few examples below:

Example

```
Private Sub Command1_click()
Label1.Visible=false
Label2.Visible=True
Text1.Text="You are correct!"

End sub
```

Example

```
Private Sub Command1_click()
Label1.Caption="Welcome"
Image1.visible=true
End sub
```

Example

```
Private Sub Command1_click()
Pictuire1.Show=true
Timer1.Enabled=True
Lable1.Caption="Start Counting"
```

End sub

Example : A Counter

- This is a counter which start counting after the user click on a command button. In this program, we insert a label, two command buttons and a Timer control.
- The label acts as a counter, one of the command buttons is to start the counter and the other one is to stop the counter.
- The Timer control is a control that is only used by the developer, it is invisible during runtime and it does not allow the user to interact with it.
- The Timer's Interval property determine how frequent the timer changes. A value of 1 is 1 milliseconds which means a value of 1000 represents 1 second.
- In this example, we set the interval to 100, which represents 0.1 second interval. In addition, the Timer's Enabled property is set to false at design time as we do not want the program to start counting immediately, the program only start counting after the the user clicks on te "Start Counting" button. You can also reset the counter using another command button.

The Code

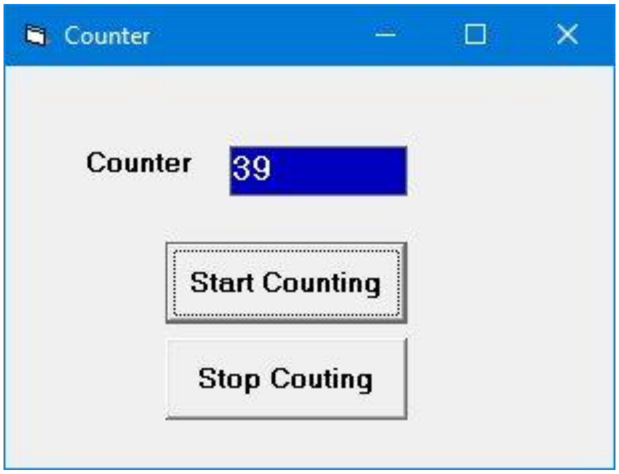
Dim n As Integer

Private Sub cmd_StartCount_Click()

```
Timer1.Enabled = True
End Sub

Private Sub cmd_Stop_Click()
Timer1.Enabled = False
End Sub

Private Sub Command1_Click()
Lbl_Display.Caption = 0
End Sub

Private Sub Timer1_Timer()
n = n + 1
Lbl_Display.Caption = n
End Sub
```

- We declare the variable n in the general area. After the Timer1 is enabled, it will add 1 to the preceding number using n=n+1 after every interval untill the user click on the "Stop Counting" button.

The Output



Example Click and Double Click

- This program display a message whether the label is being click once or click twice.
- In this program, insert a label and rename it as MyLabel and change its caption to "CLICK ME". Next, key in the following codes:

```
Private Sub MyLabel_Click()
 MyLabel.Caption = "You Click Me Once"
End Sub
Private Sub MyLabel_DblClick()
 MyLabel.Caption = "You Click Me Twice!"
End Sub
```

The out

- Running the program and click the label once, the "CLICK ME" caption will change to "You Click Me Once". If you click the label twice, the "CLICK ME" caption will change to "You Click Me Twice!".

NB: In Visual Basic, most of the syntaxes resemble the English language. Among the syntaxes are **Print, If…Then….Else….End If, For…Next, Select Case…..End Select , End** and **Exit Sub.** For example, **Print " Visual Basic"** is to display the text Visual Basic on screen and **End** is to end the program.

- Program code that involves calculations is fairly easy to write, just like what you do in mathematics. However, in order to write an event procedure that involves calculations, you need to know the basic arithmetic operators in VB as they are not exactly the same as the normal operators , except for **+** and **- .**

i. For multiplication, we use *****,
ii. for division we use **/**,
iii. for raising a number x to the power of n, we use **x ^n** and
iv. for square root, we use **Sqr(x).**

- VB offers many more advanced mathematical functions such as **Sin,** Cos, Tan and **Log,**.
- There are also two important functions that are related to arithmetic operations, i.e. the functions **Val** and **Str$** where Val is to convert text to a numerical value and Str$ is to convert numerical to a string (text).
- While the function Str$ is as important as VB can display a numeric value as string implicitly, failure to use Val will result in the wrong calculation.

Example

```
Private Sub Form_Activate()
Text3.text=text1.text+text2.text
End Sub
```

- When you run the program above and enter 12 in textbox1 and 3 in textbox2 will give you a result of 123, which is wrong.
- It is because VB treat the numbers as string and so it just joins up the two strings. On the other hand,

Example

```
Private Sub Form_Activate()
Text3.text=val(text1.text)+val(text2.text)
End Sub
```

- Running the above program code will give you the correct result, i.e., 15.

Visual Basic Data Types

- VB6 classifies the information into two major data types, they are the

    i. Numeric data types and the
    ii. Non-numeric data types.

Numeric Data Types

- Numeric data types are types of data that consist of numbers that can be computed mathematically with standard operators.
- Examples of numeric data types are

    - height,

- weight,
- share values,
- the price of goods,
- monthly bills,
- fees and others.

- In Visual Basic, numeric data are divided into 7 types, depending on the range of values they can store.
- Calculations that only involve round figures can use **Integer** or **Long integer** in the computation.
- Programs that require high precision calculation need to use **Single** and **Double** decision data types, they are also called **floating point** numbers.
- For currency calculation , you can use the currency data types.
- Lastly, if even more precision is required to perform calculations that involve many decimal points, we can use the decimal data types.
- These data types summarized below

Numeric Data Types

Type	**Storage**	**Range of Values**
Byte	1 byte	0 to 255
Integer	2 bytes	-32,768 to 32,767
Long	4 bytes	-2,147,483,648 to 2,147,483,648

Single	4 bytes	-3.402823E+38 to -1.401298E-45 for negative values 1.401298E-45 to 3.402823E+38 for positive values.
Double	8 bytes	-1.79769313486232e+308 to -4.94065645841247E-324 for negative values 4.94065645841247E-324 to 1.79769313486232e+308 for positive values.
Currency	8 bytes	-922,337,203,685,477.5808 to 922,337,203,685,477.5807
Decimal	12 bytes	+/- 79,228,162,514,264,337,593,543,950,335 if no decimal is use +/- 7.9228162514264337593543950335 (28 decimal places).

Non-numeric Data Types

- Nonnumeric data types are data that cannot be manipulated mathematically.
- Non-numeric data comprises string data types, date data types, boolean data types that store only two values (true or false), object data type and Variant data type .
- They are summarized in Table below

Nonnumeric Data Types

Data Type	Storage	Range
String(fixed length)	Length of string	1 to 65,400 characters
String(variable length)	Length + 10 bytes	0 to 2 billion characters
Date	8 bytes	January 1, 100 to December 31, 9999
Boolean	2 bytes	True or False
Object	4 bytes	Any embedded object
Variant(numeric)	16 bytes	Any value as large as Double
Variant(text)	Length+22 bytes	Same as variable-length string

Suffixes for Literals

- Literals are values that you assign to data. In some cases, we need to add a suffix behind a literal so that VB can handle the calculation more accurately.
- For example, we can use num=1.3089# for a Double type data. Some of the suffixes are displayed in Table below

Suffixes for Literals

Suffix	Data Type
&	Long
!	Single
#	Double
@	Currency

- In addition, we need to enclose string literals within two quotations and date and time literals within two # sign. Strings can contain any characters, including numbers.
- The following are few examples:

memberName="Turban, John."

TelNumber="1800-900-888-777"

LastDay=#31-Dec-00#

ExpTime=#12:00 am#

Managing Variables

- Variables are named locations in a computer program
- Variables are like mail boxes in the post office. The contents of the variables changes every now and then, just like the mail boxes.

- In term of VB, variables are areas allocated by the computer memory to hold data. Like the mail boxes, each variable must be given a name.
- To name a variable in Visual Basic, you have to follow a set of rules.

i. Variable Names

- The following are the rules when naming the variables in Visual Basic

- It must be less than 255 characters
- No spacing is allowed
- It must not begin with a number
- Period is not permitted
- Cannot use exclamation mark (!), or the characters @, &, $, #
- Cannot repeat names within the same level of scope.

Examples of valid and invalid variable names are displayed in Table below

Examples of Valid and Invalid Variable Names

Valid Name	**Invalid Name**
My_Car	My.Car
this year	1NewBoy
Long_Name_Can_beUSE	He&HisFather *& is not acceptable

Declaring Variables Explicitly

- In Visual Basic, it is a good practice to declare the variables before using them by assigning names and data types.
- They are normally declared in the general section of the codes' windows using the **Dim** statement.
- You can use any variable to hold any data , but different types of variables are designed to work efficiently with different data types . The syantax is as follows:

Dim VariableName As DataType

- If you want to declare more variables, you can declare them in separate lines or you may also combine more in one line , separating each variable with a comma, as follows:

Dim VariableName1 As DataType1, VariableName2 As DataType2,VariableName3 As DataType3

Example

Dim password As String

Dim yourName As String

Dim firstnum As Integer

Dim secondnum As Integer

Dim total As Integer

Dim doDate As Date

Dim password As String, yourName As String, firstnum As Integer

- Unlike other programming languages, Visual Basic actually doesn't require you to specifically declare a variable before it's used. If a variable isn't declared, VB will automatically declare the variable as a Variant.
- A variant is data type that can hold any type of data.
- For string declaration, there are two possible types, one for the **variable-length string** and another for the **fixed-length string**.
- For the variable-length string, just use the same format as example above. However, for the fixed-length string, you have to use the syntax as shown below:

Dim VariableName as String * n

- where n defines the number of characters the string can hold.

For example,

Dim yourName as String * 10

*yourName can holds no more than 10 Characters.

Scope of Declaration

- Other than using the Dim keyword to declare the data, you can also use other keywords to declare the data.
- Three other keywords are

    i. private ,
    ii. static and
    iii. public.

The forms are as shown below:

Private VariableName as Datatype
Static VariableName as Datatype
Public VariableName as Datatype

- The above keywords indicate the scope of the declaration.
- **Private** declares a local variable or a variable that is local to a procedure or module. However, Private is rarely used, we normally use Dim to declare a local variable.
- The **Static** keyword declares a variable that is being used multiple times, even after a procedure has been terminated.
- Most variables created inside a procedure are discarded by Visual Basic when the procedure is finished, static keyword preserves the value of a variable even after the procedure is terminated.

- **Public** is the keyword that declares a global variable, which means it can be used by all the procedures and modules of the whole program.

Constants

- Constants are different from variables in the sense that their values do not change during the running of the program.

Declaring a Constant

The syntax to declare a constant is

Constant Name As Data Type = Value

Example

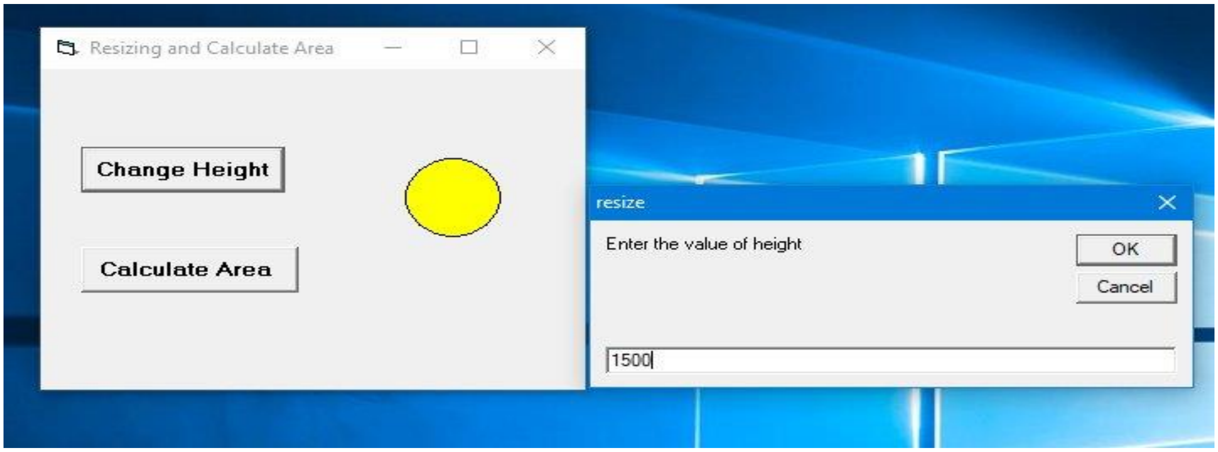
- In this example, we insert a Shape control and two command buttons. Set the shape value of the Shape control to 3 so that it becomes a circle.
- Rename one of the command buttons to CmdResize for changing the size of the circle. Rename the other command button as CmdArea for calculation of the area of the circle.
- In this program, we declare four variables and a constant in the General section.
- The varaible h is to store the value of height of the circle and the variable r is to store the value of the radius which is half of the height.

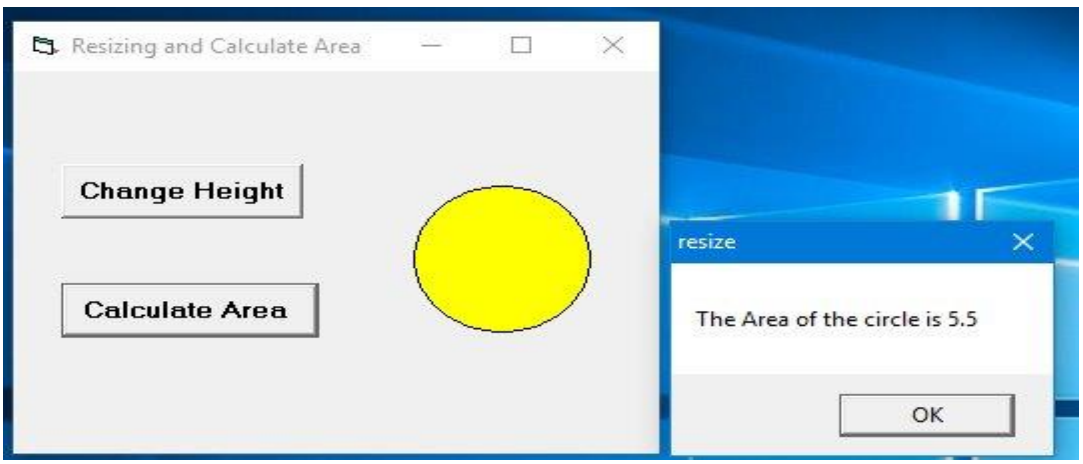
- In addtion, the variable a is to store the value of area in twip using the formula area of circle=пr^2. Besides that, the constant Pi represents п which we fixed at 3.142. Finally, the variable area is to store the value in cm by multiplying a with 0.001763889. (1 twip =0.001763889 cm)

The Code

```
Dim h, r, a, rad, area As Single
Const Pi As Single = 3.142

Private Sub CmdArea_Click()
r = h / 2
rad = r * 0.001763889
a = Pi * rad ^ 2
area = Round(a, 2)
MsgBox ("The Area of the circle is " & area)
End Sub

Private Sub CmdResize_Click()
h = InputBox("Enter the value of height")
MyShape.Height = h

End Sub
```

The Output





Assigning Values to Variables

- After declaring various variables using the Dim statements, we can assign values to those variables. The syntax of an assignment is

Variable=Expression

- The variable can be a declared variable or a control property value.
- The expression could be a mathematical expression, a number, a string, a Boolean value (true or false) and more.
- The following are some examples variable assignment:

```
firstNumber=100
secondNumber=firstNumber-99
userName="John Lyan"
userpass.Text = password
Label1.Visible = True
Command1.Visible = false
Label4.Caption = textbox1.Text
ThirdNumber = Val(usernum1.Text)
X = (3.14159 / 180) * A
```

-

Operators in Visual Basic

- Operators are symbols that are used to manipulate inputs from users and to generate results,
- We need to use various mathematical operators

- . In Visual Basic, except for + and -, the symbols for the operators are different from normal mathematical operators, as shown in Table below

Arithmetic Operators

Operator	Mathematical function	Example>
^	Exponential	2^4=16
*	Multiplication	4*3=12,
/	Division	12/4=3
Mod	Modulus (returns the remainder from an integer division)	15 Mod 4=3
\	Integer Division(discards the decimal places)	19\4=4
+ or &	String concatenation	"Visual"&"Basic"="Visual Basic"

Example

```
Private Sub Command1_Click()

Dim firstName As String
Dim secondName As String
Dim yourName As String

firstName = Text1.Text
secondName = Text2.Text
yourName = secondName +"" + firstName
```

```
Label1.Caption = yourName
End Sub
```

- In above Example, three variables are declared as string.
- For variables firstName and secondName will receive their data from the user's input into textbox1 and textbox2,
- the variable yourName will be assigned the data by combining the first two variables.
- Finally, yourName is displayed on Label1.

Example

```
Dim number1, number2,number3 as Integer
Dim total, average as variant
Private sub Form_Click()

number1=val(Text1.Text)
number2=val(Text2.Text)
number3= val(Text3.Text)
Total=number1+number2+number3
Average=Total/5
Label1.Caption=Total
Label2.Caption=Average
End Sub
```

- In the above example , three variables are declared as integer and two variables are declared as variant.
- Variant means the variable can hold any data type. The program computes the total and average of the three numbers that are entered into three text boxes.

Example : Easy Math

- This is a simple math drill program where the user enter two numbers and calculate its sum. The program will tell him whether the answer is right or wrong.
- To add some gist to the program, the user needs to enter the password before he or she can proceed.

The Code

```
Dim password As String
Dim yourName As String
Dim firstnum As Integer
Dim secondnum As Integer
Dim total As Integer
Dim doDate As Date

Private Sub Command1_Click()
If userpass.Text = password Then
Label2.Visible = True
number1.Visible = True
number2.Visible = True
sum.Visible = True
Label3.Visible = True
Command3.Visible = True
usernum1.Visible = True
usernum2.Visible = True
OK.Visible = True
Label4.Visible = True
Label4.Caption = textbox1.Text
textbox1.Visible = False
userpass.Visible = False
username.Visible = False
```

```
Label1.Visible = False
Command1.Visible = False
Else
userpass.Text = ""
userpass.SetFocus
End If

End Sub
```

3.'

```
Private Sub Form_Load()
password = "liewxun"
End Sub

Private Sub OK_Click()
firstnum = usernum1.Text
secondnum = usernum2.Text
total = sum.Text
If total = firstnum + secondnum And Val(sum.Text) <> 0 Then
correct.Visible = True
wrong.Visible = False
Else
correct.Visible = False
wrong.Visible = True
End If

End Sub
```

The Output

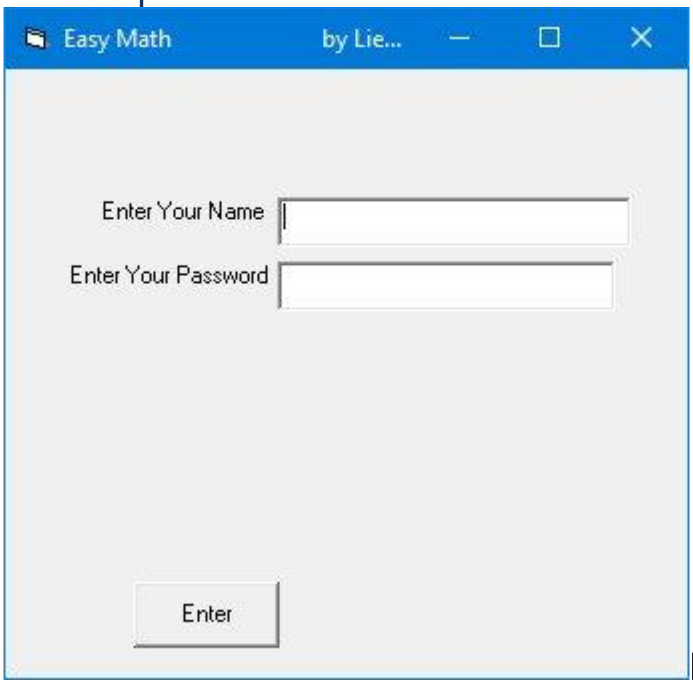
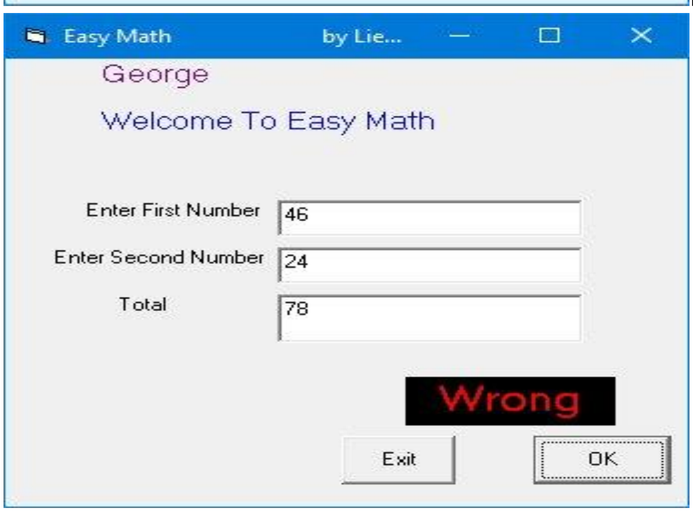


Figure 6.1 The Login dialog



Conditional Operators(Relational Operators)

- To control the VB program flow, we can use various conditional operators.

- Basically, they resemble mathematical operators.
- Conditional operators are very powerful tools, they let the VB program compare data values and then decide what action to take, whether to execute a program or terminate the program and more.
- These operators are shown in Table below

Conditional Operators

Operator	Meaning
=	Equal to
>	More than
<	Less Than
>	More than or equal
<=	Less than or equal
<>	Not Equal to

Logical Operators

- In addition to conditional operators, there are a few logical operators that offer added power to the VB programs. They are shown in Table below

Logical Operators

Operator	Description
And	Both sides must be true
Or	One side or other must be true
Xor	One side or other must be true but not both
Not	Negates true

- You can also compare strings with the operators. However, there are certain rules to follow where upper case letters are less than lowercase letters, and number are less than letters.

CONDITIONAL STATEMENTS

Using If.....Then.....Else Statements with Operators

- To effectively control the VB program flow, we shall use **If...Then...Else** statement together with the conditional operators and logical operators.

If conditions Then

VB expressions

Else

VB expressions

End If

Example

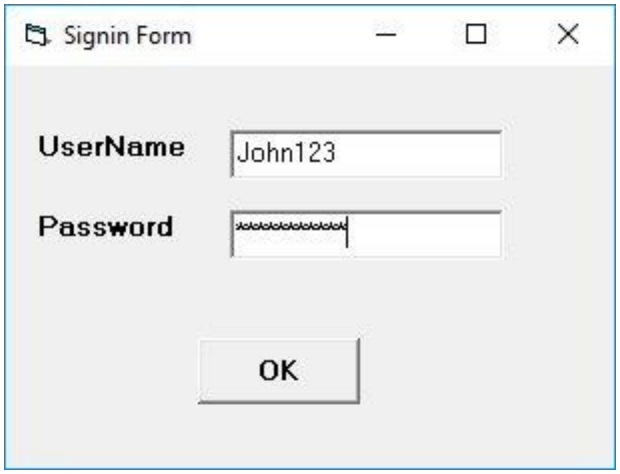
- This program simulates a sign in process.
- If the username and password are correct, sign in is successful else sign in failed.
- Start VB6 and insert two textboxes on the form, rename them UsrTxt and pwTxt, the first textbox is to accept username input and the second one for password input.
- For pwTxt, set the PasswordChr(password characters) property to * so that the password will appear as * instead of the actual character.
- We have written the code so that both username and password must be correct to enable sign in if either one of them incorrect sign in will fail.

The Code

```
Private Sub OK_Click()
Dim username, password As String
username = "John123"
password = "qwertyupi#@"

If UsrTxt.Text = username And pwTxt.Text = password Then
MsgBox ("Sign in sucessful")
ElseIf UsrTxt.Text <> username Or pwTxt.Text <> password Then

MsgBox ("Sign in failed")
End If
```

End Sub

The Output



<u>Example</u>

- This example calculate the commission based on sales volume attained. Let's say the commission structure is laid out as in the table below:

Sale Volume($)	**Commission(%)**
<5000	0

5000-9999	5
1000-14999	10
15000-19999	15
20000 and above	20

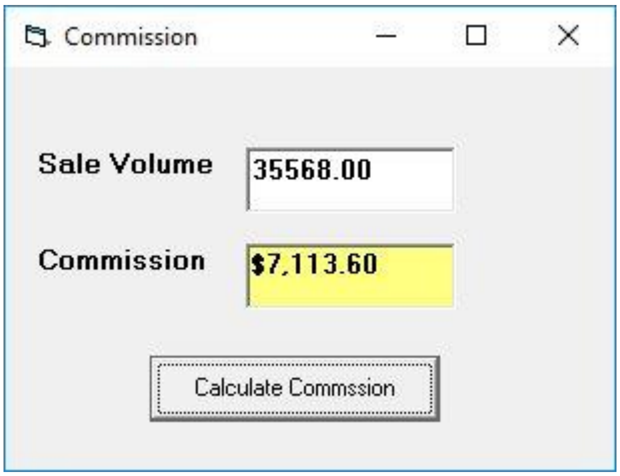
- In this example, we insert a textbox to accept sale volume input and a label to display commission. Insert a command button to trigger the calculation

The Code

```
Private Sub cmdCalComm_Click()
Dim salevol, comm As Currency
salevol = Val(TxtSaleVol.Text)
If salevol >= 5000 And salevol < 10000 Then
comm = salevol * 0.05
ElseIf salevol >= 10000 And salevol < 15000 Then
comm = salevol * 0.1
ElseIf salevol >= 15000 And salevol < 20000 Then
comm = salevol * 0.15
ElseIf salevol >= 20000 Then
comm = salevol * 0.2
Else
```

```
comm = 0
End If
LblComm.Caption = Format(comm, "$#,##0.00")
End Sub
```

The Output



Example : Guess a Number Game

- This is a guess a number game where the user key in a number and see check whether the answer is correct.
- This program will provide a hint whether the number is too small or too big.
- After a number of trial, the user should get the right answer.
- The program employ the If..Then..Else technique to check whether the entry is correct.

The Code

```
'Guess a Number
Const realNumber = 99
Dim userNumber As Integer

Private Sub EXit_Click()
End
End Sub

Private Sub OK_Click()

userNumber = entry.Text
  If userNumber > realNumber Then

 hint.Caption = "Your number is too big"
 ElseIf userNumber < realNumber Then

hint.Caption = "Your number is too small"

entry.Text = ""
entry.SetFocus
Else
```

```
hint.Caption = "Congratulation, your number is correct"

End If

End Sub
```

The IIf() Function

- The IIf function denotes immediate decision function. It provides a simple decision making process based on three arguments, as follows:

  IIf(x, y, z)

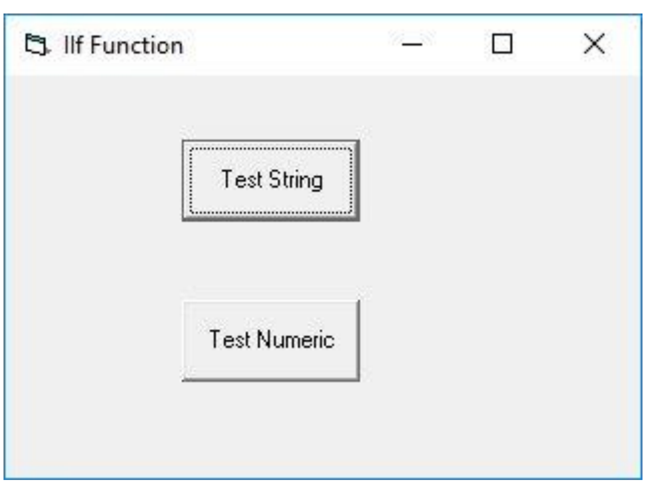
x represents a logical expression while y and z represent a numeric or a string expression.

For example, the IIF(x>y, expression 1, expression 2) function evaluates the values of x and y, if x>y. then expression 1 is true, otherwise the expression 2 is true.

Example

```
Private Sub CmdNumeric_Click()
Dim x, y, a, b, ans As Double
x = InputBox("Enter a number")
y = InputBox("Enter another number")
a = Val(x)
```

```
b = Val(y)

ans = IIf(a < b, b - a, a * b)
MsgBox ans
End Sub

Private Sub CmdString_Click()
Dim A, B, C As String
A = InputBox("Enter a word")
B = InputBox("Enter another word")
C = IIf(A < B, A, B)
MsgBox C
End Sub
```



The Interface

- If you click test string and enter the first word long and the second word short, the logical condition is true, hence the word long will be displayed, as shown in Figure 7.5.

- If you click test numeric and enter the first number 200 and the second number 40, the logical condition is false, hence the second expression will be executed, which is 20x40=800, as shown below

Select Cases

- The Select Case control structure is slightly different from the If....ElseIf control structure .The difference is that the **Select Case** control structure can handle conditions with multiple outcomes in an easier manner than the **If...Then...ElseIf** control structure.

- The syntax of the Select Case control structure is shown below:

```
Select Case expression
    Case value1
          Block of one or  more VB statements

          Case value2
          Block of one or more VB Statements

    Case Else
          Block of one or more VB Statements
End Select
```

Example

```
Dim grade As String
Private Sub Compute_Click( )
grade=txtgrade.Text
Select Case grade
Case "A"
result.Caption="High Distinction"
Case "A-"
result.Caption="Distinction"
Case "B"
result.Caption="Credit"
Case "C"
result.Caption="Pass"
Case Else
result.Caption="Fail"
End Select
End Sub
```

Example

```
Dim mark As Single
Private Sub Compute_Click()
'Examination Marks
mark = mrk.Text
Select Case mark
Case Is >= 85
comment.Caption = "Excellence"
Case Is >= 70
comment.Caption = "Good"
Case Is >= 60
comment.Caption = "Above Average"
Case Is >= 50
comment.Caption = "Average"
Case Else
comment.Caption = "Need to work harder"
End Select
End Sub
```

Example

Example above can be rewritten as follows:

```
Dim mark As Single

Private Sub Compute_Click()
'Examination Marks
mark = mrk.Text
Select Case mark
Case 0 to 49
comment.Caption ="Need to work harder"
Case 50 to 59
comment.Caption = "Average"
Case 60 to 69
comment.Caption = "Above Average"
```

```
Case 70 to 84
comment.Caption = "Good"
Case Else
comment.Caption ="Excellence"

End Select
End Sub
```

Example : Guess a Number

- In this example, we use the Select Case statement to guess a number generated randomly. The syntax 1 + Int(6 * Rnd) generates a random number from 1 to 6 inclusive

The Code

```
Dim Secret_Number As Integer


Private Sub Command1_Click()

Dim Your_Number As Integer
Your_Number = InputBox("Enter a number between 1 and 6 includisve")
Select Case Your_Number
Case Is < Secret_Number
MsgBox ("Your number is smaller than the secret number, try again!")
Case Is > Secret_Number
MsgBox ("Your number is greater than the secret number, try again!")

Case Else
Beep
MsgBox ("Your number is correct, congratulation!")
End Select

End Sub
```

```
Private Sub Form_Load()
Secret_Number = 1 + Int(6 * Rnd)

End Sub
```

Looping

- These are control structures that are used to write a Visual Basic procedure that allows the program to run repeatedly until a condition or a set of conditions is met
- Looping is a very useful feature of Visual Basic because it makes repetitive works easier.
- There are three kinds of loops in Visual Basic,
    i. **Do...Loop ,**
    ii. **For.......Next loop**
    iii. **While.....Wend Loop**

The Do Loop

The Do Loop statements have four different forms, as shown below:

a)
```
 Do While condition
  Block of one or more VB statements
Loop
```

b)
```
 Do
  Block of one or more VB statements
```

```
 Loop While condition
```

c)

```
 Do Until condition
  Block of one or more VB statements
 Loop
```

d)

```
Do
 Block of one or more VB statements
Loop Until condition
```

Example

```
Do while
counter <=1000
num.Text=counter
counter =counter+1
Loop
```

* The above example will keep on adding until counter > 1000

The above example can be rewritten as

```
Do
counter=counter+1
Loop until counter>1000
```

Exiting the Loop

Sometime we need exit to exit a loop earlier when a certain condition is fulfilled. The keyword to use is **Exit Do**.

Example

```
Dim sum, n as Integer
Private Sub Form_Activate()
List1.AddItem "n" & vbTab & "sum"
```

```
Do
n=n+1
sum=sum+n-resize
List1.AddItem n & vbTab & sum
If n=100 Then
Exit Do
End If
Loop
End Sub
```

Explanation

In the above example, we compute the summation of 1+2+3+4+……+100. In the design stage, you need to insert a ListBox into the form for displaying the output, named List1. The program uses the AddItem method to populate the ListBox. The statement List1.AddItem "n" & vbTab & "sum" will display the headings in the ListBox, where it uses the vbTab function to create a space between the headings n and sum.

The For....Next Loop

The For....Next Loop event procedure is written as follows:

```
For counter=startNumber to endNumber (Step increment)
 One or more VB statements
Next
```

Example

```
Private Sub Command1_Click()
Dim counter As Integer
For counter = 1 To 10
List1.AddItem counter
Next
End Sub
```

Example

```
Private Sub Command1_Click()

Dim counter As Integer
For counter = 1 To 1000 Step 10
counter = counter + 1
Print counter
Next
End Sub
```

Example

```
For counter=1000 to 5 step -5
counter=counter-10
If counter<50 then
Exit For
Else
Print "Keep Counting"
End If
Next
```

Example

```
Private Sub Form_Activate( )
For n=1 to 10
If n>6 then
Exit For
Else
```

```
Print n
Enf If
Next
End Sub
```

- Sometimes the user might want to get out from the loop before the whole repetitive process is executed, the command to use is **Exit For**. To exit a For….Next Loop, you can place the **Exit For** statement within the loop; and it is normally used together with the If…..Then… statement.

The While….Wend Loop

The structure of a While….Wend Loop is very similar to the Do Loop. it takes the following form:

```
While condition
 Statements
Wend
```

The above loop means that while the condition is not met, the loop will go on. The loop will end when the condition is met. Let's examine the program listed in example below

Example

```
Dim sum, n  As Integer
Private Sub Form_Activate()
List1.AddItem "n" & vbTab & "sum"
While n <> 100
n = n + 1
Sum = Sum + n
List1.AddItem n & vbTab & Sum
```

```
Wend
End Sub
```

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Internet

Using spreadsheets

- Sorting data
- Using the filter function
- Charts and graphs
- Creating charts and graphs in a worksheet
    - Formatting charts
    - setting up a page to print
    - selecting print area
    - setting up printing margins
    - setting header and footer
    - setting up page orientation
    - selecting titles to print
- importing files into a spreadsheet
- linking files

Using database

- Database
  - Definition
  - Types of databases,
    - computerized,
    - manual
- Database models,
- Database features,
- Benefits of databases
- Database management software (examples)
- Working with databases (create a database)
  - creating tables, forms, queries, reports in a database
  - setting primary and secondary keys in tables
  - setting field properties and data types
  - printing reports, forms, queries
  - sorting tables or forms
  - filtering tables or forms

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Introduction to computer networks

- Types of computer networks
    - (PAN, LAN, MAN, WAN)
- Network topologies
    - (eg bus, star, ring, mesh)
- Network reference models
    - (OSI, TCP/IP)
    - network protocols at each layer of the reference model
        - (eg HTTP, DNS, FTP, TCP, UDP, IP, Ethernet, 802.11)
- The internet
    - Comparing internet with intranet
    - Methods of internet access
        - dial-up
        - leased line
        - broadband
        - wifi
        - wimax
    - Addressing systems in computer networks
        - port numbers
        - IP addresses
            - classful and classless
            - MAC addresses
    - internet protocols
        - Differences between ipv4 and ipv6
        - configuring ip settings on computers

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Network Applications

- The world wide web (search engines)
    - web
    - website
    - search engine and
    - web browsers
- Identifying web browsers
- Using web browsers
- Listing examples of distributed applications
- Social networks (advantages and disadvantages)
- Advantages of social networks
    - Disadvantages of social networks
- Electronic mail (e- mail accounts, webmail, e-mail clients)
    - creating e-mail accounts
- Distributed applications
    - Examples of distributed applications
        - banking
        - utility bill